

Open Source Face Image Quality (OFIQ)

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Chapter 1

Open Source Face Image Quality (OFIQ) Library

1.1 Introduction

OFIQ (Open Source Face Image Quality) is a software library for computing quality aspects of a facial image. OFIQ is written in the C/C++ programming language. OFIQ is the reference implementation for the ISO/IEC 29794-5 international standard; see <https://bsi.bund.de/dok/OFIQ-e>.

1.2 License

This is the source code of OFIQ. OFIQ is a software for assessing the quality of facial image properties and potential defects. OFIQ is licensed under the MIT licenses (see text below). It includes dependencies and files that may be licensed otherwise. A documentation on the license situation of dependencies and files can be found in the table below.

1.2.1 OFIQ License

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1.2.2 License of dependencies

In the table the license situation of the files shipped with the OFIQ source is documented.

File/directory	Description	Reference	Original license
CMakeLists.txt	CMake file for building OFIQ	-	OFIQ license
README.md	Readme file	-	OFIQ license
LICENSE.md	File containing license information	-	OFIQ license
Version.txt	Contains OFIQ's current version information.	-	OFIQ license
cmake/	Folder containing cmake helper scripts	https://cmake.org/	OFIQ license
data/ofiq_config.jaxn	OFIQ's configuration file	-	OFIQ license
data/models/	Folder containing a single readme file; to this folder, model files are downloaded from the ISO portal when running one of the building scripts. Note, the readme file is subject to OFIQ's license. The license situation of the model files is documented separately in license files placed in the sub-directories after download.	-	-
data/tests/expected_results/expected_results.csv	CSV file with expected native quality scores and quality component values used for running conformance tests.	-	OFIQ license
data/tests/images/	Folder containing a single readme file; to this folder, conformance test images are downloaded from the ISO portal when running one of the building scripts. Note, the readme file is subject to OFIQ's license. The license situation of the image files is documented separately in a license file downloaded with the images.	-	-
conan/	Directory containing files that are used for the Conan package manager.	https://conan.io/	OFIQ license
OFIQlib/	Directory containing the OFIQ source code (including headers).	-	OFIQ license
doc/src/mainpage.h	Doxygen documentation of OFIQ	-	OFIQ license

File/directory	Description	Reference	Original license
doc/src/ofiq-doxxygen.cfg	Configuration for building doxygen documentation. Has been generated by doxygen and was edited afterwards. May be affected by copyleft. Was used to generate doc/refman.pdf which does not, however, need to be linked with any software compilation using OFIQ.	https://www.doxxygen.org/	mixed
doc/refman.pdf	Documentation of OFIQ. Generated by doxygen and LaTeX. The documentation does not need to be linked with any software compilation using OFIQ.	https://www.doxxygen.org/ , https://www.latex-project.org/	mixed
scripts/build.cmd	Script for building OFIQ on Windows; includes steps for installing packages via conan and downloading models and conformance test images from ISO portal.	-	OFIQ license
scripts/build.sh	Script for building OFIQ on Linux; includes steps for installing packages via conan and downloading models and conformance test images from ISO portal.	-	OFIQ license
scripts/conformance_↵ tests.cmd	Runs conformance tests on Windows	-	OFIQ license
scripts/conformance_↵ tests.sh	Runs conformance tests on Linux	-	OFIQ license
testing/	Directory containing OFIQ source code running conformance tests	-	OFIQ license
mobile/BUILD.md	building documentation	-	OFIQ license
mobile/build_for_android.↵ deploy.bat	building script	-	OFIQ license
mobile/build_for_android.↵ sh	building script	-	OFIQ license
mobile/build_for_ios.sh	building script	-	OFIQ license
mobile/apps/android/↵ OFIQMobile/build.gradle.↵ kts	gradle build script	created by gradle, no binding license	OFIQ license
mobile/apps/android/↵ OFIQMobile/gradle.↵ properties	Property file	created by gradle/Android Studio, no binding license	OFIQ license
mobile/apps/android/↵ OFIQMobile/gradlew	gradle script	contains license text	Apache 2.0 license
mobile/apps/android/↵ OFIQMobile/gradlew.bat	gradle script	contains license text	Apache 2.0 license
mobile/apps/android/↵ OFIQMobile/settings.↵ gradle.kts	gradle build script	created by gradle, no binding license	OFIQ license

File/directory	Description	Reference	Original license
mobile/apps/android/↔ OFIQMobile/app/build.↔ gradle.kts	gradle build script	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/proguard- rules.pro	proguard rules configura- tion	generated by Android-↔ SDK/Android Studio, con- tains only comments, effec- tively empty, no binding li- cense	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/↔ AndroidManifest.xml	Android app configuration	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/↔ OfiqMobileDemoApp↔ Application.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/activity/↔ CameraActivity.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/activity/↔ ImageActivity.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/activity/↔ MainActivity.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/activity/↔ ResultActivity.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/↔ CameraActivityView.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/↔ CameraPreview.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/↔ DialogWithDim↔ Background.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/↔ Dim.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/↔ ImageActivityView.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/↔ InitializeView.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/↔ OfiqImage.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/↔ OfiqScaffold.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/↔ OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/↔ Progress.kt	source file	generated by OFIQ team	OFIQ license

File/directory	Description	Reference	Original license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/ResultTable.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/ResultView.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/YesNoDialog.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/composable/ZoomableImage.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/model/MetricType.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/model/Utils.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/ui/theme/Color.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/ui/theme/Theme.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/ui/theme/Type.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/viewmodel/CameraActivityViewModel.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/viewmodel/ImageActivityViewModel.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/viewmodel/MainActivityViewModel.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/java/de/bund/bsi/ofiqmobile/viewmodel/ResultActivityViewModel.kt	source file	generated by OFIQ team	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/res/drawable/ic_launcher_background.xml	resource file	trivial file adopted from Android-SDK	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/res/drawable/ic_launcher_foreground.xml	resource file	trivial file adopted from Android-SDK	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/res/mipmap-anydpi/ic_launcher.xml	resource file	trivial file adopted from Android-SDK	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/res/mipmap-anydpi/ic_launcher_round.xml	resource file	trivial file adopted from Android-SDK	OFIQ license

File/directory	Description	Reference	Original license
mobile/apps/android/OFIQMobile/app/src/main/res/values/colors.xml	resource file	trivial file adopted from Android-SDK, modified	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/res/values/strings.xml	resource file	trivial file adopted from Android-SDK, modified	OFIQ license
mobile/apps/android/OFIQMobile/app/src/main/res/values/themes.xml	resource file	trivial file adopted from Android-SDK, modified	OFIQ license
mobile/apps/android/OFIQMobile/gradle/libs.versions.toml	version catalogue file	created by Gradle, then modified, no binding license	OFIQ license
mobile/apps/android/OFIQMobile/gradle/wrapper/gradle-wrapper.jar	executable JAR file	The license can be found in META-INF/LICENSE of the archive file	Apache 2.0 license
mobile/apps/android/OFIQMobile/gradle/wrapper/gradle-wrapper.properties	property file	trivial property file adopted from Gradle	OFIQ license
mobile/apps/iOS/OFIQMobile/ExportOptions.plist	config file	generated by XCode, no binding license	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/Info.plist	config file	generated by XCode, no binding license	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/Localizable.xcstrings	localization file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/OFIQMobile/App.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/Assets.xcassets/Contents.json	asset file	generated by XCode, no binding license	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/Assets.xcassets/AccentColor.colorset/Contents.json	asset file	generated by XCode, no binding license	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/Assets.xcassets/AppIcon.appiconset/Contents.json	asset file	generated by XCode, no binding license	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/Assets.xcassets/mainColor.colorset/Contents.json	asset file	generated by XCode, no binding license	OFIQ license

File/directory	Description	Reference	Original license
mobile/apps/iOS/OFIQMobile/OFIQMobile/model/MetricType.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/model/Rotator.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/view/AppBody.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/view/CameraView.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/view/CustomButton.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/view/Dim.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/view/ImageView.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/view/MainView.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/view/Progress.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/view/ResultTable.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/view/ResultView.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/view/TopBar.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/viewmodel/CameraViewModel.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/viewmodel/ImageViewModel.swift	source file	generated by OFIQ team	OFIQ license

File/directory	Description	Reference	Original license
mobile/apps/iOS/OFIQMobile/OFIQMobile/viewmodel/MainViewModel.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile/viewmodel/ResultViewModel.swift	source file	generated by OFIQ team	OFIQ license
mobile/apps/iOS/OFIQMobile/OFIQMobile.xcodeproj/project.pbxproj	XCode project file	generated by XCode, no binding license	OFIQ license
mobile/lib-projects/android/ofiqlib/build.gradle	gradle build script	created by gradle, no binding license	OFIQ license
mobile/lib-projects/android/ofiqlib/gradle.properties	property file	created by gradle/Android Studio, no binding license	OFIQ license
mobile/lib-projects/android/ofiqlib/gradlew	gradle script	contains license text	Apache 2.0 license
mobile/lib-projects/android/ofiqlib/gradlew.bat	gradle script	contains license text	Apache 2.0 license
mobile/lib-projects/android/ofiqlib/settings.gradle	gradle build script	created by gradle, no binding license, modified	OFIQ license
mobile/lib-projects/android/ofiqlib/gradle/libs.versions.toml	version catalogue file	created by Gradle, then modified, no binding license	OFIQ license
mobile/lib-projects/android/ofiqlib/gradle/wrapper/gradle-wrapper.jar	executable JAR file	The license can be found in META-INF/LICENSE of the archive file	Apache 2.0 license
mobile/lib-projects/android/ofiqlib/gradle/wrapper/gradle-wrapper.properties	property file	trivial property file adopted from Gradle, no binding license	OFIQ license
mobile/lib-projects/android/ofiqlib/ofiqlib/build.gradle	gradle build script	created by gradle, then modified, no binding license	OFIQ license
mobile/lib-projects/android/ofiqlib/ofiqlib/proguard-rules.pro	proguard rules configuration file	generated by Android SDK/Android Studio, contains only comments, effectively empty, no binding license	OFIQ license
mobile/lib-projects/android/ofiqlib/ofiqlib/src/main/cpp/CMakeLists.txt	CMake file	created by OFIQ team	OFIQ license
mobile/lib-projects/android/ofiqlib/ofiqlib/src/main/cpp/ofiqlib.cpp	source file	created by OFIQ team	OFIQ license
mobile/lib-projects/android/ofiqlib/ofiqlib/src/main/java/de/bund/bsi/ofiqmobile/lib/InitializeResult.kt	source file	created by OFIQ team	OFIQ license
mobile/lib-projects/android/ofiqlib/ofiqlib/src/main/java/de/bund/bsi/ofiqmobile/lib/IOfiq.kt	source file	created by OFIQ team	OFIQ license

File/directory	Description	Reference	Original license
mobile/lib-projects/android/ofiqlib/ofiqlib/src/main/java/de/bund/bsi/ofiqlib/OfiqLibNativePointer.kt	source file	created by OFIQ team	OFIQ license
mobile/lib-projects/android/ofiqlib/ofiqlib/src/main/java/de/bund/bsi/ofiqlib/OfiqLib.kt	source file	created by OFIQ team	OFIQ license
mobile/lib-projects/android/ofiqlib/ofiqlib/src/main/java/de/bund/bsi/ofiqlib/QualityMeasureAssessment.kt	source file	created by OFIQ team	OFIQ license
mobile/lib-projects/android/ofiqlib/ofiqlib/src/main/java/de/bund/bsi/ofiqlib/QualityMetrics.kt	source file	created by OFIQ team	OFIQ license
mobile/lib-projects/android/ofiqlib/ofiqlib/src/main/java/de/bund/bsi/ofiqlib/ReturnCode.kt	source file	created by OFIQ team	OFIQ license
mobile/lib-projects/iOS/ofiqlib/Podfile	CocoaPods config file	generated by cocoapod, then modified, no binding license	OFIQ license
mobile/lib-projects/iOS/ofiqlib/prefix.pch	pre-compiled header file	includes a Version.h, one-liner, no binding license	OFIQ license
mobile/lib-projects/iOS/ofiqlib/readVersion.sh	script creating Version.h	created by OFIQ team	OFIQ license
mobile/lib-projects/iOS/ofiqlib/ofiqlib/InitializeResult.h	header file	created by OFIQ team	OFIQ license
mobile/lib-projects/iOS/ofiqlib/ofiqlib/InitializeResult.mm	source file	created by OFIQ team	OFIQ license
mobile/lib-projects/iOS/ofiqlib/ofiqlib/ofiqlib.h	header file	created by OFIQ team	OFIQ license
mobile/lib-projects/iOS/ofiqlib/ofiqlib/OfiqLib.mm	source file	created by OFIQ team	OFIQ license
mobile/lib-projects/iOS/ofiqlib/ofiqlib/QualityMeasureAssessment.h	header file	created by OFIQ team	OFIQ license
mobile/lib-projects/iOS/ofiqlib/ofiqlib/QualityMeasureAssessment.mm	source file	created by OFIQ team	OFIQ license
mobile/lib-projects/iOS/ofiqlib/ofiqlib/QualityMetrics.h	header file	created by OFIQ team	OFIQ license
mobile/lib-projects/iOS/ofiqlib/ofiqlib/QualityMetrics.mm	source file	created by OFIQ team	OFIQ license
mobile/lib-projects/iOS/ofiqlib/ofiqlib/ReturnCode.h	header file	created by OFIQ team	OFIQ license
mobile/lib-projects/iOS/ofiqlib/ofiqlib.xcodeproj/project.pbxproj	XCode project file	generated by XCode, no binding license	OFIQ license
mobile/lib-projects/iOS/ofiqlib/ofiqlib.xcodeproj/project.xcworkspace/contents.xcworkspacedata	workspace description file	generated by XCode, no binding license	OFIQ license

File/directory	Description	Reference	Original license
mobile/lib-projects/i↔ OS/ofiqlib/ofiqlib.↔ xcworkspace/contents.xcworkspacedata	workspace description file	generated by XCode, no binding license	OFIQ license

1.3 Compilation

To build OFIQ you need to install Python with pip, cmake and conan.

- Python (version 3.10.12 or higher)
- [Download and install cmake](#) (version 3.26 or higher)
- [Download and install conan](#) (version 2.18.1)

1.3.1 Linux

1.3.1.1 Ubuntu 22.04 (x86_64)

Install necessary packages.

```
sudo apt-get install build-essential python3-pip
```

To install cmake (in a version 3.26) use snap (and not apt) as follows.

```
sudo snap install cmake --channel=3.26/stable --classic
```

Conan can be installed via Python with

```
pip install conan==2.18.1
```

In some cases the command above shows a warning `*"Defaulting to user installation because normal site-packages is not writeable"*`, to avoid this you can pass a `--user` argument:

```
pip install --user conan==2.18.1
```

The conan binary will be installed in `$HOME/.local/bin` and this path will be added to your `~/.profile` file. You'll need to relogin or reload the `.profile` with

```
source ~/.profile
```

In order to build OFIQ and installing all required packages via conan run the following.

```
cd path/to/OFIQ_Project/scripts
sh build.sh
```

where `path/to/OFIQ_Project/` denotes the root folder of the OFIQ source files.

This will create the following output.

file/directory	description
build/	Folder with the Linux build including the binaries <code>OFIQSampleApp</code> and <code>libofiq↔_lib.so</code> .
build/conan/	Conan cache with packages downloaded.
install_x86_64_linux/	Folder with the installation including the binaries <code>Release/bin/OFIQSampleApp</code> , <code>Release/lib/libofiq_lib.so</code> , <code>Release/lib/libonnxruntime.↔so.1.18.1</code> , and the header files in <code>Release/include/</code> .
data/models/	Model files downloaded from the ISO portal or eu-LISA during build process.
data/tests/images/	Conformance test images downloaded from the ISO portal or eu-LISA.

1.3.1.2 Ubuntu 24.04 (x86_64)

Install necessary packages.

```
sudo apt-get install build-essential python3-pip cmake python3.12-venv
```

To install conan, a virtual Python environment needs to be generated first.

```
python3 -m venv path/to/py_ofiq_env
```

where `path/to/py_ofiq_env` is the path where the python environment will be stored and `py_ofiq_env` is the name of the new environment.

Then install conan as follows.

```
source path/to/py_ofiq_env/bin/activate
pip install conan==2.18.1
```

In order to build OFIQ and installing all required packages via conan run the following.

```
source path/to/py_ofiq_env/bin/activate
cd path/to/OFIQ_Project/scripts
sh build.sh
```

where `path/to/OFIQ_Project/` denotes the root folder of the OFIQ source files.

This will create the following output.

file/directory	description
build/	Folder with the Linux build including the binaries <code>OFIQSampleApp</code> and <code>libofiq↵_lib.so</code> .
build/conan/	Conan cache with packages downloaded.
install_x86_64_linux/	Folder with the installation including the binaries <code>Release/bin/OFIQSampleApp</code> , <code>Release/lib/libofiq_lib.so</code> , <code>Release/lib/libonnxruntime.↵so.1.18.1</code> , and the header files in <code>Release/include/</code> .
data/models/	Model files downloaded from the ISO portal during build process.
data/tests/images/	Conformance test images downloaded from the ISO portal.

1.3.1.3 Ubuntu 24.04 (Raspberry Pi/ARMv8)

The following has been tested on a Raspberry Pi5 with 4GB RAM.

Install necessary packages.

```
sudo apt-get install build-essential python3-pip cmake python3.12-venv
```

To install conan, a virtual Python environment needs to be generated first.

```
python3 -m venv /path/to/py_ofiq_env
```

where `path/to/py_ofiq_env` is the path where the python environment will be stored and `py_ofiq_env` is the name of the new environment.

Then install conan as follows.

```
source /path/to/py_ofiq_env/bin/activate
pip install conan==2.18.1
```

In order to build OFIQ and installing all required packages via conan run the following.

```
source /path/to/py_ofiq_env/bin/activate
cd /path/to/OFIQ_Project/scripts
sh build.sh --os linux-arm64
```

where `path/to/OFIQ_Project/` denotes the root folder of the OFIQ source files.

This will create the following output.

file/directory	description
build/	Folder with the Linux build including the binaries OFIQSampleApp and libofiq_↵lib.so.
build/conan/	Conan cache with packages downloaded.
install_arm64_linux/	Folder with the installation including the binaries Release/bin/OFIQSampleApp, Release/lib/libofiq_lib.so, Release/lib/libonnxruntime.so.↵1.18.1, and the header files in Release/include/.
data/models/	Model files downloaded from the ISO portal during build process.
data/tests/images/	Conformance test images downloaded from the ISO portal.

1.3.2 Windows

1.3.2.1 Windows (x86_64)

The following has been tested on a Windows 10 (64 bit) installation using a **Python installation version 3.11.5** with pip package such that the `pip` command can be executed from the command prompt. Furthermore, an installation of **cmake version 3.29** has been used. As the compiler, **Microsoft's Visual Studio 2019** was used.

To install conan, run

```
pip install conan==2.18.1
```

from the command prompt.

In order to build OFIQ and install all required packages run the following.

```
cd C:\Path\To\OFIQ_Project\scripts
build.cmd
```

where `C:\Path\To\OFIQ_Project` denotes the root folder of the OFIQ source files.

This will create the following output.

directory	description
build\build_win\	Folder with the Visual Studio solution files placed and pre-compilation.
build\conan\	Conan cache with packages downloaded.
install_x86_64\	Folder with the OFIQ installation files. This includes the binaries Release\bin\OFIQSampleApp, Release\bin\libofiq_lib.so, Release\bin\libonnxruntime.so.1.18.1, and the header files in Release\include\.
data\models\	Model files downloaded from the ISO portal during build process.
data\tests\images\	Conformance test images downloaded from the ISO portal.

1.3.2.2 Windows (x86)

To compile Win32 binaries, one proceeds in the same way as for x86_64 described above. But the building script is run with the `--arch x86` argument:

```
cd C:\Path\To\OFIQ_Project\scripts\
build.cmd --arch x86
```

1.3.3 MacOS

1.3.3.1 MacOS (ARM64)

The following has been tested on macOS Sonoma Version 14.4.1 with ARM64 processor.

Install Homebrew

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Then update profiles

```
(echo; echo 'eval "$(/opt/homebrew/bin/brew shellenv)"') >> ~/.zprofile
eval "$(/opt/homebrew/bin/brew shellenv)"
```

where `/opt/homebrew/bin/brew` is the path to the homebrew executable and can vary among MacOS systems.

NOTE The two commands from above for updating profiles are output by the Homebrew installation script.

Install CMake.

```
brew install cmake
```

Install Python using Miniconda by

```
mkdir -p /path/to/miniconda3/
curl https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-arm64.sh -o
/path/to/miniconda3/miniconda.sh
bash /path/to/miniconda3/miniconda.sh -b -u -p /path/to/miniconda3
rm -rf /path/to/miniconda3/miniconda.sh
/path/to/miniconda3/bin/conda init bash
/path/to/miniconda3/bin/conda init zsh
```

where `/path/to/miniconda3` can be replaced by the path where Miniconda is installed.

Install conan.

```
$ python -m pip install conan==2.18.1
```

Finally, to build OFIQ run the following.

```
$ cd /path/to/OFIQ_Project/scripts/
$ sh build.sh --os macos
```

1.3.3.2 MacOS (x86_64)

To compile OFIQ on MacOS x86_64 one needs to edit `/path/to/OFIQ_Project/conan/conan_profile_release_macos.txt` and `/path/to/OFIQ_Project/conan/conan_profile_debug_macos.txt` first. In both files replace the line

```
arch=armv8
```

by

```
arch=x86_64
```

Then apply the same actions as for MacOS compilation on ARM64.

1.3.4 Download model files

To run OFIQ, the model files from [the ISO portal](#) (or alternatively from the [eu-LISA](#) need to be downloaded and be placed in the `./data/models/` directory so that the file structure matches the following.

Generated by Doxygen	/path/to/OFIQ-Project/data/models/expression_neutrality/*
	/path/to/OFIQ-Project/data/models/face_detection/*
	/path/to/OFIQ-Project/data/models/face_landmark_estimation/*
	/path/to/OFIQ-Project/data/models/face_occlusion_segmentation/*
	/path/to/OFIQ-Project/data/models/face_parsing/*
	/path/to/OFIQ-Project/data/models/head_pose_estimation/*
	/path/to/OFIQ-Project/data/models/no_compression_artifacts/*
	/path/to/OFIQ-Project/data/models/sharpness/*

Here, `/path/to/OFIQ-Project/` denotes the path to OFIQ's root folder.

This step is integrated in the cmake building process.

1.3.5 Download conformance test images

To run conformance tests, the conformance test images need to be downloaded from [the ISO portal](#) or (alternative from the [eu-LISA](#)) and be placed in the `./data/tests/images/` directory so that the file structure matches the following.

<code>/path/to/OFIQ-Project/data/tests/images/b-01-smile.png</code>
<code>...</code>
<code>/path/to/OFIQ-Project/data/tests/images/r-09-background.png</code>

Here, `/path/to/OFIQ-Project/` denotes the path to OFIQ's root folder. Download and extraction of conformance test images can be performed when running the cmake building process.

1.3.6 Download of external libraries

The source of the external libraries that are used by OFIQ can be downloaded along with a full OFIQ release from [eu-LISA](#).

To build OFIQ without conan (described below), we need to extract the archive so that file structure matches the following:

<code>/path/to/OFIQ-Project/extern/flatbuffers/*</code>
<code>/path/to/OFIQ-Project/extern/googletest/*</code>
<code>/path/to/OFIQ-Project/extern/gzip-hpp/*</code>
<code>/path/to/OFIQ-Project/extern/json/*</code>
<code>/path/to/OFIQ-Project/extern/magic_enum/*</code>
<code>/path/to/OFIQ-Project/extern/onnxruntime/*</code>
<code>/path/to/OFIQ-Project/extern/opencv-4.5.5/*</code>
<code>/path/to/OFIQ-Project/extern/PEGTL/*</code>

Here, `/path/to/OFIQ-Project/` denotes the path to OFIQ's root folder.

Note, download and extraction of the external libraries can be done automatically by the cmake building process described below.

1.3.7 Building without conan

To build OFIQ without conan, one can run

```
cd C:\Path\To\OFIQ-Project\scripts\
.\build.cmd --no-conan
```

on Windows or

```
cd /path/to/OFIQ-Project/scripts/
sh build.sh --no-conan
```

on Linux (both x86_64 and ARMv8) and MacOS. This will download and extract the model files, conformance test images, and external libraries from the ISO portal as described above.

To suppress download of the dependencies (e.g., when one wants to compile without an internet connection), one can run

```
cd C:\Path\To\OFIQ-Project\scripts\
.\build.cmd --no-conan --no-download
```

on Windows or

```
cd /path/to/OFIQ-Project/scripts/
sh build.sh --no-conan --no-download
```

on Linux and MacOS.

It is important to note that building without conan while suppressing download requires manual extraction of model files and external library dependencies (and perhaps conformance test images) before running the building scripts. Further note, the files have to be placed manually in the right directory as described above. The missing files can be downloaded along with the release archive available from [eu-LISA](#).

1.4 Running conformance tests

The conformance tests are executed by going to `/path/to/OFIQ_Project/scripts/` and run `conformance_tests.cmd` (Windows). `conformance_tests.sh` (Linux/x86_64). `conformance_tests.sh -os linux-arm64` (Linux/ARMv8) `conformance_tests.sh -os macos` (MacOS).

1.5 Running the sample executable

In this section, we describe how to run the sample application of OFIQ after compilation.

1.5.1 Quality assessment for a single facial image

The sample application takes an images and outputs the computed quality assessments. For example, to output the quality assessments for one of the conformance test images using OFIQ's configuration (in `./data`), run the following commands on Linux.

```
cd /path/to/OFIQ-Project/install_x86_64_linux/Release/bin/
./OFIQSampleApp -c ../../data/ofiq_config.jaxn -i
    ../../data/tests/images/b-01-smile.png
```

On Windows run the following commands.

```
cd C:\Path\To\OFIQ-Project\install_x86_64\Release\bin
.\OFIQSampleApp -c ..\..\data\ofiq_config.jaxn -i
    ..\..\data\tests\images\b-01-smile.png
```

1.5.2 Quality assessment for multiple images

To reproduce the conformance test table given in Annex A of the ISO/IEC 29794-5 international standard for all conformance test images, run the following commands on Linux.

```
cd /path/to/OFIQ-Project/install_x86_64_linux/Release/bin/
./OFIQSampleApp -c ../../../../data/ofiq_config.jaxn -i
../../../../data/tests/images/ -o table.csv
```

The result will be written in the file `/path/to/OFIQ-Project/install_x86_64_linux/Release/bin/table.csv`.

On Windows run the following commands.

```
cd C:\Path\To\OFIQ-Project\install_x86_64\Release\bin\
.\OFIQSampleApp -c ..\..\..\data\ -i
..\..\..\data\tests\images\ -o table.csv
```

The result will be written in the file `C:\Path\To\OFIQ-Project\install_x86_64\Release\bin\table.csv`.

1.5.3 Arguments

The usage pattern of the sample application is the following.

```
OFIQSampleApp
-c <directory or file path>
[-cf <config file name>]
-i <directory or image file path>
[-o <csv file path>]
```

The following table documents the usage of the sample application.

flag	argument
-c	Path to a directory containing the file <code>ofiq_config.jaxn</code> or a path to a JAXN configuration file (see doc/refman.pdf).
-cf	Name of the JAXN configuration file contained in the directory specified by the flag -c. Must be omitted if -c specifies a path to a file.
-i	Path to a directory containing facial images or a path to a facial image file. If a directory path is specified, all images in PNG and JPEG format will be processed.
-o	Path to a CSV file to where the quality assessment is written. If -o is not specified, the output is written to the standard output.

1.6 Supported platforms

OFIQ compilation has been successfully tested to compile and run on a variety of platforms. The following table gives an overview of platforms that have been successfully tested by the OFIQ development team.

platform	compiler	compiles via script?	building	remarks
Ubuntu 22.04 (x86_64)	g++ 11.4.0	yes		-
Ubuntu 24.04 (x86_64)	g++ 13.2.0	yes		-
Ubuntu 24.04 (Raspberry Pi/ARM64)	g++ 13.3.0	yes		Building script needs specification of the argument <code>-os linux-arm64</code> .

Windows 10 (x86_64/win64)	Visual Studio 2019	yes	To compile external libraries, i.e., to run compilation without conan, an installation of Visual Studio 2022 was used which was required to build the onnxruntime dependency.
Windows 10 (x86/win32)	Visual Studio 2019	yes	Building script needs specification of the argument <code>-arch x86</code> . To compile external libraries, i.e., to run compilation without conan, an installation of Visual Studio 2022 was used which was required to build the onnxruntime dependency.
MacOS (ARM64)	clang 15.0.0	yes	Building script needs specification of the argument <code>-os macos</code> .
MacOS (x86_64)	clang 15.0.0	yes	Building script needs specification of the arguments <code>-os macos</code> . Furthermore, see the details on compiling for MacOS (x86_64) above.
Android	clang 12.0.8	no	Manual compilation including linking of dependencies is required. Not supported by OFIQ's building scripts.
iOS	clang 15.0.0	no	Manual compilation including linking of dependencies is required. Not supported by OFIQ's building scripts.

OFIQ may compile on other configurations that have not been tested by the OFIQ development team. Compilation for 32-bit Linux systems, however, were not successful due to the onnxruntime being not supported by 32-bit gcc compiler. Also, compilation on Windows using gcc (for 32- nor 64-bit) did not work since onnxruntime did not build.

1.7 Precompiled binaries

Pre-compiled library binaries are available from [eu-LISA](#).

1.8 Compilation for mobile platforms

1.8.1 Android

Different build tools are required to build the ofiq libs and the demo apps, depending on the platform.

1.8.1.1 Install JDK

On Ubuntu 24.04, one may install Java by running `sudo apt install openjdk-21-jdk`.

For other platforms, various Java versions can be downloaded from [here](#). Ensure for your Java installation, that all system variables are set correctly. For example, in Windows `JAVA_HOME` must be set and `Path` should contain the directory path to `javac.exe`.

1.8.1.2 Install command-line tools

Download the Command-Line Tools for Windows from [here](#). This should be a ZIP archive, e.g., of name `commandlinetools-win-13114758_latest.zip` (Windows), `commandlinetools-linux-13114758_latest.zip` (Linux) or `commandlinetools-mac-13114758_latest.zip` (MacOS).

- Unpack the ZIP archive.
- Place the `cmdline-tools` folder in an arbitrary location, e.g., `C:\Path\To\android-sdk\cmdline-tools\` (Windows) or `/path/to/android-sdk/cmdline-tools` (Linux/MacOS)

Windows	Linux/MacOS	MacOS
Set an <code>ANDROID_HOME</code> environment variable to <code>android-sdk</code> , e.g., <code>C:\Path\To\android-sdk\</code>	Open a terminal.	Open a terminal.
Add <code>C:\Path\To\android-sdk\cmdline-tools\bin</code> to the system's <code>Path</code> variable	Run <code>export ANDROID_HOME=/path/to/android-sdk/cmdline-tools/bin</code>	Run <code>export ANDROID_HOME=/path/to/android-sdk/cmdline-tools/bin</code>
In a terminal, run <code>sdkmanager --licenses --sdk_root=C:\Path\To\android-sdk\</code> and accept licenses	Run <code>cd \$ANDROID_HOME</code>	Run <code>cd \$ANDROID_HOME</code>
	Run <code>sh sdkmanager --licenses --sdk_root=\$ANDROID_HOME</code>	Run <code>sh sdkmanager --licenses --sdk_root=\$ANDROID_HOME</code>
		Run <code>sh sdkmanager --install "cmake;3.22.1" --sdk_root=\$ANDROID_HOME</code>

NOTE: On Windows, if the `sdkmanager` complains about an incompatible Java version, one may deactivate the check by running `set SKIP_JDK_VERSION_CHECK=""` and then repeating running the `sdkmanager` with the arguments from above.

NOTE: If required, a proxy can be specified using the `--proxy=https --proxy_host=<url> --proxy_port=<port>` syntax.

1.8.1.3 Download OFIQ

Check out the repository

```
git clone https://github.com/BSI-OFIQ/OFIQ-Project
```

where we assume its path being `C:\Path\To\OFIQ-Project\` on Windows and `/path/to/OFIQ-Project/` on Linux/MacOS.

1.8.1.4 Compile OFIQ and mobile demo application

Windows	Linux/MacOS
Open a terminal	Open a terminal
Run <code>\$ cd C:\Path\To\OFIQ-Project\mobile\</code>	Run <code>\$ cd /path/to/OFIQ-Project/mobile/</code>
Run <code>.\build_for_android.bat</code>	Run <code>\$ sh build_for_android.sh</code>

After successful compilation, the path `C:\C:\Path\To\OFIQ-Project\mobile\build\` (Windows) or `/path/to/OFIQ-Project/mobile/build` (Linux/MacOS) contains the file `ofiqmobile_<version>-release-unsigned.apk` which is a demo application for OFIQ. The file can be copied, installed and then run on an Android device.

1.8.2 iOS

1.8.2.1 Required Build-Tools

To create OFIQ for iOS, you must install the following build tools.

- Xcode (version 15 or higher)
- Xcode Command Line Tools (version 15 or higher)
- Homebrew
- CocoaPods

Xcode

Xcode can be installed in various ways. The easiest way is via the app store

Xcode Command Line Tools

The command line tools can be installed with the following command:

```
$ xcode-select --install
$ sudo xcode-select -s /Applications/Xcode.app/Contents/Developer
```

Homebrew

Homebrew is required for the installation of CocoaPods.

An alternative would be the installation via Ruby-Gem.

```
$ /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

CocoaPods

Installation of CocoaPods via Homebrew:

```
$ brew install cocoapods
$ pod setup
```

1.8.2.2 Bundle Identifier and Signing

Before compiling, the bundle ID of the OFIQMobile app must be customised.

To do this, using Xcode, the project must be opened under `/path/to/OFIQ-Project/mobile/apps/iOS/OFIQMobile`.

The bundle ID is customised under `General -> Signing & Capabilities` in the OFIQMobile target.

The team must be selected in the same section.

Ensure that a certificate has been created for the app. You can create the certificate on the apple developer home-page or via Xcode.

There is an `ExportOptions.plist` file in the main directory of the OFIQMobile app. The correct `teamID` must be entered here.

You can find the `teamID` on the apple developer page. The `teamID` is a unique 10-character string.

1.8.2.3 Build

The build script is located in `/path/to/OFIQ-Project/mobile/build_for_ios.sh`. The script downloads dependencies from the web. If required, a proxy may be edited manually at the top of `build_for_ios.sh` using a text editor.

To build OFIQ, the ios framework and the OFIQMobile app, do the following.

```
$ cd /path/to/OFIQ-Project/mobile/
$ sh build_ios_lib_and_demoapp.sh
```

This will create the following output.

file/directory	description
build/	Folder with the iOS build including the binaries <code>ofiqlib.framework</code> and <code>OFIQMobile.ipa</code> .
build/framework/	Storage location for the created iOS framework.
build/OFIQMobile/	OFIQMobile App, which integrates the framework and can be used for demonstration purposes.
build/downloads/	All third party libs are stored here.

1.9 Configuration

In this section, we describe the configuration file for OFIQ. OFIQ uses a JSON-like configuration based on the `taoJSON` library. Using `taoJSON`, OFIQ reads its configuration from a `JAXN`-formatted file.

A minimal configuration working with OFIQ looks as follows.

```
{
  "config": {
```



```

"detector": "ssd",
"landmarks": "ADNet",
"measures": [
  "UnifiedQualityScore"
],
"params": {
  "detector": {
    "ssd": {
      "model_path": "models/face_detection/ssd_facedetect.caffemodel",
      "prototxt_path": "models/face_detection/ssd_facedetect.prototxt.txt",
      "confidence_thr": 0.4,
      "min_rel_face_size": 0.01,
      "padding": 0.2
    }
  },
  "landmarks": {
    "ADNet": {
      "model_path": "models/face_landmark_estimation/ADNet.onnx"
    }
  },
  "measures": {
    "UnifiedQualityScore": {
      "model_path": "models/unified_quality_score/magface_iresnet50_norm.onnx"
    },
    "HeadPose": {
      "model_path": "models/head_pose_estimation/mb1_120x120.onnx"
    },
    "FaceOcclusionSegmentation": {
      "model_path": "models/face_occlusion_segmentation/face_occlusion_segmentation_ort.onnx"
    },
    "FaceParsing": {
      "model_path": "models/face_parsing/bisenet_400.onnx"
    }
  }
}
}
}

```

OFIQ's C/C++ library provides the class [Configuration](#) which is responsible for reading and managing JAXN configurations.

Note that the model paths are specified as paths relative to the directory of the JAXN configuration file. We assume that the file above is stored in <OFIQ-SOURCE>/data.

1.9.1 Configuration of the face detector

The face detector (SSD) must be configured explicitly:

```

{
  "config": {
    "detector": "ssd",
    ...
  }
}

```

Additionally, the path to the model file and other parameters need to be configured:

```

{
  ...
  "params": {
    "detector": {
      "ssd": {
        "model_path": "models/face_detection/ssd_facedetect.caffemodel",
        "prototxt_path": "models/face_detection/ssd_facedetect.prototxt.txt",
        "confidence_thr": 0.4,
        "min_rel_face_size": 0.01,
        "padding": 0.2
      }
    },
    ...
  }
}

```

A documentation on the parameters are given in the following table.

Parameter	Description
model_path	path to the SSD model file in CAFFE format
prototxt_path	path to SSD's CAFFE prototype file
confidence_thr	minimum value for the confidence the detected faces; detected faces with a lower confidence are discarded. Note, the specified value 0.4 (fixed for OFIQ) has been determined experimentally.
min_rel_face_size	the minimum width of the face bounding boxes relative to the width w of the input image; detected faces, with a bounding box width smaller than min_rel_face_size*w are discarded. Note, the specified value 0.01 (fixed for OFIQ) has been determined experimentally.
padding	horizontal and vertical padding of the original image prior face detection. Note, the specified value 0.2 (fixed for OFIQ) has been determined experimentally.

1.9.2 Configuration of the landmark extractor

The face landmark extractor (ADNet) must be configured explicitly:

```

{
  "config": {
    ...
    "landmarks": "ADNet",
    ...
  }
}

```

Additionally, the path to the ADNet model file in ONNX format needs to be configured:

```

{
  ...
  "params": {
    ...
    "landmarks": {
      "ADNet": {
        "model_path": "models/face_landmark_estimation/ADNet.onnx"
      }
    },
    ...
  }
}

```

1.9.3 Other required configurations

As suggested by the minimal configuration file given in [Configuration](#), there are model files that need to be configured and their configurations cannot be omitted. These model files are detailed in the following table and shall be configured in the "config"."measures" environment.

Algorithm	Description
HeadPose	Head pose angles are pre-processed and used by some measures; therefore, the measure shall be configured. The path to the 3DDFAV2 model file in ONNX format should be set using the <code>model_path</code> key.
FaceOcclusionSegmentation	Face occlusion segmentation pre-processing used by some measures assessing occlusion prevention. The path to the FaceExtraction model file in ONNX format should be set using the <code>model_path</code> key. NOTE: The OFIQ development team has been permitted by the FaceExtraction authors for inclusion of the model in OFIQ without any restrictions; therefore, the referenced ONNX model file is subject to the OFIQ license agreement.
FaceParsing	Face parsing is pre-processed and used by some measures; therefore, the measure shall be configured (even if no measure is requested that uses the pre-processing result). The path to the BiSeNet model file in ONNX format should be set using the <code>model_path</code> key.

1.9.4 Requesting measures

OFIQ implements a variety of measures for assessing properties of a facial image. For a measure to be executed by OFIQ, it must be explicitly requested. For example, to only request the unified quality score measure, one can request it as follows.

```
{
  "config": {
    ...
    "measures": [
      "UnifiedQualityScore"
    ],
    ...
  }
}
```

At least one measure must be requested. An empty request list will result in OFIQ throwing an error. A full list of requestable measures and its *measure keys* can be found in the table of the [default configuration section](#).

1.9.5 Default configuration

OFIQ is the reference implementation for the ISO/IEC 29794-5 standard. To reproduce the conformance tests of the ISO/IEC 29794-5 standard one should use the (default) configuration provided by the file `<OFIQ-SOURCE>/data/ofiq_config.jaxn`; other configurations can be used; however, the resulting outputs of the quality assessment may not comply with the ISO/IEC 29794-5 standard.

The entries of the default configuration are documented in the following table. Details on the algorithms can be found in the ISO/IEC 29794-5 document. Details on the implementations can be found in the OFIQ source code. Note, the QAA identifiers listed in the table are defined in ISO/IEC 29794-5.

QAA identifier	description	config key	request key	configuration parameters	supports quality mapping config? - see details here
-	Face detector	"config">"params": "detector"		see here	-
-	Face landmark estimator	"config">"params": "landmarks"		see here	-
-	Face parsing	"config". "params". "measures". "FaceParsing"	-	see here	-
-	Face occlusion segmentation	"config". "params". "measures". "Face↔ Occlusion↔ Segmentation"	-	see here	-
-	Landmarked region	"config". "params". "measures". "FaceRegion"	-	alpha: is 0 per default and only used for internal purposes	-
0x41	Unified quality score	"config". "params". "measures". "Unified↔ QualityScore"	"config". "measures". "Unified↔ QualityScore"	model_↔ path: Path to an iResNet50 model file in ONNX format	yes
0x42	Background uniformity	"config". "params". "measures". "Background↔ Uniformity"	"config". "measures". "Background↔ Uniformity"	none	yes
0x43	Illumination uniformity	"config". "params". "measures". "Illumination↔ Uniformity"	"config". "measures". "Illumination↔ Uniformity"	none	yes
0x44	Luminance brightness	"config". "params". "measures". "Luminance"	"config". "measures". "Luminance"	none	yes
0x45	Luminance contrast	"config". "params". "measures". "Luminance"	"config". "measures". "Luminance"	none	yes
0x46	Absence of under-exposure	"config". "params". "measures". "Under↔ Exposure↔ Prevention"	"config". "measures". "Under↔ Exposure↔ Prevention"	none	yes

0x47	Absence of over-exposure	"config". "params". "measures". "Over↔ Exposure↔ Prevention"	"config". "measures". "Over↔ Exposure↔ Prevention"	none	yes
0x48	Pixel intensity variation	"config". "params". "measures". "Dynamic↔ Range"	"config". "measures". "Dynamic↔ Range"	none	yes
0x49	Sharpness	"config". "params". "measures". "Sharpness"	"config". "measures". "Sharpness"	model_↔ path: Path to the random forest model file	yes
0x4A	Absence of compression artifacts	"config". "params". "measures". "No↔ Compression↔ Artifacts"	"config". "measures". "No↔ Compression↔ Artifacts"	model_↔ path: Path to OFIQ's com- pression artifact CNN in ONNX format	yes
0x4B	Colour natural-ity	"config". "params". "measures". "NaturalColour"	"config". "measures". "NaturalColour"	none	yes
0x4C	Face unique-ness	"config". "params". "measures". "SingleFace↔ Present"	"config". "measures". "SingleFace↔ Present"	none	no
0x4D	Eyes openness	"config". "params". "measures". "EyesOpen"	"config". "measures". "EyesOpen"	none	yes
0x4E	Mouth closed-ness	"config". "params". "measures". "MouthClosed"	"config". "measures". "MouthClosed"	none	yes
0x4F	Eyes visibility	"config". "params". "measures". "EyesVisible"	"config". "measures". "EyesVisible"	none	yes
0x50	Mouth occlu-sion prevention	"config". "params". "measures". "Mouth↔ Occlusion↔ Prevention"	"config". "measures". "Mouth↔ Occlusion↔ Prevention"	none	yes
0x51	Face occlusion prevention	"config". "params". "measures". "Face↔ Occlusion↔ Prevention"	"config". "measures". "Face↔ Occlusion↔ Prevention"	none	yes

0x52	Inter-eye distance length	"config". "params". "measures". "InterEye↔ Distance"	"config". "measures". "InterEye↔ Distance"	none	yes
0x53	Size of the head in the image	"config". "params". "measures". "HeadSize"	"config". "measures". "HeadSize"	none	yes - the argument to the quality mapping is $\ x - 0.45\ $ where x is the native quality score
0x54	Leftward crop of the face image	"config">"params". "measures". "Leftward↔ CropOfThe↔ FacelImage"	"config". "measures". "CropOfThe↔ FacelImage"	none	yes
0x55	Rightward crop of the face image	"config">"params". "measures". "Rightward↔ CropOfThe↔ FacelImage"	"config". "measures". "CropOfThe↔ FacelImage"	none	yes
0x56	Margin above of the face image	"config">"params". "measures". "Margin↔ AboveOfThe↔ FacelImage"	"config". "measures". "CropOfThe↔ FacelImage"	none	yes
0x57	Margin below of the face image	"config">"params". "measures". "Margin↔ BelowOfThe↔ FacelImage"	"config". "measures". "CropOfThe↔ FacelImage"	none	yes
0x58	Pose angle yaw frontal alignment	"config">-	"config". "measures". "HeadPose"	none	no
0x59	Pose angle pitch frontal alignment	-	"config". "measures". "HeadPose"	none	no
0x5A	Pose angle roll frontal alignment	-	"config". "measures". "HeadPose"	none	no

0x5B	Expression neutrality	"config">"params". "measures". "Expression↵ Neutrality"	"config". "measures". "Expression↵ Neutrality"	cnn_model↵ _path1: Path to the CNN model <i>enet</i>↵ <i>b0_8_best</i>↵ <i>vgaf_embed</i>↵ <i>_zeroed.onnx</i> derived from here in ONNX format cnn_model↵ _path2: Path to the CNN model <i>enet_b2</i>↵ <i>8_embed</i>↵ <i>zeroed.</i>↵ <i>onnx</i> derived from here in ONNX format. adaboost↵ _model_↵ path: Path to the AdaBoost classifier model file <i>hse</i>↵ <i>1_2_C</i>↵ <i>adaboost.</i>↵ <i>yaml.gz</i> from here	yes
------	--------------------------	--	---	--	-----

0x5C	Absence of head coverings	"config">"params". "measures". "NoHeadCovering"	"config". "measures". "NoHeadCovering"	T0 - Proportion of pixels classified as head covering $\leq$ T0 will lead to a quality component value of 100 (best) T1 - Proportion of pixels classified as head covering $\geq$ T1 will lead to a quality component value of 0 (worst) w - Proportion of pixels classified as head covering in (T0,T1) will be interpolated using a sigmoid function with w as standard deviation x0 - Proportion of pixels classified as head covering in (T0,T1) will be interpolated using a sigmoid function with x0 as development point	no
------	---------------------------	---	--	---	----

1.9.6 Configuration of the quality mapping

Each measure implemented in OFIQ outputs a pair of values. The first value is called *native quality score*. The second value is called *quality component value* which is an integer between 0 (worst quality) and 100 (best quality). The quality component value is derived from the *native quality score* using a mapping function. Whether this mapping function can be configured for the measure is stated in the table of section [Default configuration](#).

Note, the OFIQ library implements hard-coded default quality mappings as a fallback.

A configurable quality mapping has the form of

$$Q(h, a, s, x, x_0, w) = h \cdot (a + s \cdot \text{sigmoid}(x, x_0, w))$$

where

$$\text{sigmoid}(x, x_0, w) = (1 + \exp((x_0 - x)/w))^{-1}.$$

Note, x is the native quality score which is mapped to the quality component value. The other symbols denote parameters that can be configured (see the example at the end of this section).

parameter	description	default value
"h"	scale factor	100
"a"	constant shift	0
"s"	signed weight for sigmoid part	1
"x0"	center point within sigmoid function; the default value has been chosen arbitrarily and should specified when a mapping is configured.	4
"w"	divisor within the sigmoid function; the default value has been chosen arbitrarily and should specified when a mapping is configured.	0.7
"round"	applies the compiler's native rounding function (<code>std::round</code>) such that only integer values are used as the quality value	true

All parameters are optional and can be omitted. In this case, the default value is chosen. Note, if a mapping results in a value not within 0 and 100, then a clipping is applied choosing the value 0 or 100 being closest to the mapped value.

For example, OFIQ's configuration for the background uniformity measure looks as follows

```
{
  "config" :
  ...
  "params": {
    ...
    "BackgroundUniformity": {
      "Sigmoid" : {
        "h": 190,
        "a": 1,
        "s": -1,
        "x0": 10,
        "w" : 100,
        "round": true
      }
    }
    ...
  }
}
```

1.10 C++ API

To use OFIQ in a C++ application one needs to include the following header file.

```
include <ofiq_lib.h>
```

In the following, we assume that the namespace [OFIQ](#) and [OFIQ_LIB](#) are used.

```
using namespace OFIQ;
using namespace OFIQ_LIB;
```

An OFIQ instance is initialized using the [Interface](#) class as follows.

```
// Get implementation pointer
auto implPtr = Interface::getImplementation();
// Initialization
auto ret = implPtr->initialize(configDir, configFile);
```

Here `configDir` is a `std::string`-representation of the path to the directory in which a JAXN configuration file of name `configFile` is stored - as documented in the [configuration section](#). Note, that the path can be absolute or relative to the path of the current working directory.

The input image is read by using the [readImage](#) function as follows

```
Image image;
ReturnStatus retStatus = readImage(imagePath, image);
```

where `imagePath` is a `std::string`-representation of a path to an image file. The representation is written to the `image` object of type [Image](#).

To compute the quality assessments, run

```
FaceImageQualityAssessment assessment;
ReturnStatus retStatus = implPtr->vectorQuality(image, assessment);
```

A successful computation is indicated by `retStatus.code` if it is of value [ReturnCode::Success](#). Then the assessment result is stored in a [FaceImageQualityAssessment](#) struct object. The obtained [FaceImageQualityAssessment](#) object has a `std::map` member which, for a specified [QualityMeasure](#) key, returns the [QualityMeasureResult](#). A [QualityMeasureResult](#) struct object contains the native quality score stored in the `rawScore` member and the quality component value stored in the `scalar` member. Note, both members are encoded as a `double` values although the `scalar` member should (on successful quality measure computation) be an integer value between 0 and 100. To check whether a [QualityMeasureResult](#) has been computed successfully, one checks if its `code` member agrees with the value [QualityMeasureReturnCode::Success](#).

As of version 1.0.3 it is possible to access pre-processing results computed during quality assessment. Therefore, the function [vectorQualityWithPreprocessingResults](#) is provided. Thus, instead of invoking [vectorQuality](#) as above, one can run

```
FaceImageQualityAssessment assessment;
FaceImageQualityPreprocessingResult preprocessingResult;
ReturnStatus retStatus = implPtr->vectorQualityWithPreprocessingResults
    (image, assessment, preprocessingResult);
```

If successful (i.e., if `retStatus` is of value [ReturnCode::Success](#)), the object `preprocessingResult` contains pre-processing results - in addition to the quality assessment result stored in `assessment`. More details can be found in the documentation of the [FaceImageQualityPreprocessingResult](#) struct.

1.11 Implementation and pre-processing workflow

Quality assessment is controlled by the implementation of the `OFIQImpl` class. A shared pointer to an `OFIQImpl` object is returned by the `Interface::getImplementation()` function. The implementation needs to be initialized once using the `OFIQImpl::initialize()` function. Note, the `OFIQImpl::initialize()` function loads all model files as specified in the input configuration into memory; thus, one should avoid creating repeated instances of the `OFIQImpl`.

After successful initialization, the implementation object can be used and one can repeatedly invoke the `OFIQImpl::vectorQuality()` function to assess the quality of a series of facial images.

The internal workflow of the `OFIQImpl::vectorQuality()` implementation is as follows.

1. Pre-processing of the input image using the `OFIQImpl::preprocess()` function.
 - (a) Face detection implemented by `SSDFaceDetector::UpdateFaces()`.
 - (b) Pose estimation implemented by `HeadPose3DDFAV2::updatePose()`.
 - (c) Landmark extraction implemented by `ADNetFaceLandmarkExtractor::updateLandmarks()`.
 - (d) Facial alignment implemented by `OFIQImpl::alignFacelImage()`.
 - (e) Face parsing implemented by `FaceParsing::UpdateMask()`.
 - (f) Face occlusion segmentation implemented by `FaceOcclusionSegmentation::UpdateMask()`.
2. Quality assessment using the `Executor::ExecuteAll()` function: For all requested measures
 - (a) its `Execute()`
 - (b) and then its `SetQualityMeasure()` functions are invoked.

1.12 Tutorial: Extending OFIQ

This section describes how to extend OFIQ by a new measure. We will choose an easy measure to illustrate the process. Therefore, we extend OFIQ by a somewhat dummy measure that assesses the subject's *non-surprisedness*. We will measure the surprisedness by the flatness of the eye-brow using the eye-brow landmarks.

Create a header file `NonSurprisedness.h` in the directory `/path/to/OFIQ-Project/OFIQlib/modules/measures/` with the following content

```
#pragma once

#include "landmarks.h"
#include "Measure.h"

namespace OFIQ_LIB::modules::measures
{
    class NonSurprisedness : public Measure
    {
    public:
        explicit NonSurprisedness(const Configuration& configuration);

        void Execute(OFIQ_LIB::Session & session) override;
    };
}
```

Also, create a source file `NonSurprisedness.cpp` in the directory `/path/to/OFIQ-Project/OFIQlib/modules/measures/src/` with the following content.

```

#include "NonSurprisedness.h"
#include "FaceMeasures.h"
#include "FaceParts.h"

using FaceMeasures = OFIQ_LIB::modules::landmarks::FaceMeasures;

namespace OFIQ_LIB::modules::measures
{
    static const auto qualityMeasure = OFIQ::QualityMeasure::NonSurprisedness;

    NonSurprisedness::NonSurprisedness(const Configuration& configuration)
        : Measure{ configuration, qualityMeasure }
    {
        // Implement me
    }

    void NonSurprisedness::Execute(OFIQ_LIB::Session & session)
    {
        // Implement me
    }
}

```

To make the source code to compile, we need to extend the enumeration `OFIQ::QualityMeasure` by the new measure. Therefore, edit the file `/path/to/OFIQ-Project/OFIQlib/include/ofiq_structs.h` and insert the item

```
NonSurprisedness
```

within the enumeration `OFIQ::QualityMeasure`. The resulting `ofiq_struct.h` could look like this.

```

...
namespace OFIQ
{
    ...
    enum class QualityMeasure
    {
        // UnifiedQualityScore
        UnifiedQualityScore = 0x41,
        ...
        // Newly inserted NonSurprisedness measure
        NonSurprisedness,
        // unknown measure
        NotSet = -1
    }
    ...
}

```

Then add the new header to `/path/to/OFIQ-Project/OFIQlib/modules/measures/AllMeasures.h`

```

...
#include "UnifiedQualityScore.h"
#include "NonSurprisedness.h"

```

and include the new measure in the file `/path/to/OFIQ-Project/OFIQlib/modules/measures/src/MeasureFactory.cpp`

```
static const std::map<OFIQ::QualityMeasure,
    std::function<std::unique_ptr<Measure>(const Configuration&)>> factoryMapping
{
    {OFIQ::QualityMeasure::SingleFacePresent, [] (const Configuration& configuration)
        { return std::make_unique<SingleFacePresent>(configuration); }},
    ...
    {OFIQ::QualityMeasure::NonSurprisedness, [] (const Configuration& configuration)
        { return std::make_unique<NonSurprisedness>(configuration); }},
}
```

If we are on Windows, we need to edit insert the new header file `NonSurprisedness.h` and the new source file `NonSurprisedness.cpp` in the file `/path/to/OFIQ-Project/OFIQLib/CMakeLists.windows.cmake`. The resulting `CMakeLists.windows.cmake` could look like this.

```
list(APPEND module_sources
    ${libImplementationSources}
    ...
    ${OFIQLIB_SOURCE_DIR}/modules/measures/src/NonSurprisedness.cpp
    ...

list(APPEND module_headers
    ${PUBLIC_HEADER_LIST}
    ...
    ${OFIQLIB_SOURCE_DIR}/modules/measures/NonSurprisedness.h
    ...
```

Likewise, if we are on Ubuntu or MacOS, we need to edit the files `CMakeLists.ubuntu.cmake` or `CMakeLists.macos.cmake`, respectively. Note, for Ubuntu or MacOS we only need to insert the file `NonSurprisedness.cpp` (and not the file `NonSurprisedness.h`).

After having finished the above steps, we may already build OFIQ by performing the steps described in the building steps. However, after compiling and running OFIQ, we may note that no output for our new no-surprisedness measure is made. Therefore, two additional steps need to be done:

1. Edit OFIQ's configuration file
2. Implement the member functions of the class `NonSurprisedness`

To edit OFIQ's default configuration file, open `/path/to/OFIQ-Project/data/ofiq_config.jaxn` and include the `NonSurprisedness` measure. The resulting `ofiq_config.jaxn` could look like this.

```
...
"config": {
    ...
    "measures": [
        ...
        "NonSurprisedness",
        ...
    ],
    ...
```

To implement the `Execute` method, we may want to compute the native quality measure first. At this point, we should define the non-surprisedness measure. For an eye-brow, we define the non-surprisedness using the difference of the y-coordinates of the northern eye-brow landmark to the y-coordinate between the left and the right eye-brow coordinate. To make the non-surprisedness independent from the facial image's resolution we divide the difference by the width of the eye-brow. the right eye-brow landmarks divided by the distance between the left and right The non-surprisedness is computed for the left and right eye-brow and then the minimum of the two eye brow's non-surprisedness as the overall non-surprisedness. The following functions computes the non-surprisedness for an eye-brow.

```
static double GetNonSurprisedness(
    const OFIQ::LandmarkPoint & north,
    const OFIQ::LandmarkPoint & left,
    const OFIQ::LandmarkPoint & right)
{
    auto middle = FaceMeasures::GetMiddle(OFIQ::Landmarks{ left, right });
    double diff = std::abs(middle.y-north.y);
    double width = FaceMeasures::GetDistance(left, right);
    return diff / width; // Results in NaN, when dividing by zero
}
```

The function `NonSurprisedness` uses functions provided by `FaceMeasures.h`. Now, we can implement the `Execute` method.

```
void NonSurprisedness::Execute(OFIQ_LIB::Session & session)
{
    auto landmarks = session.getAlignedFaceLandmarks();
    double leftNonSurprisedness = GetNonSurprisedness(landmarks.landmarks[33],
        landmarks.landmarks[38], landmarks.landmarks[35]);
    double rightNonSurprisedness = GetNonSurprisedness(landmarks.landmarks[50],
        landmarks.landmarks[46], landmarks.landmarks[44]);
    double nonSurprisedness =
        std::min(leftNonSurprisedness, rightNonSurprisedness);
    if (std::isnan(nonSurprisedness))
    {
        // If the measure is NaN, we cannot compute
        // the non-surprisedness and should indicate
        // this by 'FailureToAssess'
        SetQualityMeasure(session, qualityMeasure, nonSurprisedness,
            OFIQ::QualityMeasureReturnCode::FailureToAssess);
    }
    else
    {
        SetQualityMeasure(session, qualityMeasure, nonSurprisedness,
            OFIQ::QualityMeasureReturnCode::Success);
    }
}
```

The measure will automatically map the non-surprisedness measure to a quality values that is an integer between 0 and 100. For that, a default mapping is applied (see Section [Configuration of the quality mapping](#)). We should change the default mapping in the constructor. Therefore, we choose a mapping with the help of a sigmoid function that maps high non-surprisedness values to high quality values and vice-versa.

```
NonSurprisedness::NonSurprisedness(const Configuration& configuration)
    : Measure{ configuration, qualityMeasure }
{
    SigmoidParameters defaultValues;
    defaultValues.h = 100;
    defaultValues.x0 = 0.5;
    defaultValues.w = 1.0;
    defaultValues.round = true;
    AddSigmoid(qualityMeasure, defaultValues);
}
```

The above mapping is a fallback for the case when the mapping is not configured in the configuration file. To configure the mapping in the configuration file, edit the `ofiq_config.jaxn` so that it is of the form

```

"config": {
  ...
  "params" : {
    ...
    "measures": {
      ...
      "NonSurprisedness" : {
        "Sigmoid" : {
          "h": 100,
          "x0": 0.5,
          "w": 1.0,
          "round": true
        }
      }, ...
    }
  }
}

```

1.13 Supported image file formats

Using the function `OFIQ_LIB::readImage()` image files can be read. The same file formats are supported as those supported by the linked OpenCV compilation. Per default, when building OFIQ with Conan, OpenCV is built with the support of the following formats: BMP, PNG, JPG, JPEG2000, HDR, PBM, PGM, PNM, PPM. Per default, the following formats are not supported: TIF, RAS, RS, WEBP.

1.14 JPEG image format

There are a variety of JPEG libraries. For example, libjpeg and libjpeg-turbo. When these libraries are used to read a JPEG image from file, the resulting pixel data may not be exactly the same. Up to version 1.0.3, OFIQ was linking against libjpeg when build using the Conan approach and against libjpeg-turbo when build without Conan. This resulted in different behaviour of the OFIQSampleApp when processing JPEG images. Therefore, as of version 1.1.0, both compilation approaches -with and without Conan- are linked against libjpeg-turbo. OFIQ is using the openjpeg library (<https://www.openjpeg.org/>) as JPEG2000 image decoder. This library is used by opencv and has proven to be much faster and secure than the alternative library jasper (<https://github.com/jasper-software/jasper>).

1.15 EXIF flags

Using the function `OFIQ_LIB::readImage()` image files can be read. Some formats, like JPEG, contain EXIF information. These are supported in the same way as they are supported by the linked OpenCV compilation.

1.16 Release notes

This is OFIQ v1.2.0. The following table lists all measures and its implementation provided by this release of OFIQ. Details on the configuration and on requesting measures can be found [here](#). Note, the QAA identifiers listed in the table are defined in ISO/IEC 29794-5.

QAA identifier	Description	OFIQ implementation reference
0x41	MagFace-based unified quality score measure.	UnifiedQualityScore
0x42	Gradient-based background uniformity.	BackgroundUniformity
0x43	Illumination uniformity by summing up the minima of the histograms of the left and the right side of the face.	IlluminationUniformity
0x44	Luminance mean measure computed from the luminance histogram	Luminance
0x45	Luminance variance measure computed from the luminance histogram	Luminance
0x46	Under-exposure prevention by computing the proportion of low-intensity pixels in the luminance image to assess the absence of under-exposure	UnderExposurePrevention
0x47	Over-exposure prevention by computing the proportion of high-intensity pixels in the luminance image to assess the absence of over-exposure	OverExposurePrevention
0x48	Dynamic range computed from the luminance histogram.	DynamicRange
0x49	Sharpness assessment based on a random forest classifier trained by the OFIQ development team.	Sharpness
0x4A	Assessment of the absence of compression artifact (both JPEG and JPEG2000) based on a CNN trained by the OFIQ development team.	CompressionArtifacts
0x4B	Assessment of the naturalness of the colour based on the conversion of the RGB presentation of the image to the CIELAB colour space.	NaturalColour
0x4C	Assessment of the uniqueness of the most dominant face detected by comparing its size with the size of the second largest face detected	SingleFacePresent
0x4D	Eyes openness assessment based on computing eyes aspect ratio from eye landmarks	EyesOpen
0x4E	Mouth closed assessment based on computing a ratio from mouth landmarks	MouthClosed
0x4F	Eyes visibility assessment by measuring the coverage of the eye visibility zone with the result of face occlusion segmentation computed during pre-processing.	EyesVisible
0x50	Assessment of the absence of mouth occlusion by measuring the coverage of the mouth region from mouth landmarks with the result of face occlusion segmentation computed on pre-processing.	MouthOcclusionPrevention
0x51	Assessment of the absence of face occlusion by measuring the coverage of the landmarked region with the result of face occlusion segmentation computed during pre-processing.	FaceOcclusionPrevention
0x52	Inter-eye distance assessment based on computing the Euclidean length of eyes' centres and multiplication with the secant of the yaw angle computed during pre-processing.	InterEyeDistance
0x53	Size of the head based on computing the height of the face computed from facial landmarks with the height of the image.	HeadSize
0x54	Leftward crop of the face image	CropOfTheFaceImage

0x55	Rightward crop of the face image	CropOfTheFaceImage
0x56	Margin above of the face image	CropOfTheFaceImage
0x57	Margin below of the face image	CropOfTheFaceImage
0x58	Pose angle yaw frontal alignment based on the 3DDFAV2 .	HeadPose
0x59	Pose angle pitch frontal alignment based on the 3DDFAV2	HeadPose
0x5A	Pose angle roll frontal alignment based on the 3DDFAV2	HeadPose
0x5B	Expression neutrality estimation based on a fusion of HSEMotion with with Efficient-↔ Expression-Neutrality-Estimation .	ExpressionNeutrality
0x5C	Assessment of the absence of head coverings by counting the pixels being labeled as head covers in the mask output by the face parsing computed during pre-processing.	NoHeadCoverings

1.17 Changes

1.17.1 Version 1.2.0 (2026-03-03)

- Implemented a C interface for OFIQ library [OFIQlib\include\ofiq_libc.h](#).
- Implemented a python wrapper for OFIQ library. See the corresponding README.md.
- Implemented a web service and corresponding client web application using the python wrapper.

1.17.2 Version 1.1.2 (2026-02-05)

- In v1.0.3 using `OFIQ::Image::deepcopy`, the pixel data was interpreted as RGB. In v1.1.0 and v1.1.1 the pixel data has been interpreted as BGR which is a change in the API. To patch this, an argument `bool isRgb` has been added to the argument list of `OFIQ::Image::deepcopy` and which is `true` by default. When `true` the method `OFIQ::Image::deepcopy` interprets the specified data as RGB; otherwise, if `false`, `OFIQ::Image::deepcopy` interprets the specified pixel data as BGR.
- In v1.1.0 a new return code `OFIQ::ReturnCode::MissingConfigFileError` has been inserted in the enum. Consequently, integer casts of v1.1.0 and v1.1.1 were not compatible with integer casts of v1.↔ 0.3. In v1.1.2, the return code has been moved to the end of the enum so that integer casts of return codes from v1.0.3 match with integers casts of return codes from v1.1.2.
- Fixes compilation without conan for linux-arm64.
- Fixes of typos in documentation.

1.17.3 Version 1.1.1 (2026-01-21)

- Updates URL to external library dependency hosted by eu-LISA
- Adds a CONTRIBUTING.md
- OFIQSampleApp labels native scores clearly by a `.native` prefix

1.17.4 Version 1.1.0 (2025-10-31)

- Some runtime improvements have been implemented that affect conformance tests. In this way, the resulting OFIQ version is a conformant implementation by means of ISO/IEC 29794-5. However, it is not a reference implementation. Therefore, the minor version has been increased. The latest reference implementation is OFIQ v1.0.3.
- Provides compilation scripts and demo apps for Android and iOS
- Adds android asset-manager support (cf. <https://github.com/BSI-OFIQ/OFIQ-Project/pull/70>)
- Links with libjpeg-turbo (previously, libjpeg was used) when compiled with Conan
- Documentation that OFIQ uses openjpeg by default for JPEG2000 support
- OFIQ can now be safely accessed by multiple threads
- Switches building documentation from Conan 2.0.17 to 2.18.1
- Fixes building script so that it does not download onnxruntime when the flag `--no-download` is specified

1.17.5 Version 1.0.3 (2025-07-04)

- Added a new interface method `vectorQualityWithPreprocessingResults` that works exactly as the existing `vectorQuality` method, but additionally returns image preprocessing results. The preprocessing result types are defined in the struct `PreprocessingResultType` and include detected faces, face landmark points, face parsing segmentation mask, face occlusion mask and landmarked region mask.
- Includes a list of known issues
- Accepts <https://github.com/BSI-OFIQ/OFIQ-Project/pull/71>: Removes unneeded includes to `opencv/highgui.cpp`
- Accepts <https://github.com/BSI-OFIQ/OFIQ-Project/pull/69>: Fixes type of `Image`.↔ data
- Fixes <https://github.com/BSI-OFIQ/OFIQ-Project/issues/65>: `OFIQSampleApp` crashed when specified with inconsistent arguments

1.17.6 Version 1.0.2 (2025-04-10)

- Supports compilation for Raspberry Pi/ARMv8 - see <https://github.com/BSI-OFIQ/OFIQ-Project/pull/45>
- Fixes segmentation faults occurring on natural colour assessment in cases CIELAB values being attempted to be computed from an empty colour image
- `EyesVisible` component did not handle the cases where inter-eye distance is not computable. Now it does by resulting in `FailureToAssess`.
- `IlluminationUniformity` did not handle the case when histograms from empty subimages are to be computed. Now it does by resulting in `FailureToAssess`.
- `FailureToAssess` assessments of quality component do output a quality component value of -1 more consequently.

1.17.7 Version 1.0.1 (2025-03-11)

- `OFIQ::Image` struct had no method to import deep copies from binary data. This made it unpractical to use OFIQ with a binding such as Java/JNI. Now `OFIQ::Images` provides a method `OFIQ::Image↔::deepcopy` for importing deep copies from binary data.

This change resolves <https://github.com/BSI-OFIQ/OFIQ-Project/issues/63>.

- Code quality improvements as per Sonarqube checks (do not affect conformance tests)
- Memory management refinements
- Fixes memory allocations when `cv::copyMakeBorder` from OpenCV is invoked

1.17.8 Version 1.0.0 (2025-03-07)

First OFIQ release being the reference implementation for ISO/IEC 29794-5.

1.17.9 Version 1.0.0-RC.2 (2024-07-31)

Second release of OFIQ's release candidate. The following changes have been implemented.

- Supports compilation on MacOS
- Successfully tested that it is possible (with some effort) to compile for and be conformant with mobile devices such as Android and iOS.
- Revisions as per up coming FDIS (e.g., quality mappings, update of conformance test table, etc.)
- Fix of the default config file: Changes to the `CropOfTheFacelImage` measures did not affect the quality mappings
- Fixes link to the OFIQ-MODELS.zip archive: <https://github.com/BSI-OFIQ/OFIQ-Project/issues/12>
- Removes libgtk dependency: <https://github.com/BSI-OFIQ/OFIQ-Project/issues/18>
- Removes Lapack from ubuntu cmake file which wasn't used: <https://github.com/BSI-OFIQ/OFIQ-Project/issues/20>
- Improves readability of source code by applying a style guide to member variables: <https://github.com/BSI-OFIQ/OFIQ-Project/issues/27>
- Fixes from static code analyses, for example:
 - <https://github.com/BSI-OFIQ/OFIQ-Project/issues/28>
 - <https://github.com/BSI-OFIQ/OFIQ-Project/issues/29>
 - <https://github.com/BSI-OFIQ/OFIQ-Project/issues/30>
 - <https://github.com/BSI-OFIQ/OFIQ-Project/issues/31>
 - and others
- Removes mis-leading usage hint in the sample app: <https://github.com/BSI-OFIQ/OFIQ-Project/issues/34>
- Avoids redundant RGB conversions: <https://github.com/BSI-OFIQ/OFIQ-Project/issues/36>
- Fixes a bug on continuous OpenCV matrices: <https://github.com/BSI-OFIQ/OFIQ-Project/issues/41>
- Fixes further issues and bugs and code beautification

1.17.10 Version 1.0.0-RC.1 (2024-03-15)

Initial release of OFIQ's release candidate.

1.18 Known Issues

In the following, known issues are listed.

1.18.1 Faces detected on glasses

OFIQ uses the SSD face detector. SSD has a tendency to detect faces on glasses as, for instance, when applied to the conformance test image `r-06-pitch.png`. This affects the quality component `SingleFacePresent`. To conform with the target value of `SingleFacePresent`, implementations conformant to the ISO/IEC 29794-5 standard shall detect the same erroneous faces. Therefore, it is not possible to fix this in OFIQ v1.x.y.

Chapter 2

Namespace Index

2.1 Namespace List

Here is a list of all namespaces with brief descriptions:

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Chapter 3

Hierarchical Index

3.1 Class Hierarchy

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Chapter 4

Class Index

4.1 Class List

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Chapter 5

File Index

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Chapter 6

Namespace Documentation

6.1 cv Namespace Reference

OpenCV's namespace.

6.1.1 Detailed Description

OpenCV's namespace.

6.2 OFIQ Namespace Reference

Namespace for OFIQ API.

Classes

- struct [BoundingBox](#)
Data structure for describing bounding boxes, e.g. the face region of the faces found by a face detector.
- struct [FacelImageQualityAssessment](#)
Data structure storing the results of the different measurement computations.
- struct [FacelImageQualityPreprocessingResult](#)
Data structure storing the results of pre-processing computations.
- struct [FaceLandmarks](#)
Data structure for storing facial landmarks.
- struct [Image](#)
Struct representing a single image.
- class [Interface](#)
The interface to FACE QA implementation.
- struct [LandmarkPoint](#)
Data structure to describe the x and y coordinate of a landmark.
- struct [QualityMeasureResult](#)
Data structure to handle the results of a quality measure.
- struct [ReturnStatus](#)
A structure to contain information about a failure by the software under test.

Typedefs

- using [QualityAssessments](#) = std::map<[QualityMeasure](#), [QualityMeasureResult](#)>
Data structure that stores key-value pairs, with each entry representing a quality element and its value.
- using [Landmarks](#) = std::vector<[LandmarkPoint](#)>
container for a collection of landmarks, e.g. belonging to all the landmarks detected on a face image.

Enumerations

- enum class [PreprocessingResultType](#) {
[None](#) = 0x0 , [Faces](#) = 0x1 , [Landmarks](#) = 0x2 , [Segmentation](#) = 0x4 ,
[OcclusionMask](#) = 0x8 , [LandmarkedRegion](#) = 0x10 , [All](#) = 0x1 + 0x2 + 0x4 + 0x8 + 0x10 }
Encodes flags and mask to request preprocessing results.
- enum class [ReturnCode](#) {
[Success](#) = 0 , [ImageReadingError](#) , [ImageWritingError](#) , [MissingConfigParamError](#) ,
[UnknownConfigParamError](#) , [FaceDetectionError](#) , [FaceLandmarkExtractionError](#) , [FaceOcclusionSegmentationError](#) ,
[FaceParsingError](#) , [UnknownError](#) , [QualityAssessmentError](#) , [NotImplemented](#) ,
[MissingConfigFileError](#) }
Return codes for functions specified in this API.
- enum class [QualityMeasure](#) {
[UnifiedQualityScore](#) = 0x41 , [BackgroundUniformity](#) = 0x42 , [IlluminationUniformity](#) = 0x43 , [Luminance](#) =
-0x44 ,
[LuminanceMean](#) = 0x44 , [LuminanceVariance](#) = 0x45 , [UnderExposurePrevention](#) = 0x46 , [OverExposurePrevention](#)
= 0x47 ,
[DynamicRange](#) = 0x48 , [Sharpness](#) = 0x49 , [CompressionArtifacts](#) = 0x4a , [NaturalColour](#) = 0x4b ,
[SingleFacePresent](#) = 0x4c , [EyesOpen](#) = 0x4d , [MouthClosed](#) = 0x4e , [EyesVisible](#) = 0x4f ,
[MouthOcclusionPrevention](#) = 0x50 , [FaceOcclusionPrevention](#) = 0x51 , [InterEyeDistance](#) = 0x52 , [HeadSize](#)
= 0x53 ,
[CropOfTheFacelImage](#) = -0x54 , [LeftwardCropOfTheFacelImage](#) = 0x54 , [RightwardCropOfTheFacelImage](#) =
0x55 , [MarginAboveOfTheFacelImage](#) = 0x56 ,
[MarginBelowOfTheFacelImage](#) = 0x57 , [HeadPose](#) = -0x58 , [HeadPoseYaw](#) = 0x58 , [HeadPosePitch](#) = 0x59 ,
[HeadPoseRoll](#) = 0x5a , [ExpressionNeutrality](#) = 0x5b , [NoHeadCoverings](#) = 0x5c , [NotSet](#) = -1 }
Enums presenting the measure labels.
- enum class [QualityMeasureReturnCode](#) { [Success](#) = 0 , [FailureToAssess](#) , [NotInitialized](#) }
Return codes for QualityMeasureResult.
- enum class [FaceDetectorType](#) { [OPENCVSSD](#) , [NotSet](#) }
Enum describing the different face detector implementations.
- enum class [LandmarkType](#) { [LM_98](#) , [NotSet](#) }
Enum describing the different implementations of landmarks.

Functions

- std::ostream & [operator<<](#) (std::ostream &s, const [ReturnCode](#) &rc)
- std::string [QualityMeasureToString](#) (const [QualityMeasure](#) qm)
- std::ostream & [operator<<](#) (std::ostream &s, const [QualityMeasure](#) &qm)

6.2.1 Detailed Description

Namespace for OFIQ API.

Namespace for OFIQ API.

6.2.2 Typedef Documentation

6.2.2.1 Landmarks

```
using OFIQ::Landmarks = std::vector<LandmarkPoint>
```

container for a collection of landmarks, e.g. belonging to all the landmarks detected on a face image.

6.2.2.2 QualityAssessments

```
using OFIQ::QualityAssessments = std::map<QualityMeasure, QualityMeasureResult>
```

Data structure that stores key-value pairs, with each entry representing a quality element and its value.

6.2.3 Enumeration Type Documentation

6.2.3.1 FaceDetectorType

```
enum class OFIQ::FaceDetectorType [strong]
```

Enum describing the different face detector implementations.

Enumerator

OPENCVSSD	face detector based on the ssd implementation in opencv.
NotSet	unknown face detector

6.2.3.2 LandmarkType

```
enum class OFIQ::LandmarkType [strong]
```

Enum describing the different implementations of landmarks.

Enumerator

LM_98	Landmarks extracted with the adnet detector.
NotSet	used for iterating through the enums.

6.2.3.3 PreprocessingResultType

```
enum class OFIQ::PreprocessingResultType [strong]
```

Encodes flags and mask to request preprocessing results.

See also

[OFIQImpl::vectorQualityWithPreprocessingResults](#)

Enumerator

None	Flag to request no preprocessing data
Faces	Flag to request detected faces
Landmarks	Flag to request landmarks
Segmentation	Flag to request face parsing segmentation
OcclusionMask	Flag to request face occlusion mask
LandmarkedRegion	Flag to request landmarked region
All	Mask to request all available pre-processing data

6.2.3.4 QualityMeasure

```
enum class OFIQ::QualityMeasure [strong]
```

Enums presenting the measure labels.

Enumerator

UnifiedQualityScore	UnifiedQualityScore
BackgroundUniformity	BackgroundUniformity
IlluminationUniformity	IlluminationUniformity
Luminance	the common measure implementation for LuminanceMean, LuminanceVariance
LuminanceMean	LuminanceMean
LuminanceVariance	LuminanceVariance
UnderExposurePrevention	UnderExposurePrevention
OverExposurePrevention	OverExposurePrevention
DynamicRange	DynamicRange
Sharpness	Sharpness
CompressionArtifacts	CompressionArtifacts
NaturalColour	NaturalColour
SingleFacePresent	SingleFacePresent
EyesOpen	EyesOpen
MouthClosed	MouthClosed
EyesVisible	EyesVisible
MouthOcclusionPrevention	MouthOcclusionPrevention
FaceOcclusionPrevention	FaceOcclusionPrevention
InterEyeDistance	InterEyeDistance
HeadSize	HeadSize
CropOfTheFacelImage	CropOfTheFacelImage: common measure for {Left,Right}wardCropOfTheFacelImage, MarginAbove, and MarginBelow
LeftwardCropOfTheFacelImage	LeftwardCropOfTheFacelImage
RightwardCropOfTheFacelImage	RightwardCropOfTheFacelImage
MarginAboveOfTheFacelImage	MarginAbove
MarginBelowOfTheFacelImage	MarginBelow
HeadPose	HeadPose
HeadPoseYaw	HeadPoseYaw
HeadPosePitch	HeadPosePitch
HeadPoseRoll	HeadPoseRoll

Enumerator

ExpressionNeutrality	ExpressionNeutrality
NoHeadCoverings	NoHeadCoverings
NotSet	unknown measure

6.2.3.5 QualityMeasureReturnCode

```
enum class OFIQ::QualityMeasureReturnCode [strong]
```

Return codes for QualityMeasureResult.

Enumerator

Success	Success
FailureToAssess	Unable to assess a quality measure
NotInitialized	Quality measure is not initialized

6.2.3.6 ReturnCode

```
enum class OFIQ::ReturnCode [strong]
```

Return codes for functions specified in this API.

Enumerator

Success	Success
ImageReadingError	Failed to read an image.
ImageWritingError	failed to write an image to disk.
MissingConfigParamError	A required config parameter is missing
UnknownConfigParamError	A required config parameter is missing
FaceDetectionError	Unable to detect a face in the image
FaceLandmarkExtractionError	Unable to extract landmarks from face
FaceOcclusionSegmentationError	Unable to extract occlusion segments from face
FaceParsingError	Unable to parse face
UnknownError	Catch-all error
QualityAssessmentError	Failure to generate a quality score on the input image
NotImplemented	Function is not implemented
MissingConfigFileError	A config file is missing or the given filename is invalid

6.2.4 Function Documentation

6.2.4.1 operator<<() [1/2]

```
std::ostream & OFIQ::operator<< (
    std::ostream & s,
    const QualityMeasure & qm ) [inline]
```

6.2.4.2 operator<<() [2/2]

```
std::ostream & OFIQ::operator<< (
    std::ostream & s,
    const ReturnCode & rc ) [inline]
```

Output stream operator for a [ReturnCode](#) object.

6.2.4.3 QualityMeasureToString()

```
std::string OFIQ::QualityMeasureToString (
    const QualityMeasure qm ) [inline]
```

6.3 OFIQ_LIB Namespace Reference

Namespace for OFIQ implementations.

Namespaces

- namespace [modules](#)

Classes

- class [asset_istream](#)
- class [asset_streambuf](#)
- class [Configuration](#)
Configuration class.
- class [FaceDetectorInterface](#)
Provides the interface class to the face detector implementations.
- class [FaceLandmarkExtractorInterface](#)
Implements the base class for the face landmark extractors.
- struct [NeuronalNetworkContainer](#)
Neural network container for OFIQ's preprocessing steps.
- class [OFIQError](#)
Implementation of a custom exception.
- class [OFIQImpl](#)
Implementation of the OFIQ_LIB.
- struct [Point2i](#)
Representation of a point with integer arithmetics.
- class [PoseEstimatorInterface](#)
Implementation of the base class for integrating pose estimation algorithms capable of estimating three head orientation angles (yaw, pitch and roll) from a face image.
- class [SegmentationExtractorInterface](#)
Base class for the different implementation of segmentation algorithms.
- class [Session](#)

Typedefs

- using [DataStream](#) = [asset_istream](#)
- using [ExposureRange](#) = [std::array<int, 2>](#)
- using [EulerAngle](#) = [std::array<double, 3>](#)

Functions

- void [SetAAssetManager](#) (::AAssetManager *assetManager)
- [OFIQ_EXPORT OFIQ::ReturnStatus readImage](#) (const std::string &filename, [OFIQ::Image](#) &image)
Read image from disk.
- [OFIQ_EXPORT OFIQ::ReturnStatus readImageFromByteArray](#) (const std::vector< unsigned char > &buffer, [OFIQ::Image](#) &image)
Read image from byte array.
- double [ColorConvert](#) (double v)
Converts a color as specified in ISO/IEC 29794-5.
- double [Cubic](#) (double x, double k, double eps)
Cubic flattening function.
- void [ConvertBGRToCIELAB](#) (const cv::Mat &bgrImage, double &a, double &b)
Computes CIELAB values a^ and b^* from a BGR image.*
- cv::Mat [GetLuminanceImageFromBGR](#) (const cv::Mat &bgrImage)
Converts a BGR image to the luminance image.
- void [CalculateReferencePoints](#) (const [OFIQ::FaceLandmarks](#) &landmarks, [OFIQ::LandmarkPoint](#) &leftEyeCenter, [OFIQ::LandmarkPoint](#) &rightEyeCenter, double &interEyeDistance, double &eyeMouthDistance)
Computes the left eye center, the right eye center, the (planar) inter-eye-distance and the eye to mouth distance from facial landmarks.
- void [CalculateRegionOfInterest](#) (cv::Rect &leftRegionOfInterest, cv::Rect &rightRegionOfInterest, const [OFIQ::LandmarkPoint](#) &leftEyeCenter, const [OFIQ::LandmarkPoint](#) &rightEyeCenter, const double interEyeDistance, const double eyeMouthDistance)
Extracts regions being of interest for some measures (e.g. NaturalColour).
- void [GetNormalizedHistogram](#) (const cv::Mat &luminanceImage, const cv::Mat &maskImage, cv::Mat1f &histogram)
Computes the normalized histogram from a luminance image in 256 chunks.
- double [CalculateExposure](#) (const [Session](#) &session, const [ExposureRange](#) &exposureRange)
Helper function for some measures.
- double [ComputeBrightnessAspect](#) (const cv::Mat &luminanceImage, const cv::Mat &maskImage, const [ExposureRange](#) &exposureRange)
Helper function for some measures.
- void [makeSquareBoundingBoxWithPadding](#) (const [OFIQ::BoundingBox](#) &i_bb, const cv::Mat &i_input_image, cv::Mat &o_output_image, [OFIQ::BoundingBox](#) &o_bb, [Point2i](#) &o_translation_vector)
Some computations, especially neural networks, need a squarred image as input. This funtion consumes a boundig box and an input image. The greater parameter of width or height is used to define the side length of the new squarred bounding box. The face will be centered in the bounding box. Padding is added if needed. The squarred bounding box is used generate a new cropped image, the o_output_image. Required translations are described by the translation vector o_translation_vector.
- [OFIQ::BoundingBox makeSquareBoundingBox](#) (const [OFIQ::BoundingBox](#) &i_bb)
This function converts a non-squarred bounding box into an squarred one. The side length is defined by the greater one of height or width.
- size_t [findLargestBoundingBox](#) (const std::vector< [OFIQ::BoundingBox](#) > &faceRects)
This function returns the position of the largest bounding box (largest in terms of area) from a vector of bounding boxes.
- [OFIQ::BoundingBox findLandmarksBoundingBox](#) (const [OFIQ::Landmarks](#) &landmarks)
Find the bounding box that encloses the given landmarks.

- `cv::Mat copyToCvImage` (const `OFIQ::Image` &sourceImage, bool asGrayImage=false)
Convert images in OFIQ::Image format into the OpenCV cv::Mat format. The image can be converted from color to gray scale by setting the parameter asGrayImage to true.
- `cv::Mat alignImage` (const `OFIQ::Image` &faceImage, const `OFIQ::FaceLandmarks` &faceLandmarks, `OFIQ::FaceLandmarks` &alignedFaceLandmarks, cv::Mat &transformationMatrix)
This function transforms a face image so that the position of the eyes, nose and mouth are roughly at a pre-defined position. Face alignment is the translation, rotation and scaling of the image to do this.
- void `calculateEyeCenter` (const `OFIQ::FaceLandmarks` &faceLandmarks, `Point2f` &leftEyeCenter, `Point2f` &rightEyeCenter)
Based on face landmarks the center of the left and right eye are computed.
- `OFIQ::Image MakeGreylImage` (uint16_t width, uint16_t height)
This function generates a gray scaled image with the resolution passed by the call.
- float `tmetric` (const `OFIQ::FaceLandmarks` &faceLandmarks)
Based on the provided landmarks this function computes the distance between the point between the eyes and the chin.

6.3.1 Detailed Description

Namespace for OFIQ implementations.

Provides implementations in OFIQ.

Namespace for OFIQ implementations.

6.3.2 Typedef Documentation

6.3.2.1 DataStream

```
typedef std::ifstream OFIQ_LIB::DataStream = asset_istream
```

6.3.2.2 EulerAngle

```
using OFIQ_LIB::EulerAngle = std::array<double, 3>
```

6.3.2.3 ExposureRange

```
using OFIQ_LIB::ExposureRange = std::array<int, 2>
```

6.3.3 Function Documentation

6.3.3.1 alignImage()

```
cv::Mat OFIQ_LIB::alignImage (
    const OFIQ::Image & faceImage,
    const OFIQ::FaceLandmarks & faceLandmarks,
    OFIQ::FaceLandmarks & alignedFaceLandmarks,
    cv::Mat & transformationMatrix )
```

This function transforms a face image so that the position of the eyes, nose and mouth are roughly at a pre-defined position. Face alignment is the translation, rotation and scaling of the image to do this.

Parameters

<i>faceImage</i>	Input image.
<i>faceLandmarks</i>	Face landmarks, based on the face represented in the input image.
<i>alignedFaceLandmarks</i>	Face landmarks of the aligned face image.
<i>transformationMatrix</i>	Transformation matrix used to transform the landmarks.

Returns

cv::Mat Aligned face image with a resolution of 616x616.

6.3.3.2 CalculateExposure()

```
double OFIQ_LIB::CalculateExposure (
    const Session & session,
    const ExposureRange & exposureRange )
```

Helper function for some measures.

The function is used by [UnderExposurePrevention](#) and [OverExposurePrevention](#) class. Details can be found in the ISO/IEC 29794-5 standard.

Parameters

<i>session</i>	Session object containing the original facial image and pre-processing results computed by the OFIQImpl::preprocess() method
<i>exposureRange</i>	Range of pixels for which the aspect is computed.

Returns

Exposure computed from the inputs.

6.3.3.3 calculateEyeCenter()

```
void OFIQ_LIB::calculateEyeCenter (
    const OFIQ::FaceLandmarks & faceLandmarks,
    Point2f & leftEyeCenter,
    Point2f & rightEyeCenter )
```

Based on face landmarks the center of the left and right eye are computed.

Parameters

<i>faceLandmarks</i>	Input face landmarks.
<i>leftEyeCenter</i>	Point coordinates of the left eye center.
<i>rightEyeCenter</i>	Point coordinates of the right eye center.

6.3.3.4 CalculateReferencePoints()

```
void OFIQ_LIB::CalculateReferencePoints (
    const OFIQ::FaceLandmarks & landmarks,
    OFIQ::LandmarkPoint & leftEyeCenter,
    OFIQ::LandmarkPoint & rightEyeCenter,
    double & interEyeDistance,
    double & eyeMouthDistance )
```

Computes the left eye center, the right eye center, the (planar) inter-eye-distance and the eye to mouth distance from facial landmarks.

Parameters

in	<i>landmarks</i>	Facial landmarks
out	<i>leftEyeCenter</i>	Left eye center computed from landmarks
out	<i>rightEyeCenter</i>	Right eye center computed from landmarks
out	<i>interEyeDistance</i>	Inter-eye distance computed from landmarks (does not consider the yaw angle).
out	<i>eyeMouthDistance</i>	Distance from the eyes' midpoint to the mouth.

6.3.3.5 CalculateRegionOfInterest()

```
void OFIQ_LIB::CalculateRegionOfInterest (
    cv::Rect & leftRegionOfInterest,
    cv::Rect & rightRegionOfInterest,
    const OFIQ::LandmarkPoint & leftEyeCenter,
    const OFIQ::LandmarkPoint & rightEyeCenter,
    const double interEyeDistance,
    const double eyeMouthDistance )
```

Extracts regions being of interest for some measures (e.g. NaturalColour).

Applies a heuristic to estimate two regions being of interest for the natural colour measure. More details can be found in the ISO/IEC 29794-5 standard for the Natural colour measure.

Parameters

out	<i>leftRegionOfInterest</i>	Rectangular region corresponding to the left eye
out	<i>rightRegionOfInterest</i>	Rectangular region corresponding to the right eye
in	<i>leftEyeCenter</i>	Center of the left eye
in	<i>rightEyeCenter</i>	Center of the right eye
in	<i>interEyeDistance</i>	Planar inter-eye distance
in	<i>eyeMouthDistance</i>	Distance from the eyes' centers midpoint to the mouth

6.3.3.6 ColorConvert()

```
double OFIQ_LIB::ColorConvert (
    double v )
```

Converts a color as specified in ISO/IEC 29794-5.

Parameters

v	An intensity value between 0 (black) and 1 (white).
-----	---

Returns

If $v > 0.04045$, then $((v + 0.055)/1.055)^{2.4}$ is returned; otherwise, if $v \leq 0.04045$, then $v/12.92$ is returned.

6.3.3.7 ComputeBrightnessAspect()

```
double OFIQ_LIB::ComputeBrightnessAspect (
    const cv::Mat & luminanceImage,
    const cv::Mat & maskImage,
    const ExposureRange & exposureRange )
```

Helper function for some measures.

The function is used by [UnderExposurePrevention](#) and [OverExposurePrevention](#) class. Details can be found in the ISO/IEC 29794-5 standard.

Parameters

<i>luminanceImage</i>	luminance image.
<i>maskImage</i>	The mask on which the aspect is computed
<i>exposureRange</i>	Range of pixels for which the aspect is computed.

Returns

Brightness aspect computed from the inputs.

6.3.3.8 ConvertBGRToCIELAB()

```
void OFIQ_LIB::ConvertBGRToCIELAB (
    const cv::Mat & bgrImage,
    double & a,
    double & b )
```

Computes CIELAB values a^* and b^* from a BGR image.

Parameters

in	<i>bgrImage</i>	BGR image
out	<i>a</i>	CIELAB value a^*
out	<i>b</i>	CIELAB value b^*

6.3.3.9 copyToCvImage()

```
cv::Mat OFIQ_LIB::copyToCvImage (
    const OFIQ::Image & sourceImage,
    bool asGrayImage = false )
```

Convert images in OFIQ::Image format into the OpenCV cv::Mat format. The image can be converted from color to gray scale by setting the parameter asGrayImage to true.

Parameters

<i>sourceImage</i>	Input image.
<i>asGrayImage</i>	Switch for adding gray scale conversion.

Returns

cv::Mat Input image in cv::Mat format.

6.3.3.10 Cubic()

```
double OFIQ_LIB::Cubic (
    double x,
    double k,
    double eps )
```

Cubic flattening function.

Parameters

<i>x</i>	Argument
<i>k</i>	Argument
<i>eps</i>	Argument ϵ

Returns

If $x \leq \epsilon$, then $(k \cdot x + 16)/116$ is returned; otherwise, if $x > \epsilon$, then $\sqrt[3]{x}$ is returned.

6.3.3.11 findLandmarksBoundingBox()

```
OFIQ::BoundingBox OFIQ_LIB::findLandmarksBoundingBox (
    const OFIQ::Landmarks & landmarks )
```

Find the bounding box that encloses the given landmarks.

Parameters

<i>landmarks</i>	Face landmarks, a vector containing the landmark points.
------------------	--

Returns

OFIQ::BoundingBox Bounding box enclosing the given landmarks

6.3.3.12 findLargestBoundingBox()

```
size_t OFIQ_LIB::findLargestBoundingBox (
    const std::vector< OFIQ::BoundingBox > & faceRects )
```

This function returns the position of the largest bounding box (largest in terms of area) from a vector of bounding boxes.

Parameters

<i>faceRects</i>	Vector containing bounding boxes.
------------------	-----------------------------------

Returns

size_t Position of the largest bounding box in the vector.

6.3.3.13 GetLuminanceImageFromBGR()

```
cv::Mat OFIQ_LIB::GetLuminanceImageFromBGR (
    const cv::Mat & bgrImage )
```

Converts a BGR image to the luminance image.

The conversion is specified in the ISO/IEC 29794-5 standard and uses the function [ColorConvert\(\)](#) .

Parameters

<i>bgrImage</i>	BGR image
-----------------	-----------

Returns

Luminance image.

6.3.3.14 GetNormalizedHistogram()

```
void OFIQ_LIB::GetNormalizedHistogram (
    const cv::Mat & luminanceImage,
    const cv::Mat & maskImage,
    cv::Mat1f & histogram )
```

Computes the normalized histogram from a luminance image in 256 chunks.

Parameters

in	<i>luminanceImage</i>	Luminance image as returned by GetLuminanceImageFromBGR() .
in	<i>maskImage</i>	The histogram is computed on pixels where the values of maskImage are non-zero.
out	<i>histogram</i>	Array of length 256 where the histogram is stored.

6.3.3.15 MakeGreyImage()

```
OFIQ::Image OFIQ_LIB::MakeGreyImage (
    uint16_t width,
    uint16_t height )
```

This function generates a gray scaled image with the resolution passed by the call.

Parameters

<i>width</i>	Width of the generated image.
<i>height</i>	Height of the generated image.

Returns

OFIQ::Image Generated gray scaled image.

6.3.3.16 makeSquareBoundingBox()

```
OFIQ::BoundingBox OFIQ_LIB::makeSquareBoundingBox (
    const OFIQ::BoundingBox & i_bb )
```

This function converts a non-squared bounding box into a squared one. The side length is defined by the greater one of height or width.

Parameters

<i>i_bb</i>	Input bounding box.
-------------	---------------------

Returns

OFIQ::BoundingBox Squarred bounding box.

6.3.3.17 makeSquareBoundingBoxWithPadding()

```
void OFIQ_LIB::makeSquareBoundingBoxWithPadding (
    const OFIQ::BoundingBox & i_bb,
    const cv::Mat & i_input_image,
    cv::Mat & o_output_image,
```

```
OFIQ::BoundingBox & o_bb,
Point2i & o_translation_vector )
```

Some computations, especially neural networks, need a squarred image as input. This funtion consumes a boundig box and an input image. The greater parameter of width or height is used to define the side length of the new squarred bounding box. The face will be centered in the bounding box. Padding is added if needed. The squarred bounding box is used generate a new cropped image, the o_output_image. Required translations are described by the translation vector o_translation_vector.

Parameters

<i>i_bb</i>	Initial bounding box.
<i>i_input_image</i>	Input image.
<i>o_output_image</i>	Cropped output image. Cropping is based on the computed squarred bounding box.
<i>o_bb</i>	Squarred bounding box.
<i>o_translation_vector</i>	Translation vector.

6.3.3.18 readImage()

```
OFIQ_EXPORT OFIQ::ReturnStatus OFIQ_LIB::readImage (
    const std::string & filename,
    OFIQ::Image & image )
```

Read image from disk.

Parameters

in	<i>filename</i>	Path and file name of the image being read from disk.
out	<i>image</i>	Reference to the image object where the data is loaded to.

Returns

OFIQ::ReturnStatus

6.3.3.19 readImageFromByteArray()

```
OFIQ_EXPORT OFIQ::ReturnStatus OFIQ_LIB::readImageFromByteArray (
    const std::vector< unsigned char > & buffer,
    OFIQ::Image & image )
```

Read image from byte array.

Parameters

in	<i>buffer</i>	Data as byte array of the image being read.
out	<i>image</i>	Reference to the image object where the data is loaded to.

Returns

OFIQ::ReturnStatus

6.3.3.20 SetAAssetManager()

```
void OFIQ_LIB::SetAAssetManager (
    ::AAssetManager * assetManager )
```

6.3.3.21 tmetric()

```
float OFIQ_LIB::tmetric (
    const OFIQ::FaceLandmarks & faceLandmarks )
```

Based on the provided landmarks this function computes the distance between the point between the eyes and the chin.

Parameters

<i>faceLandmarks</i>	Input face landmarks.
----------------------	-----------------------

Returns

float Computed distance.

6.4 OFIQ_LIB::modules Namespace Reference**Namespaces**

- namespace [detectors](#)
Provides face detector implementations.
- namespace [landmarks](#)
Provides implementations of a landmark extractors.
- namespace [measures](#)
Provides measures implemented in OFIQ.
- namespace [poseEstimators](#)
Provides implementation of a head pose estimator.
- namespace [segmentations](#)
Provides segmentation-related implementations.

6.5 OFIQ_LIB::modules::detectors Namespace Reference

Provides face detector implementations.

Classes

- class [SSDFaceDetector](#)
Implementation of a face detector using the SSD face detector CNN.

6.5.1 Detailed Description

Provides face detector implementations.

6.6 OFIQ_LIB::modules::landmarks Namespace Reference

Provides implementations of a landmark extractors.

Namespaces

- namespace [adnet](#)
Namespace for ADNet-specific landmarks.

Classes

- class [ADNetFaceLandmarkExtractor](#)
Class implementing the [FaceLandmarkExtractorInterface](#) interface.
- class [FaceMeasures](#)
Provides static functions doing computations with landmarks.
- struct [LandmarkPair](#)
Data container for storing pairs of landmarks.
- class [PartExtractor](#)
Class that provides helper methods for the administration of landmarks.

Typedefs

- using [LandmarkId](#) = int
Type definition of a landmark index.
- using [LandmarkIds](#) = std::vector<[LandmarkId](#)>
Type definition of a list of landmark indices.
- using [FaceMap](#) = std::map<[FaceParts](#), [LandmarkIds](#)>
Type definition of a face map to access landmark indices for a queried face part.
- using [LandmarkIdPair](#) = std::array<[LandmarkId](#), 2>
Type definition for a pair of landmark index.
- using [LandmarkIdPairs](#) = std::vector<[LandmarkIdPair](#)>
Type definition for a list of landmark index pairs.
- using [FacePairMap](#) = std::map<[FaceParts](#), [LandmarkIdPairs](#)>
Structure defining pairs of landmark indices.

Enumerations

- enum class [FaceParts](#) {
[LEFT_EYE](#) , [RIGHT_EYE](#) , [LEFT_EYE_CORNERS](#) , [RIGHT_EYE_CORNERS](#) ,
[MOUTH_OUTER](#) , [MOUTH_INNER](#) , [FACE_CONTOUR](#) , [MOUTH_CENTER](#) ,
[CHIN](#) , [NOSETIP](#) , [FOREHEAD](#) }

Enumeration of facial landmark parts.

6.6.1 Detailed Description

Provides implementations of a landmark extractors.

Provides implementations for computations with landmarks.

Provides implementations related to facial landmarks.

6.6.2 Typedef Documentation

6.6.2.1 FaceMap

```
using OFIQ_LIB::modules::landmarks::FaceMap = std::map<FaceParts, LandmarkIds>
```

Type definition of a face map to access landmark indices for a queried face part.

6.6.2.2 FacePairMap

```
using OFIQ_LIB::modules::landmarks::FacePairMap = std::map<FaceParts, LandmarkIdPairs>
```

Structure defining pairs of landmark indices.

6.6.2.3 LandmarkId

```
using OFIQ_LIB::modules::landmarks::LandmarkId = int
```

Type definition of a landmark index.

6.6.2.4 LandmarkIdPair

```
using OFIQ_LIB::modules::landmarks::LandmarkIdPair = std::array<LandmarkId, 2>
```

Type definition for a pair of landmark index.

6.6.2.5 LandmarkIdPairs

```
using OFIQ_LIB::modules::landmarks::LandmarkIdPairs = std::vector<LandmarkIdPair>
```

Type definition for a list of landmark index pairs.

6.6.2.6 LandmarkIds

```
using OFIQ_LIB::modules::landmarks::LandmarkIds = std::vector<LandmarkId>
```

Type definition of a list of landmark indices.

6.6.3 Enumeration Type Documentation

6.6.3.1 FaceParts

```
enum class OFIQ_LIB::modules::landmarks::FaceParts [strong]
```

Enumeration of facial landmark parts.

Enumerator

LEFT_EYE	left as seen on the image, it's the persons right eye
RIGHT_EYE	right as seen on the image, it's the persons left eye
LEFT_EYE_CORNERS	two landmarks - outer, inner corner
RIGHT_EYE_CORNERS	two landmarks - outer, inner corner
MOUTH_OUTER	outer landmarks of mouth
MOUTH_INNER	inner landmarks of mouth
FACE_CONTOUR	contour of the face
MOUTH_CENTER	center of the mouth
CHIN	chin
NOSETIP	noisetip
FOREHEAD	forehead

6.7 OFIQ_LIB::modules::landmarks::adnet Namespace Reference

Namespace for ADNet-specific landmarks.

Variables

- const [LandmarkIds leftEye](#) {60,61,62,63,64,65,66,67}
Landmark indices (ADNet) of the left eye.
- const [LandmarkIds rightEye](#) {68,69,70,71,72,73,74,75}
Landmark indices (ADNet) of the right eye.
- const [LandmarkIds leftEyeCorners](#) {60,64}
Landmark indices (ADNet) of the left eyes' corners.
- const [LandmarkIds rightEyeCorners](#) {68,72}
Landmark indices (ADNet) of the right eyes' corners.
- const [LandmarkIds noisetip](#) {54}
Landmark index (ADNet) of the nose tip.
- const [LandmarkIds mouthOuter](#) {76,77,78,79,80,81,82,83,84,85,86,87}
Landmark indices (ADNet) on the mouth's outer contour.

- const [LandmarkIds](#) `mouthInner` {88,89,90,91,92,93,94,95}
Landmark indices (ADNet) on the mouth's inner lip borders.
- const [LandmarkIds](#) `contour` {0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32}
Landmark indices (ADNet) of the face contour.
- const [LandmarkIds](#) `forehead` {}
Landmark indices (ADNet) of the forehead (empty for ADNet).
- const [LandmarkIds](#) `chin` {16}
Landmark index (ADNet) of the chin.
- const [landmarks::FaceMap](#) `FaceMap`
ADNets face map definition.
- const [LandmarkIdPairs](#) `pairsLeftEye`
Pair indices of landmarks (ADNet) for the left eye.
- const [LandmarkIdPairs](#) `pairsRightEye`
Landmark index pairs (ADNet) of landmarks for the right eye.
- const [LandmarkIdPairs](#) `pairsInnerLip`
Landmark index pairs (ADNet) of inner lip pairs.
- const [LandmarkIdPairs](#) `pairsMouthCenter`
Landmark index pair (ADNet) of the inner mouth (lips) center.
- const [landmarks::FacePairMap](#) `FacePairMap`
ADNets face pair map definition.

6.7.1 Detailed Description

Namespace for ADNet-specific landmarks.

6.7.2 Variable Documentation

6.7.2.1 chin

```
const LandmarkIds OFIQ_LIB::modules::landmarks::adnet::chin {16}
```

Landmark index (ADNet) of the chin.

6.7.2.2 contour

```
const LandmarkIds OFIQ_LIB::modules::landmarks::adnet::contour {0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18}
```

Landmark indices (ADNet) of the face contour.

6.7.2.3 FaceMap

```
const landmarks::FaceMap OFIQ_LIB::modules::landmarks::adnet::FaceMap
```

Initial value:

```
{
    {FaceParts::LEFT_EYE,      leftEye      },
    {FaceParts::RIGHT_EYE,    rightEye    },
    {FaceParts::LEFT_EYE_CORNERS, leftEyeCorners},
    {FaceParts::RIGHT_EYE_CORNERS, rightEyeCorners},
    {FaceParts::MOUTH_OUTER,   mouthOuter  },
    {FaceParts::MOUTH_INNER,   mouthInner },
    {FaceParts::FACE_CONTOUR,  contour   },
    {FaceParts::CHIN,          chin         },
    {FaceParts::NOSETIP,       nosetip    },
    {FaceParts::FOREHEAD,     forehead   }
}
```

ADNets face map definition.

6.7.2.4 FacePairMap

```
const LandmarkPairs OFIQ_LIB::modules::landmarks::adnet::FacePairMap
```

Initial value:

```
{
    {FaceParts::LEFT_EYE,    pairsLeftEye    },
    {FaceParts::RIGHT_EYE,   pairsRightEye   },
    {FaceParts::MOUTH_INNER, pairsInnerLip    },
    {FaceParts::MOUTH_CENTER, pairsMouthCenter}
}
```

ADNets face pair map definition.

6.7.2.5 forehead

```
const LandmarkIds OFIQ_LIB::modules::landmarks::adnet::forehead {}
```

Landmark indices (ADNet) of the forehead (empty for ADNet).

6.7.2.6 leftEye

```
const LandmarkIds OFIQ_LIB::modules::landmarks::adnet::leftEye {60,61,62,63,64,65,66,67}
```

Landmark indices (ADNet) of the left eye.

The left eye is defined as seen on the image; it is actually the person's right eye (physically).

6.7.2.7 leftEyeCorners

```
const LandmarkIds OFIQ_LIB::modules::landmarks::adnet::leftEyeCorners {60,64}
```

Landmark indices (ADNet) of the left eyes' corners.

6.7.2.8 mouthInner

```
const LandmarkIds OFIQ_LIB::modules::landmarks::adnet::mouthInner {88,89,90,91,92,93,94,95}
```

Landmark indices (ADNet) on the mouth's inner lip borders.

6.7.2.9 mouthOuter

```
const LandmarkIds OFIQ_LIB::modules::landmarks::adnet::mouthOuter {76,77,78,79,80,81,82,83,84,85,86,87}
```

Landmark indices (ADNet) on the mouth's outer contour.

6.7.2.10 nosetip

```
const LandmarkIds OFIQ_LIB::modules::landmarks::adnet::nosetip {54}
```

Landmark index (ADNet) of the nose tip.

6.7.2.11 pairsInnerLip

```
const LandmarkIdPairs OFIQ_LIB::modules::landmarks::adnet::pairsInnerLip
```

Initial value:

```
{
    { 89, 95 },
    { 90, 94 },
    { 91, 93 }
}
```

Landmark index pairs (ADNet) of inner lip pairs.

Useful to measure closedness of mouth.

6.7.2.12 pairsLeftEye

```
const LandmarkIdPairs OFIQ_LIB::modules::landmarks::adnet::pairsLeftEye
```

Initial value:

```
{
    { 61, 67 },
    { 62, 66 },
    { 63, 65 }
}
```

Pair indices of landmarks (ADNet) for the left eye.

Useful to measure eye openness.

6.7.2.13 pairsMouthCenter

```
const LandmarkIdPairs OFIQ_LIB::modules::landmarks::adnet::pairsMouthCenter
```

Initial value:

```
{
    { 90, 94 }
}
```

Landmark index pair (ADNet) of the inner mouth (lips) center.

Useful to measure closedness of mouth.

6.7.2.14 pairsRightEye

```
const LandmarkIdPairs OFIQ_LIB::modules::landmarks::adnet::pairsRightEye
```

Initial value:

```
{
    { 69, 75 },
    { 70, 74 },
    { 71, 73 }
}
```

Landmark index pairs (ADNet) of landmarks for the right eye.

Useful to measure eye openness.

6.7.2.15 rightEye

```
const LandmarkIds OFIQ_LIB::modules::landmarks::adnet::rightEye {68,69,70,71,72,73,74,75}
```

Landmark indices (ADNet) of the right eye.

The right eye is defined as seen on the image; it is actually the person's left eye (physically).

6.7.2.16 rightEyeCorners

```
const LandmarkIds OFIQ_LIB::modules::landmarks::adnet::rightEyeCorners {68,72}
```

Landmark indices (ADNet) of the right eyes' corners.

6.8 OFIQ_LIB::modules::measures Namespace Reference

Provides measures implemented in OFIQ.

Classes

- class [BackgroundUniformity](#)
Implementation of the background uniformity measure.
- class [CompressionArtifacts](#)
Implementation of the no compression artifacts measure.
- class [CropOfTheFaceImage](#)
Implementation of the crop of the face image measure.
- class [DynamicRange](#)
Implementation of the dynamic range measure.
- class [Executor](#)
This class takes care of the computation of the measures activated.
- class [ExpressionNeutrality](#)
Provides a class implementing the expression neutrality measure.
- class [EyesOpen](#)
Implementation of the eyes open measure.
- class [EyesVisible](#)
Implementation of the eyes visible measure.
- class [FaceOcclusionPrevention](#)
Implementation of the face occlusion prevention measure.
- class [HeadPose](#)
Implementation of head pose measures.
- class [HeadSize](#)
Implementation of the head size measure.
- class [IlluminationUniformity](#)
Implementation of the illumination uniformity measure.
- class [InterEyeDistance](#)
Implementation of the inter-eye distance measure.
- class [Luminance](#)
Implementation of two luminance measures.

- class [Measure](#)
Base class for measures implemented in OFIQ.
- class [MeasureFactory](#)
Measure factor class.
- class [MouthClosed](#)
Implementation of the mouth closed measure.
- class [MouthOcclusionPrevention](#)
Implementation of the mouth occlusion prevention measure.
- class [NaturalColour](#)
Implementation of the natural colour measure.
- class [NoHeadCoverings](#)
Implementation of the no head covering measure.
- class [OverExposurePrevention](#)
Implementation of the over-exposure prevention measure.
- class [Sharpness](#)
Implementation of the sharpness measure.
- struct [SigmoidParameters](#)
Parameters of the sigmoid function based quality mapping.
- class [SingleFacePresent](#)
Implementation of the single face present measure.
- class [UnderExposurePrevention](#)
Implementation of the under-exposure prevention measure.
- class [UnifiedQualityScore](#)
Implementation of the unified quality measure.

Functions

- void [log](#) (const std::string_view &msg)
Logging function for writing debug messages to std::cout.

Variables

- static const bool [ExecutorLogActive](#) = false
This variable enables logging to std::cout for debug purposes. By default the logging is switched off.

6.8.1 Detailed Description

Provides measures implemented in OFIQ.

6.8.2 Function Documentation

6.8.2.1 log()

```
void OFIQ_LIB::modules::measures::log (
    const std::string_view & msg )
```

Logging function for writing debug messages to std::cout.

Parameters

<i>msg</i>	Message to be logged.
------------	-----------------------

6.8.3 Variable Documentation

6.8.3.1 ExecutorLogActive

```
const bool OFIQ_LIB::modules::measures::ExecutorLogActive = false [static]
```

This variable enables logging to `std::cout` for debug purposes. By default the logging is switched off.

6.9 OFIQ_LIB::modules::poseEstimators Namespace Reference

Provides implementation of a head pose estimator.

Classes

- class [HeadPose3DDFAV2](#)
Implementation of a head pose estimator.

6.9.1 Detailed Description

Provides implementation of a head pose estimator.

6.10 OFIQ_LIB::modules::segmentations Namespace Reference

Provides segmentation-related implementations.

Classes

- class [FaceOcclusionSegmentation](#)
Class managing the separation of facial parts not occluded by non-facial parts from other parts.
- class [FaceParsing](#)
Class managing the separation of facial parts not occluded by non-facial parts from other parts.

Enumerations

- enum class [SegmentClassLabels](#) {
 [background](#) , [skin](#) , [l_brow](#) , [r_brow](#) ,
 [l_eye](#) , [r_eye](#) , [eye_g](#) , [l_ear](#) ,
 [r_ear](#) , [ear_r](#) , [nose](#) , [mouth](#) ,
 [u_lip](#) , [l_lip](#) , [neck](#) , [neck_l](#) ,
 [cloth](#) , [hair](#) , [hat](#) , [face](#) }
Enum class of the different face regions that can be segmented.

6.10.1 Detailed Description

Provides segmentation-related implementations.

Namespace for implementations related to facial segmentations.

6.10.2 Enumeration Type Documentation

6.10.2.1 SegmentClassLabels

```
enum class OFIQ_LIB::modules::segmentations::SegmentClassLabels [strong]
```

Enum class of the different face regions that can be segmented.

Enumerator

background	background label
skin	skin label
l_brow	left eye brow
r_brow	right eye brow
l_eye	left eye
r_eye	right eye brow
eye_g	eye glasses
l_ear	left ear
r_ear	right eye brow
ear_r	earring
nose	nose
mouth	mouth
u_lip	upper lip
l_lip	lower lip
neck	neck
neck_l	necklace
cloth	clothing
hair	hair
hat	head covering
face	face

Chapter 7

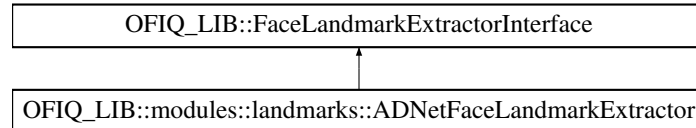
Class Documentation

7.1 OFIQ_LIB::modules::landmarks::ADNetFaceLandmarkExtractor Class Reference

Class implementing the [FaceLandmarkExtractorInterface](#) interface.

```
#include <adnet_landmarks.h>
```

Inheritance diagram for OFIQ_LIB::modules::landmarks::ADNetFaceLandmarkExtractor:



Public Member Functions

- [ADNetFaceLandmarkExtractor](#) (const [Configuration](#) &config)
Constructor.
- [~ADNetFaceLandmarkExtractor](#) () override
Destructor.

Public Member Functions inherited from [OFIQ_LIB::FaceLandmarkExtractorInterface](#)

- virtual [~FaceLandmarkExtractorInterface](#) ()=default
Destructor.
- [OFIQ::FaceLandmarks](#) [extractLandmarks](#) ([OFIQ_LIB::Session](#) &session)
Public method to extract landmarks from the image passed in the session object.

Protected Member Functions

- [OFIQ::FaceLandmarks](#) [updateLandmarks](#) ([OFIQ_LIB::Session](#) &session) override
Computes landmarks of the face detected in the session.

Private Attributes

- `std::unique_ptr< ADNetFaceLandmarkExtractorImpl > landmarkExtractor_`
Encapsulated implementation class.

7.1.1 Detailed Description

Class implementing the [FaceLandmarkExtractorInterface](#) interface.

7.1.2 Constructor & Destructor Documentation

7.1.2.1 ADNetFaceLandmarkExtractor()

```
OFIQ_LIB::modules::landmarks::ADNetFaceLandmarkExtractor::ADNetFaceLandmarkExtractor (
    const Configuration & config ) [explicit]
```

Constructor.

Parameters

<i>config</i>	Configuration object
---------------	----------------------

7.1.2.2 ~ADNetFaceLandmarkExtractor()

```
OFIQ_LIB::modules::landmarks::ADNetFaceLandmarkExtractor::~~ADNetFaceLandmarkExtractor ( )
[override]
```

Destructor.

7.1.3 Member Function Documentation

7.1.3.1 updateLandmarks()

```
OFIQ::FaceLandmarks OFIQ_LIB::modules::landmarks::ADNetFaceLandmarkExtractor::updateLandmarks
(
    OFIQ\_LIB::Session & session ) [override], [protected], [virtual]
```

Computes landmarks of the face detected in the session.

The landmarks are computed using ADNet.

Parameters

<i>session</i>	Session object containing preprocessing results used by the function to compute the landmarks.
----------------	--

Returns

Facial landmarks.

Implements [OFIQ_LIB::FaceLandmarkExtractorInterface](#).

7.1.4 Member Data Documentation**7.1.4.1 landmarkExtractor_**

```
std::unique_ptr<ADNetFaceLandmarkExtractorImpl> OFIQ_LIB::modules::landmarks::ADNetFaceLandmarkExtractor::landmarkExtractor_ [private]
```

Encapsulated implementation class.

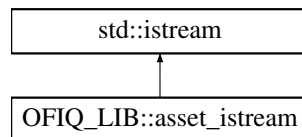
The documentation for this class was generated from the following file:

- [adnet_landmarks.h](#)

7.2 OFIQ_LIB::asset_istream Class Reference

```
#include <DataStreamAndroid.h>
```

Inheritance diagram for OFIQ_LIB::asset_istream:

**Public Member Functions**

- [asset_istream](#) (std::string const &filePath, std::ios_base::openmode mode=std::ios_base::in)
- virtual [~asset_istream](#) ()

Private Attributes

- [::AAsset * asset](#)
- [asset_streambuf streambuf](#)

7.2.1 Constructor & Destructor Documentation**7.2.1.1 asset_istream()**

```
OFIQ_LIB::asset_istream::asset_istream (
    std::string const & filePath,
    std::ios_base::openmode mode = std::ios_base::in )
```

7.2.1.2 ~asset_istream()

```
virtual OFIQ_LIB::asset_istream::~~asset_istream ( ) [virtual]
```

7.2.2 Member Data Documentation

7.2.2.1 asset

```
::AAsset* OFIQ_LIB::asset_istream::asset [private]
```

7.2.2.2 streambuf

```
asset_streambuf OFIQ_LIB::asset_istream::streambuf [private]
```

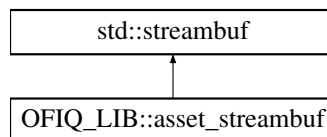
The documentation for this class was generated from the following file:

- [DataStreamAndroid.h](#)

7.3 OFIQ_LIB::asset_streambuf Class Reference

```
#include <DataStreamAndroid.h>
```

Inheritance diagram for OFIQ_LIB::asset_streambuf:



Public Member Functions

- [asset_streambuf](#) (::AAsset *asset)
- virtual [~asset_streambuf](#) ()=default

Protected Member Functions

- virtual pos_type [seekoff](#) (off_type off, std::ios_base::seekdir dir, std::ios_base::openmode which) override
- virtual pos_type [seekpos](#) (pos_type pos, std::ios_base::openmode which) override

7.3.1 Constructor & Destructor Documentation

7.3.1.1 asset_streambuf()

```
OFIQ_LIB::asset_streambuf::asset_streambuf (
    ::AAsset * asset )
```

7.3.1.2 ~asset_streambuf()

```
virtual OFIQ_LIB::asset_streambuf::~~asset_streambuf ( ) [virtual], [default]
```

7.3.2 Member Function Documentation

7.3.2.1 seekoff()

```
virtual pos_type OFIQ_LIB::asset_streambuf::seekoff (
    off_type off,
    std::ios_base::seekdir dir,
    std::ios_base::openmode which ) [override], [protected], [virtual]
```

7.3.2.2 seekpos()

```
virtual pos_type OFIQ_LIB::asset_streambuf::seekpos (
    pos_type pos,
    std::ios_base::openmode which ) [override], [protected], [virtual]
```

The documentation for this class was generated from the following file:

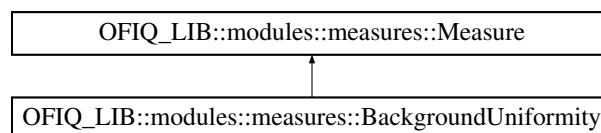
- [DataStreamAndroid.h](#)

7.4 OFIQ_LIB::modules::measures::BackgroundUniformity Class Reference

Implementation of the background uniformity measure.

```
#include <BackgroundUniformity.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::BackgroundUniformity:



Public Member Functions

- [BackgroundUniformity](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) (OFIQ_LIB::Session &session) override
Assesses uniformity of the background.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Private Attributes

- uint16_t [m_targetHeight](#) = 292
The aligned image and the face parsing mask is brought to an image of the target height before gradient computations and assessment is applied.
- uint16_t [m_targetWidth](#) = 354
The aligned image and the face parsing mask is brought to an image of the target width before gradient computations and assessment is applied.
- uint16_t [m_cropLeft](#) = 62
Crop from the left side of the aligned image (returned by [OFIQ_LIB::Session::getAlignedFace\(\)](#)) before scaling to the target dimension is applied.
- uint16_t [m_cropRight](#) = 62
Crop from the right side of the aligned image (returned by [OFIQ_LIB::Session::getAlignedFace\(\)](#)) before scaling to the target dimension is applied.
- uint16_t [m_cropTop](#) = 0
Crop from the top of the aligned image (returned by [OFIQ_LIB::Session::getAlignedFace\(\)](#)) before scaling to the target dimension is applied.
- uint16_t [m_cropBottom](#) = 210
Crop from the bottom of the aligned image (returned by [OFIQ_LIB::Session::getAlignedFace\(\)](#)) before scaling to the target dimension is applied.
- uint16_t [m_erosionKernelSize](#) = 4
Size of the erosion kernel applied to the background as per [OFIQ_LIB::modules::segmentations::FaceParsing](#) to reduce the risk that background uniformity estimation is applied to part of the subject.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from OFIQ_LIB::modules::measures::Measure

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from OFIQ_LIB::modules::measures::Measure

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.4.1 Detailed Description

Implementation of the background uniformity measure.

Uniformity of the background is measured on basis of the mean of the gradients computed on the background as per face parsing (see [OFIQ_LIB::modules::segmentations::FaceParsing](#)).

7.4.2 Constructor & Destructor Documentation

7.4.2.1 BackgroundUniformity()

```
OFIQ_LIB::modules::measures::BackgroundUniformity::BackgroundUniformity (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

configuration	Configuration object from which the measure related configuration is read.
-------------------------------	--

7.4.3 Member Function Documentation

7.4.3.1 Execute()

```
void OFIQ_LIB::modules::measures::BackgroundUniformity::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Assesses uniformity of the background.

Assessment of the background uniformity is done by computing the mean of the background as per face parsing (see [OFIQ_LIB::modules::segmentations::FaceParsing](#)).

Parameters

session	Session object computed by the OFIQImpl::preprocess() .
-------------------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

7.4.4 Member Data Documentation

7.4.4.1 m_cropBottom

```
uint16_t OFIQ_LIB::modules::measures::BackgroundUniformity::m_cropBottom = 210 [private]
```

Crop from the bottom of the aligned image (returned by [OFIQ_LIB::Session::getAlignedFace\(\)](#)) before scaling to the target dimension is applied.

7.4.4.2 m_cropLeft

```
uint16_t OFIQ_LIB::modules::measures::BackgroundUniformity::m_cropLeft = 62 [private]
```

Crop from the left side of the aligned image (returned by [OFIQ_LIB::Session::getAlignedFace\(\)](#)) before scaling to the target dimension is applied.

7.4.4.3 m_cropRight

```
uint16_t OFIQ_LIB::modules::measures::BackgroundUniformity::m_cropRight = 62 [private]
```

Crop from the right side of the aligned image (returned by [OFIQ_LIB::Session::getAlignedFace\(\)](#)) before scaling to the target dimension is applied.

7.4.4.4 m_cropTop

```
uint16_t OFIQ_LIB::modules::measures::BackgroundUniformity::m_cropTop = 0 [private]
```

Crop from the top of the aligned image (returned by [OFIQ_LIB::Session::getAlignedFace\(\)](#)) before scaling to the target dimension is applied.

7.4.4.5 m_erosionKernelSize

```
uint16_t OFIQ_LIB::modules::measures::BackgroundUniformity::m_erosionKernelSize = 4 [private]
```

Size of the erosion kernel applied to the background as per [OFIQ_LIB::modules::segmentations::FaceParsing](#) to reduce the risk that background uniformity estimation is applied to part of the subject.

7.4.4.6 m_targetHeight

```
uint16_t OFIQ_LIB::modules::measures::BackgroundUniformity::m_targetHeight = 292 [private]
```

The aligned image and the face parsing mask is brought to an image of the target height before gradient computations and assessment is applied.

7.4.4.7 m_targetWidth

```
uint16_t OFIQ_LIB::modules::measures::BackgroundUniformity::m_targetWidth = 354 [private]
```

The aligned image and the face parsing mask is brought to an image of the target width before gradient computations and assessment is applied.

The documentation for this class was generated from the following file:

- [BackgroundUniformity.h](#)

7.5 BoundingBox Struct Reference

Data structure for describing bounding boxes, e.g. the face region of the faces found by a face detector.

```
#include <ofiq_structs_c.h>
```

Public Attributes

- `int16_t xleft {-1}`
leftmost point on head, typically subject's right ear value must be on [0, imageWidth-1]
- `int16_t ytop {-1}`
high point of head, typically top of hair; value must be on [0, imageHeight-1]
- `int16_t width {-1}`
bounding box width
- `int16_t height {-1}`
bounding box height
- `FaceDetectorType faceDetector = FaceDetectorType::NotSet`
Description of the face detector used.

7.5.1 Detailed Description

Data structure for describing bounding boxes, e.g. the face region of the faces found by a face detector.

7.5.2 Member Data Documentation

7.5.2.1 faceDetector

```
FaceDetectorType BoundingBox::faceDetector = FaceDetectorType::NotSet
```

Description of the face detector used.

7.5.2.2 height

```
int16_t BoundingBox::height {-1}
```

bounding box height

7.5.2.3 width

```
int16_t BoundingBox::width {-1}
```

bounding box width

7.5.2.4 xleft

```
int16_t BoundingBox::xleft {-1}
```

leftmost point on head, typically subject's right ear value must be on [0, imageWidth-1]

7.5.2.5 ytop

```
int16_t BoundingBox::ytop {-1}
```

high point of head, typically top of hair; value must be on [0, imageHeight-1]

The documentation for this struct was generated from the following file:

- [ofiq_structs.c.h](#)

7.6 OFIQ::BoundingBox Struct Reference

Data structure for descibing bounding boxes, e.g. the face region of the faces found by a face detector.

```
#include <ofiq_structs.h>
```

Public Member Functions

- [BoundingBox](#) ()=default
Default constructor.
- [BoundingBox](#) (int16_t [xleft](#), int16_t [ytop](#), int16_t [width](#), int16_t [height](#), [FaceDetectorType](#) i_faceDetector)
Parameterized constructor.

Public Attributes

- int16_t [xleft](#) { -1 }
leftmost point on head, typically subject's right ear value must be on [0, imageWidth-1]
- int16_t [ytop](#) { -1 }
high point of head, typically top of hair; value must be on [0, imageHeight-1]
- int16_t [width](#) { -1 }
bounding box width
- int16_t [height](#) { -1 }
bounding box height
- [FaceDetectorType](#) [faceDetector](#) = [FaceDetectorType::NotSet](#)
Description of the face detector used.

7.6.1 Detailed Description

Data structure for descibing bounding boxes, e.g. the face region of the faces found by a face detector.

7.6.2 Constructor & Destructor Documentation

7.6.2.1 BoundingBox() [1/2]

```
OFIQ::BoundingBox::BoundingBox ( ) [default]
```

Default constructor.

7.6.2.2 BoundingBox() [2/2]

```
OFIQ::BoundingBox::BoundingBox (
    int16_t xleft,
    int16_t ytop,
    int16_t width,
    int16_t height,
    FaceDetectorType i_faceDetector ) [inline]
```

Parameterized constructor.

Parameters

<i>xleft</i>	x coordinate of the upper left point of the bounding box.
<i>ytop</i>	y coordinate of the upper left point of the bounding box.
<i>width</i>	width of the bounding box.
<i>height</i>	height of the bounding box.
<i>i_faceDetector</i>	used face detector.

7.6.3 Member Data Documentation

7.6.3.1 faceDetector

```
FaceDetectorType OFIQ::BoundingBox::faceDetector = FaceDetectorType::NotSet
```

Description of the face detector used.

7.6.3.2 height

```
int16_t OFIQ::BoundingBox::height { -1 }
```

bounding box height

7.6.3.3 width

```
int16_t OFIQ::BoundingBox::width { -1 }
```

bounding box width

7.6.3.4 xleft

```
int16_t OFIQ::BoundingBox::xleft { -1 }
```

leftmost point on head, typically subject's right ear value must be on [0, imageWidth-1]

7.6.3.5 ytop

```
int16_t OFIQ::BoundingBox::ytop { -1 }
```

high point of head, typically top of hair; value must be on [0, imageHeight-1]

The documentation for this struct was generated from the following file:

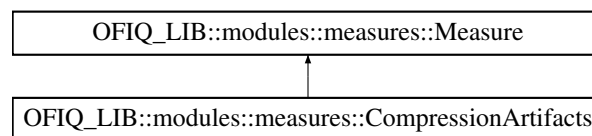
- [ofiq_structs.h](#)

7.7 OFIQ_LIB::modules::measures::CompressionArtifacts Class Reference

Implementation of the no compression artifacts measure.

```
#include <CompressionArtifacts.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::CompressionArtifacts:



Public Member Functions

- [CompressionArtifacts](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) (OFIQ_LIB::Session &session) override
Assesses absence of compression artifacts.

Public Member Functions inherited from OFIQ_LIB::modules::measures::Measure

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Private Attributes

- int [m_crop](#)
Top, right, left, and bottom margin by which the aligned image is cropped.
- int [m_dim](#)
Target dimension of cropped image being scaled before input to the CNN .
- [ONNXRuntimeEnv](#) [m_onnxRuntimeEnv](#)
Manages CNN estimations.

Additional Inherited Members

Protected Member Functions inherited from OFIQ_LIB::modules::measures::Measure

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from OFIQ_LIB::modules::measures::Measure

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from OFIQ_LIB::modules::measures::Measure

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.7.1 Detailed Description

Implementation of the no compression artifacts measure.

Assessment of the absence of compression artifact (both JPEG and JPEG2000) based on a CNN trained by the OFIQ development team.

7.7.2 Constructor & Destructor Documentation

7.7.2.1 CompressionArtifacts()

```
OFIQ_LIB::modules::measures::CompressionArtifacts::CompressionArtifacts (
    const Configuration & configuration ) [explicit]
```

Constructor.

The configuration parameter must contain the following entry:

- `params.measures.CompressionArtifacts.model_path`: Path to the CNN model file in ONNX format.

The following entries can (but do not need to) be configured.

- `params.measures.CompressionArtifacts.dim`: If configured, the value must be 248 which corresponds to the dimension of the CNN's input layer; other values will result in an error being thrown when OFIQ's CNN is invoked.
- `params.measures.CompressionArtifacts.crop`: Top, right, bottom, and left margin by which the aligned input image will be cropped before being scaled to the target dimension input of the CNN.

Parameters

<code>configuration</code>	Configuration object from which measure-related configuration is read.
----------------------------	--

Exceptions

<code>OFIQ_LIB::OFIQError</code>	if no valid model path is configured.
----------------------------------	---------------------------------------

7.7.3 Member Function Documentation

7.7.3.1 Execute()

```
void OFIQ_LIB::modules::measures::CompressionArtifacts::Execute (
    OFIQ_LIB::Session & session ) [override], [virtual]
```

Assesses absence of compression artifacts.

Assessment of the absence of compression artifact (both JPEG and JPEG2000) based on a CNN trained by the OFIQ development team.

Parameters

<code>session</code>	Session object computed by the OFIQImpl::preprocess() method.
----------------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

7.7.4 Member Data Documentation

7.7.4.1 m_crop

```
int OFIQ_LIB::modules::measures::CompressionArtifacts::m_crop [private]
```

Top, right, left, and bottom margin by which the aligned image is cropped.

The value can be configured by passing a corresponding configuration to the constructor.

Warning

The value should be 184 such that an aligned input image of dimension 616 x 616 is cropped to an image of dimension 248 x 248.

7.7.4.2 m_dim

```
int OFIQ_LIB::modules::measures::CompressionArtifacts::m_dim [private]
```

Target dimension of cropped image being scaled before input to the CNN .

The cropped image is scaled to the dimension m_dim x m_dim. The value can be configured by passing a corresponding configuration to the constructor.

Warning

The value should be 248; if configured differently, do not expect that the cropped image can be successfully be passed to the CNN.

7.7.4.3 m_onnxRuntimeEnv

```
ONNXRuntimeEnv OFIQ_LIB::modules::measures::CompressionArtifacts::m_onnxRuntimeEnv [private]
```

Manages CNN estimations.

The documentation for this class was generated from the following file:

- [CompressionArtifacts.h](#)

7.8 OFIQ_LIB::Configuration Class Reference

Configuration class.

```
#include <Configuration.h>
```

Public Member Functions

- [Configuration](#) (const std::string &configDir, const std::string &configFilename)
Constructor.
- bool [GetBool](#) (const std::string &key, bool &value) const
Accesses a boolean configuration.
- bool [GetString](#) (const std::string &key, std::string &value) const
Accesses a string configuration.
- bool [GetNumber](#) (const std::string &key, double &value) const
Accesses a double configuration.
- bool [GetStringList](#) (const std::string &key, std::vector< std::string > &value) const
Accesses an array of strings configured.
- bool [GetBool](#) (const std::string &key) const
Accesses a boolean configuration.
- std::string [GetString](#) (const std::string &key) const
Accesses a string configuration.
- double [GetNumber](#) (const std::string &key) const
Accesses a double configuration.
- std::string [getDataDir](#) () const
Access configuration directory.
- void [SetDataDir](#) (std::string_view dataDir)
Sets the configuration directory.
- std::string [GetFullPath](#) (const std::string &key) const
Accesses a full path of a string configuration.

Private Attributes

- std::map< std::string, tao::json::value, std::less<> > [parameters](#)
Map holding all configuration that can be accessed using a string key.
- std::filesystem::path [m_dataDir](#)
Path to the configuration directory.

7.8.1 Detailed Description

Configuration class.

The class consumes the [taoJSON](#) library. A configuration is read from a [JAXN](#)-formatted file.

7.8.2 Constructor & Destructor Documentation

7.8.2.1 Configuration()

```
OFIQ_LIB::Configuration::Configuration (
    const std::string & configDir,
    const std::string & configFilename )
```

Constructor.

Parameters

<i>configDir</i>	Directory from which a JAXN configuration is read. The path can be absolute or relative to the path of the current working directory.
<i>configFilename</i>	Name of the JAXN configuration file in <i>configDir</i> .

7.8.3 Member Function Documentation

7.8.3.1 GetBool() [1/2]

```
bool OFIQ_LIB::Configuration::GetBool (
    const std::string & key ) const
```

Accesses a boolean configuration.

Parameters

<i>key</i>	Key of the configuration.
------------	---------------------------

Returns

The accessed boolean configuration.

Exceptions

<i>OFIQ_LIB::OFIQError</i>	if the configuration was not successfully accessed.
----------------------------	---

7.8.3.2 GetBool() [2/2]

```
bool OFIQ_LIB::Configuration::GetBool (
    const std::string & key,
    bool & value ) const
```

Accesses a boolean configuration.

Parameters

<i>key</i>	Key of the configuration.
<i>value</i>	Boolean reference to where the configuration result is stored.

Returns

`true` if the configuration was successfully accessed; otherwise, if the configuration was not successfully accessed, the function returns `false`.

7.8.3.3 getDataDir()

```
std::string OFIQ_LIB::Configuration::getDataDir ( ) const
```

Access configuration directory.

The configuration directory has been specified in the constructor or afterwards by the [SetDataDir\(\)](#) method.

Returns

String representation of the configuration directory.

7.8.3.4 GetFullPath()

```
std::string OFIQ_LIB::Configuration::GetFullPath (
    const std::string & key ) const
```

Accesses a full path of a string configuration.

Parameters

<i>key</i>	Key of the configuration.
------------	---------------------------

Returns

The combined path of the configuration directory and accessed string configuration.

Exceptions

<i>OFIQ_LIB::OFIQError</i>	if the configuration was not successfully accessed.
----------------------------	---

7.8.3.5 GetNumber() [1/2]

```
double OFIQ_LIB::Configuration::GetNumber (
    const std::string & key ) const
```

Accesses a double configuration.

Parameters

<i>key</i>	Key of the configuration.
------------	---------------------------

Returns

The accessed double configuration.

Exceptions

<i>OFIQ_LIB::OFIQError</i>	if the configuration was not successfully accessed.
----------------------------	---

7.8.3.6 GetNumber() [2/2]

```
bool OFIQ_LIB::Configuration::GetNumber (
    const std::string & key,
    double & value ) const
```

Accesses a double configuration.

Parameters

<i>key</i>	Key of the configuration.
<i>value</i>	Reference to a double type to where the configuration result is stored.

Returns

`true` if the configuration was successfully accessed; otherwise, if the configuration was not successfully accessed, the function returns `false`.

7.8.3.7 GetString() [1/2]

```
std::string OFIQ_LIB::Configuration::GetString (
    const std::string & key ) const
```

Accesses a string configuration.

Parameters

<i>key</i>	Key of the configuration.
------------	---------------------------

Returns

The accessed string configuration.

Exceptions

<i>OFIQ_LIB::OFIQError</i>	if the configuration was not successfully accessed.
----------------------------	---

7.8.3.8 GetString() [2/2]

```
bool OFIQ_LIB::Configuration::GetString (
    const std::string & key,
    std::string & value ) const
```

Accesses a string configuration.

Parameters

<i>key</i>	Key of the configuration.
<i>value</i>	std::string reference to where the configuration result is stored.

Returns

`true` if the configuration was successfully accessed; otherwise, if the configuration was not successfully accessed, the function returns `false`.

7.8.3.9 GetStringList()

```
bool OFIQ_LIB::Configuration::GetStringList (
    const std::string & key,
    std::vector< std::string > & value ) const
```

Accesses an array of strings configured.

The string list will not be emptied; any strings read will be appended.

Parameters

<i>key</i>	Key of the configuration.
<i>value</i>	Reference to a string list to where the configuration result is stored.

Returns

`true` if the configuration was successfully accessed; otherwise, if the configuration was not successfully accessed, the function returns `false`.

7.8.3.10 SetDataDir()

```
void OFIQ_LIB::Configuration::SetDataDir (
    std::string_view dataDir )
```

Sets the configuration directory.

Parameters

<i>dataDir</i>	String representation of the configuration directory.
----------------	---

Attention

The configurations will not be updated when this method is called; it causes only an update of an internal private path member.

7.8.4 Member Data Documentation

7.8.4.1 m_dataDir

```
std::filesystem::path OFIQ_LIB::Configuration::m_dataDir [private]
```

Path to the configuration directory.

The member is set either by the constructor or by the [SetDataDir\(\)](#) method.

7.8.4.2 parameters

```
std::map<std::string, tao::json::value, std::less<> > OFIQ_LIB::Configuration::parameters
[private]
```

Map holding all configuration that can be accessed using a string key.

The documentation for this class was generated from the following file:

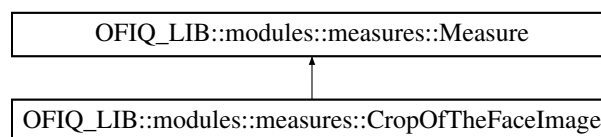
- [Configuration.h](#)

7.9 OFIQ_LIB::modules::measures::CropOfTheFacelImage Class Reference

Implementation of the crop of the face image measure.

```
#include <CropOfTheFaceImage.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::CropOfTheFacelImage:



Public Member Functions

- [CropOfTheFacelImage](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Implementation of the crop of the face image measure.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & configuration
Reference to the configuration with which the measure constructor has been invoked.

7.9.1 Detailed Description

Implementation of the crop of the face image measure.

The crop of the face images measures whether the face is centered on the input image by comparing the resolution of the image with the landmarks detected during the pre-processing.

7.9.2 Constructor & Destructor Documentation

7.9.2.1 CropOfTheFaceImage()

```
OFIQ_LIB::modules::measures::CropOfTheFaceImage::CropOfTheFaceImage (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.9.3 Member Function Documentation

7.9.3.1 Execute()

```
void OFIQ_LIB::modules::measures::CropOfTheFaceImage::Execute (
    OFIQ_LIB::Session & session ) [override], [virtual]
```

Implementation of the crop of the face image measure.

The crop of the face images measures whether the face is centered on the input image by comparing the resolution of the image with the landmarks detected during the pre-processing.

Parameters

<i>session</i>	Session object.
----------------	-----------------

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

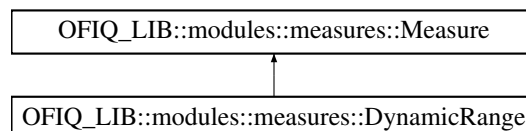
- [CropOfTheFacelImage.h](#)

7.10 OFIQ_LIB::modules::measures::DynamicRange Class Reference

Implementation of the dynamic range measure.

```
#include <DynamicRange.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::DynamicRange:



Public Member Functions

- [DynamicRange](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Assesses dynamic range.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & configuration
Reference to the configuration with which the measure constructor has been invoked.

7.10.1 Detailed Description

Implementation of the dynamic range measure.

The dynamic range is computed from the luminance histogram. of the facial image.

7.10.2 Constructor & Destructor Documentation

7.10.2.1 DynamicRange()

```
OFIQ_LIB::modules::measures::DynamicRange::DynamicRange (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which the measure-related configuration is read.
----------------------	--

7.10.3 Member Function Documentation

7.10.3.1 Execute()

```
void OFIQ_LIB::modules::measures::DynamicRange::Execute (
    OFIQ_LIB::Session & session ) [override], [virtual]
```

Assesses dynamic range.

Assessment of the dynamic range is computed from the luminance histogram.

Parameters

<i>session</i>	Session object.
----------------	-----------------

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

- [DynamicRange.h](#)

7.11 OFIQ_LIB::modules::measures::Executor Class Reference

This class takes care of the computation of the measures activated.

```
#include <Executor.h>
```

Public Member Functions

- [Executor](#) (std::vector< std::unique_ptr< [Measure](#) > > measures)
Construct a new Executor object.
- void [ExecuteAll](#) ([Session](#) &i_currentSession) const
Run the computation of the activated measures on the data of the provided session.
- const std::vector< std::unique_ptr< [Measure](#) > > & [GetMeasures](#) () const
Return the list of the activated measures.

Private Attributes

- std::vector< std::unique_ptr< [Measure](#) > > [m_measures](#)
Container for access to the measures to be computed.

7.11.1 Detailed Description

This class takes care of the computation of the measures activated.

7.11.2 Constructor & Destructor Documentation

7.11.2.1 Executor()

```
OFIQ_LIB::modules::measures::Executor::Executor (
    std::vector< std::unique_ptr< Measure > > measures ) [inline], [explicit]
```

Construct a new Executor object.

Parameters

<i>measures</i>	Provide access to the activated measures.
-----------------	---

7.11.3 Member Function Documentation

7.11.3.1 ExecuteAll()

```
void OFIQ_LIB::modules::measures::Executor::ExecuteAll (
    Session & i_currentSession ) const
```

Run the computation of the activated measures on the data of the provided session.

Parameters

<i>i_currentSession</i>	Container providing the data required for the computation of the measures.
-------------------------	--

7.11.3.2 GetMeasures()

```
const std::vector< std::unique_ptr< Measure > > & OFIQ_LIB::modules::measures::Executor::↔
GetMeasures ( ) const [inline]
```

Return the list of the activated measures.

7.11.4 Member Data Documentation

7.11.4.1 m_measures

```
std::vector<std::unique_ptr<Measure> > OFIQ_LIB::modules::measures::Executor::m_measures
[private]
```

Container for access to the measures to be computed.

The documentation for this class was generated from the following file:

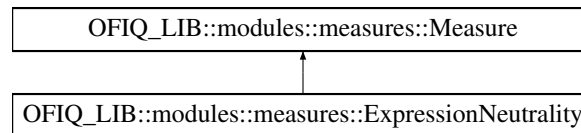
- [Executor.h](#)

7.12 OFIQ_LIB::modules::measures::ExpressionNeutrality Class Reference

Provides a class implementing the expression neutrality measure.

```
#include <ExpressionNeutrality.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::ExpressionNeutrality:



Public Member Functions

- [ExpressionNeutrality](#) (const [Configuration](#) &configuration)
Construct a new Expression Neutrality object.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Run the computation based on the data passed by the session object.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Private Attributes

- [ONNXRuntimeEnv m_onnxRuntimeEnvCNN1](#)
Instance of the enet_b0_8_best_vgaf_embed2 model. Set by ExpressionNeutrality.cnn1_model_path in the configuration file.
- [ONNXRuntimeEnv m_onnxRuntimeEnvCNN2](#)
Instance of the enet_b2_8 model. Set by ExpressionNeutrality.cnn2_model_path in the configuration file.
- std::shared_ptr< cv::ml::Boost > [m_classifier](#)
Instance of the AdaBoost classifier Set by ExpressionNeutrality.adaboost_model_path in the configuration file.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & configuration
Reference to the configuration with which the measure constructor has been invoked.

7.12.1 Detailed Description

Provides a class implementing the expression neutrality measure.

The algorithm uses the CNN models enet_b0_8_best_vgaf and enet_b2_8 from <https://github.com/HSE-asavchenko/face-emotion-recognition> as feature extractors and an AdaBoost classifier implemented in OpenCV. Inspired by Grimmer et al. [9], both CNN models have been modified to also output the embeddings of the second last layer, and have been converted to ONNX.

7.12.2 Constructor & Destructor Documentation

7.12.2.1 ExpressionNeutrality()

```
OFIQ_LIB::modules::measures::ExpressionNeutrality::ExpressionNeutrality (
    const Configuration & configuration ) [explicit]
```

Construct a new Expression Neutrality object.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.12.3 Member Function Documentation

7.12.3.1 Execute()

```
void OFIQ_LIB::modules::measures::ExpressionNeutrality::Execute (
    OFIQ_LIB::Session & session ) [override], [virtual]
```

Run the computation based on the data passed by the session object.

Parameters

<i>session</i>	Session object
----------------	----------------

Implements [OFIQ_LIB::modules::measures::Measure](#).

7.12.4 Member Data Documentation

7.12.4.1 m_classifier

```
std::shared_ptr<cv::ml::Boost> OFIQ_LIB::modules::measures::ExpressionNeutrality::m_classifier
[private]
```

Instance of the AdaBoost classifier Set by ExpressionNeutrality.adaboost_model_path in the configuration file.

7.12.4.2 m_onnxRuntimeEnvCNN1

```
ONNXRuntimeEnv OFIQ_LIB::modules::measures::ExpressionNeutrality::m_onnxRuntimeEnvCNN1 [private]
```

Instance of the enet_b0_8_best_vgaf_embed2 model. Set by ExpressionNeutrality.cnn1_model_path in the configuration file.

7.12.4.3 m_onnxRuntimeEnvCNN2

```
ONNXRuntimeEnv OFIQ_LIB::modules::measures::ExpressionNeutrality::m_onnxRuntimeEnvCNN2 [private]
```

Instance of the enet_b2_8 model. Set by ExpressionNeutrality.cnn2_model_path in the configuration file.

The documentation for this class was generated from the following file:

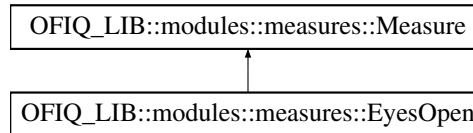
- [ExpressionNeutrality.h](#)

7.13 OFIQ_LIB::modules::measures::EyesOpen Class Reference

Implementation of the eyes open measure.

```
#include <EyesOpen.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::EyesOpen:



Public Member Functions

- [EyesOpen](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Assesses eyes openness.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from OFIQ_LIB::modules::measures::Measure

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from OFIQ_LIB::modules::measures::Measure

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.13.1 Detailed Description

Implementation of the eyes open measure.

Eyes openness is based on computing eyes aspect ratios of both eyes from eye landmarks.

7.13.2 Constructor & Destructor Documentation

7.13.2.1 EyesOpen()

```
OFIQ_LIB::modules::measures::EyesOpen::EyesOpen (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

configuration	Configuration object from which measure-related configuration is read.
-------------------------------	--

7.13.3 Member Function Documentation

7.13.3.1 Execute()

```
void OFIQ_LIB::modules::measures::EyesOpen::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Assesses eyes openness.

Eyes openness is based on computing eyes aspect ratios of both eyes from eye landmarks.

Parameters

session	Session object computed by the OFIQImpl::preprocess() method.
-------------------------	---

See also

[Session::getAlignedFaceLandmarks\(\)](#)

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

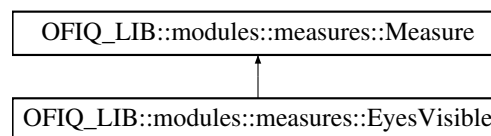
- [EyesOpen.h](#)

7.14 OFIQ_LIB::modules::measures::EyesVisible Class Reference

Implementation of the eyes visible measure.

```
#include <EyesVisible.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::EyesVisible:



Public Member Functions

- [EyesVisible](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Assesses eyes visibility.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from OFIQ_LIB::modules::measures::Measure

- void [AddSigmoid](#) (OFIQ::QualityMeasure measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) (OFIQ::QualityMeasure measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from OFIQ_LIB::modules::measures::Measure

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from OFIQ_LIB::modules::measures::Measure

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.14.1 Detailed Description

Implementation of the eyes visible measure.

Eyes visibility is assessed by measuring the coverage of the eye visibility zone with the result of face occlusion segmentation computed during pre-processing.

7.14.2 Constructor & Destructor Documentation

7.14.2.1 EyesVisible()

```
OFIQ_LIB::modules::measures::EyesVisible::EyesVisible (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.14.3 Member Function Documentation

7.14.3.1 Execute()

```
void OFIQ_LIB::modules::measures::EyesVisible::Execute (
    OFIQ_LIB::Session & session ) [override], [virtual]
```

Assesses eyes visibility.

Eyes visibility is assessed by measuring the coverage of the eye visibility zone with the result of face occlusion segmentation computed during pre-processing by [FaceOcclusionSegmentation](#). The pre-processing results are given by the `session` parameter.

Parameters

<code>session</code>	Session object computed by the OFIQImpl::preprocess() method.
----------------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

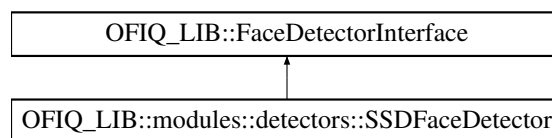
- [EyesVisible.h](#)

7.15 OFIQ_LIB::FaceDetectorInterface Class Reference

Provides the interface class to the face detector implementations.

```
#include <detectors.h>
```

Inheritance diagram for OFIQ_LIB::FaceDetectorInterface:



Public Member Functions

- virtual [~FaceDetectorInterface](#) ()=default
Destroy the Face Detector Interface object.
- `std::vector< OFIQ::BoundingBox > detectFaces (OFIQ_LIB::Session &session)`
This function detects faces in given image.

Protected Member Functions

- virtual `std::vector< OFIQ::BoundingBox > UpdateFaces (OFIQ_LIB::Session &session)=0`
This method is to be called in derived classes to perform the detection of one/more faces on the given image.

7.15.1 Detailed Description

Provides the interface class to the face detector implementations.

This class provides the base class / interface for the integration of different implementations of a face detector.

7.15.2 Constructor & Destructor Documentation

7.15.2.1 ~FaceDetectorInterface()

```
virtual OFIQ_LIB::FaceDetectorInterface::~~FaceDetectorInterface ( ) [virtual], [default]
```

Destroy the Face Detector Interface object.

7.15.3 Member Function Documentation

7.15.3.1 detectFaces()

```
std::vector< OFIQ::BoundingBox > OFIQ_LIB::FaceDetectorInterface::detectFaces (
    OFIQ_LIB::Session & session )
```

This function detects faces in given image.

Parameters

in	<i>session</i>	Session containing relevant information for the current task.
----	----------------	---

7.15.3.2 UpdateFaces()

```
virtual std::vector< OFIQ::BoundingBox > OFIQ_LIB::FaceDetectorInterface::UpdateFaces (
    OFIQ_LIB::Session & session ) [protected], [pure virtual]
```

This method is to be called in derived classes to perform the detection of one/more faces on the given image.

Parameters

<i>session</i>	Session containing relevant information for the current task.
----------------	---

Returns

std::vector<OFIQ::BoundingBox>

Implemented in [OFIQ_LIB::modules::detectors::SSDFaceDetector](#).

The documentation for this class was generated from the following file:

- [detectors.h](#)

7.16 FaceImageQualityAssessment Struct Reference

Data structure storing the results of the different measurement computations.

```
#include <ofiq_structs_c.h>
```

Public Attributes

- [QualityMeasureResult](#) * [qAssessments](#) {nullptr}
Container for storing the results of the different measure computations.
- [size_t](#) [numAssessments](#) {0}
Number of quality measure computations stored in qAssessments.
- [BoundingBox](#) [boundingBox](#)
Face region described by bounding box.

7.16.1 Detailed Description

Data structure storing the results of the different measurement computations.

Examples

```
C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h.
```

7.16.2 Member Data Documentation

7.16.2.1 boundingBox

```
BoundingBox FaceImageQualityAssessment::boundingBox
```

Face region described by bounding box.

7.16.2.2 numAssessments

```
size\_t FaceImageQualityAssessment::numAssessments {0}
```

Number of quality measure computations stored in qAssessments.

7.16.2.3 qAssessments

```
QualityMeasureResult* FaceImageQualityAssessment::qAssessments {nullptr}
```

Container for storing the results of the different measure computations.

The documentation for this struct was generated from the following file:

- [ofiq_structs_c.h](#)

7.17 OFIQ::FaceImageQualityAssessment Struct Reference

Data structure storing the results of the different measurement computations.

```
#include <ofiq_structs.h>
```

Public Member Functions

- [FaceImageQualityAssessment](#) ()=default
Default constructor.
- [FaceImageQualityAssessment](#) (const [QualityAssessments](#) &qAssessments, const [BoundingBox](#) &boundingBox)
Parameterized constructor.

Public Attributes

- [QualityAssessments](#) qAssessments
Container for storing the results of the different measure computations.
- [BoundingBox](#) boundingBox
Face region described by bounding box.

7.17.1 Detailed Description

Data structure storing the results of the different measurement computations.

7.17.2 Constructor & Destructor Documentation

7.17.2.1 FaceImageQualityAssessment() [1/2]

```
OFIQ::FaceImageQualityAssessment::FaceImageQualityAssessment ( ) [default]
```

Default constructor.

7.17.2.2 FaceImageQualityAssessment() [2/2]

```
OFIQ::FaceImageQualityAssessment::FaceImageQualityAssessment (
    const QualityAssessments & qAssessments,
    const BoundingBox & boundingBox ) [inline]
```

Parameterized constructor.

Parameters

in	<i>qAssessments</i>	
in	<i>boundingBox</i>	

7.17.3 Member Data Documentation

7.17.3.1 boundingBox

[BoundingBox](#) OFIQ::FaceImageQualityAssessment::boundingBox

Face region described by bounding box.

7.17.3.2 qAssessments

[QualityAssessments](#) OFIQ::FaceImageQualityAssessment::qAssessments

Container for storing the results of the different measure computations.

The documentation for this struct was generated from the following file:

- [ofiq_structs.h](#)

7.18 OFIQ::FaceImageQualityPreprocessingResult Struct Reference

Data structure storing the results of pre-processing computations.

```
#include <ofiq_structs.h>
```

Public Member Functions

- [FaceImageQualityPreprocessingResult](#) ()=default
Default contructor.

Public Attributes

- std::vector< [OFIQ::BoundingBox](#) > [m_faces](#)
Detected faces.
- [FaceLandmarks](#) [m_landmarks](#)
Landmarks.
- std::shared_ptr< uint8_t[] > [m_segmentationMaskPtr](#)
Segmentation mask.
- std::shared_ptr< uint8_t[] > [m_occlusionMaskPtr](#)
Occlusion mask.
- std::shared_ptr< uint8_t[] > [m_landmarkedRegionPtr](#)
Landmarked region.

7.18.1 Detailed Description

Data structure storing the results of pre-processing computations.

The members can be requested using the [OFIQImpl::vectorQualityWithPreprocessingResults](#) function. Non-requested members are empty by default.

7.18.2 Constructor & Destructor Documentation

7.18.2.1 FaceImageQualityPreprocessingResult()

```
OFIQ::FaceImageQualityPreprocessingResult::FaceImageQualityPreprocessingResult ( ) [default]
```

Default constructor.

7.18.3 Member Data Documentation

7.18.3.1 m_faces

```
std::vector<OFIQ::BoundingBox> OFIQ::FaceImageQualityPreprocessingResult::m_faces
```

Detected faces.

7.18.3.2 m_landmarkedRegionPtr

```
std::shared_ptr<uint8_t[]> OFIQ::FaceImageQualityPreprocessingResult::m_landmarkedRegionPtr
```

Landmarked region.

For each pixel at index ($y * \text{width} + x$) of an image of dimension (height,width), the value is 0 if the pixel is not part of the landmarked region mask; otherwise, the value is 1

7.18.3.3 m_landmarks

```
FaceLandmarks OFIQ::FaceImageQualityPreprocessingResult::m_landmarks
```

Landmarks.

7.18.3.4 m_occlusionMaskPtr

```
std::shared_ptr<uint8_t[]> OFIQ::FaceImageQualityPreprocessingResult::m_occlusionMaskPtr
```

Occlusion mask.

For each pixel at index ($y * \text{width} + x$) of an image of dimension (height,width), the value is 0 if the pixel is not part of the occlusion mask; otherwise, the value is 1.

7.18.3.5 m_segmentationMaskPtr

```
std::shared_ptr<uint8_t[]> OFIQ::FaceImageQualityPreprocessingResult::m_segmentationMaskPtr
```

Segmentation mask.

For each pixel (y,x) of the original image of dimension (height,width) where $y=0,\dots,\text{height}-1$ and $x=0,\dots,\text{width}-1$ the value at $y*\text{width}+x$ is a value encoding the segmentation type the pixel is assigned to. The meaning of the code is listed in the following table.

code	assignment
0	background
1	face skin
2	left eye brow
3	right eye brow
4	left eye
5	right eye
6	eyeglasses
7	left ear
8	right ear
9	earring
10	nose
11	mouth
12	upper lip
13	lower lip
14	neck
15	necklace
16	clothing
17	hair
18	head covering
19	undocumented
20	undocumented
21	undocumented
22	undocumented
23	undocumented
255	outside of the aligned face image

Attention

Other values as listed may occur as well but are not documented.

The documentation for this struct was generated from the following file:

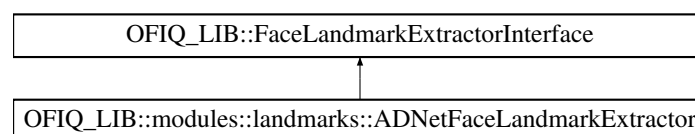
- [ofiq_structs.h](#)

7.19 OFIQ_LIB::FaceLandmarkExtractorInterface Class Reference

Implements the base class for the face landmark extractors.

```
#include <landmarks.h>
```

Inheritance diagram for OFIQ_LIB::FaceLandmarkExtractorInterface:



Public Member Functions

- virtual [~FaceLandmarkExtractorInterface](#) ()=default
Destructor.
- [OFIQ::FaceLandmarks extractLandmarks](#) ([OFIQ_LIB::Session](#) &session)
Public method to extract landmarks from the image passed in the session object.

Protected Member Functions

- virtual [OFIQ::FaceLandmarks updateLandmarks](#) ([OFIQ_LIB::Session](#) &session)=0
Internal implementation of the derived class for extracting landmarks.

7.19.1 Detailed Description

Implements the base class for the face landmark extractors.

7.19.2 Constructor & Destructor Documentation

7.19.2.1 ~FaceLandmarkExtractorInterface()

```
virtual OFIQ_LIB::FaceLandmarkExtractorInterface::~~FaceLandmarkExtractorInterface ( ) [virtual],
[default]
```

Destructor.

7.19.3 Member Function Documentation

7.19.3.1 extractLandmarks()

```
OFIQ::FaceLandmarks OFIQ_LIB::FaceLandmarkExtractorInterface::extractLandmarks (
    OFIQ_LIB::Session & session )
```

Public method to extract landmarks from the image passed in the session object.

Parameters

<i>session</i>	Data container, including the original image and preprocessed data.
----------------	---

Returns

OFIQ::FaceLandmarks

7.19.3.2 updateLandmarks()

```
virtual OFIQ::FaceLandmarks OFIQ_LIB::FaceLandmarkExtractorInterface::updateLandmarks (
    OFIQ_LIB::Session & session ) [protected], [pure virtual]
```

Internal implementation of the derived class for extracting landmarks.

Parameters

<code>session</code>	Data container, including the original image and preprocessed data.
----------------------	---

Returns

OFIQ::FaceLandmarks

Implemented in [OFIQ_LIB::modules::landmarks::ADNetFaceLandmarkExtractor](#).

The documentation for this class was generated from the following file:

- [landmarks.h](#)

7.20 FaceLandmarks Struct Reference

Data structure for storing facial landmarks.

```
#include <ofiq_structs_c.h>
```

Public Attributes

- [LandmarkType](#) type {[LandmarkType::NotSet](#)}
 - [LandmarkPoint](#) * [landmarks](#)
 - int [numLandmarks](#) = 0
- Number of landmark points stored in landmarks.*

7.20.1 Detailed Description

Data structure for storing facial landmarks.

Examples

[C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h](#).

7.20.2 Member Data Documentation

7.20.2.1 landmarks

```
LandmarkPoint* FaceLandmarks::landmarks
```

container for all detected landmarks.

7.20.2.2 numLandmarks

```
int FaceLandmarks::numLandmarks = 0
```

Number of landmark points stored in landmarks.

7.20.2.3 type

```
LandmarkType FaceLandmarks::type {LandmarkType::NotSet}
```

Enum describing the type of the landmarks.

The documentation for this struct was generated from the following file:

- [ofiq_structs_c.h](#)

7.21 OFIQ::FaceLandmarks Struct Reference

Data structure for storing facial landmarks.

```
#include <ofiq_structs.h>
```

Public Member Functions

- [FaceLandmarks](#) ()=default

Public Attributes

- [LandmarkType](#) type { [LandmarkType::NotSet](#) }
- [Landmarks](#) landmarks

7.21.1 Detailed Description

Data structure for storing facial landmarks.

7.21.2 Constructor & Destructor Documentation

7.21.2.1 FaceLandmarks()

```
OFIQ::FaceLandmarks::FaceLandmarks ( ) [default]
```

Default constructor.

7.21.3 Member Data Documentation

7.21.3.1 landmarks

[Landmarks](#) OFIQ::FaceLandmarks::landmarks

container for all detected landmarks.

7.21.3.2 type

[LandmarkType](#) OFIQ::FaceLandmarks::type { [LandmarkType::NotSet](#) }

Enum describing the type of the landmarks.

The documentation for this struct was generated from the following file:

- [ofiq_structs.h](#)

7.22 OFIQ_LIB::modules::landmarks::FaceMeasures Class Reference

Provides static functions doing computations with landmarks.

```
#include <FaceMeasures.h>
```

Public Member Functions

- [FaceMeasures](#) ()=delete
Constructor is deleted to avoid instantiations of this class.

Static Public Member Functions

- static double [InterEyeDistance](#) (const [OFIQ::FaceLandmarks](#) &faceLandmarks, double yaw)
Computes the inter-eye distance based on the specified facial landmarks and yaw angle.
- static cv::Mat [GetFaceMask](#) (const [OFIQ::FaceLandmarks](#) &faceLandmarks, const int height, const int width, const float alpha=0)
Creates a binary image of specified dimension and masks all pixels inside or on the convex hull.
- static double [GetDistance](#) (const [OFIQ::LandmarkPoint](#) &a, const [OFIQ::LandmarkPoint](#) &b)
Convenience method for computing the Euclidean distance between two landmark points.
- static double [GetDistance](#) (const [LandmarkPair](#) &pair)
Convenience method computing the Euclidean distance between two landmark points.
- static [OFIQ::LandmarkPoint](#) [GetMiddle](#) (const [OFIQ::Landmarks](#) &landmarks)
Computes the center point of the specified landmarks.
- static [OFIQ::LandmarkPoint](#) [GetMiddle](#) (const [LandmarkPair](#) &pair)
Computes the point in between two landmark points.
- static [OFIQ::LandmarkPoint](#) [GetMiddle](#) (const std::vector< [LandmarkPair](#) > &pairs)
Computes the center of the specified landmark points.
- static double [GetMaxPairDistance](#) (const [OFIQ::FaceLandmarks](#) &landmarks, [landmarks::FaceParts](#) face↔Part)
Returns this maximum of face pairs from landmarks corresponding to the specified face part.

7.22.1 Detailed Description

Provides static functions doing computations with landmarks.

7.22.2 Constructor & Destructor Documentation

7.22.2.1 FaceMeasures()

```
OFIQ_LIB::modules::landmarks::FaceMeasures::FaceMeasures ( ) [delete]
```

Constructor is deleted to avoid instantiations of this class.

7.22.3 Member Function Documentation

7.22.3.1 GetDistance() [1/2]

```
static double OFIQ_LIB::modules::landmarks::FaceMeasures::GetDistance (
    const LandmarkPair & pair ) [inline], [static]
```

Convenience method computing the Euclidean distance between two landmark points.

Parameters

<i>pair</i>	The two landmark points stored in the member <code>pair.Lower</code> and <code>pair.Upper</code> .
-------------	--

Returns

Euclidean distance.

7.22.3.2 GetDistance() [2/2]

```
static double OFIQ_LIB::modules::landmarks::FaceMeasures::GetDistance (
    const OFIQ::LandmarkPoint & a,
    const OFIQ::LandmarkPoint & b ) [static]
```

Convenience method for computing the Euclidean distance between two landmark points.

Parameters

<i>a</i>	First landmark point
<i>b</i>	Second landmark point

Returns

Euclidean distance between `a` and `b`.

7.22.3.3 GetFaceMask()

```
static cv::Mat OFIQ_LIB::modules::landmarks::FaceMeasures::GetFaceMask (
    const OFIQ::FaceLandmarks & faceLandmarks,
    const int height,
    const int width,
    const float alpha = 0 ) [static]
```

Creates a binary image of specified dimension and masks all pixels inside or on the convex hull.

All pixels on or inside the convex hull of the landmarks are set to 1; all other pixels are set to 0.

Parameters

<i>faceLandmarks</i>	Facial landmarks object
<i>height</i>	Height of the mask image
<i>width</i>	Width of the mask image
<i>alpha</i>	Should be 0; different values have only be used for NIST submissions.

Returns

Mask image

7.22.3.4 GetMaxPairDistance()

```
static double OFIQ_LIB::modules::landmarks::FaceMeasures::GetMaxPairDistance (
    const OFIQ::FaceLandmarks & landmarks,
    landmarks::FaceParts facePart ) [static]
```

Returns this maximum of face pairs from landmarks corresponding to the specified face part.

Face parts (such as mouth) consist of landmarks that have a mate. For example, the lower lip point may correspond to the upper lip point. For such face parts, the maximal separation (e.g., useful for detecting the mouth closeness or eyes openness) can be computed.

Parameters

<i>landmarks</i>	Facial landmarks
<i>facePart</i>	Face part

Returns

Maxim of face pairs

7.22.3.5 GetMiddle() [1/3]

```
static OFIQ::LandmarkPoint OFIQ_LIB::modules::landmarks::FaceMeasures::GetMiddle (
    const LandmarkPair & pair ) [inline], [static]
```

Computes the point in between two landmark points.

Parameters

<i>pair</i>	Pair of landmark points.
-------------	--------------------------

Returns

Point between the two landmark points.

7.22.3.6 GetMiddle() [2/3]

```
static OFIQ::LandmarkPoint OFIQ_LIB::modules::landmarks::FaceMeasures::GetMiddle (
    const OFIQ::Landmarks & landmarks ) [static]
```

Computes the center point of the specified landmarks.

Parameters

<i>landmarks</i>	Facial landmarks
------------------	------------------

Returns

Center point of the landmarks.

7.22.3.7 GetMiddle() [3/3]

```
static OFIQ::LandmarkPoint OFIQ_LIB::modules::landmarks::FaceMeasures::GetMiddle (
    const std::vector< LandmarkPair > & pairs ) [inline], [static]
```

Computes the center of the specified landmark points.

This is a convenience method to compute the center if the landmarks are available as a vector of landmark pairs.

Parameters

<i>pairs</i>	List of landmark pairs.
--------------	-------------------------

Returns

Center of the landmark points.

7.22.3.8 InterEyeDistance()

```
static double OFIQ_LIB::modules::landmarks::FaceMeasures::InterEyeDistance (
    const OFIQ::FaceLandmarks & faceLandmarks,
    double yaw ) [static]
```

Computes the inter-eye distance based on the specified facial landmarks and yaw angle.

If (x_0, y_0) and (x_1, y_1) are the left and right eye centres, then the following inter-eye distance is computed.

$$\text{IED} = \|(x_0, y_0) - (x_1, y_1)\|_2 \cdot \frac{1}{\cos(\alpha)}$$

where

$$\alpha = \pi \cdot \text{yaw} / 180$$

is the secant of the yaw angle.

Parameters

<i>faceLandmarks</i>	Facial landmarks
<i>yaw</i>	Yaw angle in degree

Returns

The inter-eye distance

The documentation for this class was generated from the following file:

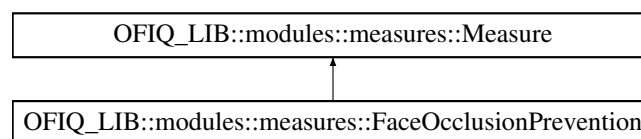
- [FaceMeasures.h](#)

7.23 OFIQ_LIB::modules::measures::FaceOcclusionPrevention Class Reference

Implementation of the face occlusion prevention measure.

```
#include <FaceOcclusionPrevention.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::FaceOcclusionPrevention:



Public Member Functions

- [FaceOcclusionPrevention](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) (OFIQ_LIB::Session &session) override
Assesses absence of face occlusion.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & configuration
Reference to the configuration with which the measure constructor has been invoked.

7.23.1 Detailed Description

Implementation of the face occlusion prevention measure.

Absence of face occlusion is assessed by measuring the coverage of the marked region with the result of face occlusion segmentation computed during pre-processing.

7.23.2 Constructor & Destructor Documentation

7.23.2.1 FaceOcclusionPrevention()

```
OFIQ_LIB::modules::measures::FaceOcclusionPrevention::FaceOcclusionPrevention (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.23.3 Member Function Documentation

7.23.3.1 Execute()

```
void OFIQ_LIB::modules::measures::FaceOcclusionPrevention::Execute (
    OFIQ_LIB::Session & session ) [override], [virtual]
```

Assesses absence of face occlusion.

Absence of face occlusion is assessed by measuring the coverage of the marked region with the result of face occlusion segmentation computed during pre-processing. Pre-processing results are passed to the method with the `session` parameter.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

See also

[FaceOcclusionSegmentation](#)

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

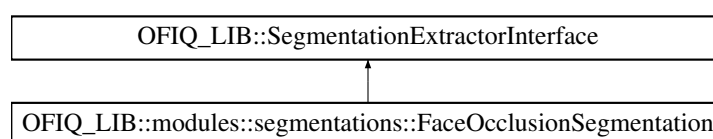
- [FaceOcclusionPrevention.h](#)

7.24 OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation Class Reference

Class managing the separation of facial parts not occluded by non-facial parts from other parts.

```
#include <FaceOcclusionSegmentation.h>
```

Inheritance diagram for OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation:



Public Member Functions

- [FaceOcclusionSegmentation](#) (const [Configuration](#) &config)
Constructor.
- [~FaceOcclusionSegmentation](#) () override=default
Destructor.

Public Member Functions inherited from [OFIQ_LIB::SegmentationExtractorInterface](#)

- virtual [~SegmentationExtractorInterface](#) ()=default
Standard destructor.
- [OFIQ::Image](#) & [GetMask](#) ([OFIQ_LIB::Session](#) &session, [modules::segmentations::SegmentClassLabels](#) faceSegment)
Get a mask of the face region requested.

Protected Member Functions

- [OFIQ::Image](#) [UpdateMask](#) ([OFIQ_LIB::Session](#) &session, [modules::segmentations::SegmentClassLabels](#) faceSegment) override
Implements face occlusion segmentation.

Protected Member Functions inherited from [OFIQ_LIB::SegmentationExtractorInterface](#)

- std::string [GetLastSessionId](#) () const
Accesses the last session id for this interface.

Private Member Functions

- cv::Mat [GetFaceOcclusionSegmentation](#) (const cv::Mat &alignedImage)
Does the actual CNN-based occlusion-aware segmentation.

Private Attributes

- [ONNXRuntimeEnv](#) [m_onnxRuntimeEnv](#)
Manages CNN computations.
- std::shared_ptr< cv::Mat > [m_segmentationImage](#)
Stores the last result computed with [UpdateMask\(\)](#).
- const std::string [m_modelConfigItem](#) = "params.measures.FaceOcclusionSegmentation.model_path"
JSON/JAXN key to access path to FaceExtraction's model file from [Configuration](#) object.
- const int [m_cropLeft](#) = 96
Cropping parameter.
- const int [m_cropRight](#) = 96
Cropping parameter.
- const int [m_cropTop](#) = 96
Cropping parameter.
- const int [m_cropBottom](#) = 96
Cropping parameter.
- const int [m_scaledWidth](#) = 224
After cropping the aligned image, the result will be scaled to a dimension of that width for being input to the CNN-based segmentation.
- const int [m_scaledHeight](#) = 224
After cropping the aligned image, the result will be scaled to a dimension of that height for being input to the CNN-based segmentation.

7.24.1 Detailed Description

Class managing the separation of facial parts not occluded by non-facial parts from other parts.

The implementation is based on a CNN from [FaceExtraction](#).

7.24.2 Constructor & Destructor Documentation

7.24.2.1 FaceOcclusionSegmentation()

```
OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::FaceOcclusionSegmentation (
    const Configuration & config ) [explicit]
```

Constructor.

Parameters

<i>config</i>	Configuration object from which some segmentation-related parameters may be read.
---------------	---

See also

[Other required configurations](#)

7.24.2.2 ~FaceOcclusionSegmentation()

```
OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::~FaceOcclusionSegmentation ( )
[override], [default]
```

Destructor.

7.24.3 Member Function Documentation

7.24.3.1 GetFaceOcclusionSegmentation()

```
cv::Mat OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::GetFaceOcclusionSegmentation
(
    const cv::Mat & alignedImage ) [private]
```

Does the actual CNN-based occlusion-aware segmentation.

Parameters

<i>alignedImage</i>	Aligned image of dimension 616 x 616 as returned by Session::getAlignedFace() .
---------------------	---

Returns

Image where a pixel belonging to non-occluded facial parts is encoded as the byte value 1 and pixels belonging to other parts are encoded by the byte value 0.

7.24.3.2 UpdateMask()

```
OFIQ::Image OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::UpdateMask (
    OFIQ_LIB::Session & session,
    modules::segmentations::SegmentClassLabels faceSegment ) [override], [protected],
[virtual]
```

Implements face occlusion segmentation.

The function is invoked by [SegmentationExtractorInterface::GetMask\(\)](#). Invokes [GetFaceOcclusionSegmentation\(\)](#) and stores its output in the private [m_segmentationImage](#) member.

Parameters

<i>session</i>	Session object containing the original facial image and pre-processing results computed by the OFIQImpl::preprocess() method.
<i>faceSegment</i>	Should be the value SegmentClassLabels::face .

Returns

Face occlusion segmentation mask.

Implements [OFIQ_LIB::SegmentationExtractorInterface](#).

7.24.4 Member Data Documentation**7.24.4.1 m_cropBottom**

```
const int OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::m_cropBottom = 96 [private]
```

Cropping parameter.

7.24.4.2 m_cropLeft

```
const int OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::m_cropLeft = 96 [private]
```

Cropping parameter.

7.24.4.3 m_cropRight

```
const int OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::m_cropRight = 96 [private]
```

Cropping parameter.

7.24.4.4 m_cropTop

```
const int OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::m_cropTop = 96 [private]
```

Cropping parameter.

7.24.4.5 m_modelConfigItem

```
const std::string OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::m_modelConfigItem = "params.measures.FaceOcclusionSegmentation.model_path" [private]
```

JSON/JAXN key to access path to FaceExtraction's model file from [Configuration](#) object.

7.24.4.6 m_onnxRuntimeEnv

```
ONNXRuntimeEnv OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::m_onnxRuntimeEnv [private]
```

Manages CNN computations.

7.24.4.7 m_scaledHeight

```
const int OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::m_scaledHeight = 224 [private]
```

After cropping the aligned image, the result will be scaled to a dimension of that height for being input to the CNN-based segmentation.

7.24.4.8 m_scaledWidth

```
const int OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::m_scaledWidth = 224 [private]
```

After cropping the aligned image, the result will be scaled to a dimension of that width for being input to the CNN-based segmentation.

7.24.4.9 m_segmentationImage

```
std::shared_ptr<cv::Mat> OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation::m_segmentationImage [private]
```

Stores the last result computed with [UpdateMask\(\)](#).

The documentation for this class was generated from the following file:

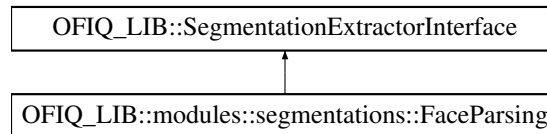
- [FaceOcclusionSegmentation.h](#)

7.25 OFIQ_LIB::modules::segmentations::FaceParsing Class Reference

Class managing the separation of facial parts not occluded by non-facial parts from other parts.

```
#include <FaceParsing.h>
```

Inheritance diagram for OFIQ_LIB::modules::segmentations::FaceParsing:



Public Member Functions

- [FaceParsing](#) (const [Configuration](#) &config)
Constructor.
- [~FaceParsing](#) () override=default
Destructor.

Public Member Functions inherited from [OFIQ_LIB::SegmentationExtractorInterface](#)

- virtual [~SegmentationExtractorInterface](#) ()=default
Standard destructor.
- [OFIQ::Image](#) & [GetMask](#) ([OFIQ_LIB::Session](#) &session, [modules::segmentations::SegmentClassLabels](#) faceSegment)
Get a mask of the face region requested.

Protected Member Functions

- [OFIQ::Image](#) [UpdateMask](#) ([OFIQ_LIB::Session](#) &session, [modules::segmentations::SegmentClassLabels](#) faceSegment) override
Implements face parsing.

Protected Member Functions inherited from [OFIQ_LIB::SegmentationExtractorInterface](#)

- std::string [GetLastSessionId](#) () const
Accesses the last session id for this interface.

Private Member Functions

- void [SetImage](#) (const [OFIQ_LIB::Session](#) &session)

Static Private Member Functions

- static std::shared_ptr< cv::Mat > [CalculateClassIds](#) (const cv::Mat &faceSegments, int imgSize)
Creates the blob being input to the face parsing CNN.

Private Attributes

- [ONNXRuntimeEnv m_onnxRuntimeEnv](#)
Manages CNN computations.
- `std::shared_ptr< cv::Mat >` [m_segmentationImage](#)
Stores the last result computed with [UpdateMask\(\)](#).
- `const std::string` [m_modelConfigItem](#) = "params.measures.FaceParsing.model_path"
JSON/JAXN key to access path to [BiSeNet](#) model in ONNX format from [Configuration](#) object.
- `const int` [m_imageSize](#) = 400
Face parsing target dimension.
- `const int` [m_cropLeft](#) = 30
Cropping parameter.
- `const int` [m_cropRight](#) = 30
Cropping parameter.
- `const int` [m_cropBottom](#) = 60
Cropping parameter.

7.25.1 Detailed Description

Class managing the separation of facial parts not occluded by non-facial parts from other parts.

Implements a [BiSeNet](#)-based face parsing. The aligned face image is cropped and then scaled to the dimension 400 x 400. All pixels of the resulting image are assigned to one of the following class.

value	class
0	background
1	face skin
2	left eye brow
3	right eye brow
4	left eye
5	right eye
6	eyeglasses
7	left ear
8	right ear
9	earring
10	nose
11	mouth
12	upper lip
13	lower lip
14	neck
15	necklace
16	clothing
17	hair
18	head covering

The result of face parsing is an image (matrix) of dimension 400 x 400 where each pixel is assigned with one of the values listed in the table from above.

7.25.2 Constructor & Destructor Documentation

7.25.2.1 FaceParsing()

```
OFIQ_LIB::modules::segmentations::FaceParsing::FaceParsing (
    const Configuration & config ) [explicit]
```

Constructor.

Parameters

<i>config</i>	Configuration object from which related parameters may be read.
---------------	---

See also

For configuration of face parsing, see [Other required configurations](#)

7.25.2.2 ~FaceParsing()

```
OFIQ_LIB::modules::segmentations::FaceParsing::~~FaceParsing ( ) [override], [default]
```

Destructor.

7.25.3 Member Function Documentation

7.25.3.1 CalculateClassIds()

```
static std::shared_ptr< cv::Mat > OFIQ_LIB::modules::segmentations::FaceParsing::Calculate←
ClassIds (
    const cv::Mat & faceSegments,
    int imgSize ) [static], [private]
```

Creates the blob being input to the face parsing CNN.

Parameters

<i>image</i>	Input image
<i>i_imgSize_one_dim</i>	Specifies the size of the blob being input to the face parsing CNN; should be 400, such that a blob of dimension 400 x 400 is created.

Returns

Blob of requested dimension.

Applies segmentation to the blob created from the input image and returns the result.

Is invoked by [SetImage\(\)](#).

Parameters

<i>faceSegments</i>	Blob being created by one of the CreateBlob functions.
<i>imgSize</i>	Specifies the size of the blob being input to the face parsing CNN; should be 400, such that a blob of dimension 400 x 400 is created.

Returns

Result of face parsing.

7.25.3.2 SetImage()

```
void OFIQ_LIB::modules::segmentations::FaceParsing::SetImage (
    const OFIQ_LIB::Session & session ) [private]
```

7.25.3.3 UpdateMask()

```
OFIQ::Image OFIQ_LIB::modules::segmentations::FaceParsing::UpdateMask (
    OFIQ_LIB::Session & session,
    modules::segmentations::SegmentClassLabels faceSegment ) [override], [protected],
[virtual]
```

Implements face parsing.

The function is invoked by [SegmentationExtractorInterface::GetMask\(\)](#). It crops the aligned face image returned by [Session::getAlignedFace\(\)](#) as configured by private member variables. The result is scaled to the dimension of 400 x 400 and passed to the [BiSeNet](#) CNN. The output is returned as face parsing.

Parameters

<i>session</i>	Session object containing the original facial image and pre-processing results computed by the OFIQImpl::preprocess() method.
<i>faceSegment</i>	Enum value encoding the requested face segment. If the value is SegmentClassLabels::face then an image of dimension 400 x 400 is returned assigning each pixel a value between 0 and 18 as in the table of FaceParsing ; otherwise a binary mask of dimension 400 x 400 is returned with the requested face segment and morphologically extended (kernel size 3) is returned.

Returns

Face parsing image of dimension 400 x 400 where each pixel is assigned with one of the values described in the table of the [OFIQ_LIB::modules::segmentations::FaceParsing](#) class documentation.

Implements [OFIQ_LIB::SegmentationExtractorInterface](#).

7.25.4 Member Data Documentation

7.25.4.1 m_cropBottom

```
const int OFIQ_LIB::modules::segmentations::FaceParsing::m_cropBottom = 60 [private]
```

Cropping parameter.

7.25.4.2 m_cropLeft

```
const int OFIQ_LIB::modules::segmentations::FaceParsing::m_cropLeft = 30 [private]
```

Cropping parameter.

7.25.4.3 m_cropRight

```
const int OFIQ_LIB::modules::segmentations::FaceParsing::m_cropRight = 30 [private]
```

Cropping parameter.

7.25.4.4 m_imageSize

```
const int OFIQ_LIB::modules::segmentations::FaceParsing::m_imageSize = 400 [private]
```

Face parsing target dimension.

7.25.4.5 m_modelConfigItem

```
const std::string OFIQ_LIB::modules::segmentations::FaceParsing::m_modelConfigItem = "params.↵  
measures.FaceParsing.model_path" [private]
```

JSON/JAXN key to access path to [BiSeNet](#) model in ONNX format from [Configuration](#) object.

7.25.4.6 m_onnxRuntimeEnv

```
ONNXRuntimeEnv OFIQ_LIB::modules::segmentations::FaceParsing::m_onnxRuntimeEnv [private]
```

Manages CNN computations.

7.25.4.7 m_segmentationImage

```
std::shared_ptr<cv::Mat> OFIQ_LIB::modules::segmentations::FaceParsing::m_segmentationImage  
[private]
```

Stores the last result computed with [UpdateMask\(\)](#).

The documentation for this class was generated from the following file:

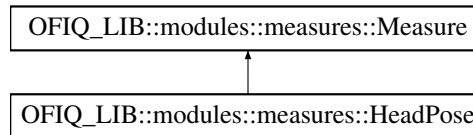
- [FaceParsing.h](#)

7.26 OFIQ_LIB::modules::measures::HeadPose Class Reference

Implementation of head pose measures.

```
#include <HeadPose.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::HeadPose:



Public Member Functions

- [HeadPose](#) (const [Configuration](#) &configuration)
Constructor for HeadPose.
- void [Execute](#) (OFIQ_LIB::Session &session) override
Assesses head pose measure.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, OFIQ::QualityMeasure measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) (OFIQ_LIB::Session &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) (OFIQ::QualityMeasure measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) (OFIQ::QualityMeasure measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.26.1 Detailed Description

Implementation of head pose measures.

Head pose measures are returned for roll, pitch and yaw face angle.

7.26.2 Constructor & Destructor Documentation

7.26.2.1 HeadPose()

```
OFIQ_LIB::modules::measures::HeadPose::HeadPose (
    const Configuration & configuration ) [explicit]
```

Constructor for HeadPose.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.26.3 Member Function Documentation

7.26.3.1 Execute()

```
void OFIQ_LIB::modules::measures::HeadPose::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Assesses head pose measure.

Quality components are computed with the help of the cosine of the native quality scores (angles).

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

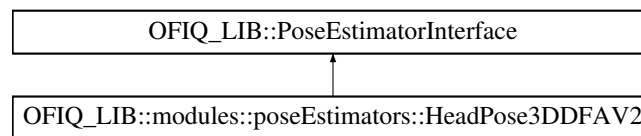
- [HeadPose.h](#)

7.27 OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2 Class Reference

Implementation of a head pose estimator.

```
#include <HeadPose3DDFAV2.h>
```

Inheritance diagram for OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2:



Public Member Functions

- [HeadPose3DDFAV2](#) (const [Configuration](#) &config)
Constructor for HeadPose3DDFAV2.
- [~HeadPose3DDFAV2](#) () override=default
Destructor.

Public Member Functions inherited from [OFIQ_LIB::PoseEstimatorInterface](#)

- virtual [~PoseEstimatorInterface](#) ()=default
Standard destructor.
- [EulerAngle](#) & [estimatePose](#) ([OFIQ_LIB::Session](#) &session)
This function estimates the three head orientation angles.

Protected Member Functions

- void [updatePose](#) ([OFIQ_LIB::Session](#) &session, [EulerAngle](#) &pose) override
Computation of the head pose.

Private Member Functions

- cv::Mat [CropImage](#) (const cv::Mat &image, const [OFIQ::BoundingBox](#) &biggestFace) const
Crop face from image. Internally the passed bounding box will be transformed to a square region.

Private Attributes

- Ort::Env [m_ortenv](#)
ONNXRuntime environment handle.
- std::unique_ptr< Ort::Session > [m_ortSession](#)
ONNXRuntime session handle.
- int64_t [m_expectedImageWidth](#) = 0
Width of the CNN used for computation, read from the loaded model.
- int64_t [m_expectedImageHeight](#) = 0
Height of the CNN used for computation, read from the loaded model.
- int64_t [m_expectedImageNumberOfChannels](#) = 0
Expected number of channels of the input image, read from the loaded model.
- int64_t [m_numberOfInputElements](#) = 0
Number of input elements of the CNN used for computation, read from the loaded model.
- std::array< int64_t, 4 > [m_inputShape](#)
inputShape of the CNN used for computation, read from the loaded model.

Static Private Attributes

- static const std::string [m_paramPoseEstimatorModel](#)
Name of the used CNN net, passed from the configuration.

Additional Inherited Members

Public Types inherited from [OFIQ_LIB::PoseEstimatorInterface](#)

- using [EulerAngle](#) = std::array<double, 3>

7.27.1 Detailed Description

Implementation of a head pose estimator.

The estimator is based on a CNN from https://github.com/cleardusk/3DDFA_V2.

7.27.2 Constructor & Destructor Documentation

7.27.2.1 HeadPose3DDFAV2()

```
OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::HeadPose3DDFAV2 (
    const Configuration & config ) [explicit]
```

Constructor for HeadPose3DDFAV2.

Parameters

<i>config</i>	Configuration from where the the path to the CNN model in ONNX format to read.
---------------	--

7.27.2.2 ~HeadPose3DDFAV2()

OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::~~HeadPose3DDFAV2 () [override], [default]

Destructor.

7.27.3 Member Function Documentation

7.27.3.1 CropImage()

```
cv::Mat OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::CropImage (
    const cv::Mat & image,
    const OFIQ::BoundingBox & biggestFace ) const [private]
```

Crop face from image. Internally the passed bounding box will be transformed to a square region.

Parameters

<i>image</i>	Input image.
<i>biggestFace</i>	Input region to be cropped.

Returns

cv::Mat Cropped face region.

7.27.3.2 updatePose()

```
void OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::updatePose (
    OFIQ_LIB::Session & session,
    EulerAngle & pose ) [override], [protected], [virtual]
```

Computation of the head pose.

Parameters

<i>session</i>	Session object containing the original facial image and pre-processing results computed.
<i>pose</i>	Estimated head pose.

Implements [OFIQ_LIB::PoseEstimatorInterface](#).

7.27.4 Member Data Documentation

7.27.4.1 m_expectedImageHeight

```
int64_t OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::m_expectedImageHeight = 0 [private]
```

Height of the CNN used for computation, read from the loaded model.

7.27.4.2 m_expectedImageNumberOfChannels

```
int64_t OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::m_expectedImageNumberOfChannels =  
0 [private]
```

Expected number of channels of the input image, read from the loaded model.

7.27.4.3 m_expectedImageWidth

```
int64_t OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::m_expectedImageWidth = 0 [private]
```

Width of the CNN used for computation, read from the loaded model.

7.27.4.4 m_inputShape

```
std::array<int64_t, 4> OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::m_inputShape [private]
```

inputShape of the CNN used for computation, read from the loaded model.

7.27.4.5 m_numberOfInputElements

```
int64_t OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::m_numberOfInputElements = 0 [private]
```

Number of input elements of the CNN used for computation, read from the loaded model.

7.27.4.6 m_ortenv

```
Ort::Env OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::m_ortenv [private]
```

ONNXRuntime environment handle.

7.27.4.7 m_ortSession

```
std::unique_ptr<Ort::Session> OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::m_ortSession  
[private]
```

ONNXRuntime session handle.

7.27.4.8 m_paramPoseEstimatorModel

```
const std::string OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2::m_paramPoseEstimatorModel  
[static], [private]
```

Name of the used CNN net, passed from the configuration.

The documentation for this class was generated from the following file:

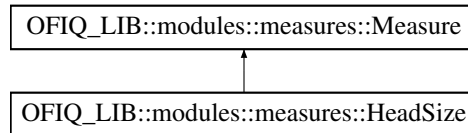
- [HeadPose3DDFAV2.h](#)

7.28 OFIQ_LIB::modules::measures::HeadSize Class Reference

Implementation of the head size measure.

```
#include <HeadSize.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::HeadSize:



Public Member Functions

- [HeadSize](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Run computation of head size measure,.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.28.1 Detailed Description

Implementation of the head size measure.

Head size measure is based on the the distance T between the midpoint between the eyes and the chin and the height of the image. Check ISO/IEC 29794-5 for more information.

7.28.2 Constructor & Destructor Documentation

7.28.2.1 HeadSize()

```
OFIQ_LIB::modules::measures::HeadSize::HeadSize (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.28.3 Member Function Documentation

7.28.3.1 Execute()

```
void OFIQ_LIB::modules::measures::HeadSize::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Run computation of head size measure,.

Parameters

<i>session</i>	Session object containing the original facial image and pre-processing results.
----------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

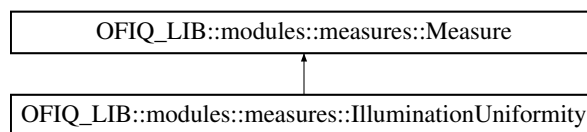
- [HeadSize.h](#)

7.29 OFIQ_LIB::modules::measures::IlluminationUniformity Class Reference

Implementation of the illumination uniformity measure.

```
#include <IlluminationUniformity.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::IlluminationUniformity:



Public Member Functions

- [IlluminationUniformity](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Assesses illumination uniformity.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.29.1 Detailed Description

Implementation of the illumination uniformity measure.

Uniformity of the illumination is measured by summing up the minima of the histograms of the left and the right side of the face.

7.29.2 Constructor & Destructor Documentation

7.29.2.1 [IlluminationUniformity\(\)](#)

```
OFIQ_LIB::modules::measures::IlluminationUniformity::IlluminationUniformity (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.29.3 Member Function Documentation

7.29.3.1 [Execute\(\)](#)

```
void OFIQ_LIB::modules::measures::IlluminationUniformity::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Assesses illumination uniformity.

Uniformity of the illumination is measured by summing up the minima of the histograms of the left and the right side of the face.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

- [IlluminationUniformity.h](#)

7.30 Image Struct Reference

Struct representing a single image.

```
#include <ofiq_structs_c.h>
```

Public Attributes

- `uint16_t width` {0}
- `uint16_t height` {0}
- `uint8_t depth` {24}
- `uint8_t * data` {nullptr}

7.30.1 Detailed Description

Struct representing a single image.

Examples

```
C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h.
```

7.30.2 Member Data Documentation

7.30.2.1 data

```
uint8_t* Image::data {nullptr}
```

Managed pointer to pixels data. Either BGR color or intensity. If `image_depth == 24` this points to 3WH bytes BGRBGRBGR... If `image_depth == 8` this points to WH bytes IIIIIII

7.30.2.2 depth

```
uint8_t Image::depth {24}
```

Number of bits per pixel. Legal values are 8 and 24.

7.30.2.3 height

```
uint16_t Image::height {0}
```

Number of pixels vertically

7.30.2.4 width

```
uint16_t Image::width {0}
```

Number of pixels horizontally

The documentation for this struct was generated from the following file:

- [ofiq_structs.c.h](#)

7.31 OFIQ::Image Struct Reference

Struct representing a single image.

```
#include <ofiq_structs.h>
```

Public Member Functions

- [Image](#) ()=default
Constructor.
- [Image](#) (const uint16_t [width](#), const uint16_t [height](#), const uint8_t [depth](#), const std::shared_ptr< uint8_t[]> &[data](#))
Constructor.
- size_t [size](#) () const
This function returns the size of the image data.
- void [deepcopy](#) (uint16_t [width](#), uint16_t [height](#), uint8_t [depth](#), const std::shared_ptr< uint8_t > &[data](#), bool isRgb=true)
Overwrites the data of the image being a deepcopy of the specified parameters.

Public Attributes

- uint16_t [width](#) { 0 }
- uint16_t [height](#) { 0 }
- uint8_t [depth](#) { 24 }
- std::shared_ptr< uint8_t[]> [data](#)

7.31.1 Detailed Description

Struct representing a single image.

7.31.2 Constructor & Destructor Documentation

7.31.2.1 Image() [1/2]

```
OFIQ::Image::Image ( ) [default]
```

Constructor.

7.31.2.2 Image() [2/2]

```
OFIQ::Image::Image (
    const uint16_t width,
    const uint16_t height,
    const uint8_t depth,
    const std::shared_ptr< uint8_t[] > & data ) [inline]
```

Constructor.

Attention

The constructor sets a shallow copy of the specified data pointer. Therefore, this constructor should no be used from binding such as Java/JNI using byte data of which memory is managed by another mechanism. Rather, for a binding such as Java/JNI, one should use the [deepcopy](#) method.

Parameters

<i>width</i>	of the image.
<i>height</i>	of the image.
<i>depth</i>	of the image
<i>data</i>	of the image.

7.31.3 Member Function Documentation

7.31.3.1 deepcopy()

```
void OFIQ::Image::deepcopy (
    uint16_t width,
    uint16_t height,
    uint8_t depth,
    const std::shared_ptr< uint8_t > & data,
    bool isRgb = true ) [inline]
```

Overwrites the data of the image being a deepcopy of the specified parameters.

This method can be used by a binding such as Java/JNI when the memory of the specified data is managed by another mechanism such as Java's garbage collector.

Parameters

in	<i>width</i>	of the image.
in	<i>height</i>	of the image.
in	<i>depth</i>	of the image
in	<i>data</i>	of the image.
in	<i>isRgb</i>	if true, then data is interpreted as RGB data; otherwise, if false, data is interpreted as BGR data.

7.31.3.2 size()

```
size_t OFIQ::Image::size ( ) const [inline]
```

This function returns the size of the image data.

7.31.4 Member Data Documentation

7.31.4.1 data

```
std::shared_ptr<uint8_t[]> OFIQ::Image::data
```

Managed pointer to raster scanned data. Either BGR color or intensity. If image_depth == 24 this points to 3WH bytes BGRBGRBGR... If image_depth == 8 this points to WH bytes IIIIII

7.31.4.2 depth

```
uint8_t OFIQ::Image::depth { 24 }
```

Number of bits per pixel. Legal values are 8 and 24.

7.31.4.3 height

```
uint16_t OFIQ::Image::height { 0 }
```

Number of pixels vertically

7.31.4.4 width

```
uint16_t OFIQ::Image::width { 0 }
```

Number of pixels horizontally

The documentation for this struct was generated from the following file:

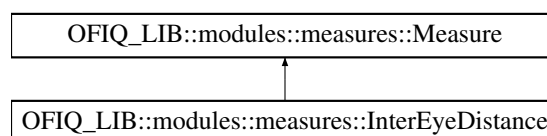
- [ofiq_structs.h](#)

7.32 OFIQ_LIB::modules::measures::InterEyeDistance Class Reference

Implementation of the inter-eye distance measure.

```
#include <InterEyeDistance.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::InterEyeDistance:



Public Member Functions

- [InterEyeDistance](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Assesses inter-eye distance.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & configuration
Reference to the configuration with which the measure constructor has been invoked.

7.32.1 Detailed Description

Implementation of the inter-eye distance measure.

Inter-eye distance assessment is based on computing the Euclidean length of both eyes' centres and multiplication with the secant of the yaw angle computed during pre-processing.

7.32.2 Constructor & Destructor Documentation

7.32.2.1 InterEyeDistance()

```
OFIQ_LIB::modules::measures::InterEyeDistance::InterEyeDistance (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.32.3 Member Function Documentation

7.32.3.1 Execute()

```
void OFIQ_LIB::modules::measures::InterEyeDistance::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Assesses inter-eye distance.

Inter-eye distance assessment is based on computing the Euclidean length of both eyes' centres and multiplication with the secant of the yaw angle computed during pre-processing.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

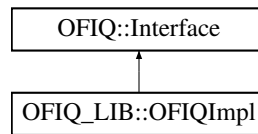
- [InterEyeDistance.h](#)

7.33 OFIQ::Interface Class Reference

The interface to FACE QA implementation.

```
#include <ofiq_lib.h>
```

Inheritance diagram for OFIQ::Interface:



Public Member Functions

- virtual [~Interface](#) ()=default

Default Destructor.

- virtual [OFIQ::ReturnStatus initialize](#) (const std::string &configDir, const std::string &configFileName)=0

This function initializes the implementation under test. The implementation under test should set all parameters.

- virtual [OFIQ::ReturnStatus scalarQuality](#) (const [OFIQ::Image](#) &face, double &quality)=0

This function takes an image and outputs a quality scalar.

- virtual [OFIQ::ReturnStatus vectorQuality](#) (const [OFIQ::Image](#) &image, [OFIQ::FacImageQualityAssessment](#) &assessments)=0

This function takes an image and outputs quality information.

- virtual [OFIQ::ReturnStatus vectorQualityWithPreprocessingResults](#) (const [OFIQ::Image](#) &image, [OFIQ::FacImageQualityAssessment](#) &assessments, [OFIQ::FacImageQualityPreprocessingResult](#) &preprocessingResult, uint32_t resultRequestsMask)=0

This function takes an image and outputs quality information and preprocessing results.

- [OFIQ_EXPORT](#) void [getVersion](#) (int &major, int &minor, int &patch) const

Access version information.

Static Public Member Functions

- static [OFIQ_EXPORT](#) std::shared_ptr< [Interface](#) > [getImplementation](#) ()

Factory method to return a shared pointer to the Interface object.

7.33.1 Detailed Description

The interface to FACE QA implementation.

Implement this interface by sub-classing this class and implementing each method therein.

7.33.2 Constructor & Destructor Documentation

7.33.2.1 ~Interface()

```
virtual OFIQ::Interface::~~Interface ( ) [virtual], [default]
```

Default Destructor.

7.33.3 Member Function Documentation

7.33.3.1 getImplementation()

```
static OFIQ_EXPORT std::shared_ptr< Interface > OFIQ::Interface::getImplementation ( ) [static]
```

Factory method to return a shared pointer to the Interface object.

This function is implemented by the submitted library and must return a shared pointer to the Interface object.

This function MUST be implemented.

Note

A possible implementation might be: `return (std::make_shared<Implementation>());`

Returns

`std::shared_ptr<Interface>` pointer to the implementation of the interface.

7.33.3.2 getVersion()

```
OFIQ_EXPORT void OFIQ::Interface::getVersion (
    int & major,
    int & minor,
    int & patch ) const
```

Access version information.

Parameters

<i>major</i>	Reference to which major version is stored.
<i>minor</i>	Reference to which minor version is stored.
<i>patch</i>	Reference to which patch version is stored.

7.33.3.3 initialize()

```
virtual OFIQ::ReturnStatus OFIQ::Interface::initialize (
    const std::string & configDir,
    const std::string & configFileName ) [pure virtual]
```

This function initializes the implementation under test. The implementation under test should set all parameters.

Parameters

in	<i>configDir</i>	string representation of the directory containing the configuration file specified by configFileName
in	<i>configFileName</i>	An string value encoding the JAXN configuration file name

Returns

OFIQ::ReturnStatus indicating if the initialization was successful.

Implemented in [OFIQ_LIB::OFIQImpl](#).

7.33.3.4 scalarQuality()

```
virtual OFIQ::ReturnStatus OFIQ::Interface::scalarQuality (
    const OFIQ::Image & face,
    double & quality ) [pure virtual]
```

This function takes an image and outputs a quality scalar.

Parameters

in	<i>face</i>	Single face image
out	<i>quality</i>	A scalar value assessment of image quality. The legal values are [0,100] So, a low value indicates high expected FNMR. A value of -1.0 indicates a failed attempt to calculate a quality score or the value is unassigned.

Returns

OFIQ::ReturnStatus

Implemented in [OFIQ_LIB::OFIQImpl](#).

7.33.3.5 vectorQuality()

```
virtual OFIQ::ReturnStatus OFIQ::Interface::vectorQuality (
    const OFIQ::Image & image,
    OFIQ::FaceImageQualityAssessment & assessments ) [pure virtual]
```

This function takes an image and outputs quality information.

Implementing functions should be performed on the largest detected face.

Parameters

in	<i>image</i>	Single face image
out	<i>assessments</i>	An ImageQualityAssessments structure. The implementation should populate 1) the bounding box and 2) those items in the QualityAssessments object that the developer chooses to implement 3) face landmarks

Returns

OFIQ::ReturnStatus

Implemented in [OFIQ_LIB::OFIQImpl](#).

7.33.3.6 vectorQualityWithPreprocessingResults()

```
virtual OFIQ::ReturnStatus OFIQ::Interface::vectorQualityWithPreprocessingResults (
    const OFIQ::Image & image,
    OFIQ::FaceImageQualityAssessment & assessments,
    OFIQ::FaceImageQualityPreprocessingResult & preprocessingResult,
    uint32_t resultRequestsMask ) [pure virtual]
```

This function takes an image and outputs quality information and preprocessing results.

Implementing functions should be performed on the largest detected face.

Parameters

in	<i>image</i>	Single face image
out	<i>assessments</i>	An ImageQualityAssessments structure. The implementation should populate 1) the bounding box and 2) those items in the QualityAssessments object that the developer chooses to implement 3) face landmarks
out	<i>preprocessingResult</i>	A container in which the preprocessing results are stored.
in	<i>resultRequestsMask</i>	A bit mask encoding the preprocessing result types to be returned.

Returns

OFIQ::ReturnStatus

See also

[FaceImageQualityPreprocessingResult](#)

Implemented in [OFIQ_LIB::OFIQImpl](#).

The documentation for this class was generated from the following file:

- [ofiq_lib.h](#)

7.34 OFIQ_LIB::modules::landmarks::LandmarkPair Struct Reference

Data container for storing pairs of landmarks.

```
#include <PartExtractor.h>
```

Public Member Functions

- [LandmarkPair](#) ([OFIQ::LandmarkPoint](#) upper, [OFIQ::LandmarkPoint](#) lower)
Parameterized constructor.

Public Attributes

- [OFIQ::LandmarkPoint Upper](#)
First Landmark.
- [OFIQ::LandmarkPoint Lower](#)
second landmark

7.34.1 Detailed Description

Data container for storing pairs of landmarks.

in some computation special landmarks are bound together via the LandmarkPair struct.

7.34.2 Constructor & Destructor Documentation

7.34.2.1 LandmarkPair()

```
OFIQ_LIB::modules::landmarks::LandmarkPair::LandmarkPair (  
    OFIQ::LandmarkPoint upper,  
    OFIQ::LandmarkPoint lower ) [inline]
```

Parameterized constructor.

Parameters

in	<i>upper</i>	LandmarkPoint of first landmark.
in	<i>lower</i>	LandmarkPoint of second landmark.

7.34.3 Member Data Documentation

7.34.3.1 Lower

```
OFIQ::LandmarkPoint OFIQ_LIB::modules::landmarks::LandmarkPair::Lower
```

second landmark

7.34.3.2 Upper

```
OFIQ::LandmarkPoint OFIQ_LIB::modules::landmarks::LandmarkPair::Upper
```

First Landmark.

The documentation for this struct was generated from the following file:

- [PartExtractor.h](#)

7.35 LandmarkPoint Struct Reference

Data structure to describe the x and y coordinate of a landmark.

```
#include <ofiq_structs_c.h>
```

Public Attributes

- `int16_t x {-1}`
x - coordinate
- `int16_t y {-1}`
y - coordinate

7.35.1 Detailed Description

Data structure to describe the x and y coordinate of a landmark.

7.35.2 Member Data Documentation

7.35.2.1 x

```
int16_t LandmarkPoint::x {-1}
```

x - coordinate

7.35.2.2 y

```
int16_t LandmarkPoint::y {-1}
```

y - coordinate

The documentation for this struct was generated from the following file:

- [ofiq_structs_c.h](#)

7.36 OFIQ::LandmarkPoint Struct Reference

Data structure to describe the x and y coordinate of a landmark.

```
#include <ofiq_structs.h>
```

Public Member Functions

- [LandmarkPoint](#) ()=default
Default constructor.
- [LandmarkPoint](#) (int16_t i_x, int16_t i_y)
Parameterized constructor.

Public Attributes

- `int16_t x {-1}`
x - coordinate
- `int16_t y {-1}`
y - coordinate

7.36.1 Detailed Description

Data structure to describe the x and y coordinate of a landmark.

7.36.2 Constructor & Destructor Documentation**7.36.2.1 LandmarkPoint() [1/2]**

```
OFIQ::LandmarkPoint::LandmarkPoint ( ) [default]
```

Default constructor.

7.36.2.2 LandmarkPoint() [2/2]

```
OFIQ::LandmarkPoint::LandmarkPoint (
    int16_t i_x,
    int16_t i_y ) [inline]
```

Parameterized constructor.

Parameters

<i>i_x</i> ← _← <i>x</i>	x - coordinate of the landmark.
<i>i_y</i> ← _← <i>y</i>	y - coordinate of the landmark.

7.36.3 Member Data Documentation**7.36.3.1 x**

```
int16_t OFIQ::LandmarkPoint::x { -1 }
```

x - coordinate

7.36.3.2 y

```
int16_t OFIQ::LandmarkPoint::y {-1 }
```

y - coordinate

The documentation for this struct was generated from the following file:

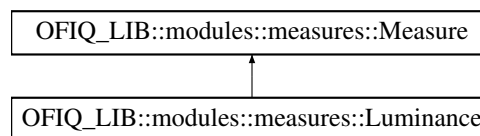
- [ofiq_structs.h](#)

7.37 OFIQ_LIB::modules::measures::Luminance Class Reference

Implementation of two luminance measures.

```
#include <Luminance.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::Luminance:



Public Member Functions

- [Luminance](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Assesses luminance mean and luminance variance measures.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from OFIQ_LIB::modules::measures::Measure

- void [AddSigmoid](#) (OFIQ::QualityMeasure measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) (OFIQ::QualityMeasure measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from OFIQ_LIB::modules::measures::Measure

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from OFIQ_LIB::modules::measures::Measure

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.37.1 Detailed Description

Implementation of two luminance measures.

On execution, two measures will be assessed: Luminance mean and luminance variance.

7.37.2 Constructor & Destructor Documentation

7.37.2.1 Luminance()

```
OFIQ_LIB::modules::measures::Luminance::Luminance (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.37.3 Member Function Documentation

7.37.3.1 Execute()

```
void OFIQ_LIB::modules::measures::Luminance::Execute (
    OFIQ_LIB::Session & session ) [override], [virtual]
```

Assesses luminance mean and luminance variance measures.

On execution, two measures will be assessed: Luminance mean and luminance variance.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

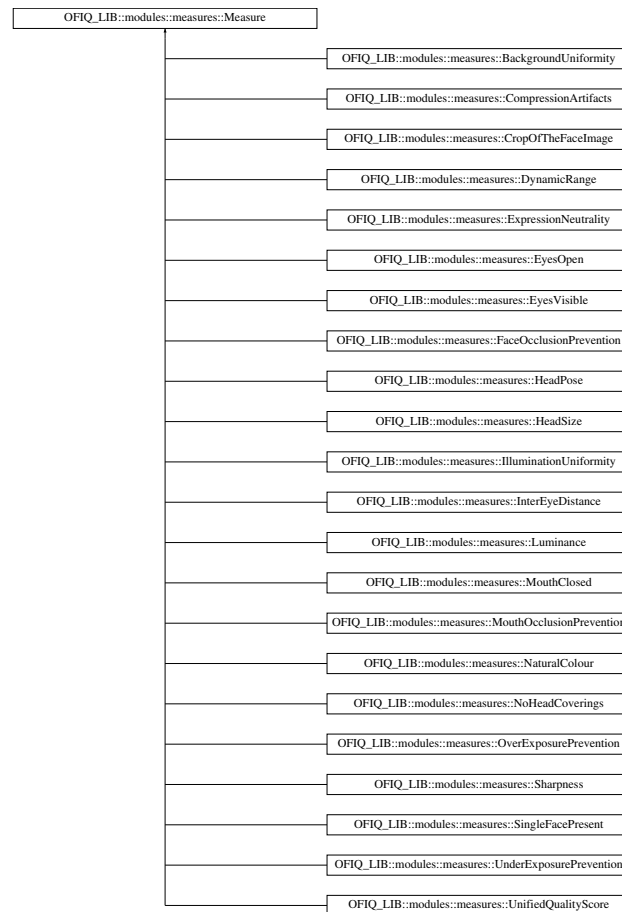
- [Luminance.h](#)

7.38 OFIQ_LIB::modules::measures::Measure Class Reference

Base class for measures implemented in OFIQ.

```
#include <Measure.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::Measure:



Public Member Functions

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual void [Execute](#) ([OFIQ_LIB::Session](#) &session)=0
Abstract quality assessment function.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Protected Member Functions

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

Static Private Member Functions

- static double [ScalarConversion](#) (double rawValue, const [SigmoidParameters](#) &par)
Applies a sigmoid-based quality mapping to a native quality score and outputs the result.
- static std::string [GetMeasureName](#) ([OFIQ::QualityMeasure](#) measure)
Returns the name of the specified measure.
- static std::string [ExpandKey](#) (std::string_view rawKey)
Expands the raw key of a measure to the key accessing its Sigmoid-based quality mapping.

Private Attributes

- std::map< std::string, [SigmoidParameters](#), std::less<> > [m_sigmoidMap](#)
Used to map the measure name to the sigmoid-based quality mapping function.
- [OFIQ::QualityMeasure](#) [m_measure](#) = [OFIQ::QualityMeasure::NotSet](#)
Value encoding the measure type.

7.38.1 Detailed Description

Base class for measures implemented in OFIQ.

7.38.2 Constructor & Destructor Documentation

7.38.2.1 Measure()

```
OFIQ_LIB::modules::measures::Measure::Measure (
    const Configuration & configuration,
    OFIQ::QualityMeasure measure ) [inline]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
<i>measure</i>	Enum of the measure.

7.38.2.2 ~Measure()

```
virtual OFIQ_LIB::modules::measures::Measure::~~Measure ( ) [virtual], [default]
```

Destructor.

7.38.3 Member Function Documentation

7.38.3.1 AddSigmoid() [1/2]

```
void OFIQ_LIB::modules::measures::Measure::AddSigmoid (
    const std::string & key,
    SigmoidParameters defaultValues ) [protected]
```

Reads sigmoid-function based quality mapping from the configuration.

The parameters are read from the configuration reference member [OFIQ_LIB::modules::measures::Measure::configuration](#). If a parameter is not configured, its default value is chosen from the `defaultValues` argument.

Parameters

<i>key</i>	Key/name of the measure of which mapping is configured.
<i>defaultValues</i>	Parameters from which default values of non-configured parameters are chosen.

7.38.3.2 AddSigmoid() [2/2]

```
void OFIQ_LIB::modules::measures::Measure::AddSigmoid (
    OFIQ::QualityMeasure measure,
    const SigmoidParameters & defaultValues ) [protected]
```

Reads sigmoid-function based quality mapping from the configuration.

The parameters are read from the configuration reference member [OFIQ_LIB::modules::measures::Measure::configuration](#). If a parameter is not configured, its default value is chosen from the `defaultValues` argument.

Parameters

<i>measure</i>	Enum value encoding the measure for which the mapping is configured.
<i>defaultValues</i>	Parameters from which default values of non-configured parameters are chosen.

7.38.3.3 Execute()

```
virtual void OFIQ_LIB::modules::measures::Measure::Execute (
    OFIQ_LIB::Session & session ) [pure virtual]
```

Abstract quality assessment function.

After quality assessment of the implemented measure, the method should invoke [SetQualityMeasure\(\)](#) to insert the result of quality assessment in `session`.

Parameters

<i>session</i>	Session object containing the original facial image and pre-processing results computed by the OFIQImpl::preprocess() method.
----------------	---

Implemented in [OFIQ_LIB::modules::measures::BackgroundUniformity](#), [OFIQ_LIB::modules::measures::CompressionArtifacts](#), [OFIQ_LIB::modules::measures::CropOfTheFacelImage](#), [OFIQ_LIB::modules::measures::DynamicRange](#), [OFIQ_LIB::modules::measures::EyesOpen](#), [OFIQ_LIB::modules::measures::EyesVisible](#), [OFIQ_LIB::modules::measures::FaceOcclusion](#), [OFIQ_LIB::modules::measures::HeadPose](#), [OFIQ_LIB::modules::measures::HeadSize](#), [OFIQ_LIB::modules::measures::IlluminationLevel](#), [OFIQ_LIB::modules::measures::InterEyeDistance](#), [OFIQ_LIB::modules::measures::Luminance](#), [OFIQ_LIB::modules::measures::MouthOcclusion](#), [OFIQ_LIB::modules::measures::MouthOcclusionPrevention](#), [OFIQ_LIB::modules::measures::NaturalColour](#), [OFIQ_LIB::modules::measures::NoHeadCoverings](#), [OFIQ_LIB::modules::measures::OverExposurePrevention](#), [OFIQ_LIB::modules::measures::Sharpness](#), [OFIQ_LIB::modules::measures::SingleFacePresent](#), [OFIQ_LIB::modules::measures::UnifiedQualityScore](#) and [OFIQ_LIB::modules::measures::UnifiedQualityScore](#).

7.38.3.4 ExecuteScalarConversion() [1/2]

```
double OFIQ_LIB::modules::measures::Measure::ExecuteScalarConversion (
    const std::string & key,
    double rawValue ) [protected]
```

Maps a native quality score to a quality component value.

Parameters

<i>key</i>	Key/name of the measure used to read parameters from a private map member.
<i>rawValue</i>	Native quality score.

Returns

Quality component value.

7.38.3.5 ExecuteScalarConversion() [2/2]

```
double OFIQ_LIB::modules::measures::Measure::ExecuteScalarConversion (
    OFIQ::QualityMeasure measure,
    double rawValue ) [protected]
```

Maps a native quality score to a quality component value.

Parameters

<i>measure</i>	Enum value of the measure used to read parameters from a private map member.
<i>rawValue</i>	Native quality score.

Returns

Quality component value.

7.38.3.6 ExpandKey()

```
static std::string OFIQ_LIB::modules::measures::Measure::ExpandKey (
    std::string_view rawKey ) [static], [private]
```

Expands the raw key of a measure to the key accessing its Sigmoid-based quality mapping.

Parameters

<i>rawKey</i>	representation of the measure (e.g., "BackgroundUniformity").
---------------	---

Returns

std::string representation of the key accessing the quality mapping function configuration (e.g., "params.↔ measures.BackgroundUniformity.Sigmoid").

7.38.3.7 GetMeasureName()

```
static std::string OFIQ_LIB::modules::measures::Measure::GetMeasureName (
    OFIQ::QualityMeasure measure ) [static], [private]
```

Returns the name of the specified measure.

Parameters

<i>measure</i>	Enum value of a measure.
----------------	--------------------------

Returns

std::string representation of the requested measure.

7.38.3.8 GetName()

```
virtual std::string OFIQ_LIB::modules::measures::Measure::GetName ( ) const [virtual]
```

Returns the name of the measure.

Unless overwritten, the member `m_measure` is passed to the private `GetMeasureName()` method and the result is returned.

Returns

std::string representation of the measures.

7.38.3.9 GetQualityMeasure()

```
virtual OFIQ::QualityMeasure OFIQ_LIB::modules::measures::Measure::GetQualityMeasure ( ) const
[virtual]
```

Returns an enum encoding the measure.

Returns

Enum encoding the measure.

7.38.3.10 ScalarConversion()

```
static double OFIQ_LIB::modules::measures::Measure::ScalarConversion (
    double rawValue,
    const SigmoidParameters & par ) [inline], [static], [private]
```

Applies a sigmoid-based quality mapping to a native quality score and outputs the result.

Before output, the result is checked if it is below 0 or above

1. If it is below 0, then 0 is returned. If it is above 100, then 100 is returned.

Parameters

<i>rawValue</i>	Native quality score.
<i>par</i>	Parameters of the sigmoid-based quality mapping.

Returns

The mapped quality value.

7.38.3.11 SetQualityMeasure()

```
void OFIQ_LIB::modules::measures::Measure::SetQualityMeasure (
    OFIQ_LIB::Session & session,
    OFIQ::QualityMeasure measure,
    double rawValue,
    OFIQ::QualityMeasureReturnCode code )
```

Inserts the result of a quality assessment in the session object.

The method [ExecuteScalarConversion\(\)](#) is invoked to map the native quality score to its quality component value.

Parameters

<i>session</i>	Session object containing the original facial image and pre-processing results computed by the OFIQImpl::preprocess() method.
<i>measure</i>	Enum value specifying this measure.
<i>rawValue</i>	Native quality score
<i>code</i>	Value indicating whether the quality assessment was computed successfully or otherwise (e.g., failureToAssess).

7.38.3.12 Sigmoid()

```
static double OFIQ_LIB::modules::measures::Measure::Sigmoid (
    double x,
    double x0,
    double w ) [inline], [static], [protected]
```

Sigmoid function.

Parameters

x	Native quality score
$x0$	Non-zero center point
w	Divisor

Returns

$$(1 + \exp((x0 - x)/w))^{-1}.$$

7.38.4 Member Data Documentation

7.38.4.1 configuration

```
const Configuration& OFIQ_LIB::modules::measures::Measure::configuration [protected]
```

Reference to the configuration with which the measure constructor has been invoked.

7.38.4.2 m_measure

```
OFIQ::QualityMeasure OFIQ_LIB::modules::measures::Measure::m_measure = OFIQ::QualityMeasure::NotSet
[private]
```

Value encoding the measure type.

The value is set to [QualityMeasure::NotSet](#) by default which effectively corresponds to a non-specified measure.

7.38.4.3 m_sigmoidMap

```
std::map<std::string, SigmoidParameters, std::less<> > OFIQ_LIB::modules::measures::Measure↵
::m_sigmoidMap [private]
```

Used to map the measure name to the sigmoid-based quality mapping function.

The documentation for this class was generated from the following file:

- [Measure.h](#)

7.39 OFIQ_LIB::modules::measures::MeasureFactory Class Reference

Measure factor class.

```
#include <MeasureFactory.h>
```

Public Member Functions

- [MeasureFactory](#) ()=delete

Static Public Member Functions

- static `std::unique_ptr< Measure > CreateMeasure` (const [OFIQ::QualityMeasure](#) measure, const [Configuration](#) &configuration)

Requests the creation of a measure implementation.

7.39.1 Detailed Description

Measure factor class.

7.39.2 Constructor & Destructor Documentation

7.39.2.1 MeasureFactory()

```
OFIQ_LIB::modules::measures::MeasureFactory::MeasureFactory ( ) [delete]
```

7.39.3 Member Function Documentation

7.39.3.1 CreateMeasure()

```
static std::unique_ptr< Measure > OFIQ_LIB::modules::measures::MeasureFactory::CreateMeasure (
    const OFIQ::QualityMeasure measure,
    const Configuration & configuration ) [static]
```

Requests the creation of a measure implementation.

Parameters

<i>measure</i>	Enum value encoding the requested measure.
<i>configuration</i>	Configuration from which measure parameters are read.

Attention

The function returns `nullptr` if the request of a measure is not implemented by the function.

If any constructor of a requested measures throws something, the throw is forwarded to this function.

The documentation for this class was generated from the following file:

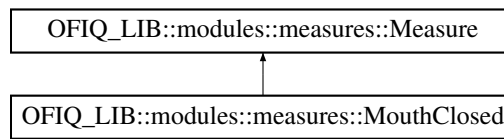
- [MeasureFactory.h](#)

7.40 OFIQ_LIB::modules::measures::MouthClosed Class Reference

Implementation of the mouth closed measure.

```
#include <MouthClosed.h>
```


Inheritance diagram for OFIQ_LIB::modules::measures::MouthClosed:



Public Member Functions

- [MouthClosed](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Assesses mouth closeness.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & [configuration](#)

Reference to the configuration with which the measure constructor has been invoked.

7.40.1 Detailed Description

Implementation of the mouth closed measure.

Mouth closed assessment based on computing a ratio from mouth landmarks.

7.40.2 Constructor & Destructor Documentation

7.40.2.1 MouthClosed()

```
OFIQ_LIB::modules::measures::MouthClosed::MouthClosed (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.40.3 Member Function Documentation

7.40.3.1 Execute()

```
void OFIQ_LIB::modules::measures::MouthClosed::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Assesses mouth closeness.

Mouth closed assessment based on computing a ratio from mouth landmarks.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

See also

[Session::getAlignedFaceLandmarks\(\)](#)

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

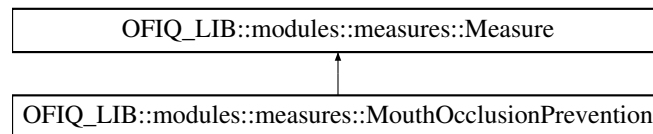
- [MouthClosed.h](#)

7.41 OFIQ_LIB::modules::measures::MouthOcclusionPrevention Class Reference

Implementation of the mouth occlusion prevention measure.

```
#include <MouthOcclusionPrevention.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::MouthOcclusionPrevention:



Public Member Functions

- [MouthOcclusionPrevention](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) (OFIQ_LIB::Session &session) override
Assesses absence of mouth occlusion.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, OFIQ::QualityMeasure measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual OFIQ::QualityMeasure [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) (OFIQ_LIB::Session &session, OFIQ::QualityMeasure measure, double rawValue, OFIQ::QualityMeasureReturnCode code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) (OFIQ::QualityMeasure measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) (OFIQ::QualityMeasure measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.41.1 Detailed Description

Implementation of the mouth occlusion prevention measure.

Absence of mouth occlusion is assessed by measuring the coverage of the mouth region from mouth landmarks with the result of face occlusion segmentation computed on pre-processing.

7.41.2 Constructor & Destructor Documentation

7.41.2.1 MouthOcclusionPrevention()

```
OFIQ_LIB::modules::measures::MouthOcclusionPrevention::MouthOcclusionPrevention (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.41.3 Member Function Documentation

7.41.3.1 Execute()

```
void OFIQ_LIB::modules::measures::MouthOcclusionPrevention::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Assesses absence of mouth occlusion.

Absence of mouth occlusion is assessed by measuring the coverage of the mouth region from mouth landmarks with the result of face occlusion segmentation computed on pre-processing. Pre-processing results are passed to the method with the `session` parameter.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

See also

[FaceOcclusionSegmentation](#)
[Session::getAlignedFaceLandmarks\(\)](#)

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

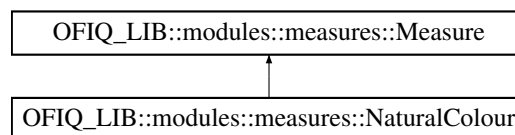
- [MouthOcclusionPrevention.h](#)

7.42 OFIQ_LIB::modules::measures::NaturalColour Class Reference

Implementation of the natural colour measure.

```
#include <NaturalColour.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::NaturalColour:



Public Member Functions

- [NaturalColour](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Assesses natural colourness.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Private Member Functions

- `cv::Mat CreateMaskedImage` (const `OFIQ::FaceLandmarks` &landmarks, const `cv::Mat` &cvImage) const
Creates a mask image from the convex full of the specified landmarks.
- `cv::Mat ReducelImageToRegionOfInterest` (const `cv::Mat` &maskedImage, const `cv::Rect` &leftRegionOfInterest, const `cv::Rect` &rightRegionOfInterest) const
Extracts two rectangular regions from an image and returns its concatenation.
- `double CalculateScore` (double meanChannelA, double meanChannelB) const
Combines two CIELAB values a^ and b^* to computed the native quality score.*

Additional Inherited Members

Protected Member Functions inherited from `OFIQ_LIB::modules::measures::Measure`

- `void AddSigmoid` (`OFIQ::QualityMeasure` measure, const `SigmoidParameters` &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- `void AddSigmoid` (const `std::string` &key, `SigmoidParameters` defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- `double ExecuteScalarConversion` (`OFIQ::QualityMeasure` measure, double rawValue)
Maps a native quality score to a quality component value.
- `double ExecuteScalarConversion` (const `std::string` &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from `OFIQ_LIB::modules::measures::Measure`

- `static double Sigmoid` (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from `OFIQ_LIB::modules::measures::Measure`

- `const Configuration` & configuration
Reference to the configuration with which the measure constructor has been invoked.

7.42.1 Detailed Description

Implementation of the natural colour measure.

Assessment of the naturalness of the colour based on the conversion of the RGB presentation of the image to the CIELAB colour space.

7.42.2 Constructor & Destructor Documentation

7.42.2.1 NaturalColour()

```
OFIQ_LIB::modules::measures::NaturalColour::NaturalColour (
    const Configuration & configuration ) [explicit]
```

Constructor.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.42.3 Member Function Documentation

7.42.3.1 CalculateScore()

```
double OFIQ_LIB::modules::measures::NaturalColour::CalculateScore (
    double meanChannelA,
    double meanChannelB ) const [private]
```

Combines two CIELAB values a^* and b^* to computed the native quality score.

If a^* and b^* are both larger than or equals to 0, then the following formula is applied

$$D = \sqrt{\max(\max(0, 5 - a^*), \max(0, a^* - 25))^2 + \max(\max(0, 5 - b^*), \max(b^* - 35))^2}$$

and D is returned; otherwise, the value 100 is returned.

Parameters

<i>meanChannelA</i>	The CIELAB value a^* input value.
<i>meanChannelB</i>	The CIELAB value b^* input value.

Returns

Native quality score

7.42.3.2 CreateMaskedImage()

```
cv::Mat OFIQ_LIB::modules::measures::NaturalColour::CreateMaskedImage (
    const OFIQ::FaceLandmarks & landmarks,
    const cv::Mat & cvImage ) const [private]
```

Creates a mask image from the convex full of the specified landmarks.

Parameters

<i>landmarks</i>	Facial landmarks.
<i>cvImage</i>	The mask image returned has the same dimension as <i>cvImage</i> .

Returns

Mask image

7.42.3.3 Execute()

```
void OFIQ_LIB::modules::measures::NaturalColour::Execute (
    OFIQ_LIB::Session & session ) [override], [virtual]
```

Assesses natural colourness.

Assessment of the naturalness of the colour based on the conversion of the RGB presentation of the image to the CIELAB colour space.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

7.42.3.4 ReduceImageToRegionOfInterest()

```
cv::Mat OFIQ_LIB::modules::measures::NaturalColour::ReduceImageToRegionOfInterest (
    const cv::Mat & maskedImage,
    const cv::Rect & leftRegionOfInterest,
    const cv::Rect & rightRegionOfInterest ) const [private]
```

Extracts two rectangular regions from an image and returns its concatenation.

Parameters

<i>maskedImage</i>	The input image from which the two regions are extracted.
<i>leftRegionOfInterest</i>	First region
<i>rightRegionOfInterest</i>	Second region

Returns

Concatenation of the requested regions; the first columns correspond to `rightRegionOfInterest` and the last columns correspond to `leftRegionOfInterest`.

Attention

An error occurs if the height of the two requested regions differ.

The documentation for this class was generated from the following file:

- [NaturalColour.h](#)

7.43 OFIQ_LIB::NeuronalNetworkContainer Struct Reference

Neural network container for OFIQ's preprocessing steps.

```
#include <NeuronalNetworkContainer.h>
```


Public Member Functions

- [NeuronalNetworkContainer](#) (std::shared_ptr< [FaceDetectorInterface](#) > [faceDetector](#), std::shared_ptr< [FaceLandmarkExtractorInterface](#) > [landmarkExtractor](#), std::shared_ptr< [SegmentationExtractorInterface](#) > [segmentationExtractor](#), std::shared_ptr< [PoseEstimatorInterface](#) > [poseEstimator](#), std::shared_ptr< [SegmentationExtractorInterface](#) > [faceOcclusionExtractor](#))

Constructor.

Public Attributes

- std::shared_ptr< [FaceDetectorInterface](#) > [faceDetector](#)
Pointer to a [FaceDetectorInterface](#) .
- std::shared_ptr< [FaceLandmarkExtractorInterface](#) > [landmarkExtractor](#)
Pointer to a [FaceLandmarkExtractorInterface](#) .
- std::shared_ptr< [SegmentationExtractorInterface](#) > [segmentationExtractor](#)
Pointer to a [SegmentationExtractorInterface](#) .
- std::shared_ptr< [PoseEstimatorInterface](#) > [poseEstimator](#)
Pointer to a [SegmentationExtractorInterface](#) .
- std::shared_ptr< [SegmentationExtractorInterface](#) > [faceOcclusionExtractor](#)
Pointer to a [SegmentationExtractorInterface](#) .

7.43.1 Detailed Description

Neural network container for OFIQ's preprocessing steps.

7.43.2 Constructor & Destructor Documentation

7.43.2.1 NeuronalNetworkContainer()

```
OFIQ_LIB::NeuronalNetworkContainer::NeuronalNetworkContainer (
    std::shared_ptr< FaceDetectorInterface > faceDetector,
    std::shared_ptr< FaceLandmarkExtractorInterface > landmarkExtractor,
    std::shared_ptr< SegmentationExtractorInterface > segmentationExtractor,
    std::shared_ptr< PoseEstimatorInterface > poseEstimator,
    std::shared_ptr< SegmentationExtractorInterface > faceOcclusionExtractor ) [inline]
```

Constructor.

Parameters

<i>faceDetector</i>	Implementation of a FaceDetectorInterface
<i>landmarkExtractor</i>	Implementation of a FaceLandmarkExtractorInterface
<i>segmentationExtractor</i>	Implementation of a SegmentationExtractorInterface . A pointer to an object instantiated from the FaceParsing class would be valid.
<i>poseEstimator</i>	Implementation of a PoseEstimatorInterface
<i>faceOcclusionExtractor</i>	Implementation of a SegmentationExtractorInterface . A pointer to an object instantiated from the FaceOcclusionSegmentation class would be valid.

7.43.3 Member Data Documentation

7.43.3.1 faceDetector

```
std::shared_ptr<FaceDetectorInterface> OFIQ_LIB::NeuronalNetworkContainer::faceDetector
```

Pointer to a [FaceDetectorInterface](#) .

7.43.3.2 faceOcclusionExtractor

```
std::shared_ptr<SegmentationExtractorInterface> OFIQ_LIB::NeuronalNetworkContainer::faceOcclusionExtractor
```

Pointer to a [SegmentationExtractorInterface](#) .

A pointer to an object instantiated from the [FaceOcclusionSegmentation](#) class would be valid.

7.43.3.3 landmarkExtractor

```
std::shared_ptr<FaceLandmarkExtractorInterface> OFIQ_LIB::NeuronalNetworkContainer::landmarkExtractor
```

Pointer to a [FaceLandmarkExtractorInterface](#) .

7.43.3.4 poseEstimator

```
std::shared_ptr<PoseEstimatorInterface> OFIQ_LIB::NeuronalNetworkContainer::poseEstimator
```

Pointer to a [SegmentationExtractorInterface](#) .

7.43.3.5 segmentationExtractor

```
std::shared_ptr<SegmentationExtractorInterface> OFIQ_LIB::NeuronalNetworkContainer::segmentationExtractor
```

Pointer to a [SegmentationExtractorInterface](#) .

A pointer to an object instantiated from the [FaceParsing](#) class would be valid.

The documentation for this struct was generated from the following file:

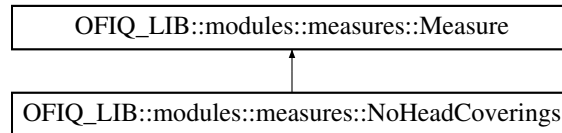
- [NeuronalNetworkContainer.h](#)

7.44 OFIQ_LIB::modules::measures::NoHeadCoverings Class Reference

Implementation of the no head covering measure.

```
#include <NoHeadCoverings.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::NoHeadCoverings:



Public Member Functions

- [NoHeadCoverings](#) (const [Configuration](#) &configuration)
Constructor.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Assesses no head covering.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Private Attributes

- double [m_t0](#)
Lower threshold.
- double [m_t1](#)
Upper threshold.
- double [m_w](#)
Standard deviation used in sigmoid function.
- double [m_x0](#)
Development point used in sigmoid function.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & configuration
Reference to the configuration with which the measure constructor has been invoked.

7.44.1 Detailed Description

Implementation of the no head covering measure.

The face parsing pre-processing assigns all pixels on the aligned image to one class each encoded by a value between 0 and 19 (inclusively). The values 16 and 18 encode the class *clothing* and *head covering*, respectively. Assessment of no head covering is done on the base of counting all pixels classified as clothing and head covering on the upper part of the aligned facial image and dividing it by the number of all pixels in the aligned image. The ratio is the native quality score. If it exceeds a configurable threshold, a quality the quality component value 0 is used; otherwise, if the ratio is below (or equals) the threshold, a quality of 100 is used.

See also

[FaceParsing](#)

7.44.2 Constructor & Destructor Documentation

7.44.2.1 NoHeadCoverings()

```
OFIQ_LIB::modules::measures::NoHeadCoverings::NoHeadCoverings (
    const Configuration & configuration ) [explicit]
```

Constructor.

The configuration object can optionally configure the threshold using the `params.measures.NoHeadCoverings.threshold` key which is 0.02 by default.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.44.3 Member Function Documentation

7.44.3.1 Execute()

```
void OFIQ_LIB::modules::measures::NoHeadCoverings::Execute (
    OFIQ_LIB::Session & session ) [override], [virtual]
```

Assesses no head covering.

The face parsing pre-processing assigns all pixels on the aligned image to one class each encoded by a value between 0 and 19 (inclusively). The values 16 and 18 encode the class *clothing* and *head covering*, respectively. Assessment of no head covering is done on the base of counting all pixels classified as clothing and head covering on the upper part of the aligned facial image and dividing it by the number of all pixels in the aligned image. The ratio is the native quality score. If it exceeds a configurable threshold, a quality the quality component value 0 is used; otherwise, if the ratio is below (or equals) the threshold, a quality of 100 is used.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

See also

[FaceParsing](#)

Implements [OFIQ_LIB::modules::measures::Measure](#).

7.44.4 Member Data Documentation

7.44.4.1 m_t0

```
double OFIQ_LIB::modules::measures::NoHeadCoverings::m_t0 [private]
```

Lower threshold.

If the native quality score (number of pixels classified as head covering divided by the number of total number of pixels in the aligned image), is below (or equals) this threshold, then a quality of 100 (best) is used;

7.44.4.2 m_t1

```
double OFIQ_LIB::modules::measures::NoHeadCoverings::m_t1 [private]
```

Upper threshold.

If the native quality score (number of pixels classified as head covering divided by the number of total number of pixels in the aligned image), is below (or equals) this threshold, then a quality of 0 (worst) is used;

7.44.4.3 m_w

```
double OFIQ_LIB::modules::measures::NoHeadCoverings::m_w [private]
```

Standard deviation used in sigmoid function.

If the native quality score is between (m_t0,m_t1), then the quality component value is interpoalted using a sigmoid function with a standard deviation specified by m_w.

7.44.4.4 m_x0

```
double OFIQ_LIB::modules::measures::NoHeadCoverings::m_x0 [private]
```

Development point used in sigmoid function.

If the native quality score is between (m_t0,m_t1), then the quality component value is interpoalted using a sigmoid function with a deveelopment point specified by m_x0.

The documentation for this class was generated from the following file:

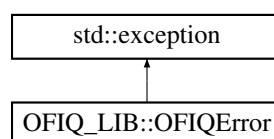
- [NoHeadCoverings.h](#)

7.45 OFIQ_LIB::OFIQError Class Reference

Implementation of a custom exception.

```
#include <OFIQError.h>
```

Inheritance diagram for OFIQ_LIB::OFIQError:



Public Member Functions

- [OFIQError](#) ([OFIQ::ReturnCode](#) returnCode, std::string_view message)
Contructor.
- const char * [what](#) () const noexcept override
Getter to the message, overwriting the what method of the base class.
- [OFIQ::ReturnCode](#) [whatCode](#) () const noexcept
Getter to the ReturnCode of the QFIQError.

Private Attributes

- [OFIQ::ReturnCode m_returnCode](#)
Member storing the ReturnCode.
- `std::string` [m_message](#)
Member, storing the message passed in the constructor.
- `std::string` [m_extendedMessage](#)
The extended message merges the ReturnCode and the message into one string.

7.45.1 Detailed Description

Implementation of a custom exception.

This exception is derived from the standard exception.

7.45.2 Constructor & Destructor Documentation

7.45.2.1 OFIQError()

```
OFIQ_LIB::OFIQError::OFIQError (
    OFIQ::ReturnCode returnCode,
    std::string_view message )
```

Contructor.

Parameters

<i>returnCode</i>	Based on the OFIQ::ReturnCode (see OFIQ::ReturnCode).
<i>message</i>	Message that will be attached to exception.

7.45.3 Member Function Documentation

7.45.3.1 what()

```
const char * OFIQ_LIB::OFIQError::what ( ) const [inline], [override], [noexcept]
```

Getter to the message, overwriting the what method of the base class.

Returns

`const char*` Pointer to the extended message.

7.45.3.2 whatCode()

```
OFIQ::ReturnCode OFIQ_LIB::OFIQError::whatCode ( ) const [inline], [noexcept]
```

Getter to the ReturnCode of the QFIQError.

Returns

[OFIQ::ReturnCode](#)

7.45.4 Member Data Documentation

7.45.4.1 m_extendedMessage

```
std::string OFIQ_LIB::OFIQError::m_extendedMessage [private]
```

The extended message merges the ReturnCode and the message into one string.

7.45.4.2 m_message

```
std::string OFIQ_LIB::OFIQError::m_message [private]
```

Member, storing the message passed in the constructor.

7.45.4.3 m_returnCode

```
OFIQ::ReturnCode OFIQ_LIB::OFIQError::m_returnCode [private]
```

Member storing the ReturnCode.

The documentation for this class was generated from the following file:

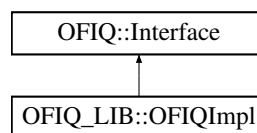
- [OFIQError.h](#)

7.46 OFIQ_LIB::OFIQImpl Class Reference

Implementation of the OFIQ_LIB.

```
#include <ofiq_lib_impl.h>
```

Inheritance diagram for OFIQ_LIB::OFIQImpl:



Public Member Functions

- [OFIQImpl](#) ()=default
Constructor.
- [~OFIQImpl](#) () override=default
Destructor.
- [OFIQ::ReturnStatus initialize](#) (const std::string &configDir, const std::string &configValue) override
Initialize the lib by reading the configuration file.
- [OFIQ::ReturnStatus scalarQuality](#) (const [OFIQ::Image](#) &face, double &quality) override
Compute an overall quality score for the image provided.
- [OFIQ::ReturnStatus vectorQuality](#) (const [OFIQ::Image](#) &image, [OFIQ::FacelImageQualityAssessment](#) &assessments) override
Run the computation of all measures set in the configuration.
- [OFIQ::ReturnStatus vectorQualityWithPreprocessingResults](#) (const [OFIQ::Image](#) &image, [OFIQ::FacelImageQualityAssessment](#) &assessments, [OFIQ::FacelImageQualityPreprocessingResult](#) &preprocessingResult, uint32_t result←RequestsMask=static_cast< int >(OFIQ::PreprocessingResultType::All)) override
Run the computation of all measures set in the configuration and access pre-precessing result.

Public Member Functions inherited from OFIQ::Interface

- virtual [~Interface](#) ()=default
Default Destructor.
- [OFIQ_EXPORT](#) void [getVersion](#) (int &major, int &minor, int &patch) const
Access version information.

Private Member Functions

- std::unique_ptr< [OFIQ_LIB::modules::measures::Executor](#) > [CreateExecutor](#) ()
Create a Executor object.
- void [CreateNetworks](#) ()
Create a NeuronalNetworkContainer.
- [OFIQ::ReturnStatus](#) [preprocess](#) ([Session](#) &session)
Perform the preprocessing.
- [OFIQ::ReturnStatus](#) [performAssessment](#) ([Session](#) &session)
Perform the assessment.
- void [alignFacelImage](#) ([Session](#) &session) const
Perform the face alignment.
- void [computeAlignedFaceLuminanceImage](#) ([Session](#) &session) const
Compute luminance image for aligned face image.
- [OFIQ::ReturnStatus](#) [getPreprocessingResults](#) (const [Session](#) &session, [OFIQ::FacelImageQualityPreprocessingResult](#) &preprocessingResult, uint32_t resultRequestsMask) const
Processes and image and outputs its quality assessment; optionally, if requested, pre-processing data can be output by the function.

Private Attributes

- std::unique_ptr< [OFIQ_LIB::modules::measures::Executor](#) > [m_executorPtr](#)
Pointer to the executor instance, see [OFIQ_LIB::modules::measures::Executor](#).
- std::unique_ptr< [Configuration](#) > [config](#)
Pointer to the cinfiguration.
- std::unique_ptr< [NeuronalNetworkContainer](#) > [networks](#)
Pointer to the different neural network instances, used during the preprocessing.

Additional Inherited Members

Static Public Member Functions inherited from OFIQ::Interface

- static [OFIQ_EXPORT](#) std::shared_ptr< [Interface](#) > [getImplementation](#) ()
Factory method to return a shared pointer to the Interface object.

7.46.1 Detailed Description

Implementation of the OFIQ_LIB.

7.46.2 Constructor & Destructor Documentation

7.46.2.1 OFIQImpl()

```
OFIQ_LIB::OFIQImpl::OFIQImpl ( ) [default]
```

Constructor.

7.46.2.2 ~OFIQImpl()

```
OFIQ_LIB::OFIQImpl::~~OFIQImpl ( ) [override], [default]
```

Destructor.

7.46.3 Member Function Documentation

7.46.3.1 alignFacelImage()

```
void OFIQ_LIB::OFIQImpl::alignFaceImage (
    Session & session ) const [private]
```

Perform the face alignment.

Parameters

<i>session</i>	Session object containing the original facial image and pre-processing results computed by the OFIQImpl::preprocess() method
----------------	--

7.46.3.2 computeAlignedFaceLuminanceImage()

```
void OFIQ_LIB::OFIQImpl::computeAlignedFaceLuminanceImage (
    Session & session ) const [private]
```

Compute luminance image for aligned face image.

Parameters

<i>session</i>	Session object containing the original facial image and pre-processing results computed by the OFIQImpl::preprocess() method
----------------	--

7.46.3.3 CreateExecutor()

```
std::unique_ptr< OFIQ_LIB::modules::measures::Executor > OFIQ_LIB::OFIQImpl::CreateExecutor (
) [private]
```

Create a Executor object.

Returns

std::unique_ptr<OFIQ_LIB::modules::measures::Executor>

7.46.3.4 CreateNetworks()

```
void OFIQ_LIB::OFIQImpl::CreateNetworks ( ) [private]
```

Create a NeuronalNetworkContainer.

7.46.3.5 getPreprocessingResults()

```
OFIQ::ReturnStatus OFIQ_LIB::OFIQImpl::getPreprocessingResults (
    const Session & session,
    OFIQ::FaceImageQualityPreprocessingResult & preprocessingResult,
    uint32_t resultRequestsMask ) const [private]
```

Processes and image and outputs its quality assessment; optionally, if requested, pre-processing data can be output by the function.

Parameters

in	<i>session</i>	Session object containing the original facial image
out	<i>preprocessingResult</i>	Structure in which requested pre-processing data is stored
in	<i>resultRequestsMask</i>	Mask encoding the requested pre-processing results

See also

[FaceImageQualityPreprocessingResult](#)

7.46.3.6 initialize()

```
OFIQ::ReturnStatus OFIQ_LIB::OFIQImpl::initialize (
    const std::string & configDir,
    const std::string & configValue ) [override], [virtual]
```

Initialize the lib by reading the configuration file.

Parameters

<i>configDir</i>	Path to the configuration file.
<i>configValue</i>	Name of the configuration file.

Returns

OFIQ::ReturnStatus

Implements [OFIQ::Interface](#).

7.46.3.7 performAssessment()

```
OFIQ::ReturnStatus OFIQ_LIB::OFIQImpl::performAssessment (
    Session & session ) [private]
```

Perform the assessment.

Parameters

<i>session</i>	Session object containing the original facial image and pre-processing results computed by the OFIQImpl::preprocess() method
----------------	--

7.46.3.8 preprocess()

```
OFIQ::ReturnStatus OFIQ_LIB::OFIQImpl::preprocess (
    Session & session ) [private]
```

Perform the preprocessing.

Parameters

<i>session</i>	Session object containing the original facial image for which the preprocessing will be performed. The pre-processing results will be stored in the passed Session object.
----------------	--

7.46.3.9 scalarQuality()

```
OFIQ::ReturnStatus OFIQ_LIB::OFIQImpl::scalarQuality (
    const OFIQ::Image & face,
    double & quality ) [override], [virtual]
```

Compute an overall quality score for the image provided.

The overall quality score will be equal to the measure `ualityMeasure::UnifiedQualityScore` if it is activated. Otherwise, the overall quality score will be the mean of all active measure scores.

Parameters

in	<i>face</i>	Input image.
out	<i>quality</i>	Computed UnifiedQualityScore.

Returns

OFIQ::ReturnStatus

Implements [OFIQ::Interface](#).

7.46.3.10 vectorQuality()

```
OFIQ::ReturnStatus OFIQ_LIB::OFIQImpl::vectorQuality (
    const OFIQ::Image & image,
    OFIQ::FaceImageQualityAssessment & assessments ) [override], [virtual]
```

Run the computation of all measures set in the configuration.

Parameters

in	<i>image</i>	Input image.
out	<i>assessments</i>	Container to store the resulting scores.

Returns

OFIQ::ReturnStatus

Implements [OFIQ::Interface](#).

7.46.3.11 vectorQualityWithPreprocessingResults()

```
OFIQ::ReturnStatus OFIQ_LIB::OFIQImpl::vectorQualityWithPreprocessingResults (
    const OFIQ::Image & image,
    OFIQ::FaceImageQualityAssessment & assessments,
    OFIQ::FaceImageQualityPreprocessingResult & preprocessingResult,
    uint32_t resultRequestsMask = static_cast< int > (OFIQ::PreprocessingResultType::All)
) [override], [virtual]
```

Run the computation of all measures set in the configuration and access pre-processing result.

Parameters

in	<i>image</i>	Input image.
out	<i>assessments</i>	Container to store the resulting scores.
out	<i>preprocessingResult</i>	Container to store preprocessing results.
in	<i>resultRequestsMask</i>	Mask encoding the pre-processing data being requested.

Returns

OFIQ::ReturnStatus

See also

[PreprocessingResultType](#)

Implements [OFIQ::Interface](#).

7.46.4 Member Data Documentation

7.46.4.1 config

```
std::unique_ptr<Configuration> OFIQ_LIB::OFIQImpl::config [private]
```

Pointer to the cinfiguration.

7.46.4.2 m_executorPtr

```
std::unique_ptr<OFIQ_LIB::modules::measures::Executor> OFIQ_LIB::OFIQImpl::m_executorPtr  
[private]
```

Pointer to the executor instance, see [OFIQ_LIB::modules::measures::Executor](#).

7.46.4.3 networks

```
std::unique_ptr<NeuronalNetworkContainer> OFIQ_LIB::OFIQImpl::networks [private]
```

Pointer to the different neural network instances, used during the preprocessing.

The documentation for this class was generated from the following file:

- [ofiq_lib_impl.h](#)

7.47 ONNXRuntimeEnv Class Reference

Helper class to manage the ONNXRuntime session object.

```
#include <ONNXRTSegmentation.h>
```

Public Member Functions

- [ONNXRuntimeEnv](#) ()=default
Constructor.
- [~ONNXRuntimeEnv](#) ()=default
Destructor.
- void [initialize](#) (const std::vector< uint8_t > &i_modelData, int64_t i_imageWidth, int64_t i_imageHeight)
Public method to generate an ONNXRuntime session object.
- size_t [getNumberOfOutputNodes](#) () const
Get the number of output nodes (results) based on the loaded model.
- std::vector< Ort::Value > [run](#) (std::vector< float > &i_netInput)
Perform the computation.

Private Member Functions

- void [init_session](#) (const std::vector< uint8_t > &i_model_data, int64_t i_imageWidth, int64_t i_imageHeight)
Private method to generate an ONNXRuntime session object.

Private Attributes

- Ort::Env [m_ortenv](#)
Handle to the ONNXRuntime environment variable.
- Ort::MemoryInfo [m_memoryInfo](#) = Ort::MemoryInfo::CreateCpu(OrtDeviceAllocator, OrtMemTypeCPU)
ONNXRuntime variable to setup the tensors used in ONNXRuntime.
- std::array< int64_t, 4 > [m_inputShape](#)
Description of the shape of the input data expected by the model.
- std::unique_ptr< Ort::Session > [m_ortSession](#)
Handle to the ONNXRuntime session.

7.47.1 Detailed Description

Helper class to manage the ONNXRuntime session object.

Helper class to manage the ONNXRuntime session object. Details can be found on the ONNXRuntime documentation <https://onnxruntime.ai/docs/get-started/with-cpp.html>.

7.47.2 Constructor & Destructor Documentation

7.47.2.1 ONNXRuntimeEnv()

```
ONNXRuntimeEnv::ONNXRuntimeEnv ( ) [default]
```

Constructor.

7.47.2.2 ~ONNXRuntimeEnv()

```
ONNXRuntimeEnv::~ONNXRuntimeEnv ( ) [default]
```

Destructor.

7.47.3 Member Function Documentation

7.47.3.1 getNumberOfOutputNodes()

```
size_t ONNXRuntimeEnv::getNumberOfOutputNodes ( ) const
```

Get the number of output nodes (results) based on the loaded model.

Returns

size_t number of output nodes (results).

7.47.3.2 init_session()

```
void ONNXRuntimeEnv::init_session (
    const std::vector< uint8_t > & i_model_data,
    int64_t i_imageWidth,
    int64_t i_imageHeight ) [private]
```

Private method to generate an ONNXRuntime session object.

Parameters

<i>i_model_data</i>	Model data loaded from file.
<i>i_imageWidth</i>	Width of the input image as expected by the model.
<i>i_imageHeight</i>	Height of the input image as expected by the model.

7.47.3.3 initialize()

```
void ONNXRuntimeEnv::initialize (
    const std::vector< uint8_t > & i_modelData,
    int64_t i_imageWidth,
    int64_t i_imageHeight )
```

Public method to generate an ONNXRuntime session object.

Parameters

<i>i_modelData</i>	Model data loaded from file.
<i>i_imageWidth</i>	Width of the input image as expected by the model.
<i>i_imageHeight</i>	Height of the input image as expected by the model.

7.47.3.4 run()

```
std::vector< Ort::Value > ONNXRuntimeEnv::run (
    std::vector< float > & i_netInput )
```

Perform the computation.

Parameters

<i>i_netInput</i>	Input to the neural net.
-------------------	--------------------------

Returns

std::vector<Ort::Value> Result of the neural net computation.

7.47.4 Member Data Documentation**7.47.4.1 m_inputShape**

```
std::array<int64_t, 4> ONNXRuntimeEnv::m_inputShape [private]
```

Description of the shape of the input data expected by the model.

7.47.4.2 m_memoryInfo

```
Ort::MemoryInfo ONNXRuntimeEnv::m_memoryInfo = Ort::MemoryInfo::CreateCpu(OrtDeviceAllocator,
OrtMemTypeCPU) [private]
```

ONNXRuntime variable to setup the tensors used in ONNXRuntime.

7.47.4.3 m_ortenv

```
Ort::Env ONNXRuntimeEnv::m_ortenv [private]
```

Handle to the ONNXRuntime environment variable.

7.47.4.4 m_ortSession

```
std::unique_ptr<Ort::Session> ONNXRuntimeEnv::m_ortSession [private]
```

Handle to the ONNXRuntime session.

The documentation for this class was generated from the following file:

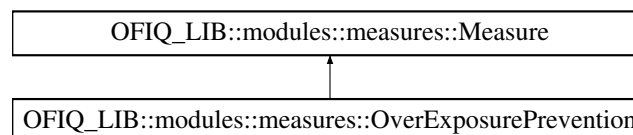
- [ONNXRTSegmentation.h](#)

7.48 OFIQ_LIB::modules::measures::OverExposurePrevention Class Reference

Implementation of the over-exposure prevention measure.

```
#include <OverExposurePrevention.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::OverExposurePrevention:



Public Member Functions

- [OverExposurePrevention](#) (const [Configuration](#) &configuration)
Constructor a new Over Exposure Prevention.
- void [Execute](#) (OFIQ_LIB::Session &session) override
Run the computation of the over-exposure prevention measure.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & configuration
Reference to the configuration with which the measure constructor has been invoked.

7.48.1 Detailed Description

Implementation of the over-exposure prevention measure.

The representation of a face is considered is light if it has a high proportion of pixels that have a high luminance value i.e. hot spots

7.48.2 Constructor & Destructor Documentation

7.48.2.1 OverExposurePrevention()

```
OFIQ_LIB::modules::measures::OverExposurePrevention::OverExposurePrevention (
    const Configuration & configuration ) [explicit]
```

Constructor a new Over Exposure Prevention.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.48.3 Member Function Documentation

7.48.3.1 Execute()

```
void OFIQ_LIB::modules::measures::OverExposurePrevention::Execute (
    OFIQ_LIB::Session & session ) [override], [virtual]
```

Run the computation of the over-exposure prevention measure.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

- [OverExposurePrevention.h](#)

7.49 OFIQ_LIB::modules::landmarks::PartExtractor Class Reference

Class that provides helper methods for the administration of landmarks.

```
#include <PartExtractor.h>
```

Static Public Member Functions

- static [OFIQ::Landmarks](#) [getFacePart](#) (const [OFIQ::FaceLandmarks](#) &faceLandmarks, [FaceParts](#) part)
Extract the landmarks that correspondend to the requested face part out of a set of provided landmarks.
- static `std::vector< LandmarkPair >` [getPairsForPart](#) (const [OFIQ::FaceLandmarks](#) &faceLandmarks, [FaceParts](#) part)
Get LandmarkPairs for a face part.

7.49.1 Detailed Description

Class that provides helper methods for the administration of landmarks.

7.49.2 Member Function Documentation

7.49.2.1 getFacePart()

```
static OFIQ::Landmarks OFIQ_LIB::modules::landmarks::PartExtractor::getFacePart (
    const OFIQ::FaceLandmarks & faceLandmarks,
    FaceParts part ) [static]
```

Extract the landmarks that correspondend to the requested face part out of a set of provided landmarks.

Parameters

in	<i>faceLandmarks</i>	Landmarks to be filtered.
	<i>part</i>	Face part of interest.

Returns

OFIQ::Landmarks Filtered landmarks that belong to the requested face part.

7.49.2.2 getPairsForPart()

```
static std::vector< LandmarkPair > OFIQ_LIB::modules::landmarks::PartExtractor::getPairsForPart (
    const OFIQ::FaceLandmarks & faceLandmarks,
    FaceParts part ) [static]
```

Get LandmarkPairs for a face part.

LandmarkPairs might be used to compute a distance between upper and lower landmark.

Parameters

<i>faceLandmarks</i>	Set of face landmarks.
<i>part</i>	Face part of interest.

Returns

std::vector<LandmarkPair>

The documentation for this class was generated from the following file:

- [PartExtractor.h](#)

7.50 Point2f Struct Reference

Representation of a point with floating point arithmetics.

```
#include <utils.h>
```

Public Attributes

- float *x*
- float *y*

7.50.1 Detailed Description

Representation of a point with floating point arithmetics.

7.50.2 Member Data Documentation

7.50.2.1 x

```
float Point2f::x
```

7.50.2.2 y

```
float Point2f::y
```

The documentation for this struct was generated from the following file:

- [utils.h](#)

7.51 OFIQ_LIB::Point2i Struct Reference

Representation of a point with integer arithmetics.

```
#include <utils.h>
```

Public Attributes

- [int x](#)
- [int y](#)

7.51.1 Detailed Description

Representation of a point with integer arithmetics.

7.51.2 Member Data Documentation

7.51.2.1 x

```
int OFIQ_LIB::Point2i::x
```

7.51.2.2 y

```
int OFIQ_LIB::Point2i::y
```

The documentation for this struct was generated from the following file:

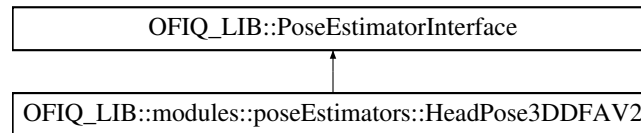
- [utils.h](#)

7.52 OFIQ_LIB::PoseEstimatorInterface Class Reference

Implementation of the base class for integrating pose estimation algorithms capable of estimating three head orientation angles (yaw, pitch and roll) from a face image.

```
#include <poseEstimators.h>
```

Inheritance diagram for OFIQ_LIB::PoseEstimatorInterface:



Public Types

- using [EulerAngle](#) = std::array<double, 3>

Public Member Functions

- virtual [~PoseEstimatorInterface](#) ()=default
Standard destructor.
- [EulerAngle](#) & [estimatePose](#) ([OFIQ_LIB::Session](#) &session)
This function estimates the three head orientation angles.

Protected Member Functions

- virtual void [updatePose](#) ([OFIQ_LIB::Session](#) &session, [EulerAngle](#) &pose)=0
Call to estimate the head orientations. Has to be implemented in the derived class.

Private Attributes

- std::string [m_lastSessionId](#)
id of the session that has been used in the latest request, for internal use.
- [EulerAngle](#) [m_pose](#)
Container for storing the estimated head orientations.

7.52.1 Detailed Description

Implementation of the base class for integrating pose estimation algorithms capable of estimating three head orientation angles (yaw, pitch and roll) from a face image.

7.52.2 Member Typedef Documentation

7.52.2.1 EulerAngle

```
using OFIQ_LIB::PoseEstimatorInterface::EulerAngle = std::array<double, 3>
```

The first entry encodes yaw, the second encodes pitch, and the third encodes roll.

7.52.3 Constructor & Destructor Documentation

7.52.3.1 ~PoseEstimatorInterface()

```
virtual OFIQ_LIB::PoseEstimatorInterface::~PoseEstimatorInterface ( ) [virtual], [default]
```

Standard destructor.

7.52.4 Member Function Documentation

7.52.4.1 estimatePose()

```
EulerAngle & OFIQ_LIB::PoseEstimatorInterface::estimatePose (
    OFIQ_LIB::Session & session )
```

This function estimates the three head orientation angles.

Parameters

<i>session</i>	Session object containing the original facial image and pre-processing results computed by the OFIQImpl::preprocess() method
----------------	--

7.52.4.2 updatePose()

```
virtual void OFIQ_LIB::PoseEstimatorInterface::updatePose (
    OFIQ_LIB::Session & session,
    EulerAngle & pose ) [protected], [pure virtual]
```

Call to estimate the head orientations. Has to be implemented in the derived class.

Parameters

<i>session</i>	Containing the input image for the estimation.
<i>pose</i>	Return the estimated pose.

Implemented in [OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2](#).

7.52.5 Member Data Documentation

7.52.5.1 m_lastSessionId

```
std::string OFIQ_LIB::PoseEstimatorInterface::m_lastSessionId [private]
```

id of the session that has been used in the latest request, for internal use.

7.52.5.2 m_pose

`EulerAngle` OFIQ_LIB::PoseEstimatorInterface::m_pose [private]

Container for storing the estimated head orientations.

The documentation for this class was generated from the following file:

- [poseEstimators.h](#)

7.53 PreprocessingResult Struct Reference

```
#include <ofiq_structs_c.h>
```

Public Attributes

- [BoundingBox](#) * [faces](#)
Bounding boxes of detected faces.
- `size_t` [numFaces](#)
Number of detected faces.
- [FaceLandmarks](#) [landmarks](#)
Facial landmarks.
- [Image](#) [segmentationMask](#)
Segmentation mask.
- [Image](#) [occlusionMask](#)
Occlusion mask.
- [Image](#) [landmarkedRegionMask](#)
Landmarked region.

7.53.1 Detailed Description

Examples

[C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h.](#)

7.53.2 Member Data Documentation

7.53.2.1 faces

`BoundingBox*` `PreprocessingResult::faces`

Bounding boxes of detected faces.

7.53.2.2 landmarkedRegionMask

`Image` `PreprocessingResult::landmarkedRegionMask`

Landmarked region.

For each pixel at index ($y * \text{width} + x$) of an image of dimension (height,width), the value is 0 if the pixel is not part of the landmarked region mask; otherwise, the value is 1

7.53.2.3 landmarks

`FaceLandmarks` `PreprocessingResult::landmarks`

Facial landmarks.

7.53.2.4 numFaces

`size_t` `PreprocessingResult::numFaces`

Number of detected faces.

7.53.2.5 occlusionMask

`Image` `PreprocessingResult::occlusionMask`

Occlusion mask.

For each pixel at index ($y * \text{width} + x$) of an image of dimension (height,width), the value is 0 if the pixel is not part of the occlusion mask; otherwise, the value is 1.

7.53.2.6 segmentationMask

`Image` `PreprocessingResult::segmentationMask`

Segmentation mask.

For each pixel at index ($y * \text{width} + x$) of an image of dimension (height,width), the value is assigned a segment type from the following table.

code	segment type
0	background
1	face skin
2	left eye brow
3	right eye brow
4	left eye
5	right eye
6	eyeglasses
7	left ear
8	right ear
9	earring
10	nose
11	mouth
12	upper lip
13	lower lip
14	neck
15	necklace
16	clothing
17	hair
18	head covering
19	undocumented
20	undocumented
21	undocumented
22	undocumented
23	undocumented
255	outside of the aligned face image

The documentation for this struct was generated from the following file:

- [ofiq_structs.c.h](#)

7.54 OFIQ::QualityMeasureResult Struct Reference

Data structure to handle the results of a quality measure.

```
#include <ofiq_structs.h>
```

Public Member Functions

- [QualityMeasureResult](#) ()=default
Default constructor.
- [QualityMeasureResult](#) (double [rawScore](#), double [scalar](#)=-1, [QualityMeasureReturnCode](#) code=[QualityMeasureReturnCode::No](#))
Parameterized constructor.

Public Attributes

- double `rawScore` { -1 }
Raw value as computed by the quality measure implementation.
- double `scalar` { -1 }
A scalar value from the interval [0,100] Higher values mean higher quality. A value of -1.0 indicates a failed attempt to calculate a quality score or the value is unassigned.
- `QualityMeasureReturnCode` `code` { `QualityMeasureReturnCode::NotInitialized` }
Return status code.

7.54.1 Detailed Description

Data structure to handle the results of a quality measure.

7.54.2 Constructor & Destructor Documentation

7.54.2.1 QualityMeasureResult() [1/2]

```
OFIQ::QualityMeasureResult::QualityMeasureResult ( ) [default]
```

Default constructor.

7.54.2.2 QualityMeasureResult() [2/2]

```
OFIQ::QualityMeasureResult::QualityMeasureResult (
    double rawScore,
    double scalar = -1,
    QualityMeasureReturnCode code = QualityMeasureReturnCode::NotInitialized ) [inline]
```

Parameterized constructor.

Parameters

in	<code>rawScore</code>	Computed raw score.
in	<code>scalar</code>	Computed scalar score.
in	<code>code</code>	QualityMeasureReturnCode describing the state of the computation.

7.54.3 Member Data Documentation

7.54.3.1 code

```
QualityMeasureReturnCode OFIQ::QualityMeasureResult::code { QualityMeasureReturnCode::NotInitialized
}
```

Return status code.

7.54.3.2 rawScore

```
double OFIQ::QualityMeasureResult::rawScore { -1 }
```

Raw value as computed by the quality measure implementation.

7.54.3.3 scalar

```
double OFIQ::QualityMeasureResult::scalar { -1 }
```

A scalar value from the interval [0,100] Higher values mean higher quality. A value of -1.0 indicates a failed attempt to calculate a quality score or the value is unassigned.

The documentation for this struct was generated from the following file:

- [ofiq_structs.h](#)

7.55 QualityMeasureResult Struct Reference

Data structure to handle the results of a quality measure.

```
#include <ofiq_structs_c.h>
```

Public Attributes

- [QualityMeasure](#) [measure](#) {[QualityMeasure::NotSet](#)}
- double [rawScore](#) {-1}
Raw value as computed by the quality measure implementation.
- double [scalar](#) {-1}
A scalar value from the interval [0,100] Higher values mean higher quality. A value of -1.0 indicates a failed attempt to calculate a quality score or the value is unassigned.
- [QualityMeasureReturnCode](#) [code](#) {[QualityMeasureReturnCode::NotInitialized](#)}
Return status code.

7.55.1 Detailed Description

Data structure to handle the results of a quality measure.

7.55.2 Member Data Documentation

7.55.2.1 code

```
QualityMeasureReturnCode QualityMeasureResult::code {QualityMeasureReturnCode::NotInitialized}
```

Return status code.

7.55.2.2 measure

```
QualityMeasure QualityMeasureResult::measure {QualityMeasure::NotSet}
```

7.55.2.3 rawScore

```
double QualityMeasureResult::rawScore {-1}
```

Raw value as computed by the quality measure implementation.

7.55.2.4 scalar

```
double QualityMeasureResult::scalar {-1}
```

A scalar value from the interval [0,100] Higher values mean higher quality. A value of -1.0 indicates a failed attempt to calculate a quality score or the value is unassigned.

The documentation for this struct was generated from the following file:

- [ofiq_structs_c.h](#)

7.56 OFIQ::ReturnStatus Struct Reference

A structure to contain information about a failure by the software under test.

```
#include <ofiq_structs.h>
```

Public Member Functions

- [ReturnStatus](#) ()=default
Default constructor.
- [ReturnStatus](#) (const [ReturnCode](#) code, const std::string &info="")
Parameterized constructor.

Public Attributes

- [ReturnCode](#) code { [ReturnCode::UnknownError](#) }
Return status code.
- std::string info
Optional information string.

7.56.1 Detailed Description

A structure to contain information about a failure by the software under test.

An object of this class allows the software to return some information from a function call. The string within this object can be optionally set to provide more information for debugging etc. The status code will be set by the function to Success on success, or one of the other codes on failure.

7.56.2 Constructor & Destructor Documentation

7.56.2.1 ReturnStatus() [1/2]

```
OFIQ::ReturnStatus::ReturnStatus ( ) [default]
```

Default constructor.

7.56.2.2 ReturnStatus() [2/2]

```
OFIQ::ReturnStatus::ReturnStatus (
    const ReturnCode code,
    const std::string & info = "" ) [inline]
```

Parameterized constructor.

Parameters

in	<i>code</i>	The return status code; required.
in	<i>info</i>	The optional information string.

7.56.3 Member Data Documentation

7.56.3.1 code

```
ReturnCode OFIQ::ReturnStatus::code { ReturnCode::UnknownError }
```

Return status code.

7.56.3.2 info

```
std::string OFIQ::ReturnStatus::info
```

Optional information string.

The documentation for this struct was generated from the following file:

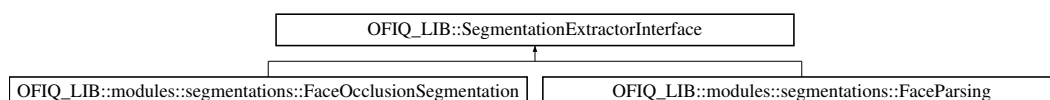
- [ofiq_structs.h](#)

7.57 OFIQ_LIB::SegmentationExtractorInterface Class Reference

Base class for the different implementation of segmentation algorithms.

```
#include <segmentations.h>
```

Inheritance diagram for OFIQ_LIB::SegmentationExtractorInterface:



Public Member Functions

- virtual [~SegmentationExtractorInterface](#) ()=default
Standard destructor.
- [OFIQ::Image](#) & [GetMask](#) ([OFIQ_LIB::Session](#) &session, [modules::segmentations::SegmentClassLabels](#) faceSegment)
Get a mask of the face region requested.

Protected Member Functions

- virtual [OFIQ::Image](#) [UpdateMask](#) ([OFIQ_LIB::Session](#) &session, [modules::segmentations::SegmentClassLabels](#) faceSegment)=0
Segmentation call that has to be implemented in the derived class.
- [std::string](#) [GetLastSessionId](#) () const
Accesses the last session id for this interface.

Private Attributes

- [std::string](#) [m_lastSessionId](#)
id of the session that has been used in the latest request, for internal use.
- [std::map](#)< [modules::segmentations::SegmentClassLabels](#), [OFIQ::Image](#) > [m_masks](#)
Container for storing the segmented face region masks.

7.57.1 Detailed Description

Base class for the different implementation of segmentation algorithms.

Base class for the FaceParsing (see [OFIQ_LIB::modules::segmentations::FaceParsing](#)) and FaceOcclusion↔ Segmentation (see [OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation](#))

7.57.2 Constructor & Destructor Documentation

7.57.2.1 ~SegmentationExtractorInterface()

```
virtual OFIQ_LIB::SegmentationExtractorInterface::~~SegmentationExtractorInterface ( ) [virtual],
[default]
```

Standard destructor.

7.57.3 Member Function Documentation

7.57.3.1 GetLastSessionId()

```
std::string OFIQ_LIB::SegmentationExtractorInterface::GetLastSessionId ( ) const [inline],
[protected]
```

Accesses the last session id for this interface.

Returns

Session id

7.57.3.2 GetMask()

```
OFIQ::Image & OFIQ_LIB::SegmentationExtractorInterface::GetMask (
    OFIQ_LIB::Session & session,
    modules::segmentations::SegmentClassLabels faceSegment )
```

Get a mask of the face region requested.

Parameters

<i>session</i>	Object containing the relevant data information on the input image.
<i>faceSegment</i>	Enum of the face region that is requested.

Returns

OFIQ::Image& Reference on the mask of the face region image.

7.57.3.3 UpdateMask()

```
virtual OFIQ::Image OFIQ_LIB::SegmentationExtractorInterface::UpdateMask (
    OFIQ_LIB::Session & session,
    modules::segmentations::SegmentClassLabels faceSegment ) [protected], [pure virtual]
```

Segmentation call that has to be implemented in the derived class.

Parameters

<i>session</i>	Object containing the relevant data information on the input image.
<i>faceSegment</i>	Enum of the face region that is requested

Returns

OFIQ::Image Segmented face region mask.

Implemented in [OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation](#), and [OFIQ_LIB::modules::segmentations::FacePar](#)

7.57.4 Member Data Documentation

7.57.4.1 m_lastSessionId

```
std::string OFIQ_LIB::SegmentationExtractorInterface::m_lastSessionId [private]
```

id of the session that has been used in the latest request, for internal use.

7.57.4.2 m_masks

```
std::map<modules::segmentations::SegmentClassLabels, OFIQ::Image> OFIQ_LIB::Segmentation↔
ExtractorInterface::m_masks [private]
```

Container for storing the segmented face region masks.

The documentation for this class was generated from the following file:

- [segmentations.h](#)

7.58 OFIQ_LIB::Session Class Reference

```
#include <Session.h>
```

Public Member Functions

- [Session](#) (const [OFIQ::Image](#) &image, [OFIQ::FacImageQualityAssessment](#) &assessment)
Construct a new Session object.
- const [OFIQ::Image](#) & [image](#) () const
Access reference to the input image, connected to this session.
- [OFIQ::FacImageQualityAssessment](#) & [assessment](#) ()
Access reference to the FacImageQualityAssessment object, connected to this session.
- const std::string & [Id](#) () const
Access to the id connected to this session.
- void [setDetectedFaces](#) (const std::vector< [OFIQ::BoundingBox](#) > &i_boundingBoxes)
Set the Detected Faces.
- std::vector< [OFIQ::BoundingBox](#) > [getDetectedFaces](#) () const
Get the Detected Faces.
- void [setPose](#) (const [EulerAngle](#) &i_pose)
Set the Pose of the input image.
- [EulerAngle](#) [getPose](#) () const
Get the Pose of the input image.
- void [setLandmarks](#) (const [OFIQ::FaceLandmarks](#) &i_landmarks)
Set the Landmarks detected on the input image.
- [OFIQ::FaceLandmarks](#) [getLandmarks](#) () const
Get the Landmarks detected on the input image.
- void [setAlignedFaceLandmarks](#) (const [OFIQ::FaceLandmarks](#) &i_landmarks)
Set the Aligned Face Landmarks detected on the aligned image.
- [OFIQ::FaceLandmarks](#) [getAlignedFaceLandmarks](#) () const
Get the Aligned Face Landmarks detected on the aligned image.
- void [setAlignedFaceTransformationMatrix](#) (const cv::Mat &i_transformationMatrix)
Set the Aligned Face Transformation Matrix.
- cv::Mat [getAlignedFaceTransformationMatrix](#) () const
Get the Aligned Face Transformation Matrix.
- void [setAlignedFace](#) (const cv::Mat &i_alignedFace)
Assigns the aligned face image.
- cv::Mat [getAlignedFace](#) () const
Get the Aligned Face object.
- void [setAlignedFaceLuminance](#) (const cv::Mat &i_alignedFaceLuminance)
Assigns the luminance image of the aligned face.
- cv::Mat [getAlignedFaceLuminance](#) () const
Get luminance image of the aligned face.
- void [setAlignedFaceLandmarkedRegion](#) (const cv::Mat &i_alignedFaceRegion)
Set the Aligned Face Landmarked Region.
- cv::Mat [getAlignedFaceLandmarkedRegion](#) () const
Get the Aligned Face Landmarked Region.
- void [setFaceParsingImage](#) (const cv::Mat &i_parsingImage)
Set the Face Parsing Image, see [OFIQ_LIB::modules::segmentations::FaceParsing](#).
- cv::Mat [getFaceParsingImage](#) () const
Get the Face Parsing Image, see [OFIQ_LIB::modules::segmentations::FaceParsing](#).
- void [setFaceOcclusionSegmentationImage](#) (const cv::Mat &i_segmentationImage)
Set the Face Occlusion Segmentation Image, see [OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation](#).
- cv::Mat [getFaceOcclusionSegmentationImage](#) () const
Get the Face Occlusion Segmentation Image, see [OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation](#).

Private Member Functions

- `std::string GenerateId () const`
Method for generating uuid's for the session.

Private Attributes

- `const OFIQ::Image & m_image`
Reference to the input image, connected to this session.
- `OFIQ::FaceImageQualityAssessment & m_assessment`
Refernce to the FaceImageQualityAssessment object, connected to this session.
- `std::vector< OFIQ::BoundingBox > m_detectedFaces`
Container for the faces found on the input image.
- `EulerAngle m_pose`
Container for storing the pose information.
- `OFIQ::FaceLandmarks m_landmarks`
Container for storing the landmark information.
- `OFIQ::FaceLandmarks m_alignedFaceLandmarks`
Container for storing the landmark information of the aligned image.
- `cv::Mat m_alignedFaceTransformationMatrix`
Container for storing the transformation matrix that led to the aligned image.
- `cv::Mat m_alignedFace`
Container for storing the aligned image.
- `cv::Mat m_alignedFaceLuminance`
Aligned face image converte to luminance/gray image.
- `cv::Mat m_alignedFacelandmarkedRegion`
Container for storing the landmarks of the aligned face image.
- `cv::Mat m_faceParsingImage`
Container for storing the segmented face image.
- `cv::Mat m_faceOcclusionSegmentationImage`
Container for storing the result of the face occlusion segmented image.
- `std::string m_id`
Container for storing the id of the session.

7.58.1 Detailed Description

@brief The session class is the data container used to distribute the image and additional data,

including the data computed during the pre-processing.

One instance of this class contains the relevant face information used for the computation of the activated measures. Most information is acquired during the pre-processing where the detection of the facial landmarks, the aligned image, etc. is computed.

7.58.2 Constructor & Destructor Documentation

7.58.2.1 Session()

```
OFIQ_LIB::Session::Session (
    const OFIQ::Image & image,
    OFIQ::FaceImageQualityAssessment & assessment ) [inline]
```

Construct a new Session object.

Parameters

<i>image</i>	Input image that shall be analysed.
<i>assessment</i>	Container to store the computed measures.

7.58.3 Member Function Documentation

7.58.3.1 `assessment()`

```
OFIQ::FaceImageQualityAssessment & OFIQ_LIB::Session::assessment ( ) [inline]
```

Access reference to the FaceImageQualityAssessment object, connected to this session.

Returns

quality assessment object reference.

7.58.3.2 `GenerateId()`

```
std::string OFIQ_LIB::Session::GenerateId ( ) const [private]
```

Method for generating uuid's for the session.

Returns

std::string

7.58.3.3 `getAlignedFace()`

```
cv::Mat OFIQ_LIB::Session::getAlignedFace ( ) const
```

Get the Aligned Face object.

Returns

cv::Mat

7.58.3.4 `getAlignedFaceLandmarkedRegion()`

```
cv::Mat OFIQ_LIB::Session::getAlignedFaceLandmarkedRegion ( ) const
```

Get the Aligned Face Landmarked Region.

Returns

cv::Mat

7.58.3.5 getAlignedFaceLandmarks()

```
OFIQ::FaceLandmarks OFIQ_LIB::Session::getAlignedFaceLandmarks ( ) const
```

Get the Aligned Face Landmarks detected on the aligned image.

Returns

OFIQ::FaceLandmarks

7.58.3.6 getAlignedFaceLuminance()

```
cv::Mat OFIQ_LIB::Session::getAlignedFaceLuminance ( ) const
```

Get luminance image of the aligned face.

Returns

cv::Mat

7.58.3.7 getAlignedFaceTransformationMatrix()

```
cv::Mat OFIQ_LIB::Session::getAlignedFaceTransformationMatrix ( ) const
```

Get the Aligned Face Transformation Matrix.

Returns

cv::Mat

7.58.3.8 getDetectedFaces()

```
std::vector< OFIQ::BoundingBox > OFIQ_LIB::Session::getDetectedFaces ( ) const
```

Get the Detected Faces.

Returns

std::vector<OFIQ::BoundingBox> Return the bounding boxes of faces found on the image.

7.58.3.9 getFaceOcclusionSegmentationImage()

```
cv::Mat OFIQ_LIB::Session::getFaceOcclusionSegmentationImage ( ) const
```

Get the Face Occlusion Segmentation Image, see [OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation](#))

Returns

cv::Mat

7.58.3.10 getFaceParsingImage()

```
cv::Mat OFIQ_LIB::Session::getFaceParsingImage ( ) const
```

Get the Face Parsing Image, see [OFIQ_LIB::modules::segmentations::FaceParsing](#)).

Returns

cv::Mat

7.58.3.11 getLandmarks()

```
OFIQ::FaceLandmarks OFIQ_LIB::Session::getLandmarks ( ) const
```

Get the Landmarks detected on the input image.

Returns

OFIQ::FaceLandmarks

7.58.3.12 getPose()

```
EulerAngle OFIQ_LIB::Session::getPose ( ) const
```

Get the Pose of the input image.

Returns

EulerAngle Pose of the ipnut image.

7.58.3.13 Id()

```
const std::string & OFIQ_LIB::Session::Id ( ) const [inline]
```

Access to the id connected to this session.

Returns

const std::string& Reference to the id of this session.

7.58.3.14 image()

```
const OFIQ::Image & OFIQ_LIB::Session::image ( ) const [inline]
```

Acess reference to the input image, connected to this session.

Returns

input image reference.

7.58.3.15 setAlignedFace()

```
void OFIQ_LIB::Session::setAlignedFace (
    const cv::Mat & i_alignedFace )
```

Assigns the aligned face image.

Parameters

<i>i_alignedFace</i>	
----------------------	--

7.58.3.16 setAlignedFaceLandmarkedRegion()

```
void OFIQ_LIB::Session::setAlignedFaceLandmarkedRegion (
    const cv::Mat & i_alignedFaceRegion )
```

Set the Aligned Face Landmarked Region.

Parameters

<i>i_alignedFaceRegion</i>	
----------------------------	--

7.58.3.17 setAlignedFaceLandmarks()

```
void OFIQ_LIB::Session::setAlignedFaceLandmarks (
    const OFIQ::FaceLandmarks & i_landmarks )
```

Set the Aligned Face Landmarks detected on the aligned image.

Parameters

<i>i_landmarks</i>	
--------------------	--

7.58.3.18 setAlignedFaceLuminance()

```
void OFIQ_LIB::Session::setAlignedFaceLuminance (
    const cv::Mat & i_alignedFaceLuminance )
```

Assigns the luminance image of the aligned face.

Parameters

<i>i_alignedFace</i>	
----------------------	--

7.58.3.19 setAlignedFaceTransformationMatrix()

```
void OFIQ_LIB::Session::setAlignedFaceTransformationMatrix (
    const cv::Mat & i_transformationMatrix )
```

Set the Aligned Face Transformation Matrix.

Parameters

<i>i_transformationMatrix</i>	
-------------------------------	--

7.58.3.20 setDetectedFaces()

```
void OFIQ_LIB::Session::setDetectedFaces (
    const std::vector< OFIQ::BoundingBox > & i_boundingBoxes )
```

Set the Detected Faces.

Parameters

<i>i_boundingBoxes</i>	Vector of face bounding boxes found by a face detector.
------------------------	---

7.58.3.21 setFaceOcclusionSegmentationImage()

```
void OFIQ_LIB::Session::setFaceOcclusionSegmentationImage (
    const cv::Mat & i_segmentationImage )
```

Set the Face Occlusion Segmentation Image, see [OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation](#))

Parameters

<i>i_segmentationImage</i>	
----------------------------	--

7.58.3.22 setFaceParsingImage()

```
void OFIQ_LIB::Session::setFaceParsingImage (
    const cv::Mat & i_parsingImage )
```

Set the Face Parsing Image, see [OFIQ_LIB::modules::segmentations::FaceParsing](#)).

Parameters

<i>i_parsingImage</i>	
-----------------------	--

7.58.3.23 setLandmarks()

```
void OFIQ_LIB::Session::setLandmarks (
    const OFIQ::FaceLandmarks & i_landmarks )
```

Set the Landmarks detected on the input image.

Parameters

<i>i_landmarks</i>	
--------------------	--

7.58.3.24 setPose()

```
void OFIQ_LIB::Session::setPose (
    const EulerAngle & i_pose )
```

Set the Pose of the input image.

Parameters

<i>i_pose</i>	
---------------	--

7.58.4 Member Data Documentation**7.58.4.1 m_alignedFace**

```
cv::Mat OFIQ_LIB::Session::m_alignedFace [private]
```

Container for storing the aligned image.

7.58.4.2 m_alignedFacelandmarkedRegion

```
cv::Mat OFIQ_LIB::Session::m_alignedFacelandmarkedRegion [private]
```

Container for storing the landmarks of the aligned face image.

7.58.4.3 m_alignedFaceLandmarks

```
OFIQ::FaceLandmarks OFIQ_LIB::Session::m_alignedFaceLandmarks [private]
```

Container for storing the landmark information of the aligned image.

7.58.4.4 m_alignedFaceLuminance

```
cv::Mat OFIQ_LIB::Session::m_alignedFaceLuminance [private]
```

Aligned face image convert to luminance/gray image.

7.58.4.5 m_alignedFaceTransformationMatrix

```
cv::Mat OFIQ_LIB::Session::m_alignedFaceTransformationMatrix [private]
```

Container for storing the transformation matrix that led to the aligned image.

7.58.4.6 m_assessment

```
OFIQ::FaceImageQualityAssessment& OFIQ_LIB::Session::m_assessment [private]
```

Reference to the FaceImageQualityAssessment object, connected to this session.

7.58.4.7 m_detectedFaces

```
std::vector<OFIQ::BoundingBox> OFIQ_LIB::Session::m_detectedFaces [private]
```

Container for the faces found on the input image.

7.58.4.8 m_faceOcclusionSegmentationImage

```
cv::Mat OFIQ_LIB::Session::m_faceOcclusionSegmentationImage [private]
```

Container for storing the result of the face occlusion segmented image.

7.58.4.9 m_faceParsingImage

```
cv::Mat OFIQ_LIB::Session::m_faceParsingImage [private]
```

Container for storing the segmented face image.

7.58.4.10 m_id

```
std::string OFIQ_LIB::Session::m_id [private]
```

Container for storing the id of the session.

7.58.4.11 m_image

```
const OFIQ::Image& OFIQ_LIB::Session::m_image [private]
```

Reference to the input image, connected to this session.

7.58.4.12 m_landmarks

```
OFIQ::FaceLandmarks OFIQ_LIB::Session::m_landmarks [private]
```

Container for storing the landmark information.

7.58.4.13 m_pose

`EulerAngles` OFIQ_LIB::Session::m_pose [private]

Container for storing the pose information.

The documentation for this class was generated from the following file:

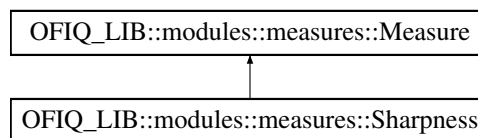
- [Session.h](#)

7.59 OFIQ_LIB::modules::measures::Sharpness Class Reference

Implementation of the sharpness measure.

```
#include <Sharpness.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::Sharpness:



Public Member Functions

- [Sharpness](#) (const [Configuration](#) &configuration)
Construct a new Sharpness object.
- void [Execute](#) (OFIQ_LIB::Session &session) override
Run computation of the sharpness measure.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, OFIQ::QualityMeasure measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual OFIQ::QualityMeasure [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) (OFIQ_LIB::Session &session, OFIQ::QualityMeasure measure, double rawValue, OFIQ::QualityMeasureReturnCode code)
Inserts the result of a quality assessment in the session object.

Private Member Functions

- void [GetCroppedImages](#) (const [Session](#) &session, cv::Mat &faceCrop, cv::Mat &maskCrop, bool useAligned, float faceRegionAlpha) const
Get the cropped face region.
- cv::Mat [GetClassifierFocusFeatures](#) (const cv::Mat &image, const cv::Mat &mask, bool applyBlur) const
Computation of the input features using different edge detectors.

Private Attributes

- std::string [m_modelFile](#)
Name of the random forest model, extracted from the configuration file.
- std::shared_ptr< cv::ml::RTrees > [m_rtree](#)
Instance of the random forest model.
- bool [m_useAligned](#)
The sharpness measure can be computed on the aligned or the original image. useAligned set to true will run the computation on the aligned image. The member is read from the configuration file.
- double [m_faceRegionAlpha](#)
For faceRegionAlpha = 0, the algorithm uses the inner face region. For faceRegionAlpha = 0.85, the algorithm uses the extended face region as specified for the FaceOcclusionin FRVT Quality.
- int [m_numTrees](#)
This member stores the number of trees used for the random forest. Internal use only.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.59.1 Detailed Description

Implementantation of the sharpness measure.

This quality component can be used to efficiently choose the better focused face portrait among several face samples of the same biometric capture subject. It should not be used to perform an absolute sharpness assessment if only one sample is available

7.59.2 Constructor & Destructor Documentation

7.59.2.1 Sharpness()

```
OFIQ_LIB::modules::measures::Sharpness::Sharpness (
    const Configuration & configuration ) [explicit]
```

Construct a new Sharpness object.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.59.3 Member Function Documentation

7.59.3.1 Execute()

```
void OFIQ_LIB::modules::measures::Sharpness::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Run computation of the sharpness measure.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

7.59.3.2 GetClassifierFocusFeatures()

```
cv::Mat OFIQ_LIB::modules::measures::Sharpness::GetClassifierFocusFeatures (
    const cv::Mat & image,
    const cv::Mat & mask,
    bool applyBlur ) const [private]
```

Computation of the input features using different edge detectors.

Parameters

<i>image</i>	Input image.
<i>mask</i>	Input region of the face.
<i>applyBlur</i>	Whether or not rub a GaussianBlur before the edge detection.

Returns

cv::Mat Container storing the results of the different edge detectors.

7.59.3.3 GetCroppedImages()

```
void OFIQ_LIB::modules::measures::Sharpness::GetCroppedImages (
    const Session & session,
    cv::Mat & faceCrop,
    cv::Mat & maskCrop,
    bool useAligned,
    float faceRegionAlpha ) const [private]
```

Get the cropped face region.

Parameters

<i>session</i>	Data container.
<i>faceCrop</i>	Computed crop of the face.
<i>maskCrop</i>	Mask used for the cropping. Will be computed in the method.
<i>useAligned</i>	Switch for using the aligned image.
<i>faceRegionAlpha</i>	Enlarge the face region by passing this parameter.

7.59.4 Member Data Documentation**7.59.4.1 m_faceRegionAlpha**

```
double OFIQ_LIB::modules::measures::Sharpness::m_faceRegionAlpha [private]
```

For faceRegionAlpha = 0, the algorithm uses the inner face region. For faceRegionAlpha = 0.85, the algorithm uses the extended face region as specified for the FaceOcclusionin FRVT Quality.

7.59.4.2 m_modelFile

```
std::string OFIQ_LIB::modules::measures::Sharpness::m_modelFile [private]
```

Name of the random forest model, extracted from the configuration file.

7.59.4.3 m_numTrees

```
int OFIQ_LIB::modules::measures::Sharpness::m_numTrees [private]
```

This member stores the number of trees used for the random forest. Internal use only.

7.59.4.4 m_rtree

```
std::shared_ptr<cv::ml::RTrees> OFIQ_LIB::modules::measures::Sharpness::m_rtree [private]
```

Instance of the random forest model.

7.59.4.5 m_useAligned

```
bool OFIQ_LIB::modules::measures::Sharpness::m_useAligned [private]
```

The sharpness measure can be computed on the aligned or the original image. useAligned set to true will run the computation on the aligned image. The member is read from the configuration file.

The documentation for this class was generated from the following file:

- [Sharpness.h](#)

7.60 OFIQ_LIB::modules::measures::SigmoidParameters Struct Reference

Parameters of the sigmoid function based quality mapping.

```
#include <Measure.h>
```

Public Member Functions

- [SigmoidParameters](#) ()
Default constructor.
- void [setInverse](#) ()
Sets this quality mapping to a smaller-is-better variant.
- void [Reset](#) ()
Resets the members of the quality mapping to their default values.

Public Attributes

- double [h](#)
Scale factor.
- double [a](#)
Constant shift.
- double [s](#)
Signed weight for sigmoid part.
- double [x0](#)
Center point in sigmoid part.
- double [w](#)
Divisor in sigmoid part.
- bool [round](#)
Flag controlling if the compiler's native rounding function (`std::round`) is applied.

7.60.1 Detailed Description

Parameters of the sigmoid function based quality mapping.

A sigmoid-based quality mapping is the following function

$$Q(x) = h \cdot (a + s \cdot \text{sigmoid}(x, x_0, w))$$

where

$$\text{sigmoid}(x, x_0, w) = (1 + \exp((x_0 - x)/w))^{-1}.$$

Q can be used to map a native quality score x to a value between 0 and 100. The other symbols denote parameters that can be configured using the struct.

7.60.2 Constructor & Destructor Documentation

7.60.2.1 SigmoidParameters()

```
OFIQ_LIB::modules::measures::SigmoidParameters::SigmoidParameters ( ) [inline]
```

Default constructor.

After construction, all members are set to their default values.

7.60.3 Member Function Documentation

7.60.3.1 Reset()

```
void OFIQ_LIB::modules::measures::SigmoidParameters::Reset ( ) [inline]
```

Resets the members of the quality mapping to their default values.

7.60.3.2 setInverse()

```
void OFIQ_LIB::modules::measures::SigmoidParameters::setInverse ( ) [inline]
```

Sets this quality mapping to a smaller-is-better variant.

If the parameters a is 0 and s is, then this quality mapping is in larger-is-better-semantics. For those mappings, the method can be used to set the mapping to its smaller-is-better counterpart by setting a to 1 and s to -1. This is used by some measures to conveniently set a quality mapping.

7.60.4 Member Data Documentation

7.60.4.1 a

```
double OFIQ_LIB::modules::measures::SigmoidParameters::a
```

Constant shift.

The default value is 0.

7.60.4.2 h

```
double OFIQ_LIB::modules::measures::SigmoidParameters::h
```

Scale factor.

The default value is 100.

7.60.4.3 round

```
bool OFIQ_LIB::modules::measures::SigmoidParameters::round
```

Flag controlling if the compiler's native rounding function (`std::round`) is applied.

The default value is `true`.

7.60.4.4 s

```
double OFIQ_LIB::modules::measures::SigmoidParameters::s
```

Signed weight for sigmoid part.

The default value is 1.

7.60.4.5 w

```
double OFIQ_LIB::modules::measures::SigmoidParameters::w
```

Divisor in sigmoid part.

The default value of 0.7 has been chosen arbitrarily and should specified when a mapping is configured.

7.60.4.6 x0

```
double OFIQ_LIB::modules::measures::SigmoidParameters::x0
```

Center point in sigmoid part.

The default value of 4 has been chosen arbitrarily and should specified when a mapping is configured.

The documentation for this struct was generated from the following file:

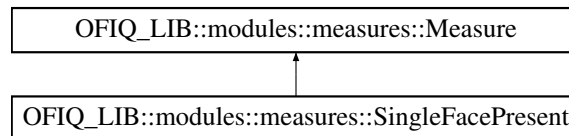
- [Measure.h](#)

7.61 OFIQ_LIB::modules::measures::SingleFacePresent Class Reference

Implementation of the single face present measure.

```
#include <SingleFacePresent.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::SingleFacePresent:



Public Member Functions

- [SingleFacePresent](#) (const [Configuration](#) &configuration)
Construct a new Single Face Present object.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Run computation of the single face present analysis.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.61.1 Detailed Description

Implementation of the single face present measure.

Single face present measure shall ensure that only one face is visible on the image.

7.61.2 Constructor & Destructor Documentation

7.61.2.1 SingleFacePresent()

```
OFIQ_LIB::modules::measures::SingleFacePresent::SingleFacePresent (
    const Configuration & configuration ) [explicit]
```

Construct a new Single Face Present object.

Parameters

configuration	Configuration object from which measure-related configuration is read.
-------------------------------	--

7.61.3 Member Function Documentation

7.61.3.1 Execute()

```
void OFIQ_LIB::modules::measures::SingleFacePresent::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Run computation of the single face present analysis.

Parameters

session	Session object containing the original facial image and pre-processing results computed by the OFIQImpl::preprocess() method
-------------------------	--

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

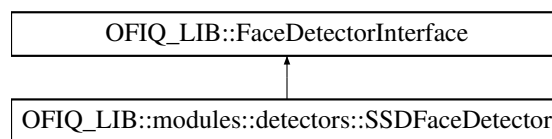
- [SingleFacePresent.h](#)

7.62 OFIQ_LIB::modules::detectors::SSDFaceDetector Class Reference

Implementation of a face detector using the SSD face detector CNN.

```
#include <opencv_ssd_face_detector.h>
```

Inheritance diagram for OFIQ_LIB::modules::detectors::SSDFaceDetector:



Public Member Functions

- [SSDFaceDetector](#) (const [Configuration](#) &config)
Constructor a new SSDFaceDetector.
- [~SSDFaceDetector](#) () override=default
Destructor of the SSDFaceDetector.

Public Member Functions inherited from [OFIQ_LIB::FaceDetectorInterface](#)

- virtual [~FaceDetectorInterface](#) ()=default
Destroy the Face Detector Interface object.
- std::vector< [OFIQ::BoundingBox](#) > [detectFaces](#) ([OFIQ_LIB::Session](#) &session)
This function detects faces in given image.

Protected Member Functions

- std::vector< [OFIQ::BoundingBox](#) > [UpdateFaces](#) ([OFIQ_LIB::Session](#) &session) override
Implementation of the face detection method.

Private Attributes

- std::shared_ptr< cv::dnn::Net > [m_dnnNet](#) {nullptr}
Instance of an opencv dnn::Net.
- std::mutex [m_dnnNetMutex](#)
Mutex preventing the opencv dnn::Net to crash during parallel call.
- double [m_confidenceThreshold](#)
Confidence threshold used for the face detection. The value is read from the configuration file.
- double [m_padding](#)
*Add padding around the image (faceImage.width * padding; faceImage.height * padding;)*
- double [m_minimalRelativeFaceSize](#)
Filter threshold for removing to small face found on the image. This value is read from the configuration file.

7.62.1 Detailed Description

Implementation of a face detector using the SSD face detector CNN.

7.62.2 Constructor & Destructor Documentation

7.62.2.1 SSDFaceDetector()

```
OFIQ_LIB::modules::detectors::SSDFaceDetector::SSDFaceDetector (
    const Configuration & config ) [explicit]
```

Constructor a new SSDFaceDetector.

Parameters

<i>config</i>	
---------------	--

7.62.2.2 ~SSDFaceDetector()

```
OFIQ_LIB::modules::detectors::SSDFaceDetector::~~SSDFaceDetector ( ) [override], [default]
```

Destructor of the SSDFaceDetector.

7.62.3 Member Function Documentation

7.62.3.1 UpdateFaces()

```
std::vector< OFIQ::BoundingBox > OFIQ_LIB::modules::detectors::SSDFaceDetector::UpdateFaces (
    OFIQ\_LIB::Session & session ) [override], [protected], [virtual]
```

Implementation of the face detection method.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

Returns

std::vector<OFIQ::BoundingBox> Bounding boxes of the detected faces

Implements [OFIQ_LIB::FaceDetectorInterface](#).

7.62.4 Member Data Documentation

7.62.4.1 m_confidenceThreshold

```
double OFIQ_LIB::modules::detectors::SSDFaceDetector::m_confidenceThreshold [private]
```

Confidence threshold used for the face detection. The value is read from the configuration file.

7.62.4.2 m_dnnNet

```
std::shared_ptr<cv::dnn::Net> OFIQ_LIB::modules::detectors::SSDFaceDetector::m_dnnNet {nullptr}
[private]
```

Instance of an opencv dnn::Net.

7.62.4.3 m_dnnNetMutex

```
std::mutex OFIQ_LIB::modules::detectors::SSDFaceDetector::m_dnnNetMutex [private]
```

Mutex preventing the opencv dnn::Net to crash during parallel call.

7.62.4.4 m_minimalRelativeFaceSize

```
double OFIQ_LIB::modules::detectors::SSDFaceDetector::m_minimalRelativeFaceSize [private]
```

Filter threshold for removing to small face found on the image. This value is read from the configuration file.

7.62.4.5 m_padding

```
double OFIQ_LIB::modules::detectors::SSDFaceDetector::m_padding [private]
```

Add padding around the image (faceImage.width * padding; faceImage.height * padding;)

The documentation for this class was generated from the following file:

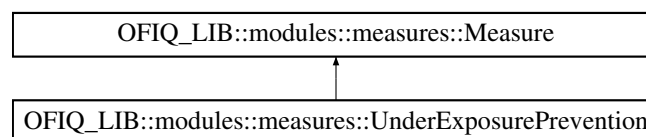
- [opencv_ssd_face_detector.h](#)

7.63 OFIQ_LIB::modules::measures::UnderExposurePrevention Class Reference

Implementation of the under-exposure prevention measure.

```
#include <UnderExposurePrevention.h>
```

Inheritance diagram for OFIQ_LIB::modules::measures::UnderExposurePrevention:



Public Member Functions

- [UnderExposurePrevention](#) (const [Configuration](#) &configuration)
Constructor a new Under Exposure Prevention object.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Run the computation of the under-exposure prevention measure.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & configuration
Reference to the configuration with which the measure constructor has been invoked.

7.63.1 Detailed Description

Implementation of the under-exposure prevention measure.

The representation of a face is considered too dark if it has a high proportion of pixels that have a low luminance value.

7.63.2 Constructor & Destructor Documentation

7.63.2.1 UnderExposurePrevention()

```
OFIQ_LIB::modules::measures::UnderExposurePrevention::UnderExposurePrevention (
    const Configuration & configuration ) [explicit]
```

Constructor a new Under Exposure Prevention object.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.63.3 Member Function Documentation

7.63.3.1 Execute()

```
void OFIQ_LIB::modules::measures::UnderExposurePrevention::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Run the computation of the under-exposure prevention measure.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

The documentation for this class was generated from the following file:

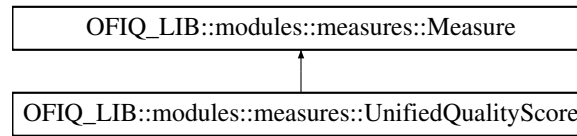
- [UnderExposurePrevention.h](#)

7.64 OFIQ_LIB::modules::measures::UnifiedQualityScore Class Reference

Implementation of the unified quality measure.

```
#include <UnifiedQualityScore.h>
```


Inheritance diagram for OFIQ_LIB::modules::measures::UnifiedQualityScore:



Public Member Functions

- [UnifiedQualityScore](#) (const [Configuration](#) &configuration)
Construct a new Unified Quality Score object.
- void [Execute](#) ([OFIQ_LIB::Session](#) &session) override
Run the computation on the measure.

Public Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- [Measure](#) (const [Configuration](#) &configuration, [OFIQ::QualityMeasure](#) measure)
Constructor.
- virtual [~Measure](#) ()=default
Destructor.
- virtual std::string [GetName](#) () const
Returns the name of the measure.
- virtual [OFIQ::QualityMeasure](#) [GetQualityMeasure](#) () const
Returns an enum encoding the measure.
- void [SetQualityMeasure](#) ([OFIQ_LIB::Session](#) &session, [OFIQ::QualityMeasure](#) measure, double rawValue, [OFIQ::QualityMeasureReturnCode](#) code)
Inserts the result of a quality assessment in the session object.

Private Attributes

- [ONNXRuntimeEnv](#) m_onnxRuntimeEnv
Instance of the neural network (iResNet50 model M).

Additional Inherited Members

Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- void [AddSigmoid](#) ([OFIQ::QualityMeasure](#) measure, const [SigmoidParameters](#) &defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- void [AddSigmoid](#) (const std::string &key, [SigmoidParameters](#) defaultValues)
Reads sigmoid-function based quality mapping from the configuration.
- double [ExecuteScalarConversion](#) ([OFIQ::QualityMeasure](#) measure, double rawValue)
Maps a native quality score to a quality component value.
- double [ExecuteScalarConversion](#) (const std::string &key, double rawValue)
Maps a native quality score to a quality component value.

Static Protected Member Functions inherited from [OFIQ_LIB::modules::measures::Measure](#)

- static double [Sigmoid](#) (double x, double x0, double w)
Sigmoid function.

Protected Attributes inherited from [OFIQ_LIB::modules::measures::Measure](#)

- const [Configuration](#) & [configuration](#)
Reference to the configuration with which the measure constructor has been invoked.

7.64.1 Detailed Description

Implementation of the unified quality measure.

The quality score refers to the requirements in clause 5.4.8 in ISO/IEC 19794-5:2011 frontal image type (relevant for EU-EES implementing decision 2019/329) and in clause 7.7 in ISO/IEC 39794-1 (relevant for UC1 specified in ICAO Document 9303)

7.64.2 Constructor & Destructor Documentation

7.64.2.1 UnifiedQualityScore()

```
OFIQ_LIB::modules::measures::UnifiedQualityScore::UnifiedQualityScore (
    const Configuration & configuration ) [explicit]
```

Construct a new Unified Quality Score object.

Parameters

<i>configuration</i>	Configuration object from which measure-related configuration is read.
----------------------	--

7.64.3 Member Function Documentation

7.64.3.1 Execute()

```
void OFIQ_LIB::modules::measures::UnifiedQualityScore::Execute (
    OFIQ\_LIB::Session & session ) [override], [virtual]
```

Run the computation on the measure.

The algorithm uses a iResNet50 model M from <https://github.com/IrvingMeng/MagFace> trained on MS1MV2 with MagFace loss without DDP parallelisation. The algorithm takes as input the image I output by the alignment algorithm.

Parameters

<i>session</i>	Session object computed by the OFIQImpl::preprocess() method.
----------------	---

Implements [OFIQ_LIB::modules::measures::Measure](#).

7.64.4 Member Data Documentation

7.64.4.1 m_onnxRuntimeEnv

[ONNXRuntimeEnv](#) OFIQ_LIB::modules::measures::UnifiedQualityScore::m_onnxRuntimeEnv [private]

Instance of the neural network (iResNet50 model M).

The documentation for this class was generated from the following file:

- [UnifiedQualityScore.h](#)

Chapter 8

File Documentation

8.1 mainpage.h File Reference

This header file is for generating the doxygen documentation for OFIQ.

8.1.1 Detailed Description

This header file is for generating the doxygen documentation for OFIQ.

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OFIQ development team

8.2 mainpage.h

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```
00001
01229 #pragma once
```

8.3 ofiq_lib.h File Reference

Class describing the interface to the OFIQ.

```
#include <cstdint>
#include <string>
#include <vector>
#include <ofiq_structs.h>
```

Classes

- class [OFIQ::Interface](#)
The interface to FACE QA implementation.

Namespaces

- namespace [OFIQ](#)
Namespace for OFIQ API.

Macros

- `#define` [OFIQ_EXPORT](#)

Enumerations

- enum class [OFIQ::PreprocessingResultType](#) {
[OFIQ::None](#) = 0x0 , [OFIQ::Faces](#) = 0x1 , [OFIQ::Landmarks](#) = 0x2 , [OFIQ::Segmentation](#) = 0x4 ,
[OFIQ::OcclusionMask](#) = 0x8 , [OFIQ::LandmarkedRegion](#) = 0x10 , [OFIQ::All](#) = 0x1 + 0x2 + 0x4 + 0x8 + 0x10
 }
Encodes flags and mask to request preprocessing results.

8.3.1 Detailed Description

Class describing the interface to the OFIQ.

Utilities for manipulating the OFIC C API (ofiq_libc) data structures.

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Author

OFIQ development team

8.3.2 Macro Definition Documentation**8.3.2.1 OFIQ_EXPORT**

```
#define OFIQ_EXPORT
```

8.4 ofiq_lib.h

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```
00001
00027 #ifndef OFIQ_LIB_H
00028 #define OFIQ_LIB_H
00029
00030 #include <cstdint>
00031 #include <string>
00032 #include <vector>
00033
00034 #include <ofiq_structs.h>
00035
00036 #ifdef _WIN32
00037 #   ifdef OFIQ_EXPORTS
00038 #       define OFIQ_EXPORT __declspec(dllexport)
00039 #   else
00040 #       define OFIQ_EXPORT __declspec(dllimport)
00041 #   endif
00042 #else
00043 #   define OFIQ_EXPORT
00044 #endif
00045
00049 namespace OFIQ
00050 {
00056     enum class PreprocessingResultType
00057     {
00058         // NOTE: Don't use one-line comments here; otherwise
```

```

00059         // Doxygen does not transfer parse them.
00060
00064         None = 0x0,
00065
00069         Faces = 0x1,
00070
00074         Landmarks = 0x2,
00075
00079         Segmentation = 0x4,
00080
00084         OcclusionMask = 0x8,
00085
00089         LandmarkedRegion = 0x10,
00090
00094         All = 0x1 + 0x2 + 0x4 + 0x8 + 0x10
00095     };
00096
00104     class Interface
00105     {
00106     public:
00111         virtual ~Interface() = default;
00112
00125         virtual OFIQ::ReturnStatus
00126             initialize(const std::string& configDir, const std::string& configFileName) = 0;
00127
00140         virtual OFIQ::ReturnStatus scalarQuality(const OFIQ::Image& face, double& quality) = 0;
00141
00160         virtual OFIQ::ReturnStatus vectorQuality(
00161             const OFIQ::Image& image, OFIQ::FaceImageQualityAssessment& assessments) = 0;
00162
00189         virtual OFIQ::ReturnStatus vectorQualityWithPreprocessingResults(
00190             const OFIQ::Image& image,
00191             OFIQ::FaceImageQualityAssessment& assessments,
00192             OFIQ::FaceImageQualityPreprocessingResult& preprocessingResult,
00193             uint32_t resultRequestsMask) = 0;
00194
00210         OFIQ_EXPORT static std::shared_ptr<Interface> getImplementation();
00211
00218         OFIQ_EXPORT void getVersion(int& major, int& minor, int& patch) const;
00219
00220     };
00221 }
00222
00223 #endif /* OFIQ_LIB_H */

```

8.5 ofiq_lib_impl.h File Reference

Implementation of the OFIQ_LIB.

```

#include "Configuration.h"
#include "Executor.h"
#include "ofiq_lib.h"
#include "NeuronalNetworkContainer.h"

```

Classes

- class [OFIQ_LIB::OFIQImpl](#)
Implementation of the OFIQ_LIB.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

8.5.1 Detailed Description

Implementation of the OFIQ_LIB.

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Author

OFIQ development team

8.6 ofiq_lib_impl.h

[Go to the documentation of this file.](#)

```
00001
00027 #ifndef OFIQ_LIB_IMPL_H
00028 #define OFIQ_LIB_IMPL_H
00029
00030 #include "Configuration.h"
00031 #include "Executor.h"
00032 #include "ofiq_lib.h"
00033 #include "NeuronalNetworkContainer.h"
00034
00038 namespace OFIQ_LIB
00039 {
00044     class OFIQImpl : public OFIQ::Interface
00045     {
00046     public:
00051         OFIQImpl() = default;
00052
00057         ~OFIQImpl() override = default;
00058
00066         OFIQ::ReturnStatus
00067             initialize(const std::string& configDir, const std::string& configValue) override;
00068
00077         OFIQ::ReturnStatus scalarQuality(const OFIQ::Image& face, double& quality) override;
00078
00086         OFIQ::ReturnStatus vectorQuality(
00087             const OFIQ::Image& image, OFIQ::FaceImageQualityAssessment& assessments) override;
00088
00102         OFIQ::ReturnStatus vectorQualityWithPreprocessingResults(
00103             const OFIQ::Image& image,
00104             OFIQ::FaceImageQualityAssessment& assessments,
00105             OFIQ::FaceImageQualityPreprocessingResult& preprocessingResult,
00106             uint32_t resultRequestsMask = static_cast<int>(OFIQ::PreprocessingResultType::All))
00107             override;
00108     private:
00113         std::unique_ptr<OFIQ_LIB::modules::measures::Executor> m_executorPtr;
00114
00119         std::unique_ptr<Configuration> config;
```

```

00120
00125     std::unique_ptr<NeuronalNetworkContainer> networks;
00126
00132     std::unique_ptr<OFIQ_LIB::modules::measures::Executor> CreateExecutor();
00133
00138     void CreateNetworks();
00139
00147     OFIQ::ReturnStatus preprocess(Session& session);
00148
00156     OFIQ::ReturnStatus performAssessment(Session& session);
00157
00165     void alignFaceImage(Session& session) const;
00166
00174     void computeAlignedFaceLuminanceImage(Session& session) const;
00175
00184     OFIQ::ReturnStatus getPreprocessingResults(
00185         const Session& session,
00186         OFIQ::FaceImageQualityPreprocessingResult& preprocessingResult,
00187         uint32_t resultRequestsMask) const;
00188     };
00189 }
00190
00191 #endif /* OFIQ_LIB_IMPL_H */

```

8.7 ofiq_libc.h File Reference

```
#include <ofiq_structs_c.h>
```

Macros

- `#define OFIQ_C_EXPORT`

Functions

- `OFIQ_C_EXPORT ReturnStatus OFIQInitialize (OFIQHandle *handle, const char *configDir, const char *configFileName)`
Initializes the OFIQ library with the specified configuration file.
- `OFIQ_C_EXPORT ReturnStatus OFIQCleanup (OFIQHandle *handle)`
Cleans up resources initialized by OFIQInitialize().
- `OFIQ_C_EXPORT ReturnStatus OFIQReadImage (const char *filename, Image &image)`
Reads an image from the specified file into the provided Image object.
- `OFIQ_C_EXPORT ReturnStatus OFIQReadImageFromByteArray (const unsigned char *imageData, size_t dataLength, Image &image)`
Reads an image from a byte array into the provided Image object.
- `OFIQ_C_EXPORT ReturnStatus OFIQFreeImage (Image *image)`
Frees the memory allocated for the specified Image object.
- `OFIQ_C_EXPORT ReturnStatus OFIQScalarQuality (OFIQHandle handle, const Image &face, double &quality)`
This function computes a scalar image quality measure for the given image.
- `OFIQ_C_EXPORT ReturnStatus OFIQVectorQuality (OFIQHandle handle, const Image &faceImage, FaceImageQualityAssessment &qualityAssessment)`
Performs a face image quality assessment using various quality measurement methods.
- `OFIQ_C_EXPORT ReturnStatus OFIQVectorQualityExt (OFIQHandle handle, const Image &faceImage, FaceImageQualityAssessment &qualityAssessment, PreprocessingResult &preprocessingResult)`
Performs a face image quality assessment by using various quality measurement methods. Additionally to the OFIQVectorQuality method this function returns preprocessing results such as.

- [OFIQ_C_EXPORT](#) [ReturnStatus](#) [OFIQFreeFacelImageQualityAssessment](#) ([FacelImageQualityAssessment](#) *qa)
Frees the memory allocated for the specified FacelImageQualityAssessment object.
- [OFIQ_C_EXPORT](#) [ReturnStatus](#) [OFIQFreeFaceLandmarks](#) ([FaceLandmarks](#) *landmarks)
Frees the memory allocated for the specified FaceLandmarks data.
- [OFIQ_C_EXPORT](#) [ReturnStatus](#) [OFIQFreePreprocessingResult](#) ([PreprocessingResult](#) *preprocRes)
Frees the memory allocated for the specified PreprocessingResult object.
- [OFIQ_C_EXPORT](#) void [OFIQGetVersion](#) (int &major, int &minor, int &patch)
- [OFIQ_C_EXPORT](#) const char * [OFIQGetQualityMeasureName](#) ([QualityMeasure](#) measure)
Gets the name of a quality measure as a string.

8.7.1 Macro Definition Documentation

8.7.1.1 OFIQ_C_EXPORT

```
#define OFIQ_C_EXPORT
```

Examples

[C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h.](#)

8.7.2 Function Documentation

8.7.2.1 OFIQCleanup()

```
OFIQ_C_EXPORT ReturnStatus OFIQCleanup (
    OFIQHandle * handle )
```

Cleans up resources initialized by OFIQInitialize().

This function cleans up or releases any resources associated with the specified handle. It should be called after initialization to properly terminate and free resources used by the OFIQ library.

Parameters

in	handle	The handle obtained from a successful call to ofiqInitialize(). Must not be NULL.
----	--------	---

Returns

[ReturnStatus](#)

- [ReturnStatus.code](#) == [ReturnCode::Success](#): If the cleanup was successful.
- Other error code: Indicates an error occurred during cleanup (e.g., invalid handle).

See also

[OFIQInitialize\(\)](#)

Examples

[C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h.](#)

8.7.2.2 OFIQFreeFacelImageQualityAssessment()

```
OFIQ_C_EXPORT ReturnStatus OFIQFreeFaceImageQualityAssessment (
    FaceImageQualityAssessment * qa )
```

Frees the memory allocated for the specified FaceImageQualityAssessment object.

This function releases any resources or memory associated with the provided FaceImageQualityAssessment object that was previously created by function OFIQVectorQuality() or OFIQVectorQualityExt().

Parameters

<i>qa</i>	The pointer to the FaceImageQualityAssessment object whose memory is to be freed.
-----------	---

Returns

ReturnStatus Indicates whether the operation was successful

- ReturnStatus.code == ReturnCode::Success: If the object has been properly freed.
- Other error code: Indicates an error occurred, such as failure to free resources.

Examples

```
C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h.
```

8.7.2.3 OFIQFreeFaceLandmarks()

```
OFIQ_C_EXPORT ReturnStatus OFIQFreeFaceLandmarks (
    FaceLandmarks * landmarks )
```

Frees the memory allocated for the especified FaceLandmarks data.

This function releases memory associated with face landmarks, preventing memory leaks. It's essential to ensure that the provided pointer is valid and not null before calling this function.

Parameters

<i>landmarks</i>	A pointer to the FaceLandmarks object whose memory should be freed.
------------------	---

Returns

ReturnStatus Indicates whether the operation was successful

- ReturnStatus.code == ReturnCode::Success: If the object has been properly freed.
- Other error code: Indicates an error occurred, such as failure to free resources.

Examples

```
C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h.
```

8.7.2.4 OFIQFreeImage()

```
OFIQ_C_EXPORT ReturnStatus OFIQFreeImage (
    Image * image )
```

Frees the memory allocated for the specified Image object.

This function releases any resources or memory associated with the provided Image object that was previously created by functions like OFIQReadImage().

Parameters

<i>image</i>	The pointer to the Image object whose memory is to be freed.
--------------	--

Returns

ReturnStatus

- ReturnStatus.code == ReturnCode::Success: If the operation was successful and the image has been properly freed.
- Other error code: Indicates an error occurred, such as an invalid handle or failure to free resources.

Examples

```
C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h.
```

8.7.2.5 OFIQFreePreprocessingResult()

```
OFIQ_C_EXPORT ReturnStatus OFIQFreePreprocessingResult (
    PreprocessingResult * preprocRes )
```

Frees the memory allocated for the specified PreprocessingResult object.

This function is used to free the resources associated with a preprocessing result. It ensures that any memory allocated for the result is properly deallocated.

Parameters

<i>preprocRes</i>	A pointer to the PreprocessingResult object whose resources are to be freed. The caller must ensure this pointer is valid and not NULL to avoid undefined behavior.
-------------------	---

Returns

ReturnStatus Indicating whether the operation was successful

- ReturnStatus.code == ReturnCode::Success: If the object has been properly freed.
- Other error code: Indicates an error occurred, such as failure to free resources.

Examples

```
C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h.
```

8.7.2.6 OFIQGetQualityMeasureName()

```
OFIQ_C_EXPORT const char * OFIQGetQualityMeasureName (
    QualityMeasure measure )
```

Gets the name of a quality measure as a string.

This function returns the name of the specified quality measure as a null-terminated string. The returned string should not be modified or freed by the caller.

Parameters

<i>measure</i>	The QualityMeasure value for which to get the name.
----------------	---

Returns

A pointer to a const char array containing the name of the quality measure.

Examples

[C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h](#).

8.7.2.7 OFIQGetVersion()

```
OFIQ_C_EXPORT void OFIQGetVersion (
    int & major,
    int & minor,
    int & patch )
```

Examples

[C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h](#).

8.7.2.8 OFIQInitialize()

```
OFIQ_C_EXPORT ReturnStatus OFIQInitialize (
    OFIQHandle * handle,
    const char * configDir,
    const char * configFileName )
```

Initializes the OFIQ library with the specified configuration file.

This function initializes the OFIQ system using the given configuration directory and configuration file name. The handle parameter will be populated with the initialized handle that can be used for subsequent operations.

Parameters

in, out	<i>handle</i>	A pointer to a OFIQHandle where the initialized handle will be stored.
in	<i>configDir</i>	The path to the directory containing the configuration file.
in	<i>configFileName</i>	The name of the configuration file to use for initialization.

Returns

ReturnStatus

- ReturnStatus.code == ReturnCode::Success: If the initialization was successful.
- Other error code: Indicates an error occurred during initialization (e.g., invalid path, missing configuration file).

See also

OFIQCleanup()

Examples

[C:/Path/To/eesfm_cpp/OFIQLib/include/ofiq_libc.h](#).

8.7.2.9 OFIQReadImage()

```
OFIQ_C_EXPORT ReturnStatus OFIQReadImage (  
    const char * filename,  
    Image & image )
```

Reads an image from the specified file into the provided Image object.

This function loads image data from the given filename and stores it in the provided Image object.

Parameters

in	<i>filename</i>	The path to the image file to read.
out	<i>image</i>	The Image object that will be populated with the loaded image data. This method allocates memory for the Image.data, to free this memory the method OFIQFreeImage() should be used.

Returns

ReturnStatus

- ReturnStatus.code == ReturnCode::Success: If the image has been read from the file and successfully decoded.
- Other error code: Indicates an error occurred (e.g., file not found, invalid format).

See also

OFIQFreeImage()

Examples

[C:/Path/To/eesfm_cpp/OFIQLib/include/ofiq_libc.h](#).

8.7.2.10 OFIQReadImageFromByteArray()

```
OFIQ_C_EXPORT ReturnStatus OFIQReadImageFromByteArray (
    const unsigned char * imageData,
    size_t dataLength,
    Image & image )
```

Reads an image from a byte array into the provided Image object.

This function reads image data from the specified byte array and populates the provided Image object with the loaded data. The byte array is expected to contain valid image data in a supported format.

Parameters

in	<i>imageData</i>	Pointer to the byte array containing the image data to be read.
in	<i>dataLength</i>	Size of the byte array in bytes.
out	<i>image</i>	Reference to the Image object that will be populated with the loaded image data. This method allocates memory for the Image.data, to free this memory the method OFIQFreeImage() should be used.

Returns

ReturnStatus

- ReturnStatus.code == ReturnCode::Success: If the image has been read from the byte array and successfully decoded.
- Other error code: Indicates an error occurred during the process, such as invalid data format or insufficient memory.

See also

OFIQFreeImage()

ReturnCode

Examples

[C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h.](#)

8.7.2.11 OFIQScalarQuality()

```
OFIQ_C_EXPORT ReturnStatus OFIQScalarQuality (
    OFIQHandle handle,
    const Image & face,
    double & quality )
```

This function computes a scalar image quality measure for the given image.

Parameters

in	<i>handle</i>	A handle to OFIQ instance.
in	<i>face</i>	Single face image
out	<i>quality</i>	A scalar value assessment of image quality. The values are in range [0,100] So, a low value indicates high expected FNMR. A value of -1.0 indicates a failed attempt to calculate a quality score or the value is unassigned.

Generated by Doxygen

Returns

ReturnStatus

- ReturnStatus.code == ReturnCode::Success: If the cleanup was successful.
- Other error code: Indicates an error occurred during image analysis.

Examples

[C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h](#).

8.7.2.12 OFIQVectorQuality()

```
OFIQ_C_EXPORT ReturnStatus OFIQVectorQuality (
    OFIQHandle handle,
    const Image & faceImage,
    FaceImageQualityAssessment & qualityAssessment )
```

Performs a face image quality assessment using various quality measurement methods.

Parameters

in	<i>handle</i>	A handle to OFIQ instance.
in	<i>faceImage</i>	Single face image
out	<i>qualityAssessment</i>	Results of face image quality assessment containing a bounding box of the largest face detected and results of various quality measurements. This method allocates memory for the FaceImageQualityAssessment object, to free the memory the method OFIQFreeFaceImageQualityAssessment() should be used.

Returns

ReturnStatus

- ReturnStatus.code == ReturnCode::Success: If the cleanup was successful.
- Other error code: Indicates an error occurred during image analysis.

See also

FaceImageQualityAssessment

OFIQFreeFaceImageQualityAssessment()

Examples

[C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h](#).

8.7.2.13 OFIQVectorQualityExt()

```
OFIQ_C_EXPORT ReturnStatus OFIQVectorQualityExt (
    OFIQHandle handle,
    const Image & faceImage,
    FaceImageQualityAssessment & qualityAssessment,
    PreprocessingResult & preprocessingResult )
```

Performs a face image quality assessment by using various quality measurement methods. Additionally to the OFIQVectorQuality method this function returns preprocessing results such as.

- bounding boxes of all faces detected in the processed image
- face landmarks of the biggest (most prominent) face
- face masks and face segmentation images

Parameters

in	<i>handle</i>	A handle to OFIQ instance.
in	<i>faceImage</i>	Single face image
out	<i>qualityAssessment</i>	Results of face image quality assessment containing a bounding box of the largest face detected and results of various quality measurements. This method allocates memory for the <code>FacelImageQualityAssessment</code> object, to free the memory the method <code>OFIQFreeFacelImageQualityAssessment()</code> should be used.
out	<i>preprocessingResult</i>	Preprocessing results containing bounding boxes of all faces detected in the processed image, face landmarks of the most prominent face, a mask of the face region defined by face landmarks, a face occlusion mask and a face segmentation image.

Returns

ReturnStatus Indicating whether the operation was successful

- ReturnStatus.code == ReturnCode::Success: If the cleanup was successful.
- Other error code: Indicates an error occurred during image analysis.

See also

[FacelImageQualityAssessment](#)
[PreprocessingResult](#)
[OFIQFreeFacelImageQualityAssessment\(\)](#)
[OFIQFreePreprocessingResult\(\)](#)

Examples

[C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h.](#)

8.8 ofiq_libc.h

[Go to the documentation of this file.](#)

```

00001
00027 #ifndef OFIQ_LIBC_H
00028 #define OFIQ_LIBC_H
00029
00030 #ifdef _WIN32
00031 #   ifdef OFIQ_EXPORTS
00032 #       define OFIQ_C_EXPORT __declspec(dllexport)
00033 #   else
00034 #       define OFIQ_C_EXPORT __declspec(dllimport)
00035 #   endif
00036 #else
00037 #   define OFIQ_C_EXPORT
00038 #endif
00039
00040 #ifdef __cplusplus
00041 extern "C"
00042 {

```

```

00043 #endif
00044
00045 #include <ofiq_structs_c.h>
00046
00066     OFIQ_C_EXPORT ReturnStatus
00067         OFIQInitialize(OFIQHandle* handle, const char* configDir, const char* configFileName);
00068
00085     OFIQ_C_EXPORT ReturnStatus OFIQCleanup(OFIQHandle* handle);
00086
00105     OFIQ_C_EXPORT ReturnStatus OFIQReadImage(const char* filename, Image& image);
00106
00129     OFIQ_C_EXPORT ReturnStatus
00130         OFIQReadImageFromByteArray(const unsigned char* imageData, size_t dataLength, Image& image);
00131
00146     OFIQ_C_EXPORT ReturnStatus OFIQFreeImage(Image* image);
00147
00161     OFIQ_C_EXPORT ReturnStatus
00162         OFIQScalarQuality(OFIQHandle handle, const Image& face, double& quality);
00163
00182     OFIQ_C_EXPORT ReturnStatus OFIQVectorQuality(
00183         OFIQHandle handle, const Image& faceImage, FaceImageQualityAssessment& qualityAssessment);
00184
00213     OFIQ_C_EXPORT ReturnStatus OFIQVectorQualityExt(
00214         OFIQHandle handle,
00215         const Image& faceImage,
00216         FaceImageQualityAssessment& qualityAssessment,
00217         PreprocessingResult& preprocessingResult);
00218
00232     OFIQ_C_EXPORT ReturnStatus OFIQFreeFaceImageQualityAssessment(FaceImageQualityAssessment* qa);
00233
00247     OFIQ_C_EXPORT ReturnStatus OFIQFreeFaceLandmarks(FaceLandmarks* landmarks);
00248
00262     OFIQ_C_EXPORT ReturnStatus OFIQFreePreprocessingResult(PreprocessingResult* preprocRes);
00263
00283     OFIQ_C_EXPORT void OFIQGetVersion(int& major, int& minor, int& patch);
00284
00294     OFIQ_C_EXPORT const char* OFIQGetQualityMeasureName(QualityMeasure measure);
00295
00296 #ifdef __cplusplus
00297 } // extern "C"
00298 #endif
00299
00300 #endif /* OFIQ_LIBC_H */

```

8.9 ofiq_libc_utils.h File Reference

```

#include <ofiq_structs_c.h>
#include <ofiq_structs.h>
#include <string>
#include <string_view>

```

Functions

- const char * [getQualityMeasureName](#) (QualityMeasure measure)
- ReturnStatus [createReturnStatus](#) (ReturnCode code, std::string_view str="")
- ReturnStatus [copyImageToCapi](#) (const OFIQ::Image &ofiqImage, Image &capImage)
- ReturnStatus [copyImageFromCapi](#) (const Image &capImage, OFIQ::Image &ofiqImage)
- ReturnStatus [copyQualityAssessmentToCapi](#) (const OFIQ::FaceImageQualityAssessment &ofiqQA, FaceImageQualityAssessment &qa)
- ReturnStatus [copyPreprocessingResultToCapi](#) (const OFIQ::FaceImageQualityPreprocessingResult &ofiqPreprocRes, const uint16_t imageWidth, const uint16_t imageHeight, PreprocessingResult &preprocRes)
- QualityMeasure [QualityMeasureToCapi](#) (OFIQ::QualityMeasure measure)
- QualityMeasureReturnCode [QualityMeasureReturnCodeToCapi](#) (OFIQ::QualityMeasureReturnCode qmCode)

8.9.1 Function Documentation

8.9.1.1 copyImageFromCApi()

```
ReturnStatus copyImageFromCApi (
    const Image & capiImage,
    OFIQ::Image & ofiqImage )
```

8.9.1.2 copyImageToCApi()

```
ReturnStatus copyImageToCApi (
    const OFIQ::Image & ofiqImage,
    Image & capiImage )
```

8.9.1.3 copyPreprocessingResultToCApi()

```
ReturnStatus copyPreprocessingResultToCApi (
    const OFIQ::FaceImageQualityPreprocessingResult & ofiqPreprocRes,
    const uint16_t imageWidth,
    const uint16_t imageHeight,
    PreprocessingResult & preprocRes )
```

8.9.1.4 copyQualityAssessmentToCApi()

```
ReturnStatus copyQualityAssessmentToCApi (
    const OFIQ::FaceImageQualityAssessment & ofiqQA,
    FaceImageQualityAssessment & qa )
```

8.9.1.5 createReturnStatus()

```
ReturnStatus createReturnStatus (
    ReturnCode code,
    std::string_view str = "" )
```

8.9.1.6 getQualityMeasureName()

```
const char * getQualityMeasureName (
    QualityMeasure measure )
```

8.9.1.7 QualityMeasureReturnCodeToCApi()

```
QualityMeasureReturnCode QualityMeasureReturnCodeToCApi (
    OFIQ::QualityMeasureReturnCode qmCode )
```

8.9.1.8 QualityMeasureToCapi()

```
QualityMeasure QualityMeasureToCapi (
    OFIQ::QualityMeasure measure )
```

8.10 ofiq_libc_utils.h

[Go to the documentation of this file.](#)

```
00001
00027 #ifndef OFIQ_LIBC_UTILS_H
00028 #define OFIQ_LIBC_UTILS_H
00029
00030 #include <ofiq_structs_c.h>
00031 #include <ofiq_structs.h>
00032 #include <string>
00033 #include <string_view>
00034
00035 const char* getQualityMeasureName(QualityMeasure measure);
00036
00037 ReturnStatus createReturnStatus(ReturnCode code, std::string_view str = "");
00038
00039 ReturnStatus copyImageToCapi(const OFIQ::Image& ofiqImage, Image& capiImage);
00040 ReturnStatus copyImageFromCapi(const Image& capiImage, OFIQ::Image& ofiqImage);
00041
00042 ReturnStatus copyQualityAssessmentToCapi(
00043     const OFIQ::FaceImageQualityAssessment& ofiqQA,
00044     FaceImageQualityAssessment& qa);
00045
00046 ReturnStatus copyPreprocessingResultToCapi(
00047     const OFIQ::FaceImageQualityPreprocessingResult& ofiqPreprocRes,
00048     const uint16_t imageWidth,
00049     const uint16_t imageHeight,
00050     PreprocessingResult& preprocRes);
00051
00052 QualityMeasure QualityMeasureToCapi(
00053     OFIQ::QualityMeasure measure);
00054
00055 QualityMeasureReturnCode QualityMeasureReturnCodeToCapi(
00056     OFIQ::QualityMeasureReturnCode qmCode);
00057
00058 #endif /* OFIQ_LIBC_UTILS_H */
```

8.11 ofiq_structs.h File Reference

Provides several helper classes, enums and functions used in the OFIQ framework.

```
#include <cstdint>
#include <cstring>
#include <iostream>
#include <map>
#include <memory>
#include <string>
#include <utility>
#include <vector>
```

Classes

- struct [OFIQ::Image](#)
Struct representing a single image.
- struct [OFIQ::ReturnStatus](#)
A structure to contain information about a failure by the software under test.

- struct [OFIQ::QualityMeasureResult](#)
Data structure to handle the results of a quality measure.
- struct [OFIQ::BoundingBox](#)
Data structure for describing bounding boxes, e.g. the face region of the faces found by a face detector.
- struct [OFIQ::LandmarkPoint](#)
Data structure to describe the x and y coordinate of a landmark.
- struct [OFIQ::FaceLandmarks](#)
Data structure for storing facial landmarks.
- struct [OFIQ::FacelImageQualityAssessment](#)
Data structure storing the results of the different measurement computations.
- struct [OFIQ::FacelImageQualityPreprocessingResult](#)
Data structure storing the results of pre-processing computations.

Namespaces

- namespace [OFIQ](#)
Namespace for OFIQ API.

Typedefs

- using [OFIQ::QualityAssessments](#) = `std::map<QualityMeasure, QualityMeasureResult>`
Data structure that stores key-value pairs, with each entry representing a quality element and its value.
- using [OFIQ::Landmarks](#) = `std::vector<LandmarkPoint>`
container for a collection of landmarks, e.g. belonging to all the landmarks detected on a face image.

Enumerations

- enum class [OFIQ::ReturnCode](#) {
[OFIQ::Success](#) = 0 , [OFIQ::ImageReadingError](#) , [OFIQ::ImageWritingError](#) , [OFIQ::MissingConfigParamError](#) ,
[OFIQ::UnknownConfigParamError](#) , [OFIQ::FaceDetectionError](#) , [OFIQ::FaceLandmarkExtractionError](#) ,
[OFIQ::FaceOcclusionSegmentationError](#) ,
[OFIQ::FaceParsingError](#) , [OFIQ::UnknownError](#) , [OFIQ::QualityAssessmentError](#) , [OFIQ::NotImplemented](#) ,
[OFIQ::MissingConfigFileError](#) }
Return codes for functions specified in this API.
- enum class [OFIQ::QualityMeasure](#) {
[OFIQ::UnifiedQualityScore](#) = 0x41 , [OFIQ::BackgroundUniformity](#) = 0x42 , [OFIQ::IlluminationUniformity](#) = 0x43 , [OFIQ::Luminance](#) = -0x44 ,
[OFIQ::LuminanceMean](#) = 0x44 , [OFIQ::LuminanceVariance](#) = 0x45 , [OFIQ::UnderExposurePrevention](#) = 0x46 , [OFIQ::OverExposurePrevention](#) = 0x47 ,
[OFIQ::DynamicRange](#) = 0x48 , [OFIQ::Sharpness](#) = 0x49 , [OFIQ::CompressionArtifacts](#) = 0x4a ,
[OFIQ::NaturalColour](#) = 0x4b ,
[OFIQ::SingleFacePresent](#) = 0x4c , [OFIQ::EyesOpen](#) = 0x4d , [OFIQ::MouthClosed](#) = 0x4e , [OFIQ::EyesVisible](#) = 0x4f ,
[OFIQ::MouthOcclusionPrevention](#) = 0x50 , [OFIQ::FaceOcclusionPrevention](#) = 0x51 , [OFIQ::InterEyeDistance](#) = 0x52 , [OFIQ::HeadSize](#) = 0x53 ,
[OFIQ::CropOfTheFacelImage](#) = -0x54 , [OFIQ::LeftwardCropOfTheFacelImage](#) = 0x54 , [OFIQ::RightwardCropOfTheFacelImage](#) = 0x55 , [OFIQ::MarginAboveOfTheFacelImage](#) = 0x56 ,
[OFIQ::MarginBelowOfTheFacelImage](#) = 0x57 , [OFIQ::HeadPose](#) = -0x58 , [OFIQ::HeadPoseYaw](#) = 0x58 ,
[OFIQ::HeadPosePitch](#) = 0x59 ,
[OFIQ::HeadPoseRoll](#) = 0x5a , [OFIQ::ExpressionNeutrality](#) = 0x5b , [OFIQ::NoHeadCoverings](#) = 0x5c ,
[OFIQ::NotSet](#) = -1 }

Enums presenting the measure labels.

- enum class `OFIQ::QualityMeasureReturnCode` { `OFIQ::Success` = 0 , `OFIQ::FailureToAssess` , `OFIQ::NotInitialized` }

Return codes for QualityMeasureResult.

- enum class `OFIQ::FaceDetectorType` { `OFIQ::OPENCVSSD` , `OFIQ::NotSet` }

Enum describing the different face detector implementations.

- enum class `OFIQ::LandmarkType` { `OFIQ::LM_98` , `OFIQ::NotSet` }

Enum describing the different implementations of landmarks.

Functions

- `std::ostream & OFIQ::operator<<` (`std::ostream &s`, const `ReturnCode` &rc)
- `std::string OFIQ::QualityMeasureToString` (const `QualityMeasure` qm)
- `std::ostream & OFIQ::operator<<` (`std::ostream &s`, const `QualityMeasure` &qm)

8.11.1 Detailed Description

PROvides several helper classes, enums and functions used in the OFIQ framework.

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Author

OFIQ development team

8.12 ofiq_structs.h

[Go to the documentation of this file.](#)

```

00001
00027 #ifndef OFIQ_STRUCTS_H
00028 #define OFIQ_STRUCTS_H
00029
00030 #include <stdint>
00031 #include <cstring>
00032 #include <stdint>
00033 #include <iostream>
00034 #include <map>
00035 #include <memory>
00036 #include <string>
00037 #include <utility>
00038 #include <vector>
00039
00043 namespace OFIQ
00044 {
00049     struct Image
00050     {
00051
00053         uint16_t width{ 0 };
00055         uint16_t height{ 0 };
00057         uint8_t depth{ 24 };
00062         std::shared_ptr<uint8_t[]> data;
00063
00067         Image() = default;
00068
00082         Image(const uint16_t width, const uint16_t height, const uint8_t depth, const
std::shared_ptr<uint8_t[]>& data)
00083             : width{ width },
00084               height{ height },
00085               depth{ depth },
00086               data{ data }
00087         {
00088         }
00089
00091         size_t size() const { return (static_cast<size_t>(width) * height * (depth / 8)); }
00092
00107         void deepcopy(uint16_t width, uint16_t height, uint8_t depth,
00108             const std::shared_ptr<uint8_t[]>& data, bool isRgb = true )
00109         {
00110             this->width = width;
00111             this->height = height;
00112             this->depth = depth;
00113             size_t size = this->size();
00114             this->data.reset(new uint8_t[size], std::default_delete<uint8_t[]>());
00115             memcpy(this->data.get(), data.get(), size);
00116             if ( isRgb && depth == 24 )
00117             {
00118                 auto ptr = this->data.get();
00119                 for (size_t idx = 0; idx < size; idx += 3)
00120                 {
00121                     std::swap(ptr[idx], ptr[idx+2]);
00122                 }
00123             }
00124         }
00125     };
00126
00127
00132     enum class ReturnCode
00133     {
00135         Success = 0,
00137         ImageReadingError,
00139         ImageWritingError,
00141         MissingConfigParamError,
00143         UnknownConfigParamError,
00145         FaceDetectionError,
00147         FaceLandmarkExtractionError,
00149         FaceOcclusionSegmentationError,
00151         FaceParsingError,
00153         UnknownError,
00155         QualityAssessmentError,
00157         NotImplemented,
00159         MissingConfigFileError
00160     };
00161
00163     inline std::ostream& operator<<(std::ostream& s, const ReturnCode& rc)
00164     {
00165         switch (rc)
00166         {
00167             case ReturnCode::Success:
00168                 return (s << "Success");
00169             case ReturnCode::UnknownError:

```



```

00170         return (s « "Unknown Error");
00171     case ReturnCode::QualityAssessmentError:
00172         return (s « "Failure to generate a quality score on the input image");
00173     case ReturnCode::NotImplemented:
00174         return (s « "Function is not implemented");
00175     default:
00176         return (s « "Undefined error");
00177     }
00178 }
00179
00192 struct ReturnStatus
00193 {
00195     ReturnCode code{ ReturnCode::UnknownError };
00197     std::string info;
00198
00203     ReturnStatus() = default;
00204
00214     ReturnStatus(const ReturnCode code, const std::string& info = "")
00215         : code{code},
00216           info{info}
00217     {
00218     }
00219 };
00220
00223 enum class QualityMeasure
00224 {
00226     UnifiedQualityScore = 0x41,
00228     BackgroundUniformity = 0x42,
00230     IlluminationUniformity = 0x43,
00232     Luminance = -0x44,
00234     LuminanceMean = 0x44,
00236     LuminanceVariance = 0x45,
00238     UnderExposurePrevention = 0x46,
00240     OverExposurePrevention = 0x47,
00242     DynamicRange = 0x48,
00244     Sharpness = 0x49,
00246     CompressionArtifacts = 0x4a,
00248     NaturalColour = 0x4b,
00250     SingleFacePresent = 0x4c,
00252     EyesOpen = 0x4d,
00254     MouthClosed = 0x4e,
00256     EyesVisible = 0x4f,
00258     MouthOcclusionPrevention = 0x50,
00260     FaceOcclusionPrevention = 0x51,
00262     InterEyeDistance = 0x52,
00264     HeadSize = 0x53,
00266     CropOfTheFaceImage = -0x54,
00268     LeftwardCropOfTheFaceImage = 0x54,
00270     RightwardCropOfTheFaceImage = 0x55,
00272     MarginAboveOfTheFaceImage = 0x56,
00274     MarginBelowOfTheFaceImage = 0x57,
00276     HeadPose = -0x58,
00278     HeadPoseYaw = 0x58,
00280     HeadPosePitch = 0x59,
00282     HeadPoseRoll = 0x5a,
00284     ExpressionNeutrality = 0x5b,
00286     NoHeadCoverings = 0x5c,
00288     NotSet = -1
00289 };
00290
00295 enum class QualityMeasureReturnCode
00296 {
00298     Success = 0,
00300     FailureToAssess,
00302     NotInitialized
00303 };
00304
00309 struct QualityMeasureResult
00310 {
00312     double rawScore{ -1 };
00316     double scalar{ -1 };
00318     QualityMeasureReturnCode code{ QualityMeasureReturnCode::NotInitialized };
00319
00324     QualityMeasureResult() = default;
00325
00333     QualityMeasureResult(double rawScore, double scalar = -1, QualityMeasureReturnCode code =
QualityMeasureReturnCode::NotInitialized)
00334         : rawScore{rawScore},
00335           scalar{scalar},
00336           code{code}
00337     {
00338     }
00339 };
00340
00346 using QualityAssessments = std::map<QualityMeasure, QualityMeasureResult>;
00347
00352 enum class FaceDetectorType

```

```

00353     {
00354         OPENCVSSD,
00355         NotSet
00356     };
00357
00358     struct BoundingBox
00359     {
00360         int16_t xleft{ -1 };
00361         int16_t ytop{ -1 };
00362         int16_t width{ -1 };
00363         int16_t height{ -1 };
00364
00365         FaceDetectorType faceDetector = FaceDetectorType::NotSet;
00366
00367         BoundingBox() = default;
00368
00369         BoundingBox(int16_t xleft, int16_t ytop, int16_t width, int16_t height, FaceDetectorType
i_faceDetector)
00370             : xleft{xleft},
00371               ytop{ytop},
00372               width{width},
00373               height{height},
00374               faceDetector{i_faceDetector}
00375         {
00376         }
00377     };
00378
00379     struct LandmarkPoint
00380     {
00381         int16_t x{ -1 };
00382         int16_t y{-1 };
00383
00384         LandmarkPoint() = default;
00385
00386         LandmarkPoint(int16_t i_x, int16_t i_y)
00387             : x{i_x},
00388               y{i_y}
00389         {
00390         }
00391     };
00392
00393     using Landmarks = std::vector<LandmarkPoint>;
00394
00395     enum class LandmarkType
00396     {
00397         LM_98,
00398         NotSet
00399     };
00400
00401     struct FaceLandmarks
00402     {
00403         LandmarkType type{ LandmarkType::NotSet };
00404         Landmarks landmarks;
00405
00406         FaceLandmarks() = default;
00407     };
00408
00409     struct FaceImageQualityAssessment
00410     {
00411         QualityAssessments qAssessments;
00412
00413         BoundingBox boundingBox;
00414
00415         FaceImageQualityAssessment() = default;
00416
00417         FaceImageQualityAssessment(
00418             const QualityAssessments& qAssessments, const BoundingBox& boundingBox)
00419             : qAssessments{qAssessments},
00420               boundingBox{boundingBox}
00421         {
00422         }
00423     };
00424
00425     struct FaceImageQualityPreprocessingResult
00426     {
00427         std::vector<OFIQ::BoundingBox> m_faces;
00428
00429         FaceLandmarks m_landmarks;
00430
00431         std::shared_ptr<uint8_t[]> m_segmentationMaskPtr;
00432
00433         std::shared_ptr<uint8_t[]> m_occlusionMaskPtr;
00434     };

```

```

00588         std::shared_ptr<uint8_t[]> m_landmarkedRegionPtr;
00589
00593         FaceImageQualityPreprocessingResult() = default;
00594     };
00595
00596     inline std::string QualityMeasureToString(const QualityMeasure qm)
00597     {
00598         switch (qm)
00599         {
00600             case QualityMeasure::UnifiedQualityScore:           return "UnifiedQualityScore";
00601             case QualityMeasure::BackgroundUniformity:         return "BackgroundUniformity";
00602             case QualityMeasure::IlluminationUniformity:        return "IlluminationUniformity";
00603             case QualityMeasure::Luminance:                    return "Luminance";
00604             case QualityMeasure::LuminanceMean:                return "LuminanceMean";
00605             case QualityMeasure::LuminanceVariance:            return "LuminanceVariance";
00606             case QualityMeasure::UnderExposurePrevention:       return "UnderExposurePrevention";
00607             case QualityMeasure::OverExposurePrevention:        return "OverExposurePrevention";
00608             case QualityMeasure::DynamicRange:                  return "DynamicRange";
00609             case QualityMeasure::Sharpness:                     return "Sharpness";
00610             case QualityMeasure::NaturalColour:                 return "NaturalColour";
00611             case QualityMeasure::SingleFacePresent:             return "SingleFacePresent";
00612             case QualityMeasure::EyesOpen:                      return "EyesOpen";
00613             case QualityMeasure::MouthClosed:                   return "MouthClosed";
00614             case QualityMeasure::EyesVisible:                   return "EyesVisible";
00615             case QualityMeasure::MouthOcclusionPrevention:       return "MouthOcclusionPrevention";
00616             case QualityMeasure::FaceOcclusionPrevention:        return "FaceOcclusionPrevention";
00617             case QualityMeasure::InterEyeDistance:              return "InterEyeDistance";
00618             case QualityMeasure::HeadSize:                      return "HeadSize";
00619             case QualityMeasure::CropOfTheFaceImage:            return "CropOfTheFaceImage";
00620             case QualityMeasure::HeadPose:                      return "HeadPose";
00621             case QualityMeasure::ExpressionNeutrality:          return "ExpressionNeutrality";
00622             case QualityMeasure::NoHeadCoverings:               return "NoHeadCoverings";
00623             case QualityMeasure::NotSet:                        return "NotSet";
00624
00625             // the following names were modified (they don't correspond to the enum names) to
00626             // match the names used in ISO/IEC 25722 standard.
00627             case QualityMeasure::CompressionArtifacts:          return "NoCompressionArtifacts";
00628             case QualityMeasure::LeftwardCropOfTheFaceImage:    return "LeftwardCropOfFace";
00629             case QualityMeasure::RightwardCropOfTheFaceImage:   return "RightwardCropOfFace";
00630             case QualityMeasure::MarginAboveOfTheFaceImage:      return "MarginAboveTheFace";
00631             case QualityMeasure::MarginBelowOfTheFaceImage:      return "MarginBelowTheFace";
00632             case QualityMeasure::HeadPoseYaw:                   return "HeadPoseYawFrontal";
00633             case QualityMeasure::HeadPosePitch:                  return "HeadPosePitchFrontal";
00634             case QualityMeasure::HeadPoseRoll:                  return "HeadPoseRollFrontal";
00635
00636             default:                                             return "UnknownQualityMeasure";
00637         }
00638     }
00639
00640     inline std::ostream& operator<<(std::ostream& s, const QualityMeasure& qm)
00641     {
00642         return (s << QualityMeasureToString(qm));
00643     }
00644 }
00645
00646 #endif /* OFIQ_STRUCTS_H */

```

8.13 ofiq_structs_c.h File Reference

Provides several helper classes, enums and functions used in the OFIQ framework.

```

#include <stdint.h>
#include <stddef.h>

```

Classes

- struct [Image](#)
Struct representing a single image.
- struct [QualityMeasureResult](#)
Data structure to handle the results of a quality measure.
- struct [BoundingBox](#)

Data structure for describing bounding boxes, e.g. the face region of the faces found by a face detector.

- struct [LandmarkPoint](#)

Data structure to describe the x and y coordinate of a landmark.

- struct [FaceLandmarks](#)

Data structure for storing facial landmarks.

- struct [FaceImageQualityAssessment](#)

Data structure storing the results of the different measurement computations.

- struct [PreprocessingResult](#)

Typedefs

- using [OFIQHandle](#) = void*

Handle type for an OFIQ implementation instance.

- using [ReturnStatus](#) = [ReturnCode](#)

A structure containing return status - a code and a information message under test.

Enumerations

- enum class [ReturnCode](#) {
[Success](#) = 0 , [ImageReadingError](#) , [ImageWritingError](#) , [MissingConfigFileError](#) ,
[MissingConfigParamError](#) , [UnknownConfigParamError](#) , [FaceDetectionError](#) , [FaceLandmarkExtractionError](#)
, [FaceOcclusionSegmentationError](#) , [FaceParsingError](#) , [UnknownError](#) , [QualityAssessmentError](#) ,
[NotImplemented](#) }

Return codes for functions specified in this API.

- enum class [QualityMeasure](#) {
[UnifiedQualityScore](#) = 0x41 , [BackgroundUniformity](#) = 0x42 , [IlluminationUniformity](#) = 0x43 , [Luminance](#) =
-0x44 ,
[LuminanceMean](#) = 0x44 , [LuminanceVariance](#) = 0x45 , [UnderExposurePrevention](#) = 0x46 , [OverExposurePrevention](#)
= 0x47 ,
[DynamicRange](#) = 0x48 , [Sharpness](#) = 0x49 , [CompressionArtifacts](#) = 0x4a , [NaturalColour](#) = 0x4b ,
[SingleFacePresent](#) = 0x4c , [EyesOpen](#) = 0x4d , [MouthClosed](#) = 0x4e , [EyesVisible](#) = 0x4f ,
[MouthOcclusionPrevention](#) = 0x50 , [FaceOcclusionPrevention](#) = 0x51 , [InterEyeDistance](#) = 0x52 , [HeadSize](#)
= 0x53 ,
[CropOfTheFacelImage](#) = -0x54 , [LeftwardCropOfTheFacelImage](#) = 0x54 , [RightwardCropOfTheFacelImage](#) =
0x55 , [MarginAboveOfTheFacelImage](#) = 0x56 ,
[MarginBelowOfTheFacelImage](#) = 0x57 , [HeadPose](#) = -0x58 , [HeadPoseYaw](#) = 0x58 , [HeadPosePitch](#) = 0x59 ,
[HeadPoseRoll](#) = 0x5a , [ExpressionNeutrality](#) = 0x5b , [NoHeadCoverings](#) = 0x5c , [NotSet](#) = -1 }

Enumeration of the measure labels.

- enum class [QualityMeasureReturnCode](#) { [Success](#) = 0 , [FailureToAssess](#) , [NotInitialized](#) }

Return codes for QualityMeasureResult.

- enum class [FaceDetectorType](#) { [OPENCVSSD](#) , [NotSet](#) }

Enum containing different face detector types.

- enum class [LandmarkType](#) { [LM_98](#) , [NotSet](#) }

container for a collection of landmarks, e.g. belonging to all the landmarks detected on a face image.

Variables

- constexpr size_t [STATUS_INFO_LENGTH](#) = 512

8.13.1 Detailed Description

Provides several helper classes, enums and functions used in the OFIQ framework.

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Author

OFIQ development team

8.13.2 Typedef Documentation

8.13.2.1 OFIQHandle

```
using OFIQHandle = void*
```

Handle type for an OFIQ implementation instance.

8.13.2.2 ReturnStatus

```
using ReturnStatus = ReturnCode
```

A structure containing return status - a code and a information message under test.

An object of this class allows the software to return some information from a function call. The string within this object can be optionally set to provide more information for debugging etc. The status code will be set by the function to Success on success, or one of the other codes on failure.

8.13.3 Enumeration Type Documentation

8.13.3.1 FaceDetectorType

```
enum class FaceDetectorType [strong]
```

Enum containing different face detector types.

Enumerator

OPENCVSSD	face detector based on the ssd implementation in opencv.
NotSet	unknown face detector

8.13.3.2 LandmarkType

```
enum class LandmarkType [strong]
```

container for a collection of landmarks, e.g. belonging to all the landmarks detected on a face image.

Enum describing the different implementations of landmarks.

Enumerator

LM_98	Landmarks extracted with the adnet detector.
NotSet	used for iterating through the enums.

8.13.3.3 QualityMeasure

```
enum class QualityMeasure [strong]
```

Enumeration of the measure labels.

Enumerator

UnifiedQualityScore	UnifiedQualityScore
BackgroundUniformity	BackgroundUniformity
IlluminationUniformity	IlluminationUniformity
Luminance	the common measure implementation for LuminanceMean, LuminanceVariance
LuminanceMean	LuminanceMean
LuminanceVariance	LuminanceVariance
UnderExposurePrevention	UnderExposurePrevention
OverExposurePrevention	OverExposurePrevention
DynamicRange	DynamicRange
Sharpness	Sharpness
CompressionArtifacts	CompressionArtifacts
NaturalColour	NaturalColour
SingleFacePresent	SingleFacePresent
EyesOpen	EyesOpen
MouthClosed	MouthClosed
EyesVisible	EyesVisible
MouthOcclusionPrevention	MouthOcclusionPrevention
FaceOcclusionPrevention	FaceOcclusionPrevention
InterEyeDistance	InterEyeDistance
HeadSize	HeadSize
CropOfTheFacelImage	CropOfTheFacelImage: common measure for {Left,Right}wardCropOfTheFacelImage, MarginAbove, and MarginBelow

Enumerator

LeftwardCropOfTheFacelImage	LeftwardCropOfTheFacelImage
RightwardCropOfTheFacelImage	RightwardCropOfTheFacelImage
MarginAboveOfTheFacelImage	MarginAbove
MarginBelowOfTheFacelImage	MarginBelow
HeadPose	HeadPose
HeadPoseYaw	HeadPoseYaw
HeadPosePitch	HeadPosePitch
HeadPoseRoll	HeadPoseRoll
ExpressionNeutrality	ExpressionNeutrality
NoHeadCoverings	NoHeadCoverings
NotSet	unknown measure

8.13.3.4 QualityMeasureReturnCode

```
enum class QualityMeasureReturnCode [strong]
```

Return codes for QualityMeasureResult.

Enumerator

Success	Success
FailureToAssess	Unable to assess a quality measure
NotInitialized	Quality measure is not initialized

8.13.3.5 ReturnCode

```
enum class ReturnCode [strong]
```

Return codes for functions specified in this API.

Enumerator

Success	Success
ImageReadingError	Failed to read an image.
ImageWritingError	failed to write an image to disk.
MissingConfigFileError	A config file is missing or the given filename is invalid
MissingConfigParamError	A required config parameter is missing
UnknownConfigParamError	A required config parameter is missing
FaceDetectionError	Unable to detect a face in the image
FaceLandmarkExtractionError	Unable to extract landmarks from face
FaceOcclusionSegmentationError	Unable to extract occlusion segments from face
FaceParsingError	Unable to parse face
UnknownError	Catch-all error
QualityAssessmentError	Failure to generate a quality score on the input image
NotImplemented	Function is not implemented

8.13.4 Variable Documentation

8.13.4.1 STATUS_INFO LENGHT

```
constexpr size_t STATUS_INFO LENGHT = 512 [constexpr]
```

8.14 ofiq_structs_c.h

[Go to the documentation of this file.](#)

```
00001
00027 #ifndef OFIQ_STRUCTS_C_H
00028 #define OFIQ_STRUCTS_C_H
00029
00030 #include <stdint.h>
00031 #include <stddef.h>
00032
00036 using OFIQHandle = void*;
00037
00042 enum class ReturnCode
00043 {
00045     Success = 0,
00047     ImageReadingError,
00049     ImageWritingError,
00051     MissingConfigFileError,
00053     MissingConfigParamError,
00055     UnknownConfigParamError,
00057     FaceDetectionError,
00059     FaceLandmarkExtractionError,
00061     FaceOcclusionSegmentationError,
00063     FaceParsingError,
00065     UnknownError,
00067     QualityAssessmentError,
00069     NotImplemented
00070 };
00071
00072 constexpr size_t STATUS_INFO LENGHT = 512;
00073
00086 using ReturnStatus = ReturnCode;
00087 //struct ReturnStatus
00088 //{
00089 //    /** @brief Return status code */
00090 //    ReturnCode code{ReturnCode::UnknownError};
00091 //    /** @brief Optional information string */
00092 //    char info[STATUS_INFO LENGHT]{};
00093 //};
00094
00099 struct Image
00100 {
00102     uint16_t width{0};
00104     uint16_t height{0};
00106     uint8_t depth{24};
00111     uint8_t* data{nullptr};
00112 };
00113
00116 enum class QualityMeasure
00117 {
00119     UnifiedQualityScore = 0x41,
00121     BackgroundUniformity = 0x42,
00123     IlluminationUniformity = 0x43,
00125     Luminance = -0x44,
00127     LuminanceMean = 0x44,
00129     LuminanceVariance = 0x45,
00131     UnderExposurePrevention = 0x46,
00133     OverExposurePrevention = 0x47,
00135     DynamicRange = 0x48,
00137     Sharpness = 0x49,
00139     CompressionArtifacts = 0x4a,
00141     NaturalColour = 0x4b,
00143     SingleFacePresent = 0x4c,
00145     EyesOpen = 0x4d,
00147     MouthClosed = 0x4e,
00149     EyesVisible = 0x4f,
00151     MouthOcclusionPrevention = 0x50,
00153     FaceOcclusionPrevention = 0x51,
00155     InterEyeDistance = 0x52,
00157     HeadSize = 0x53,
00160     CropOfTheFaceImage = -0x54,
```



```

00162     LeftwardCropOfTheFaceImage = 0x54,
00164     RightwardCropOfTheFaceImage = 0x55,
00166     MarginAboveOfTheFaceImage = 0x56,
00168     MarginBelowOfTheFaceImage = 0x57,
00170     HeadPose = -0x58,
00172     HeadPoseYaw = 0x58,
00174     HeadPosePitch = 0x59,
00176     HeadPoseRoll = 0x5a,
00178     ExpressionNeutrality = 0x5b,
00180     NoHeadCoverings = 0x5c,
00182     NotSet = -1
00183 };
00184
00185 enum class QualityMeasureReturnCode
00186 {
00187     Success = 0,
00188     FailureToAssess,
00189     NotInitialized
00190 };
00191
00192 struct QualityMeasureResult
00193 {
00194     QualityMeasure measure{QualityMeasure::NotSet};
00195
00196     double rawScore{-1};
00197     double scalar{-1};
00198     QualityMeasureReturnCode code{QualityMeasureReturnCode::NotInitialized};
00199 };
00200
00201 enum class FaceDetectorType
00202 {
00203     OPENCVSSD,
00204     NotSet
00205 };
00206
00207 struct BoundingBox
00208 {
00209     int16_t xleft{-1};
00210     int16_t ytop{-1};
00211     int16_t width{-1};
00212     int16_t height{-1};
00213
00214     FaceDetectorType faceDetector = FaceDetectorType::NotSet;
00215 };
00216
00217 struct LandmarkPoint
00218 {
00219     int16_t x{-1};
00220     int16_t y{-1};
00221 };
00222 // using Landmarks = LandmarkPoint*;
00223
00224 enum class LandmarkType
00225 {
00226     LM_98,
00227     NotSet
00228 };
00229
00230 struct FaceLandmarks
00231 {
00232     LandmarkType type{LandmarkType::NotSet};
00233     LandmarkPoint* landmarks;
00234     int numLandmarks = 0;
00235 };
00236
00237 struct FaceImageQualityAssessment
00238 {
00239     QualityMeasureResult* qAssessments{nullptr};
00240
00241     size_t numAssessments{0};
00242     BoundingBox boundingBox;
00243 };
00244
00245 struct PreprocessingResult
00246 {
00247     BoundingBox* faces;
00248
00249     size_t numFaces;
00250     FaceLandmarks landmarks;
00251     Image segmentationMask;

```

```
00380
00386     Image occlusionMask;
00387
00393     Image landmarkedRegionMask;
00394
00395 };
00396 #endif /* OFIQ_STRUCTS_C_H */
```

8.15 AllDetectors.h File Reference

Provides the include statements to all classes derived from [FaceDetectorInterface](#).

```
#include "opencv_ssd_face_detector.h"
```

8.15.1 Detailed Description

Provides the include statements to all classes derived from [FaceDetectorInterface](#).

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Author

OFIQ development team

8.16 AllDetectors.h

[Go to the documentation of this file.](#)

```
00001
00028 #include "opencv_ssd_face_detector.h"
```

8.17 detectors.h File Reference

Provides the interface class to the face detector implementations.

```
#include "ofiq_lib.h"
#include "Session.h"
```

Classes

- class [OFIQ_LIB::FaceDetectorInterface](#)
Provides the interface class to the face detector implementations.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

8.17.1 Detailed Description

Provides the interface class to the face detector implementations.

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Author

OFIQ development team

8.18 detectors.h

[Go to the documentation of this file.](#)

```

00001
00028 #pragma once
00029
00030 #include "ofiq_lib.h"
00031 #include "Session.h"
00032
00036 namespace OFIQ_LIB
00037 {
00038
00043     class FaceDetectorInterface
00044     {
00045     public:
00050         virtual ~FaceDetectorInterface() = default;
00051
00058         std::vector<OFIQ::BoundingBox> detectFaces (OFIQ_LIB::Session& session);
00059
00060     protected:
00068         virtual std::vector<OFIQ::BoundingBox> UpdateFaces (OFIQ_LIB::Session& session) = 0;
00069     };
00070 }
```

8.19 opencv_ssd_face_detector.h File Reference

Implementation of a face detector using the SSD face detector CNN.

```

#include "Configuration.h"
#include "detectors.h"
#include <opencv2/dnn.hpp>
#include <mutex>
```

Classes

- class [OFIQ_LIB::modules::detectors::SSDFaceDetector](#)
Implementation of a face detector using the SSD face detector CNN.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::detectors](#)
Provides face detector implementations.

8.19.1 Detailed Description

Implementation of a face detector using the SSD face detector CNN.

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Author

OFIQ development team

8.20 opencv_ssd_face_detector.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "Configuration.h"
00030 #include "detectors.h"
00031 #include <opencv2/dnn.hpp>
00032 #include <mutex>
00033
00034
00038 namespace OFIQ_LIB::modules::detectors
00039 {
00040
00044     class SSDFaceDetector : public OFIQ_LIB::FaceDetectorInterface
00045     {
00046     public:
00052         explicit SSDFaceDetector(const Configuration& config);
00053
00058         ~SSDFaceDetector() override = default;
00059
00060     protected:
00068         std::vector<OFIQ::BoundingBox> UpdateFaces(OFIQ_LIB::Session& session) override;
00069
00070
00071     private:
00076         std::shared_ptr<cv::dnn::Net> m_dnnNet{nullptr};
00077
00082         std::mutex m_dnnNetMutex;
00083
00088         double m_confidenceThreshold;
00089
00094         double m_padding;
00095
00100         double m_minimalRelativeFaceSize;
00101     };
00102 }
```

8.21 adnet_FaceMap.h File Reference

Provides definitions of landmark indices to access specific parts of ADNet landmarks.

```
#include "FaceParts.h"
#include <array>
#include <map>
#include <vector>
```

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::landmarks](#)
Provides implementations of a landmark extractors.
- namespace [OFIQ_LIB::modules::landmarks::adnet](#)
Namespace for ADNet-specific landmarks.

Variables

- const [LandmarkIds OFIQ_LIB::modules::landmarks::adnet::leftEye](#) {60,61,62,63,64,65,66,67}
Landmark indices (ADNet) of the left eye.
- const [LandmarkIds OFIQ_LIB::modules::landmarks::adnet::rightEye](#) {68,69,70,71,72,73,74,75}
Landmark indices (ADNet) of the right eye.
- const [LandmarkIds OFIQ_LIB::modules::landmarks::adnet::leftEyeCorners](#) {60,64}
Landmark indices (ADNet) of the left eyes' corners.
- const [LandmarkIds OFIQ_LIB::modules::landmarks::adnet::rightEyeCorners](#) {68,72}
Landmark indices (ADNet) of the right eyes' corners.
- const [LandmarkIds OFIQ_LIB::modules::landmarks::adnet::nosetip](#) {54}
Landmark index (ADNet) of the nose tip.
- const [LandmarkIds OFIQ_LIB::modules::landmarks::adnet::mouthOuter](#) {76,77,78,79,80,81,82,83,84,85,86,87}
Landmark indices (ADNet) on the mouth's outer contour.
- const [LandmarkIds OFIQ_LIB::modules::landmarks::adnet::mouthInner](#) {88,89,90,91,92,93,94,95}
Landmark indices (ADNet) on the mouth's inner lip borders.
- const [LandmarkIds OFIQ_LIB::modules::landmarks::adnet::contour](#) {0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22}
Landmark indices (ADNet) of the face contour.
- const [LandmarkIds OFIQ_LIB::modules::landmarks::adnet::forehead](#) {}
Landmark indices (ADNet) of the forehead (empty for ADNet).
- const [LandmarkIds OFIQ_LIB::modules::landmarks::adnet::chin](#) {16}
Landmark index (ADNet) of the chin.
- const [landmarks::FaceMap OFIQ_LIB::modules::landmarks::adnet::FaceMap](#)
ADNets face map definition.
- const [LandmarkIdPairs OFIQ_LIB::modules::landmarks::adnet::pairsLeftEye](#)
Pair indices of landmarks (ADNet) for the left eye.
- const [LandmarkIdPairs OFIQ_LIB::modules::landmarks::adnet::pairsRightEye](#)
Landmark index pairs (ADNet) of landmarks for the right eye.
- const [LandmarkIdPairs OFIQ_LIB::modules::landmarks::adnet::pairsInnerLip](#)
Landmark index pairs (ADNet) of inner lip pairs.
- const [LandmarkIdPairs OFIQ_LIB::modules::landmarks::adnet::pairsMouthCenter](#)
Landmark index pair (ADNet) of the inner mouth (lips) center.
- const [landmarks::FacePairMap OFIQ_LIB::modules::landmarks::adnet::FacePairMap](#)
ADNets face pair map definition.

8.21.1 Detailed Description

Provides definitions of landmark indices to access specific parts of ADNet landmarks.

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The definitions provided by this header were taken from <https://arxiv.org/pdf/2109.05721.pdf> Appendix A, Figure 6.

Author

OFIQ development team

8.22 adnet_FaceMap.h

[Go to the documentation of this file.](#)

```
00001
00033 #pragma once
00034
00035 #include "FaceParts.h"
00036 #include <array>
00037 #include <map>
00038 #include <vector>
00039
00043 namespace OFIQ_LIB::modules::landmarks::adnet
00044 {
00049     const LandmarkIds leftEye{60, 61, 62, 63, 64, 65, 66, 67};
00050
00055     const LandmarkIds rightEye{68, 69, 70, 71, 72, 73, 74, 75};
00056
00060     const LandmarkIds leftEyeCorners{60, 64};
00061
00065     const LandmarkIds rightEyeCorners{68, 72};
00066
00070     const LandmarkIds nosetip{54};
00071
00075     const LandmarkIds mouthOuter{76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87};
00076
00080     const LandmarkIds mouthInner{88, 89, 90, 91, 92, 93, 94, 95};
00081
00085     const LandmarkIds
00086     contour{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32};
00086
00090     const LandmarkIds forehead{};
00091
00095     const LandmarkIds chin{16};
00096
00100     const landmarks::FaceMap FaceMap{
```

```

00101         {FaceParts::LEFT_EYE,          leftEye      },
00102         {FaceParts::RIGHT_EYE,         rightEye     },
00103         {FaceParts::LEFT_EYE_CORNERS,   leftEyeCorners},
00104         {FaceParts::RIGHT_EYE_CORNERS,   rightEyeCorners},
00105         {FaceParts::MOUTH_OUTER,        mouthOuter    },
00106         {FaceParts::MOUTH_INNER,        mouthInner    },
00107         {FaceParts::FACE_CONTOUR,       contour      },
00108         {FaceParts::CHIN,               chin         },
00109         {FaceParts::NOSETIP,            nosetip      },
00110         {FaceParts::FOREHEAD,           forehead     },
00111     };
00112
00117     const LandmarkIdPairs pairsLeftEye{
00118         {61, 67},
00119         {62, 66},
00120         {63, 65}
00121     };
00122
00127     const LandmarkIdPairs pairsRightEye{
00128         {69, 75},
00129         {70, 74},
00130         {71, 73}
00131     };
00132
00137     const LandmarkIdPairs pairsInnerLip{
00138         {89, 95},
00139         {90, 94},
00140         {91, 93}
00141     };
00142
00147     const LandmarkIdPairs pairsMouthCenter{
00148         {90, 94}
00149     };
00150
00154     const landmarks::FacePairMap FacePairMap{
00155         {FaceParts::LEFT_EYE,    pairsLeftEye    },
00156         {FaceParts::RIGHT_EYE,   pairsRightEye   },
00157         {FaceParts::MOUTH_INNER, pairsInnerLip    },
00158         {FaceParts::MOUTH_CENTER, pairsMouthCenter}
00159     };
00160 }

```

8.23 adnet_landmarks.h File Reference

Provides the [ADNetFaceLandmarkExtractor](#) class.

```

#include <memory>
#include "Configuration.h"
#include "detectors.h"
#include "landmarks.h"

```

Classes

- class [OFIQ_LIB::modules::landmarks::ADNetFaceLandmarkExtractor](#)
Class implementing the [FaceLandmarkExtractorInterface](#) interface.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::landmarks](#)
Provides implementations of a landmark extractors.

8.23.1 Detailed Description

Provides the [ADNetFaceLandmarkExtractor](#) class.

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Author

OFIQ development team

8.24 adnet_landmarks.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include <memory>
00031 #include "Configuration.h"
00032 #include "detectors.h"
00033 #include "landmarks.h"
00034
00038 namespace OFIQ_LIB::modules::landmarks
00039 {
00040
00042     class ADNetFaceLandmarkExtractorImpl;
00043
00049     class ADNetFaceLandmarkExtractor : public FaceLandmarkExtractorInterface
00050     {
00051     public:
00056         explicit ADNetFaceLandmarkExtractor(const Configuration& config);
00057
00061         ~ADNetFaceLandmarkExtractor() override;
00062
00063     protected:
00071         OFIQ::FaceLandmarks updateLandmarks(OFIQ_LIB::Session& session) override;
00072
00073     private:
00074
00078         std::unique_ptr<ADNetFaceLandmarkExtractorImpl> landmarkExtractor_;
00079     };
00080 }
```

8.25 AllLandmarks.h File Reference

Provides the include statements to all classes derived from [FaceLandmarkExtractorInterface](#).

```
#include "adnet_landmarks.h"
```

8.25.1 Detailed Description

Provides the include statements to all classes derived from [FaceLandmarkExtractorInterface](#).

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Author

OFIQ development team

8.26 AllLandmarks.h

[Go to the documentation of this file.](#)

```
00001
00029 #include "adnet_landmarks.h"
```

8.27 FaceMeasures.h File Reference

Provides a class implementing two luminance measures.

```
#include "ofiq_lib.h"
#include "PartExtractor.h"
#include <opencv2/opencv.hpp>
```

Classes

- class [OFIQ_LIB::modules::landmarks::FaceMeasures](#)
Provides static functions doing computations with landmarks.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::landmarks](#)
Provides implementations of a landmark extractors.

8.27.1 Detailed Description

Provides a class implementing two luminance measures.

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Author

OFIQ development team

8.28 FaceMeasures.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "ofiq_lib.h"
00030 #include "PartExtractor.h"
00031 #include <opencv2/opencv.hpp>
00032
00036 namespace OFIQ_LIB::modules::landmarks
00037 {
00041     class FaceMeasures
00042     {
00043     public:
00047         FaceMeasures() = delete;
00048
00066         static double InterEyeDistance(const OFIQ::FaceLandmarks& faceLandmarks, double yaw);
00067
00078         static cv::Mat GetFaceMask
00079         (const OFIQ::FaceLandmarks& faceLandmarks, const int height, const int width,
00080          const float alpha = 0);
00081
00088         static double GetDistance(const OFIQ::LandmarkPoint& a, const OFIQ::LandmarkPoint& b);
00089
```

```

00096     static double GetDistance(const LandmarkPair& pair)
00097     {
00098         return GetDistance(pair.Lower, pair.Upper);
00099     }
00100
00106     static OFIQ::LandmarkPoint GetMiddle(const OFIQ::Landmarks& landmarks);
00107
00113     static OFIQ::LandmarkPoint GetMiddle(const LandmarkPair& pair)
00114     {
00115         return GetMiddle(OFIQ::Landmarks{pair.Lower, pair.Upper});
00116     }
00117
00125     static OFIQ::LandmarkPoint GetMiddle(const std::vector<LandmarkPair>& pairs)
00126     {
00127         std::vector<OFIQ::LandmarkPoint> points;
00128         for (auto pair : pairs)
00129         {
00130             points.push_back(GetMiddle(pair));
00131         }
00132         return GetMiddle(points);
00133     }
00134
00147     static double GetMaxPairDistance(
00148         const OFIQ::FaceLandmarks& landmarks,
00149         landmarks::FaceParts facePart);
00150 };
00151 }

```

8.29 FaceParts.h File Reference

Provides several helper classes, enums and functions used in the OFIQ framework.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::landmarks](#)
Provides implementations of a landmark extractors.

Typedefs

- using [OFIQ_LIB::modules::landmarks::LandmarkId](#) = int
Type definition of a landmark index.
- using [OFIQ_LIB::modules::landmarks::LandmarkIds](#) = std::vector<[LandmarkId](#)>
Type definition of a list of landmark indices.
- using [OFIQ_LIB::modules::landmarks::FaceMap](#) = std::map<[FaceParts](#), [LandmarkIds](#)>
Type definition of a face map to access landmark indices for a queried face part.
- using [OFIQ_LIB::modules::landmarks::LandmarkIdPair](#) = std::array<[LandmarkId](#), 2>
Type definition for a pair of landmark index.
- using [OFIQ_LIB::modules::landmarks::LandmarkIdPairs](#) = std::vector<[LandmarkIdPair](#)>
Type definition for a list of landmark index pairs.
- using [OFIQ_LIB::modules::landmarks::FacePairMap](#) = std::map<[FaceParts](#), [LandmarkIdPairs](#)>
Structure defining pairs of landmark indices.

Enumerations

- enum class OFIQ_LIB::modules::landmarks::FaceParts {
 OFIQ_LIB::modules::landmarks::LEFT_EYE , OFIQ_LIB::modules::landmarks::RIGHT_EYE , OFIQ_LIB::modules::landmarks::
 , OFIQ_LIB::modules::landmarks::RIGHT_EYE_CORNERS ,
 OFIQ_LIB::modules::landmarks::MOUTH_OUTER , OFIQ_LIB::modules::landmarks::MOUTH_INNER ,
 OFIQ_LIB::modules::landmarks::FACE_CONTOUR , OFIQ_LIB::modules::landmarks::MOUTH_CENTER
 ,
 OFIQ_LIB::modules::landmarks::CHIN , OFIQ_LIB::modules::landmarks::NOSETIP , OFIQ_LIB::modules::landmarks::FOREHEAD
 }

Enumeration of facial landmark parts.

8.29.1 Detailed Description

Provides several helper classes, enums and functions used in the OFIQ framework.

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Author

OFIQ development team

8.30 FaceParts.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00033 namespace OFIQ_LIB::modules::landmarks
00034 {
00038     enum class FaceParts
00039     {
00041         LEFT_EYE,
00043         RIGHT_EYE,
00045         LEFT_EYE_CORNERS,
00047         RIGHT_EYE_CORNERS,
00049         MOUTH_OUTER,
00051         MOUTH_INNER,
00053         FACE_CONTOUR,
00055         MOUTH_CENTER,
00057         CHIN,
```

```

00059         NOSETIP,
00061         FOREHEAD
00062     };
00063
00067     using LandmarkId = int;
00068
00072     using LandmarkIds = std::vector<LandmarkId>;
00073
00078     using FaceMap = std::map<FaceParts, LandmarkIds>;
00079
00083     using LandmarkIdPair = std::array<LandmarkId, 2>;
00084
00088     using LandmarkIdPairs = std::vector<LandmarkIdPair>;
00089
00093     using FacePairMap = std::map<FaceParts, LandmarkIdPairs>;
00094 }

```

8.31 landmarks.h File Reference

Provides the base class for the implementation of face landmark extractors.

```

#include "ofiq_lib.h"
#include "Session.h"

```

Classes

- class [OFIQ_LIB::FaceLandmarkExtractorInterface](#)
Implements the base class for the face landmark extractors.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

8.31.1 Detailed Description

Provides the base class for the implementation of face landmark extractors.

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Author

OFIQ development team

8.32 landmarks.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028
00029 #include "ofiq_lib.h"
00030 #include "Session.h"
00031
00035 namespace OFIQ_LIB
00036 {
00041     class FaceLandmarkExtractorInterface
00042     {
00043     public:
00048         virtual ~FaceLandmarkExtractorInterface() = default;
00049
00056         OFIQ::FaceLandmarks extractLandmarks(OFIQ_LIB::Session& session);
00057
00058     protected:
00065         virtual OFIQ::FaceLandmarks updateLandmarks(OFIQ_LIB::Session& session) = 0;
00066     };
00067 }
```

8.33 PartExtractor.h File Reference

Provides helper class for face landmark handling.

```

#include "ofiq_lib.h"
#include "FaceParts.h"
```

Classes

- struct [OFIQ_LIB::modules::landmarks::LandmarkPair](#)
Data container for storing pairs of landmarks.
- class [OFIQ_LIB::modules::landmarks::PartExtractor](#)
Class that provides helper methods for the administration of landmarks.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::landmarks](#)
Provides implementations of a landmark extractors.

8.33.1 Detailed Description

Provides helper class for face landmark handling.

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Author

OFIQ development team

8.34 PartExtractor.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028
00029 #include "ofiq_lib.h"
00030 #include "FaceParts.h"
00031
00035 namespace OFIQ_LIB::modules::landmarks
00036 {
00037
00042     enum class FaceParts;
00043
00048     struct LandmarkPair
00049     {
00054         OFIQ::LandmarkPoint Upper;
00055
00060         OFIQ::LandmarkPoint Lower;
00061
00068         LandmarkPair(OFIQ::LandmarkPoint upper, OFIQ::LandmarkPoint lower) : Upper{upper},
Lower{lower}
00069     {
00070     }
00071 };
00072
00077 class PartExtractor
00078 {
00079 public:
00088     static OFIQ::Landmarks getFacePart(const OFIQ::FaceLandmarks& faceLandmarks, FaceParts part);
00089
00098     static std::vector<LandmarkPair> getPairsForPart(const OFIQ::FaceLandmarks& faceLandmarks,
FaceParts part);
00099 };
00100 }

```

8.35 AllMeasures.h File Reference

Provides all classes derived from the [OFIQ_LIB::modules::measures::Measure](#) class.


```
#include "BackgroundUniformity.h"
#include "CompressionArtifacts.h"
#include "CropOfTheFaceImage.h"
#include "DynamicRange.h"
#include "ExpressionNeutrality.h"
#include "EyesOpen.h"
#include "EyesVisible.h"
#include "FaceOcclusionPrevention.h"
#include "FaceOcclusionSegmentation.h"
#include "FaceParsing.h"
#include "HeadPose.h"
#include "HeadSize.h"
#include "IlluminationUniformity.h"
#include "InterEyeDistance.h"
#include "Luminance.h"
#include "MouthClosed.h"
#include "MouthOcclusionPrevention.h"
#include "NaturalColour.h"
#include "NoHeadCoverings.h"
#include "OverExposurePrevention.h"
#include "Sharpness.h"
#include "SingleFacePresent.h"
#include "UnderExposurePrevention.h"
#include "UnifiedQualityScore.h"
```

8.35.1 Detailed Description

Provides all classes derived from the [OFIQ_LIB::modules::measures::Measure](#) class.

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Author

OFIQ development team

8.36 AllMeasures.h

[Go to the documentation of this file.](#)

```
00001
00029 #include "BackgroundUniformity.h"
00030 #include "CompressionArtifacts.h"
00031 #include "CropOfTheFaceImage.h"
00032 #include "DynamicRange.h"
00033 #include "ExpressionNeutrality.h"
00034 #include "EyesOpen.h"
00035 #include "EyesVisible.h"
00036 #include "FaceOcclusionPrevention.h"
00037 #include "FaceOcclusionSegmentation.h"
00038 #include "FaceParsing.h"
00039 #include "HeadPose.h"
00040 #include "HeadSize.h"
00041 #include "IlluminationUniformity.h"
00042 #include "InterEyeDistance.h"
00043 #include "Luminance.h"
00044 #include "MouthClosed.h"
00045 #include "MouthOcclusionPrevention.h"
00046 #include "NaturalColour.h"
00047 #include "NoHeadCoverings.h"
00048 #include "OverExposurePrevention.h"
00049 #include "Sharpness.h"
00050 #include "SingleFacePresent.h"
00051 #include "UnderExposurePrevention.h"
00052 #include "UnifiedQualityScore.h"
```

8.37 BackgroundUniformity.h File Reference

Provides a class implementing the background uniformity measure.

```
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::BackgroundUniformity](#)
Implementation of the background uniformity measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.37.1 Detailed Description

Provides a class implementing the background uniformity measure.

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Author

OFIQ development team

8.38 BackgroundUniformity.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include "Measure.h"
00031
00035 namespace OFIQ_LIB::modules::measures
00036 {
00043     class BackgroundUniformity : public Measure
00044     {
00045     public:
00050         explicit BackgroundUniformity(
00051             const Configuration& configuration);
00052
00060         void Execute(OFIQ_LIB::Session & session) override;
00061
00062     private:
00067         uint16_t m_targetHeight = 292;
00068
00073         uint16_t m_targetWidth = 354;
00074
00080         uint16_t m_cropLeft = 62;
00081
00087         uint16_t m_cropRight = 62;
00088
00094         uint16_t m_cropTop = 0;
00095
00101         uint16_t m_cropBottom = 210;
00102
00109         uint16_t m_erosionKernelSize = 4;
00110     };
00111 }
```

8.39 CompressionArtifacts.h File Reference

Provides a class implementing the no compression artifact measure.

```
#include "landmarks.h"
#include "Measure.h"
#include <ONNXRTSegmentation.h>
```

Classes

- class [OFIQ_LIB::modules::measures::CompressionArtifacts](#)
Implementation of the no compression artifacts measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.39.1 Detailed Description

Provides a class implemtenting the no compression artifact measure.

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Author

OFIQ development team

8.40 CompressionArtifacts.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include "landmarks.h"
00031 #include "Measure.h"
00032 #include <ONNXRTSegmentation.h>
00033
00037 namespace OFIQ_LIB::modules::measures
00038 {
00045     class CompressionArtifacts : public Measure
00046     {
00047     public:
00068         explicit CompressionArtifacts(const Configuration& configuration);
00069
00078         void Execute(OFIQ_LIB::Session& session) override;
00079
00080     private:
00087         int m_crop;
00088
00096         int m_dim;
00097
00101         ONNXRuntimeEnv m_onnxRuntimeEnv;
00102     };
00103 }
```

8.41 CropOfTheFacelImage.h File Reference

Provides a class implementing the crop of the face image measure.

```
#include "landmarks.h"  
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::CropOfTheFacelImage](#)
Implementation of the crop of the face image measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.41.1 Detailed Description

Provides a class implementing the crop of the face image measure.

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Author

OFIQ development team

8.42 CropOfTheFaceImage.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028
00029 #include "landmarks.h"
00030 #include "Measure.h"
00031
00035 namespace OFIQ_LIB::modules::measures
00036 {
00042     class CropOfTheFaceImage : public Measure
00043     {
00044     public:
00049         explicit CropOfTheFaceImage(const Configuration& configuration);
00050
00057         void Execute(OFIQ_LIB::Session & session) override;
00058     };
00059 }
```

8.43 DynamicRange.h File Reference

Provides a class implementing the dynamic range measure.

```

#include "landmarks.h"
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::DynamicRange](#)
Implementation of the dynamic range measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.43.1 Detailed Description

Provides a class implementing the dynamic range measure.

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Author

OFIQ development team

8.44 DynamicRange.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include "landmarks.h"
00031 #include "Measure.h"
00032
00036 namespace OFIQ_LIB::modules::measures
00037 {
00043     class DynamicRange : public Measure
00044     {
00045     public:
00050         explicit DynamicRange(
00051             const Configuration& configuration);
00052
00058         void Execute(OFIQ_LIB::Session & session) override;
00059     };
00060 }
```

8.45 Executor.h File Reference

This class takes care of the computation of the measures activated.

```
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::Executor](#)

This class takes care of the computation of the measures activated.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

Functions

- void [OFIQ_LIB::modules::measures::log](#) (const std::string_view &msg)
Logging function for writing debug messages to std::cout.

Variables

- static const bool [OFIQ_LIB::modules::measures::ExecutorLogActive](#) = false
This variable enables logging to std::cout for debug purposes. By default the logging is switched off.

8.45.1 Detailed Description

This class takes care of the computation of the measures activated.

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Author

OFIQ development team

8.46 Executor.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028
00029 #include "Measure.h"
00030
00034 namespace OFIQ_LIB::modules::measures
00035 {
00039     const static bool ExecutorLogActive = false;
00040
00046     void log(const std::string_view& msg);
00047
00051     class Executor
00052     {
00053     public:
00059         explicit Executor(std::vector<std::unique_ptr<Measure>> measures)
00060             : m_measures{std::move(measures)}
00061         {
00062         }
00063
00069         void ExecuteAll(Session & i_currentSession) const;
00070
00075         const std::vector<std::unique_ptr<Measure>>& GetMeasures() const { return m_measures; }
00076
00077     private:
00082         std::vector<std::unique_ptr<Measure>> m_measures;
00083     };
00084 }

```

8.47 ExpressionNeutrality.h File Reference

Provides a class implementing the expression neutrality measure.

```

#include "landmarks.h"
#include "Measure.h"
#include <ONNXRTSegmentation.h>

```

Classes

- class [OFIQ_LIB::modules::measures::ExpressionNeutrality](#)
Provides a class implementing the expression neutrality measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.47.1 Detailed Description

Provides a class implementing the expression neutrality measure.

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Author

OFIQ development team

8.48 ExpressionNeutrality.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include "landmarks.h"
00031 #include "Measure.h"
00032 #include <ONNXRTSegmentation.h>
00033
00037 namespace OFIQ_LIB::modules::measures
00038 {
00047     class ExpressionNeutrality : public Measure
00048     {
00049     public:
00054         explicit ExpressionNeutrality(
00055             const Configuration& configuration);
00056
00062         void Execute(OFIQ_LIB::Session& session) override;
00063
00064     private:
00069         ONNXRuntimeEnv m_onnxRuntimeEnvCNN1;
00070
00075         ONNXRuntimeEnv m_onnxRuntimeEnvCNN2;
00076
00081         std::shared_ptr<cv::ml::Boost> m_classifier;
00082     };
00083 }
```

8.49 EyesOpen.h File Reference

Provides a class implementing the eyes open measure.

```
#include "landmarks.h"
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::EyesOpen](#)
Implementation of the eyes open measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.49.1 Detailed Description

Provides a class implementing the eyes open measure.

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Author

OFIQ development team

8.50 EyesOpen.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include "landmarks.h"
00031 #include "Measure.h"
00032
00036 namespace OFIQ_LIB::modules::measures
00037 {
00043     class EyesOpen : public Measure
00044     {
00045     public:
00050         explicit EyesOpen(const Configuration& configuration);
00051
00060         void Execute(OFIQ_LIB::Session & session) override;
00061     };
00062 }
```

8.51 EyesVisible.h File Reference

Provides a class implementing the eyes visible measure.

```
#include "landmarks.h"
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::EyesVisible](#)
Implementation of the eyes visible measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.51.1 Detailed Description

Provides a class implementing the eyes visible measure.

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Author

OFIQ development team

8.52 EyesVisible.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028
00029 #include "landmarks.h"
00030 #include "Measure.h"
00031
00035 namespace OFIQ_LIB::modules::measures
00036 {
00043     class EyesVisible : public Measure
00044     {
00045     public:
00050         explicit EyesVisible(const Configuration& configuration);
00051
00061         void Execute(OFIQ_LIB::Session & session) override;
00062     };
00063 }
```

8.53 FaceOcclusionPrevention.h File Reference

Provides a class implementing the face occlusion prevention measure.

```

#include "landmarks.h"
#include "Measure.h"
#include <ONNXRTSegmentation.h>
```

Classes

- class [OFIQ_LIB::modules::measures::FaceOcclusionPrevention](#)
Implementation of the face occlusion prevention measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.53.1 Detailed Description

Provides a class implementing the face occlusion prevention measure.

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Author

OFIQ development team

8.54 FaceOcclusionPrevention.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028
00029 #include "landmarks.h"
00030 #include "Measure.h"
00031 #include <ONNXRTSegmentation.h>
00032
00036 namespace OFIQ_LIB::modules::measures
00037 {
00044     class FaceOcclusionPrevention : public Measure
00045     {
00046     public:
00051         explicit FaceOcclusionPrevention(
00052             const Configuration& configuration);
00053
00064         void Execute(OFIQ_LIB::Session & session) override;
00065     };
00066 }

```

8.55 HeadPose.h File Reference

Provides a class implementing head pose measures.

```
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::HeadPose](#)
Implementation of head pose measures.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.55.1 Detailed Description

Provides a class implementing head pose measures.

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Author

OFIQ development team

8.56 HeadPose.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "Measure.h"
00030
00034 namespace OFIQ_LIB::modules::measures
00035 {
00040     class HeadPose : public Measure
00041     {
00042     public:
00047         explicit HeadPose(
00048             const Configuration& configuration);
00049
00057         void Execute(OFIQ_LIB::Session & session) override;
00058     };
00059 }
```

8.57 HeadSize.h File Reference

Provides a class implementing the head size measure.

```
#include "landmarks.h"
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::HeadSize](#)
Implementation of the head size measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.57.1 Detailed Description

Provides a class implementing the head size measure.

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Author

OFIQ development team

8.58 HeadSize.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "landmarks.h"
00030 #include "Measure.h"
00031
00035 namespace OFIQ_LIB::modules::measures
00036 {
00042     class HeadSize : public Measure
00043     {
00044     public:
00049         explicit HeadSize(
00050             const Configuration& configuration);
00051
00057         void Execute(OFIQ_LIB::Session & session) override;
00058     };
00059 }
```

8.59 IlluminationUniformity.h File Reference

Provides a class implementing the illumination uniformity measure.

```
#include "landmarks.h"
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::IlluminationUniformity](#)
Implementation of the illumination uniformity measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.59.1 Detailed Description

Provides a class implementing the illumination uniformity measure.

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Author

OFIQ development team

8.60 IlluminationUniformity.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include "landmarks.h"
00031 #include "Measure.h"
00032
00036 namespace OFIQ_LIB::modules::measures
00037 {
00044     class IlluminationUniformity : public Measure
00045     {
00046     public:
00051         explicit IlluminationUniformity(const Configuration& configuration);
00052
00061         void Execute(OFIQ_LIB::Session & session) override;
00062     };
00063 }
```

8.61 InterEyeDistance.h File Reference

Provides a class implementing the inter-eye distance measure.

```
#include "landmarks.h"
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::InterEyeDistance](#)
Implementation of the inter-eye distance measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.61.1 Detailed Description

Provides a class implementing the inter-eye distance measure.

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Author

OFIQ development team

8.62 InterEyeDistance.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "landmarks.h"
00030 #include "Measure.h"
00031
00035 namespace OFIQ_LIB::modules::measures
00036 {
00043     class InterEyeDistance : public Measure
00044     {
00045     public:
00050         explicit InterEyeDistance(const Configuration& configuration);
00051
00060         void Execute(OFIQ_LIB::Session & session) override;
00061     };
00062 }
```

8.63 Luminance.h File Reference

Provides a class implementing two luminance measures.

```
#include "landmarks.h"
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::Luminance](#)
Implementation of two luminance measures.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.63.1 Detailed Description

Provides a class implementing two luminance measures.

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Author

OFIQ development team

8.64 Luminance.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028
00029 #include "landmarks.h"
00030 #include "Measure.h"
00031
00035 namespace OFIQ_LIB::modules::measures
00036 {
00042     class Luminance : public Measure
00043     {
00044     public:
00049         explicit Luminance(const Configuration& configuration);
00050
00058         void Execute(OFIQ_LIB::Session & session) override;
00059     };
00060 }
```

8.65 Measure.h File Reference

Provides the base class for all measures implemented in OFIQ.

```

#include "Configuration.h"
#include "ofiq_lib.h"
#include "Session.h"
#include <math.h>
```

Classes

- struct [OFIQ_LIB::modules::measures::SigmoidParameters](#)
Parameters of the sigmoid function based quality mapping.
- class [OFIQ_LIB::modules::measures::Measure](#)
Base class for measures implemented in OFIQ.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.65.1 Detailed Description

Provides the base class for all measures implemented in OFIQ.

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Author

OFIQ development team

8.66 Measure.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "Configuration.h"
00030 #include "ofiq_lib.h"
00031 #include "Session.h"
00032 #ifndef _WIN32
00033 #    include <math.h>
00034 #endif
00035
00039 namespace OFIQ_LIB::modules::measures
00040 {
00056     struct SigmoidParameters
00057     {
00063         SigmoidParameters() { Reset(); }
00064
00069         double h;
00070
00075         double a;
00076
00081         double s;
00082
00088         double x0;
00089
00095         double w;
00096
00102         bool round;
00103
00112         void setInverse()
00113         {
00114             a = 1;
00115             s = -1;
00116         }
00117
00121         void Reset()
00122         {
00123             h = 100;
00124             a = 0;
00125             s = 1;
00126             x0 = 4;
00127             w = 0.7;
00128             round = true;
00129         }
00130     };
00131 }
```

```

00135     class Measure
00136     {
00137     public:
00143         Measure(const Configuration& configuration,
00144                 OFIQ::QualityMeasure measure)
00145             : configuration{configuration}, m_measure(measure)
00146         {
00147         }
00148
00158         virtual void Execute(OFIQ_LIB::Session& session) = 0;
00159
00163         virtual ~Measure() = default;
00164
00172         virtual std::string GetName() const;
00173
00178         virtual OFIQ::QualityMeasure GetQualityMeasure() const;
00179
00193         void SetQualityMeasure(OFIQ_LIB::Session& session, OFIQ::QualityMeasure measure, double
rawValue, OFIQ::QualityMeasureReturnCode code);
00194
00195     protected:
00203         static double Sigmoid(double x, double x0, double w)
00204         {
00205             return 1.0 / (1 + exp((x0 - x) / w));
00206         }
00207
00220         void AddSigmoid(OFIQ::QualityMeasure measure, const SigmoidParameters& defaultValues);
00221
00234         void AddSigmoid(const std::string& key, SigmoidParameters defaultValues);
00235
00243         double ExecuteScalarConversion(OFIQ::QualityMeasure measure, double rawValue);
00244
00252         double ExecuteScalarConversion(const std::string& key, double rawValue);
00253
00258         const Configuration& configuration;
00259
00260     private:
00271         static double ScalarConversion(double rawValue, const SigmoidParameters& par)
00272         {
00273             double scalarScore = par.h * (par.a + par.s * Sigmoid(rawValue, par.x0, par.w));
00274             if (par.round)
00275                 scalarScore = round(scalarScore);
00276             if (scalarScore < 0.0)
00277             {
00278                 scalarScore = 0.0;
00279             }
00280             else if (scalarScore > 100.0)
00281             {
00282                 scalarScore = 100.0;
00283             }
00284             return scalarScore;
00285         }
00286
00291         std::map<std::string, SigmoidParameters, std::less<> m_sigmoidMap;
00292
00298         static std::string GetMeasureName(OFIQ::QualityMeasure measure);
00299
00307         static std::string ExpandKey(std::string_view rawKey);
00308
00314         OFIQ::QualityMeasure m_measure = OFIQ::QualityMeasure::NotSet;
00315     };
00316 }

```

8.67 MeasureFactory.h File Reference

Provides a class for requesting creation of measure implementations.

```

#include "Configuration.h"
#include "ofiq_lib.h"
#include "Measure.h"
#include "Session.h"

```

Classes

- class [OFIQ_LIB::modules::measures::MeasureFactory](#)
Measure factor class.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.67.1 Detailed Description

Provides a class for requesting creation of measure implementations.

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Author

OFIQ development team

8.68 MeasureFactory.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include "Configuration.h"
00031 #include "ofiq_lib.h"
00032 #include "Measure.h"
00033 #include "Session.h"
00034
00038 namespace OFIQ_LIB::modules::measures
00039 {
00043     class MeasureFactory
00044     {
00045     public:
00046         // Avoids instantiation from this class
00047         MeasureFactory() = delete;
00048
00059         static std::unique_ptr<Measure> CreateMeasure(
00060             const OFIQ::QualityMeasure measure,
00061             const Configuration& configuration);
00062     };
00063 }
```


8.69 MouthClosed.h File Reference

Provides a class implementing the mouth closed measure.

```
#include "landmarks.h"
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::MouthClosed](#)
Implementation of the mouth closed measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.69.1 Detailed Description

Provides a class implementing the mouth closed measure.

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Author

OFIQ development team

8.70 MouthClosed.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "landmarks.h"
00030 #include "Measure.h"
00031
00035 namespace OFIQ_LIB::modules::measures
00036 {
00042     class MouthClosed : public Measure
00043     {
00044     public:
00049         explicit MouthClosed(const Configuration& configuration);
00050
00059         void Execute(OFIQ_LIB::Session& session) override;
00060     };
00061 }
```

8.71 MouthOcclusionPrevention.h File Reference

Provides a class implementing the mouth occlusion prevention measure.

```
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::MouthOcclusionPrevention](#)
Implementation of the mouth occlusion prevention measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.71.1 Detailed Description

Provides a class implementing the mouth occlusion prevention measure.

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Author

OFIQ development team

8.72 MouthOcclusionPrevention.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "Measure.h"
00030
00034 namespace OFIQ_LIB::modules::measures
00035 {
00042     class MouthOcclusionPrevention : public Measure
00043     {
00044     public:
00049         explicit MouthOcclusionPrevention(const Configuration& configuration);
00050
00062         void Execute(OFIQ_LIB::Session & session) override;
00063     };
00064 }
```

8.73 NaturalColour.h File Reference

Provides a class implementing the natural colour measure.

```
#include "landmarks.h"
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::NaturalColour](#)
Implementation of the natural colour measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.73.1 Detailed Description

Provides a class implementing the natural colour measure.

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Author

OFIQ development team

8.74 NaturalColour.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "landmarks.h"
00030 #include "Measure.h"
00031
00035 namespace OFIQ_LIB::modules::measures
00036 {
00043     class NaturalColour : public Measure
00044     {
00045     public:
00050         explicit NaturalColour(const Configuration& configuration);
00051
00060         void Execute(OFIQ_LIB::Session & session) override;
00061
00062     private:
00069         cv::Mat CreateMaskedImage(const OFIQ::FaceLandmarks& landmarks, const cv::Mat& cvImage) const;
00070
00081         cv::Mat ReduceImageToRegionOfInterest(
00082             const cv::Mat& maskedImage,
00083             const cv::Rect& leftRegionOfInterest,
00084             const cv::Rect& rightRegionOfInterest) const;
00085
00099         double CalculateScore(double meanChannelA, double meanChannelB) const;
00100     };
00101 }
```

8.75 NoHeadCoverings.h File Reference

Provides a class implementing the no head covering measure.

```
#include "Measure.h"
#include "segmentations.h"
```

Classes

- class [OFIQ_LIB::modules::measures::NoHeadCoverings](#)
Implementation of the no head covering measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.75.1 Detailed Description

Provides a class implementing the no head covering measure.

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Author

OFIQ development team

8.76 NoHeadCoverings.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028
00029 #include "Measure.h"
00030 #include "segmentations.h"
00031
00035 namespace OFIQ_LIB::modules::measures
00036 {
00051     class NoHeadCoverings : public Measure
00052     {
00053     public:
00062         explicit NoHeadCoverings(const Configuration& configuration);
00063
00080         void Execute(OFIQ_LIB::Session & session) override;
00081
00082     private:
00089         double m_t0;
00090
00097         double m_tl;
00098
00105         double m_w;
00106
00113         double m_x0;
00114     };
00115 }
```

8.77 OverExposurePrevention.h File Reference

Provides a class implementing the background uniformity measure.

```
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::OverExposurePrevention](#)
Implementation of the over-exposure prevention measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.77.1 Detailed Description

Provides a class implementing the background uniformity measure.

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Author

OFIQ development team

8.78 OverExposurePrevention.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "Measure.h"
00030
00031
00035 namespace OFIQ_LIB::modules::measures
00036 {
00042     class OverExposurePrevention : public Measure
00043     {
00044     public:
00049         explicit OverExposurePrevention(const Configuration& configuration);
00050
00056         void Execute(OFIQ_LIB::Session & session) override;
00057     };
00058 }
```

8.79 Sharpness.h File Reference

Provides a class implementing the sharpness measure.

```
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::Sharpness](#)
Implementation of the sharpness measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.79.1 Detailed Description

Provides a class implementing the sharpness measure.

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Author

OFIQ development team

8.80 Sharpness.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "Measure.h"
00030
00034 namespace OFIQ_LIB::modules::measures
00035 {
00043     class Sharpness : public Measure
00044     {
00045     public:
00050         explicit Sharpness(const Configuration& configuration);
00051
00057         void Execute(OFIQ_LIB::Session & session) override;
00058
00059     private:
00060
00064         std::string m_modelFile;
00065
00070         std::shared_ptr<cv::ml::RTrees> m_rtree;
00071
00077         bool m_useAligned;
00078
```



```

00084         double m_faceRegionAlpha;
00085
00090         int m_numTrees;
00091
00101         void GetCroppedImages(
00102             const Session& session,
00103             cv::Mat& faceCrop,
00104             cv::Mat& maskCrop,
00105             bool useAligned,
00106             float faceRegionAlpha) const;
00107
00116         cv::Mat GetClassifierFocusFeatures(const cv::Mat& image, const cv::Mat& mask, bool applyBlur)
00117             const;
00117     };
00118 }

```

8.81 SingleFacePresent.h File Reference

Provides a class implementing the single face present measure.

```

#include "detectors.h"
#include "Measure.h"

```

Classes

- class [OFIQ_LIB::modules::measures::SingleFacePresent](#)
Implementation of the single face present measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.81.1 Detailed Description

Provides a class implementing the single face present measure.

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Author

OFIQ development team

8.82 SingleFacePresent.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "detectors.h"
00030 #include "Measure.h"
00031
00035 namespace OFIQ_LIB::modules::measures
00036 {
00041     class SingleFacePresent : public Measure
00042     {
00043     public:
00048         explicit SingleFacePresent(const Configuration& configuration);
00049
00056         void Execute(OFIQ_LIB::Session & session) override;
00057     };
00058 }
```

8.83 UnderExposurePrevention.h File Reference

Provides a class implementing the under-exposure prevention measure.

```
#include "Measure.h"
```

Classes

- class [OFIQ_LIB::modules::measures::UnderExposurePrevention](#)
Implementation of the under-exposure prevention measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.83.1 Detailed Description

Provides a class implementing the under-exposure prevention measure.

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Author

OFIQ development team

8.84 UnderExposurePrevention.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "Measure.h"
00030
00034 namespace OFIQ_LIB::modules::measures
00035 {
00041     class UnderExposurePrevention : public Measure
00042     {
00043     public:
00048         explicit UnderExposurePrevention(const Configuration& configuration);
00049
00055         void Execute(OFIQ_LIB::Session & session) override;
00056     };
00057 }
```

8.85 UnifiedQualityScore.h File Reference

Provides a class implementing the unified quality measure.

```
#include "landmarks.h"
#include "Measure.h"
#include <opencv2/dnn.hpp>
#include <ONNXRTSegmentation.h>
```

Classes

- class [OFIQ_LIB::modules::measures::UnifiedQualityScore](#)
Implementation of the unified quality measure.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::measures](#)
Provides measures implemented in OFIQ.

8.85.1 Detailed Description

Provides a class implementing the unified quality measure.

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Author

OFIQ development team

8.86 UnifiedQualityScore.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028
00029 #include "landmarks.h"
00030 #include "Measure.h"
00031 #include <opencv2/dnn.hpp>
00032 #include <ONNXRTSegmentation.h>
00033
00037 namespace OFIQ_LIB::modules::measures
00038 {
00046     class UnifiedQualityScore : public Measure
00047     {
00048     public:
00053         explicit UnifiedQualityScore(const Configuration& configuration);
00054
00064         void Execute(OFIQ_LIB::Session & session) override;
00065
00066     private:
00071         ONNXRuntimeEnv m_onnxRuntimeEnv;
00072     };
00073 }
```

8.87 AllPoseEstimators.h File Reference

Provides the include statements to all classes derived from [PoseEstimatorInterface](#).

```
#include "HeadPose3DDFAV2.h"
```

8.87.1 Detailed Description

Provides the include statements to all classes derived from [PoseEstimatorInterface](#).

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Author

OFIQ development team

8.88 AllPoseEstimators.h

[Go to the documentation of this file.](#)

```
00001
00029 #include "HeadPose3DDFAV2.h"
```

8.89 HeadPose3DDFAV2.h File Reference

Provides a class implementing a head pose estimator based on https://github.com/cleardusk/3DDFA_V2.

```
#include "Configuration.h"
#include "poseEstimators.h"
#include <onnxruntime_cxx_api.h>
#include <opencv2/core/mat.hpp>
```

Classes

- class [OFIQ_LIB::modules::poseEstimators::HeadPose3DDFAV2](#)
Implementation of a head pose estimator.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::poseEstimators](#)
Provides implementation of a head pose estimator.

8.89.1 Detailed Description

Provides a class implementing a head pose estimator based on https://github.com/cleardusk/3DDFA_V2.

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Author

OFIQ development team

8.90 HeadPose3DDFAV2.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028
00029 #include "Configuration.h"
00030 #include "poseEstimators.h"
00031 #include <onnxruntime_cxx_api.h>
00032 #include <opencv2/core/mat.hpp>
00033
00038 namespace OFIQ_LIB::modules::poseEstimators
00039 {
00044     class HeadPose3DDFAV2 : public PoseEstimatorInterface
00045     {
00046     public:
00053         explicit HeadPose3DDFAV2(const Configuration& config);
00054
00058         ~HeadPose3DDFAV2() override = default;
00059
00060     protected:
00068         void updatePose(OFIQ_LIB::Session& session, EulerAngle& pose) override;
00069
00070     private:
00074         static const std::string m_paramPoseEstimatorModel;
00075
00079         Ort::Env m_ortenv;
00080
00084         std::unique_ptr<Ort::Session> m_ortSession;
00085
00089         int64_t m_expectedImageWidth = 0;
00090
00094         int64_t m_expectedImageHeight = 0;
00095
00099         int64_t m_expectedImageNumberOfChannels = 0;
00100
00104         int64_t m_numberOfInputElements = 0;
00105
00109         std::array<int64_t, 4> m_inputShape;
00110
00118         cv::Mat CropImage(const cv::Mat& image, const OFIQ::BoundingBox& biggestFace) const;
00119     };
00120 }
```

8.91 poseEstimators.h File Reference

Base class for the different implementation of pose estimation algorithms.

```

#include "ofiq_lib.h"
#include "Session.h"
#include <array>
```

Classes

- class [OFIQ_LIB::PoseEstimatorInterface](#)

Implementation of the base class for integrating pose estimation algorithms capable of estimating three head orientation angles (yaw, pitch and roll) from a face image.

Namespaces

- namespace [OFIQ_LIB](#)

Namespace for OFIQ implementations.

8.91.1 Detailed Description

Base class for the different implementation of pose estimation algorithms.

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Author

OFIQ development team

8.92 poseEstimators.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "ofiq_lib.h"
00030 #include "Session.h"
00031 #include <array>
00032
00036 namespace OFIQ_LIB
00037 {
00038
00043     class PoseEstimatorInterface
00044     {
00045     public:
00049         using EulerAngle = std::array<double, 3>;
00050
00055         virtual ~PoseEstimatorInterface() = default;
00056
00064         EulerAngle& estimatePose(OFIQ_LIB::Session& session);
00065
00066     protected:
00073         virtual void updatePose(OFIQ_LIB::Session& session, EulerAngle& pose) = 0;
00074
00075     private:
00080         std::string m_lastSessionId;
00081
00086         EulerAngle m_pose;
00087     };
00088 }
```


8.93 FaceOcclusionSegmentation.h File Reference

Provides a class for segmenting the facial part not occluded by any non-facial parts from an image.

```
#include "Configuration.h"
#include "segmentations.h"
#include <ONNXRTSegmentation.h>
```

Classes

- class [OFIQ_LIB::modules::segmentations::FaceOcclusionSegmentation](#)
Class managing the separation of facial parts not occluded by non-facial parts from other parts.

Namespaces

- namespace [cv](#)
OpenCV's namespace.
- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::segmentations](#)
Provides segmentation-related implementations.

8.93.1 Detailed Description

Provides a class for segmenting the facial part not occluded by any non-facial parts from an image.

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Author

OFIQ development team

8.94 FaceOcclusionSegmentation.h

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```

00001
00028 #pragma once
00029
00030 #include "Configuration.h"
00031 #include "segmentations.h"
00032 #include <ONNXRTSegmentation.h>
00033
00037 namespace cv
00038 {
00042     class Mat;
00043 }
00044
00048 namespace OFIQ_LIB::modules::segmentations
00049 {
00056     class FaceOcclusionSegmentation : public SegmentationExtractorInterface
00057     {
00058     public:
00065         explicit FaceOcclusionSegmentation(const Configuration& config);
00070         ~FaceOcclusionSegmentation() override = default;
00071
00072     protected:
00073         OFIQ::Image UpdateMask(
00093             OFIQ_LIB::Session& session, modules::segmentations::SegmentClassLabels faceSegment)
00094         override;
00095
00096     private:
00097
00105         cv::Mat GetFaceOcclusionSegmentation(const cv::Mat& alignedImage);
00106
00110         ONNXRuntimeEnv m_onnxRuntimeEnv;
00111
00117         std::shared_ptr<cv::Mat> m_segmentationImage;
00118
00123         const std::string m_modelConfigItem = "params.measures.FaceOcclusionSegmentation.model_path";
00124
00128         const int m_cropLeft = 96;
00129
00133         const int m_cropRight = 96;
00134
00138         const int m_cropTop = 96;
00139
00143         const int m_cropBottom = 96;
00144
00150         const int m_scaledWidth = 224;
00151
00157         const int m_scaledHeight = 224;
00158
00159     };
00160 }

```

8.95 FaceParsing.h File Reference

Provides a class implementing the face parsing pre-processing.

```

#include "Configuration.h"
#include "segmentations.h"
#include <ONNXRTSegmentation.h>

```

Classes

- class [OFIQ_LIB::modules::segmentations::FaceParsing](#)
Class managing the separation of facial parts not occluded by non-facial parts from other parts.

Namespaces

- namespace `cv`
OpenCV's namespace.
- namespace `OFIQ_LIB`
Namespace for OFIQ implementations.
- namespace `OFIQ_LIB::modules`
- namespace `OFIQ_LIB::modules::segmentations`
Provides segmentation-related implementations.

8.95.1 Detailed Description

Provides a class implementing the face parsing pre-processing.

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Author

OFIQ development team

8.96 FaceParsing.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include "Configuration.h"
00031 #include "segmentations.h"
00032
00033 #include <ONNXRTSegmentation.h>
00034
00038 namespace cv
00039 {
00043     class Mat;
00044 }
00045
00049 namespace OFIQ_LIB::modules::segmentations
00050 {
00083     class FaceParsing : public SegmentationExtractorInterface
00084     {
00085     public:
```

```

00092         explicit FaceParsing(const Configuration& config);
00093
00097         ~FaceParsing() override = default;
00098
00099     protected:
00100         OFIQ::Image UpdateMask(
00126             OFIQ_LIB::Session& session, modules::segmentations::SegmentClassLabels faceSegment)
00127         override;
00128
00129     private:
00130
00134         ONNXRuntimeEnv m_onnxRuntimeEnv;
00135
00141         std::shared_ptr<cv::Mat> m_segmentationImage;
00142
00148         const std::string m_modelConfigItem = "params.measures.FaceParsing.model_path";
00149
00153         const int m_imageSize = 400;
00154
00158         const int m_cropLeft = 30;
00159
00163         const int m_cropRight = 30;
00164
00168         const int m_cropBottom = 60;
00169
00178         //static cv::Mat CreateBlob(const cv::Mat& image, int i_imageSize_one_dim);
00179
00191         static std::shared_ptr<cv::Mat> CalculateClassIds(
00192             const cv::Mat& faceSegments,
00193             int imgSize);
00194
00195         /**
00196          * @brief Derives the private member \link segmentationImage\endlink
00197          * from the facial image data provided by the session object.
00198          * @details Implements CNN processing step of \link
00199 OFIQ_LIB::modules::segmentations::FaceParsing::UpdateMask()
00199          * UpdateMask()\endlink.
00200          * @param session Session object containing the original facial image and pre-processing
00201 results
00201          * computed by the \link OFIQ_LIB::OFIQImpl::performPreprocessing()
00202          * OFIQImpl::performPreprocessing()\endlink method.
00203          */
00204         void SetImage(const OFIQ_LIB::Session& session);
00205     };
00206 }

```

8.97 ONNXRTSegmentation.h File Reference

Helper class to manage the ONNXRuntime session object.

```

#include <vector>
#include <opencv2/opencv.hpp>
#include <onnxruntime_cxx_api.h>

```

Classes

- class [ONNXRuntimeEnv](#)
Helper class to manage the ONNXRuntime session object.

8.97.1 Detailed Description

Helper class to manage the ONNXRuntime session object.

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Author

OFIQ development team

8.98 ONNXRTSegmentation.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include <vector>
00030
00031 #include <opencv2/opencv.hpp>
00032 #include <onnxruntime_cxx_api.h>
00033
00039 class ONNXRuntimeEnv
00040 {
00041 private:
00042
00047     Ort::Env m_ortenv;
00048
00053     Ort::MemoryInfo m_memoryInfo = Ort::MemoryInfo::CreateCpu(OrtDeviceAllocator, OrtMemTypeCPU);
00054
00059     std::array<int64_t, 4> m_inputShape;
00060
00065     std::unique_ptr<Ort::Session> m_ortSession;
00066
00074     void init_session(const std::vector<uint8_t>& i_model_data, int64_t i_imageWidth, int64_t
i_imageHeight);
00075
00076
00077 public:
00082     ONNXRuntimeEnv() = default;
00083
00088     ~ONNXRuntimeEnv() = default;
00089
00097     void initialize(
00098         const std::vector<uint8_t>& i_modelData, int64_t i_imageWidth, int64_t i_imageHeight);
00099
00105     size_t getNumberOfOutputNodes() const;
00106
00113     std::vector<Ort::Value> run( std::vector<float>& i_netInput);
00114
00115 };
```

8.99 segmentations.h File Reference

Base class for the different implementation of segmentation algorithms.

```
#include "ofiq_lib.h"
#include "Session.h"
```

Classes

- class [OFIQ_LIB::SegmentationExtractorInterface](#)
Base class for the different implementation of segmentation algorithms.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.
- namespace [OFIQ_LIB::modules](#)
- namespace [OFIQ_LIB::modules::segmentations](#)
Provides segmentation-related implementations.

Enumerations

- enum class [OFIQ_LIB::modules::segmentations::SegmentClassLabels](#) {
[OFIQ_LIB::modules::segmentations::background](#) , [OFIQ_LIB::modules::segmentations::skin](#) , [OFIQ_LIB::modules::segmentations::r_brow](#) ,
[OFIQ_LIB::modules::segmentations::l_eye](#) , [OFIQ_LIB::modules::segmentations::r_eye](#) , [OFIQ_LIB::modules::segmentations::l_ear](#) ,
[OFIQ_LIB::modules::segmentations::r_ear](#) , [OFIQ_LIB::modules::segmentations::ear_r](#) , [OFIQ_LIB::modules::segmentations::mouth](#) ,
[OFIQ_LIB::modules::segmentations::u_lip](#) , [OFIQ_LIB::modules::segmentations::l_lip](#) , [OFIQ_LIB::modules::segmentations::neck_l](#) ,
[OFIQ_LIB::modules::segmentations::neck_r](#) , [OFIQ_LIB::modules::segmentations::cloth](#) , [OFIQ_LIB::modules::segmentations::hair](#) , [OFIQ_LIB::modules::segmentations::face](#) }
Enum class of the different face regions that can be segmented.

8.99.1 Detailed Description

Base class for the different implementation of segmentation algorithms.

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Author

OFIQ development team

8.100 segmentations.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028
00029 #include "ofiq_lib.h"
00030 #include "Session.h"
00031
00035 namespace OFIQ_LIB
00036 {
00040     namespace modules::segmentations
00041     {
00045         enum class SegmentClassLabels
00046         {
00050             background,
00054             skin,
00058             l_brow,
00062             r_brow,
00066             l_eye,
00070             r_eye,
00074             eye_g,
00078             l_ear,
00082             r_ear,
00086             ear_r,
00090             nose,
00094             mouth,
00098             u_lip,
00102             l_lip,
00106             neck,
00110             neck_l,
00114             cloth,
00118             hair,
00122             hat,
00126             face
00127         };
00128     }
00129
00137     class SegmentationExtractorInterface
00138     {
00139     public:
00144         virtual ~SegmentationExtractorInterface() = default;
00145
00153         OFIQ::Image& GetMask(
00154             OFIQ_LIB::Session& session, modules::segmentations::SegmentClassLabels faceSegment);
00155
00156     protected:
00157
00165         virtual OFIQ::Image UpdateMask(
00166             OFIQ_LIB::Session& session,
00167             modules::segmentations::SegmentClassLabels faceSegment) = 0;
00168
00173         std::string GetLastSessionId() const { return m_lastSessionId; };
00174
00175     private:
00180         std::string m_lastSessionId;
00185         std::map<modules::segmentations::SegmentClassLabels, OFIQ::Image> m_masks;
00186     };
00187 }

```

8.101 Configuration.h File Reference

Provides a configuration class for handling configurations.

```

#include <map>
#include <string>
#include <filesystem>
#include <tao/json/forward.hpp>
#include <tao/json/value.hpp>

```

Classes

- class [OFIQ_LIB::Configuration](#)
Configuration class.

Namespaces

- namespace [OFIQ_LIB](#)

Namespace for OFIQ implementations.

8.101.1 Detailed Description

Provides a configuration class for handling configurations.

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Author

OFIQ development team

8.102 Configuration.h

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```
00001
00029 #pragma once
00030
00031 #include <map>
00032 #include <string>
00033 #include <filesystem>
00034
00035 #include <tao/json/forward.hpp>
00036 #include <tao/json/value.hpp>
00037
00041 namespace OFIQ_LIB
00042 {
00049     class Configuration
00050     {
00051     public:
00058         Configuration(const std::string& configDir, const std::string& configFilename);
00059
00068         bool GetBool(const std::string& key, bool& value) const;
00069
00078         bool GetString(const std::string& key, std::string& value) const;
00079
00089         bool GetNumber(const std::string& key, double& value) const;
00090
00102         bool GetStringList(const std::string& key, std::vector<std::string>& value) const;
00103
00110         bool GetBool(const std::string& key) const;
00111
```



```
00118         std::string GetString(const std::string& key) const;
00119
00126         double GetNumber(const std::string& key) const;
00127
00135         std::string getDataDir() const;
00136
00143         void setDataDir(std::string_view dataDir);
00144
00151         std::string GetFullPath(const std::string& key) const;
00152
00153     private:
00157         std::map<std::string, tao::json::value, std::less<> > parameters;
00158
00164         std::filesystem::path m_dataDir;
00165     };
00166 }
```

8.103 DataStream.h File Reference

Platform-specific file interface selector for loading models as streams.

8.103.1 Detailed Description

Platform-specific file interface selector for loading models as streams.

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Author

OFIQ development team

8.104 DataStream.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #if defined(ANDROID)
00031     #include "DataStreamAndroid.h"
00032 #elif defined(_WIN32)
00033     #include "DataStreamDesktop.h"
00034 #elif defined(__linux__)
00035     #include "DataStreamDesktop.h"
00036 #elif defined(__APPLE__)
00037     #include <TargetConditionals.h>
00038     #if TARGET_OS_IPHONE
00039         #include "DataStreamIOS.h"
00040     #else
00041         #include "DataStreamDesktop.h"
00042     #endif
00043 #else
00044     #error "Undefined operating system"
00045 #endif
```

8.105 DataStreamAndroid.h File Reference

Provides the Android-specific file interface implementation for streaming model data.

```
#include <istream>
#include <streambuf>
#include <ios>
```

Classes

- class [OFIQ_LIB::asset_streambuf](#)
- class [OFIQ_LIB::asset_istream](#)

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

Typedefs

- using [OFIQ_LIB::DataStream](#) = [asset_istream](#)

Functions

- void [OFIQ_LIB::SetAAssetManager](#) (::AAssetManager *assetManager)

8.105.1 Detailed Description

Provides the Android-specific file interface implementation for streaming model data.

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Author

OFIQ development team

8.106 DataStreamAndroid.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include <istream>
00031 #include <streambuf>
00032 #include <ios>
00033
00034 struct AAsset;
00035 struct AAssetDir;
00036 struct AAssetManager;
00037
00038 namespace OFIQ_LIB
00039 {
00040     void SetAAssetManager(::AAssetManager* assetManager);
00041
00042     class asset_streambuf: public std::streambuf
00043     {
00044     public:
00045         asset_streambuf(::AAsset* asset);
00046         virtual ~asset_streambuf() = default;
00047     protected:
00048         virtual pos_type seekoff(off_type off, std::ios_base::seekdir dir, std::ios_base::openmode
which) override;
00049         virtual pos_type seekpos(pos_type pos, std::ios_base::openmode which) override;
00050     };
00051
00052     class asset_istream: public std::istream
00053     {
00054     public:
00055         asset_istream(std::string const& filePath, std::ios_base::openmode mode = std::ios_base::in);
00056         virtual ~asset_istream();
00057     private:
00058         ::AAsset* asset;
00059         asset_streambuf streambuf;
00060     };
00061
00062     using DataStream = asset_istream;
00063 }
```

8.107 DataStreamDesktop.h File Reference

Provides the Desktop-specific file interface implementation for streaming model data.

```
#include <fstream>
```

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

8.107.1 Detailed Description

Provides the Desktop-specific file interface implementation for streaming model data.

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Author

OFIQ development team

8.108 DataStreamDesktop.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include <fstream>
00031
00032 namespace OFIQ_LIB
00033 {
00034     using DataStream = std::ifstream;
00035 }
```

8.109 DataStreamIOS.h File Reference

Provides the iOS-specific file interface implementation for streaming model data.

```
#include <fstream>
```

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

8.109.1 Detailed Description

Provides the iOS-specific file interface implementation for streaming model data.

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Author

OFIQ development team

8.110 DataStreamIOS.h

[Go to the documentation of this file.](#)

```
00001
00028 #pragma once
00029
00030 #include <fstream>
00031
00032 namespace OFIQ_LIB
00033 {
00034     using DataStream = std::ifstream;
00035 }
```

8.111 image_io.h File Reference

Provides helper functions for reading/writing images from/to disk.

```
#include "ofiq_lib.h"
```

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

Functions

- [OFIQ_EXPORT OFIQ::ReturnStatus OFIQ_LIB::readImage](#) (const std::string &filename, [OFIQ::Image](#) &image)
Read image from disk.
- [OFIQ_EXPORT OFIQ::ReturnStatus OFIQ_LIB::readImageFromByteArray](#) (const std::vector< unsigned char > &buffer, [OFIQ::Image](#) &image)
Read image from byte array.

8.111.1 Detailed Description

Provides helper functions for reading/writing images from/to disk.

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Author

OFIQ development team

8.112 image_io.h

[Go to the documentation of this file.](#)

```
00001
00027 #ifndef OFIQ_LIB_IMAGE_IO_H
00028 #define OFIQ_LIB_IMAGE_IO_H
00029
00030 #include "ofiq_lib.h"
00031
00032 namespace OFIQ_LIB {
00033
00044     OFIQ_EXPORT OFIQ::ReturnStatus
00045         readImage(const std::string& filename, OFIQ::Image& image);
00046
00054     OFIQ_EXPORT OFIQ::ReturnStatus
00055         readImageFromByteArray(const std::vector<unsigned char>& buffer, OFIQ::Image& image);
00056 }
00057
00058 #endif
```

8.113 image_utils.h File Reference

Provides image utility functions such as color conversion, luminance computation etc.

```
#include "ofiq_lib.h"
#include "Session.h"
#include <opencv2/imgcodecs.hpp>
#include <opencv2/imgproc.hpp>
```

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

Typedefs

- using [OFIQ_LIB::ExposureRange](#) = std::array<int, 2>

Functions

- double [OFIQ_LIB::ColorConvert](#) (double v)
Converts a color as specified in ISO/IEC 29794-5.
- double [OFIQ_LIB::Cubic](#) (double x, double k, double eps)
Cubic flattening function.
- void [OFIQ_LIB::ConvertBGRToCIELAB](#) (const cv::Mat &bgrImage, double &a, double &b)
Computes CIELAB values a^ and b^* from a BGR image.*
- cv::Mat [OFIQ_LIB::GetLuminanceImageFromBGR](#) (const cv::Mat &bgrImage)
Converts a BGR image to the luminance image.
- void [OFIQ_LIB::CalculateReferencePoints](#) (const [OFIQ::FaceLandmarks](#) &landmarks, [OFIQ::LandmarkPoint](#) &leftEyeCenter, [OFIQ::LandmarkPoint](#) &rightEyeCenter, double &interEyeDistance, double &eyeMouthDistance)
Computes the left eye center, the right eye center, the (planar) inter-eye-distance and the eye to mouth distance from facial landmarks.

- void [OFIQ_LIB::CalculateRegionOfInterest](#) (cv::Rect &leftRegionOfInterest, cv::Rect &rightRegionOfInterest, const [OFIQ::LandmarkPoint](#) &leftEyeCenter, const [OFIQ::LandmarkPoint](#) &rightEyeCenter, const double interEyeDistance, const double eyeMouthDistance)

Extracts regions being of interest for some measures (e.g. NaturalColour).

- void [OFIQ_LIB::GetNormalizedHistogram](#) (const cv::Mat &luminanceImage, const cv::Mat &maskImage, cv↵::Mat1f &histogram)

Computes the normalized histogram from a luminance image in 256 chunks.

- double [OFIQ_LIB::CalculateExposure](#) (const [Session](#) &session, const [ExposureRange](#) &exposureRange)

Helper function for some measures.

- double [OFIQ_LIB::ComputeBrightnessAspect](#) (const cv::Mat &luminanceImage, const cv::Mat &maskImage, const [ExposureRange](#) &exposureRange)

Helper function for some measures.

8.113.1 Detailed Description

Provides image utility functions such as color conversion, luminance computation etc.

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Author

OFIQ development team

8.114 image_utils.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028
00029 #include "ofiq_lib.h"
00030 #include "Session.h"
00031 #include <opencv2/imgcodecs.hpp>
00032 #include <opencv2/imgproc.hpp>
00033
00037 namespace OFIQ_LIB {
00038
00039     using ExposureRange = std::array<int, 2>;
00040
00047     double ColorConvert (double v);
00048
```



```

00057     double Cubic(double x, double k, double eps);
00058
00065     void ConvertBGRToCIELAB(const cv::Mat& bgrImage, double& a, double& b);
00066
00074     cv::Mat GetLuminanceImageFromBGR(const cv::Mat& bgrImage );
00075
00085     void CalculateReferencePoints(const OFIQ::FaceLandmarks& landmarks,
00086                                 OFIQ::LandmarkPoint& leftEyeCenter,
00087                                 OFIQ::LandmarkPoint& rightEyeCenter,
00088                                 double& interEyeDistance,
00089                                 double& eyeMouthDistance);
00090
00103     void CalculateRegionOfInterest(cv::Rect& leftRegionOfInterest,
00104                                    cv::Rect& rightRegionOfInterest,
00105                                    const OFIQ::LandmarkPoint& leftEyeCenter,
00106                                    const OFIQ::LandmarkPoint& rightEyeCenter,
00107                                    const double interEyeDistance, const double eyeMouthDistance);
00108
00116     void GetNormalizedHistogram(const cv::Mat& luminanceImage, const cv::Mat& maskImage, cv::MatI_f&
00117                                histogram);
00117
00131     double CalculateExposure(const Session& session, const ExposureRange& exposureRange);
00132
00144     double ComputeBrightnessAspect(
00145         const cv::Mat& luminanceImage, const cv::Mat& maskImage, const ExposureRange& exposureRange);
00146 }

```

8.115 NeuronalNetworkContainer.h File Reference

Provides a neural network container structure.

```

#include "detectors.h"
#include "landmarks.h"
#include "segmentations.h"
#include "poseEstimators.h"

```

Classes

- struct [OFIQ_LIB::NeuronalNetworkContainer](#)
Neural network container for OFIQ's preprocessing steps.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

8.115.1 Detailed Description

Provides a neural network container structure.

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Author

OFIQ development team

8.116 NeuronalNetworkContainer.h

[Go to the documentation of this file.](#)

```

00001
00027 #pragma once
00028 #include "detectors.h"
00029 #include "landmarks.h"
00030 #include "segmentations.h"
00031 #include "poseEstimators.h"
00032
00036 namespace OFIQ_LIB
00037 {
00041     struct NeuronalNetworkContainer
00042     {
00055         NeuronalNetworkContainer(
00056             std::shared_ptr<FaceDetectorInterface> faceDetector,
00057             std::shared_ptr<FaceLandmarkExtractorInterface> landmarkExtractor,
00058             std::shared_ptr<SegmentationExtractorInterface> segmentationExtractor,
00059             std::shared_ptr<PoseEstimatorInterface> poseEstimator,
00060             std::shared_ptr<SegmentationExtractorInterface> faceOcclusionExtractor
00061         )
00062             : faceDetector{faceDetector},
00063               landmarkExtractor{landmarkExtractor},
00064               segmentationExtractor{segmentationExtractor},
00065               poseEstimator{poseEstimator},
00066               faceOcclusionExtractor{faceOcclusionExtractor}
00067         {
00068         }
00069
00073         std::shared_ptr<FaceDetectorInterface> faceDetector;
00074
00078         std::shared_ptr<FaceLandmarkExtractorInterface> landmarkExtractor;
00079
00085         std::shared_ptr<SegmentationExtractorInterface> segmentationExtractor;
00086
00090         std::shared_ptr<PoseEstimatorInterface> poseEstimator;
00091
00097         std::shared_ptr<SegmentationExtractorInterface> faceOcclusionExtractor;
00098     };
00099 }

```

8.117 OFIQError.h File Reference

Provides a class for the error handling within the QFIQ.

```

#include "ofiq_lib.h"
#include <string_view>

```

Classes

- class [OFIQ_LIB::OFIQError](#)
Implementation of a custom exception.

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

8.117.1 Detailed Description

Provides a class for the error handling within the QFIQ.

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Author

OFIQ development team

8.118 OFIQError.h

[Go to the documentation of this file.](#)

```
00001
00027 #pragma once
00028 #include "ofiq_lib.h"
00029 #include <string_view>
00030
00034 namespace OFIQ_LIB
00035 {
00040     class OFIQError : public std::exception
00041     {
00042     public:
00049         OFIQError(OFIQ::ReturnCode returnCode, std::string_view message);
00050
00056         const char* what() const noexcept override { return m_extendedMessage.c_str(); }
00057
00063         OFIQ::ReturnCode whatCode() const noexcept { return m_returnCode; }
00064
00065     private:
00070         OFIQ::ReturnCode m_returnCode;
00071
00076         std::string m_message;
00077
00082         std::string m_extendedMessage;
00083     };
00084 }
```

8.119 Session.h File Reference

The session class is the data container used to distribute the image and additional data, including the data computed during the pre-processing.

```
#include "ofiq_lib.h"  
#include <opencv2/opencv.hpp>
```

Classes

- class [OFIQ_LIB::Session](#)

Namespaces

- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

Typedefs

- using [OFIQ_LIB::EulerAngle](#) = std::array<double, 3>

8.119.1 Detailed Description

The session class is the data container used to distribute the image and additional data, including the data computed during the pre-processing.

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Author

OFIQ development team

8.120 Session.h

[Go to the documentation of this file.](#)

```

00001
00028 #pragma once
00029
00030 #include "ofiq_lib.h"
00031 #include <opencv2/opencv.hpp>
00032
00036 namespace OFIQ_LIB
00037 {
00041     struct NeuronalNetworkContainer;
00042
00043     using EulerAngle = std::array<double, 3>;
00044
00051     class Session
00052     {
00053     public:
00054
00061         Session(const OFIQ::Image& image, OFIQ::FaceImageQualityAssessment& assessment)
00062             : m_image{image},
00063               m_assessment{assessment},
00064               m_id{GenerateId()}
00065         {
00066         }
00067
00072         const OFIQ::Image& image() const { return m_image; }
00073
00078         OFIQ::FaceImageQualityAssessment& assessment() { return m_assessment; }
00079
00085         const std::string& Id() const { return m_id; }
00086
00087         // use the session object as data container
00088
00094         void setDetectedFaces(const std::vector<OFIQ::BoundingBox>& i_boundingBoxes);
00095
00101         std::vector<OFIQ::BoundingBox> getDetectedFaces() const;
00102
00108         void setPose(const EulerAngle& i_pose);
00109
00115         EulerAngle getPose() const;
00116
00122         void setLandmarks(const OFIQ::FaceLandmarks& i_landmarks);
00123
00129         OFIQ::FaceLandmarks getLandmarks() const;
00130
00131
00137         void setAlignedFaceLandmarks(const OFIQ::FaceLandmarks& i_landmarks);
00138
00144         OFIQ::FaceLandmarks getAlignedFaceLandmarks() const;
00145
00151         void setAlignedFaceTransformationMatrix(const cv::Mat & i_transformationMatrix);
00152
00153
00159         cv::Mat getAlignedFaceTransformationMatrix() const;
00160
00167         void setAlignedFace(const cv::Mat & i_alignedFace);
00168
00174         cv::Mat getAlignedFace() const;
00175
00182         void setAlignedFaceLuminance(const cv::Mat& i_alignedFaceLuminance);
00189         cv::Mat getAlignedFaceLuminance() const;
00190
00196         void setAlignedFaceLandmarkedRegion(const cv::Mat & i_alignedFaceRegion);
00197
00203         cv::Mat getAlignedFaceLandmarkedRegion() const;
00204
00210         void setFaceParsingImage(const cv::Mat& i_parsingImage);
00211
00217         cv::Mat getFaceParsingImage() const;
00218
00224         void setFaceOcclusionSegmentationImage(const cv::Mat& i_segmentationImage);
00225
00231         cv::Mat getFaceOcclusionSegmentationImage() const;
00232
00233     private:
00238         const OFIQ::Image& m_image;
00239
00244         OFIQ::FaceImageQualityAssessment& m_assessment;
00249         std::vector<OFIQ::BoundingBox> m_detectedFaces;
00250
00255         EulerAngle m_pose;
00256
00261         OFIQ::FaceLandmarks m_landmarks;

```

```

00262
00267     OFIQ::FaceLandmarks m_alignedFaceLandmarks;
00268
00273     cv::Mat m_alignedFaceTransformationMatrix;
00274
00279     cv::Mat m_alignedFace;
00280
00285     cv::Mat m_alignedFaceLuminance;
00290     cv::Mat m_alignedFaceLandmarkedRegion;
00291
00296     cv::Mat m_faceParsingImage;
00297
00302     cv::Mat m_faceOcclusionSegmentationImage;
00303
00309     std::string GenerateId() const;
00310
00315     std::string m_id;
00316 };
00317 }

```

8.121 utils.h File Reference

Helper functions used by several classes.

```
#include "ofiq_lib.h"
```

Classes

- struct [Point2f](#)
Representation of a point with floating point arithmetics.
- struct [OFIQ_LIB::Point2i](#)
Representation of a point with integer arithmetics.

Namespaces

- namespace [cv](#)
OpenCV's namespace.
- namespace [OFIQ_LIB](#)
Namespace for OFIQ implementations.

Functions

- void [OFIQ_LIB::makeSquareBoundingBoxWithPadding](#) (const [OFIQ::BoundingBox](#) &i_bb, const cv::Mat &i_input_image, cv::Mat &o_output_image, [OFIQ::BoundingBox](#) &o_bb, [Point2i](#) &o_translation_vector)
Some computations, especially neural networks, need a squarred image as input. This funtion consumes a boundig box and an input image. The greater parameter of width or height is used to define the side length of the new squarred bounding box. The face will be centered in the bounding box. Padding is added if needed. The squarred bounding box is used generate a new cropped image, the o_output_image. Required translations are described by the translation vector o_translation_vector.
- [OFIQ::BoundingBox](#) [OFIQ_LIB::makeSquareBoundingBox](#) (const [OFIQ::BoundingBox](#) &i_bb)
This function converts a non-squarred bounding box into an squarred one. The side length is defined by the greater one of height or width.
- size_t [OFIQ_LIB::findLargestBoundingBox](#) (const std::vector< [OFIQ::BoundingBox](#) > &faceRects)
This function returns the position of the largest bounding box (largest in terms of area) from a vector of bounding boxes.
- [OFIQ::BoundingBox](#) [OFIQ_LIB::findLandmarksBoundingBox](#) (const [OFIQ::Landmarks](#) &landmarks)

Find the bounding box that encloses the given landmarks.

- `cv::Mat OFIQ_LIB::copyToCvImage` (const `OFIQ::Image` &sourceImage, bool asGrayImage=false)

Convert images in OFIQ::Image format into the OpenCV cv::Mat format. The image can be converted from color to gray scale by setting the parameter asGrayImage to true.

- `cv::Mat OFIQ_LIB::alignImage` (const `OFIQ::Image` &faceImage, const `OFIQ::FaceLandmarks` &faceLandmarks, `OFIQ::FaceLandmarks` &alignedFaceLandmarks, cv::Mat &transformationMatrix)

This function transforms a face image so that the position of the eyes, nose and mouth are roughly at a pre-defined position. Face alignment is the translation, rotation and scaling of the image to do this.

- void `OFIQ_LIB::calculateEyeCenter` (const `OFIQ::FaceLandmarks` &faceLandmarks, `Point2f` &leftEyeCenter, `Point2f` &rightEyeCenter)

Based on face landmarks the center of the left and right eye are computed.

- `OFIQ::Image OFIQ_LIB::MakeGreyImage` (uint16_t width, uint16_t height)

This function generates a gray scaled image with the resolution passed by the call.

- float `OFIQ_LIB::tmetric` (const `OFIQ::FaceLandmarks` &faceLandmarks)

Based on the provided landmarks this function computes the distance between the point between the eyes and the chin.

8.121.1 Detailed Description

Helper functions used by several classes.

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Author

OFIQ development team

8.122 utils.h

[Go to the documentation of this file.](#)

```

00001
00027 #ifndef OFIQ_LIB_UTILS_H
00028 #define OFIQ_LIB_UTILS_H
00029
00030 #include "ofiq_lib.h"
00031
00035 namespace cv
00036 {
00040     class Mat;
00041 }
00042
00047 struct Point2f
00048 {
00049     float x;
00050     float y;
00051 };
00052
00053
00054
00055 namespace OFIQ_LIB
00056 {
00061     struct Point2i
00062     {
00063         int x;
00064         int y;
00065     };
00066
00081     void makeSquareBoundingBoxWithPadding(
00082         const OFIQ::BoundingBox& i_bb,
00083         const cv::Mat& i_input_image,
00084         cv::Mat& o_output_image,
00085         OFIQ::BoundingBox& o_bb,
00086         Point2i & o_translation_vector
00087     );
00088
00095     OFIQ::BoundingBox makeSquareBoundingBox(
00096         const OFIQ::BoundingBox& i_bb);
00097
00104     size_t findLargestBoundingBox(
00105         const std::vector<OFIQ::BoundingBox>& faceRects);
00106
00113     OFIQ::BoundingBox findLandmarksBoundingBox(
00114         const OFIQ::Landmarks& landmarks);
00115
00123     cv::Mat copyToCvImage(const OFIQ::Image& sourceImage, bool asGrayImage = false);
00124
00134     cv::Mat alignImage(
00135         const OFIQ::Image& faceImage,
00136         const OFIQ::FaceLandmarks& faceLandmarks,
00137         OFIQ::FaceLandmarks& alignedFaceLandmarks,
00138         cv::Mat& transformationMatrix);
00139
00147     void calculateEyeCenter(
00148         const OFIQ::FaceLandmarks& faceLandmarks,
00149         Point2f& leftEyeCenter,
00150         Point2f& rightEyeCenter);
00151
00159     OFIQ::Image MakeGreyImage(uint16_t width, uint16_t height);
00160
00167     float tmetric(const OFIQ::FaceLandmarks& faceLandmarks);
00168 }
00169
00170 #endif

```


Chapter 9

Examples

9.1 C:/Path/To/eesfm_cpp/OFIQlib/include/ofiq_libc.h

Retrieves the version information of the OFIQ library.

Retrieves the version information of the OFIQ library. This function provides the version number of the OFIQ library in the form of three integers: major, minor, and patch. These values follow semantic versioning principles, where:

- Major version increases with incompatible API changes.
- Minor version increases with backward-compatible feature additions.
- Patch version increases with backward-compatible bug fixes.

Parameters

out	<i>major</i>	The major version number will be stored here.
out	<i>minor</i>	The minor version number will be stored here.
out	<i>patch</i>	The patch or build number will be stored here.

```
int major, minor, patch; OFIQGetVersion(major, minor, patch); printf("Using OFIQ library version %d.%d.%d\n", major, minor, patch);
```

```
#ifndef OFIQ_LIBC_H
#define OFIQ_LIBC_H

#ifdef _WIN32
#   ifdef OFIQ_EXPORTS
#       define OFIQ_C_EXPORT __declspec(dllexport)
#   else
#       define OFIQ_C_EXPORT __declspec(dllimport)
#   endif
#else
#   define OFIQ_C_EXPORT
#endif

#ifdef __cplusplus
extern "C"
{
#endif

#include <ofiq_structs_c.h>

    OFIQ_C_EXPORT ReturnStatus
```

```

    OFIQInitialize(OFIQHandle* handle, const char* configDir, const char* configFileName);

OFIQ_C_EXPORT ReturnStatus OFIQCleanup(OFIQHandle* handle);

OFIQ_C_EXPORT ReturnStatus OFIQReadImage(const char* filename, Image& image);

OFIQ_C_EXPORT ReturnStatus
    OFIQReadImageFromByteArray(const unsigned char* imageData, size_t dataLength, Image& image);

OFIQ_C_EXPORT ReturnStatus OFIQFreeImage(Image* image);

OFIQ_C_EXPORT ReturnStatus
    OFIQScalarQuality(OFIQHandle handle, const Image& face, double& quality);

OFIQ_C_EXPORT ReturnStatus OFIQVectorQuality(
    OFIQHandle handle, const Image& faceImage, FaceImageQualityAssessment& qualityAssessment);

OFIQ_C_EXPORT ReturnStatus OFIQVectorQualityExt(
    OFIQHandle handle,
    const Image& faceImage,
    FaceImageQualityAssessment& qualityAssessment,
    PreprocessingResult& preprocessingResult);

OFIQ_C_EXPORT ReturnStatus OFIQFreeFaceImageQualityAssessment(FaceImageQualityAssessment* qa);

OFIQ_C_EXPORT ReturnStatus OFIQFreeFaceLandmarks(FaceLandmarks* landmarks);

OFIQ_C_EXPORT ReturnStatus OFIQFreePreprocessingResult(PreprocessingResult* preprocRes);

OFIQ_C_EXPORT void OFIQGetVersion(int& major, int& minor, int& patch);

OFIQ_C_EXPORT const char* OFIQGetQualityMeasureName(QualityMeasure measure);

#ifdef __cplusplus
} // extern "C"
#endif

#endif /* OFIQ_LIBC_H */

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